CompTIA Server+

Troubleshooting Server Storage Related Issues

- **Introduction**
- **Lab Topology**
- **Exercise 1 - Common Storage Problems**
- **Exercise 2 - Tools and Techniques for Troubleshooting**
- **Review**

Introduction

Server+
Read Errors
Write Errors
Disk Cleanup
Defragment and Optimize Drives
System File Checker
Event Viewer

Welcome to the **Troubleshooting Server Storage Related Issues** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Just like all other aspects of computing, storage is fallible. In this module, some common issues that happen with storage and some tools that can be used to help troubleshoot and diagnose issues will be discussed.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Common Storage Problems
- Exercise 2 - Tools and Techniques for Troubleshooting

After completing this module, you should be able to:

- Know about Read/Write Errors due to Permissions

- Illustrate the Use of Disk Cleanup for Insufficient Disk Space Errors
- Use Defragment and Optimize Drives for Slow Performance of Disks
- Use Component Services for Unable to Access Logical Drive Errors
- Explore Page / Swap / Scratch File or Partition Issues
- Describe Caching Issues
- Use the Check Disk (chkdsk) Tool
- Explore the System File Checker (SFC) Tool
- Use the Event Viewer
- Know about Monitoring Tools

Exam Objectives

The following exam objectives are covered in this module:

4.3 Given a scenario, troubleshoot storage problems

- Common problems
- Causes of common problems
- Tools and techniques

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABDM01 -** (Windows Server 2022 - Domain Member Server)
- **PLABWIN10 -** (Windows 10 - Domain Member Workstation)
- **PLABALMA -** (Alma Linux 8.7 - Stand-alone Linux Workstation)
- **PLABUBUNTU -** (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Common Storage Problems

Storage problems can become costly to users. They can lead to access issues and loss of productivity and service availability.

In this exercise, common storage problems and areas that users should be looking to address them will be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Know about Read/Write Errors due to Permissions
- Illustrate the Use of Disk Cleanup for Insufficient Disk Space Errors
- Use Defragment and Optimize Drives for Slow Performance of Disks
- Use Component Services for Unable to Access Logical Drive Errors
- Explore Page / Swap / Scratch File or Partition Issues
- Describe Caching Issues

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)
- **PLABUBUNTU -** (Linux Ubuntu - Stand-alone Linux Server)

Task 1 - Read/Write Errors due to Permissions

A write error is created when a disk or storage medium's write operation fails for various reasons, including insufficient space, corrupt disks, or write protection. When a write operation fails, several types of write error messages are delivered depending on the underlying reason for the failure.

In this task, you will learn about permissions that can cause the inability to read and write to the disk.

Step 1

Connect to **PLABDC01**.

Minimize the **Server Manager** window.

Click the **File Explorer** icon on the **Taskbar**.

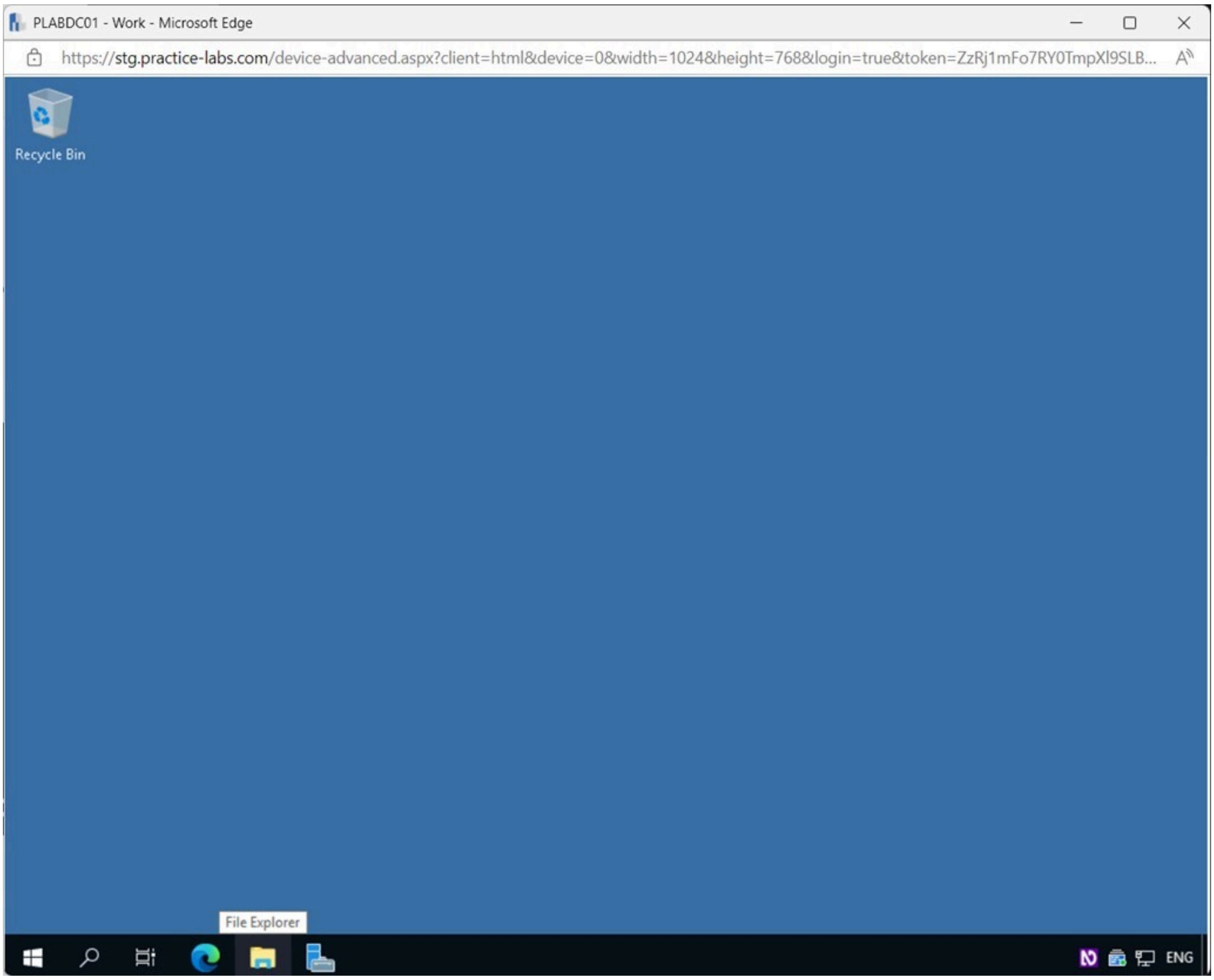

Figure 1.1 Screenshot of PLABDC01: Displaying opening File Explorer from the Taskbar.

Step 2

In the **File Explorer** window, click **Desktop** on the left pane.

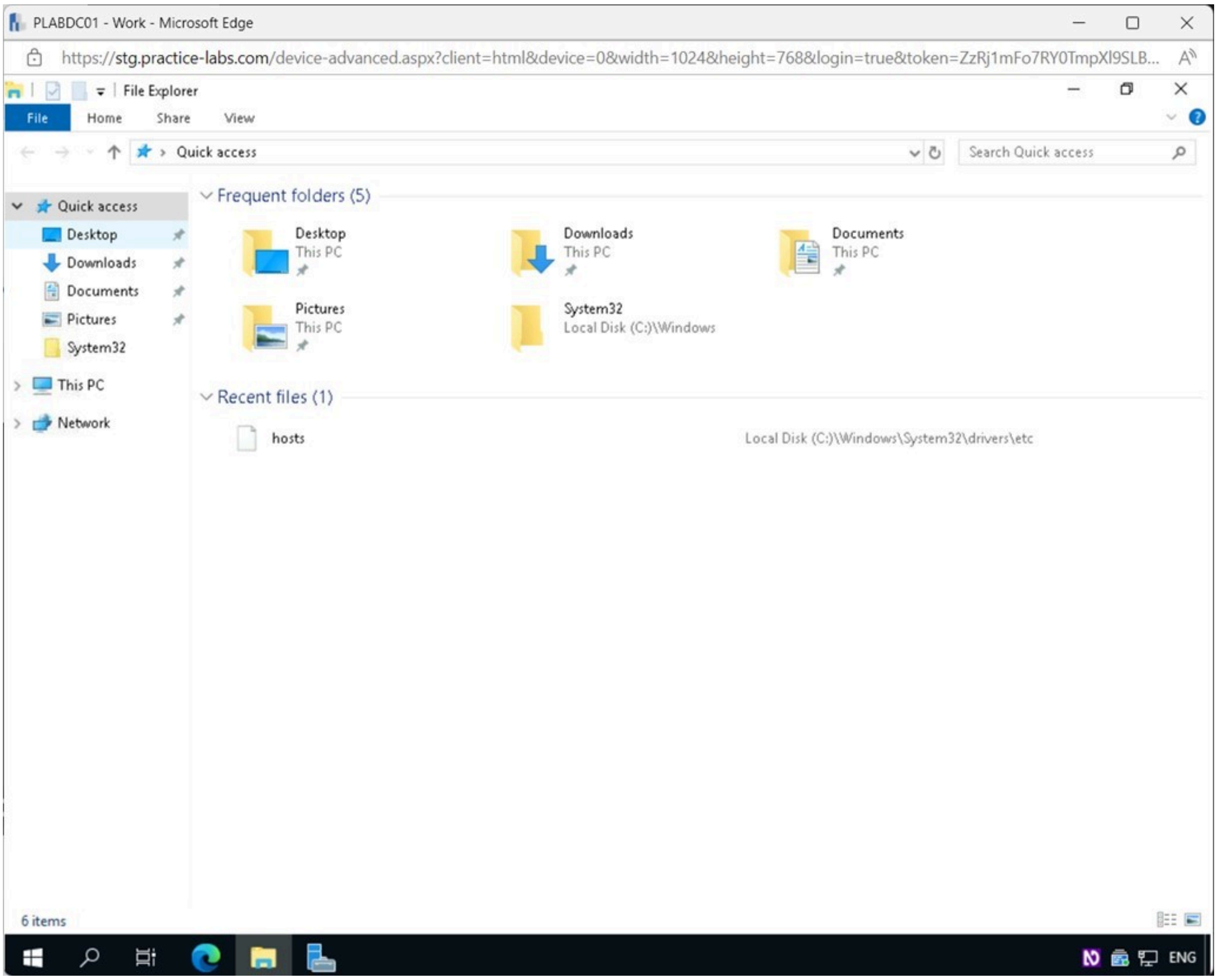

Figure 1.2 Screenshot of PLABDC01: Displaying selecting Desktop in the File Explorer window.

Step 3

In the right details pane, right-click anywhere on an empty space and select **New** > **Folder**.

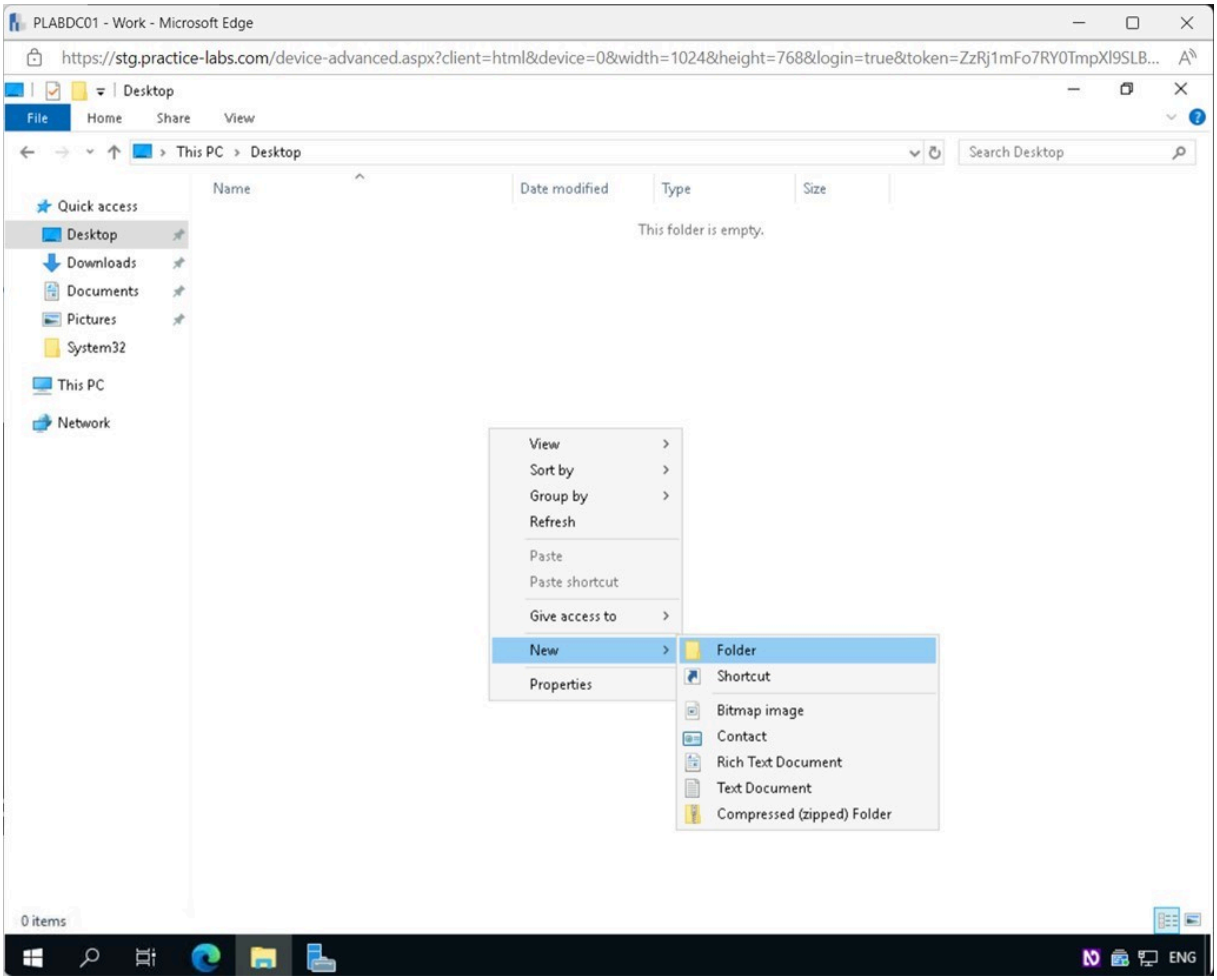

Figure 1.3 Screenshot of PLABDC01: Displaying the File Explorer - Desktop window. New > Folder menu options are selected from the right-click context menu.

Step 4

Right-click on the **New folder** and select **Properties.**

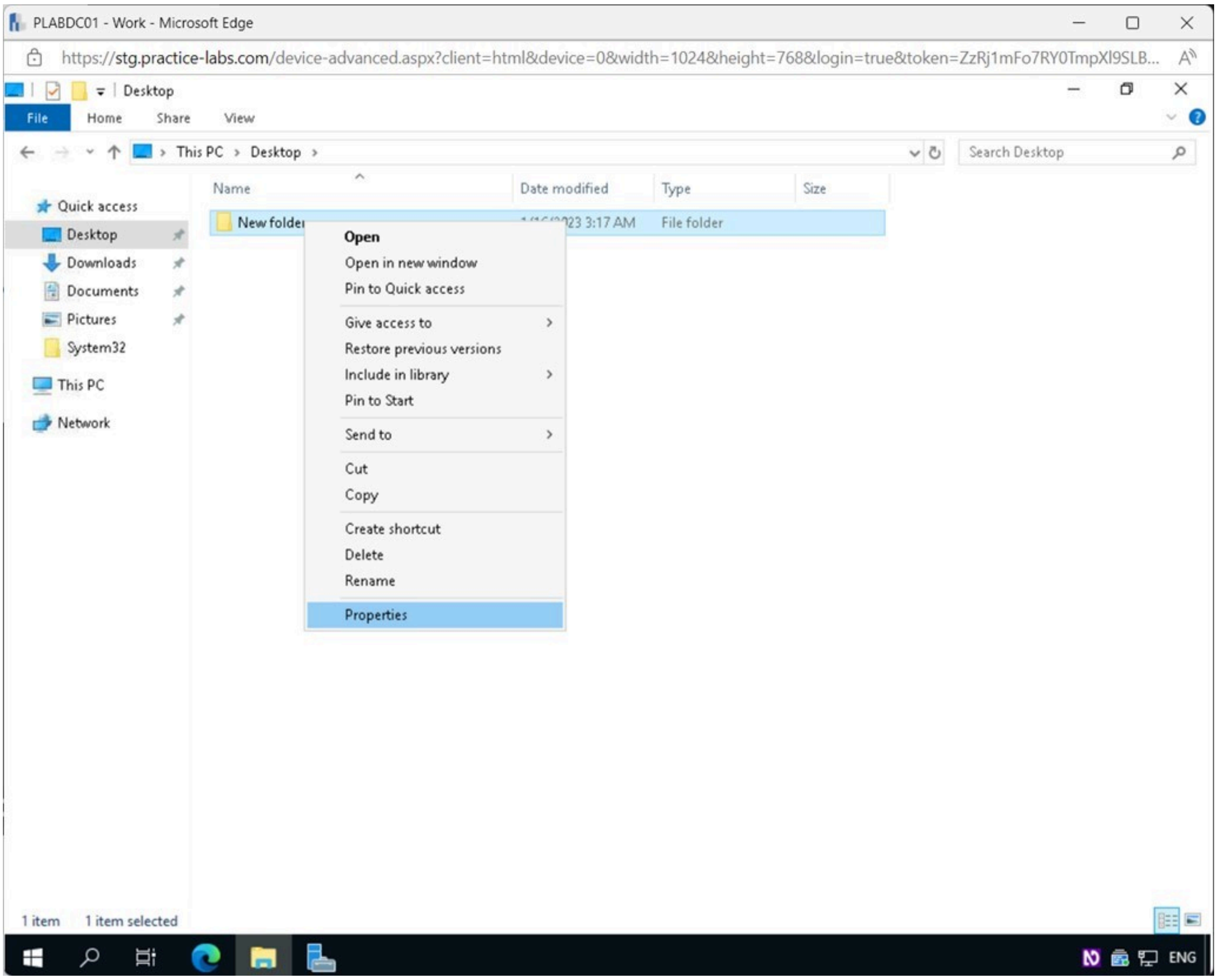

Figure 1.4 Screenshot of PLABDC01: Displaying right-clicking on the New Folder and selecting Properties in the File Explorer - Desktop window.

Step 5

In the **New folder Properties** dialog box, click the **Sharing** tab.

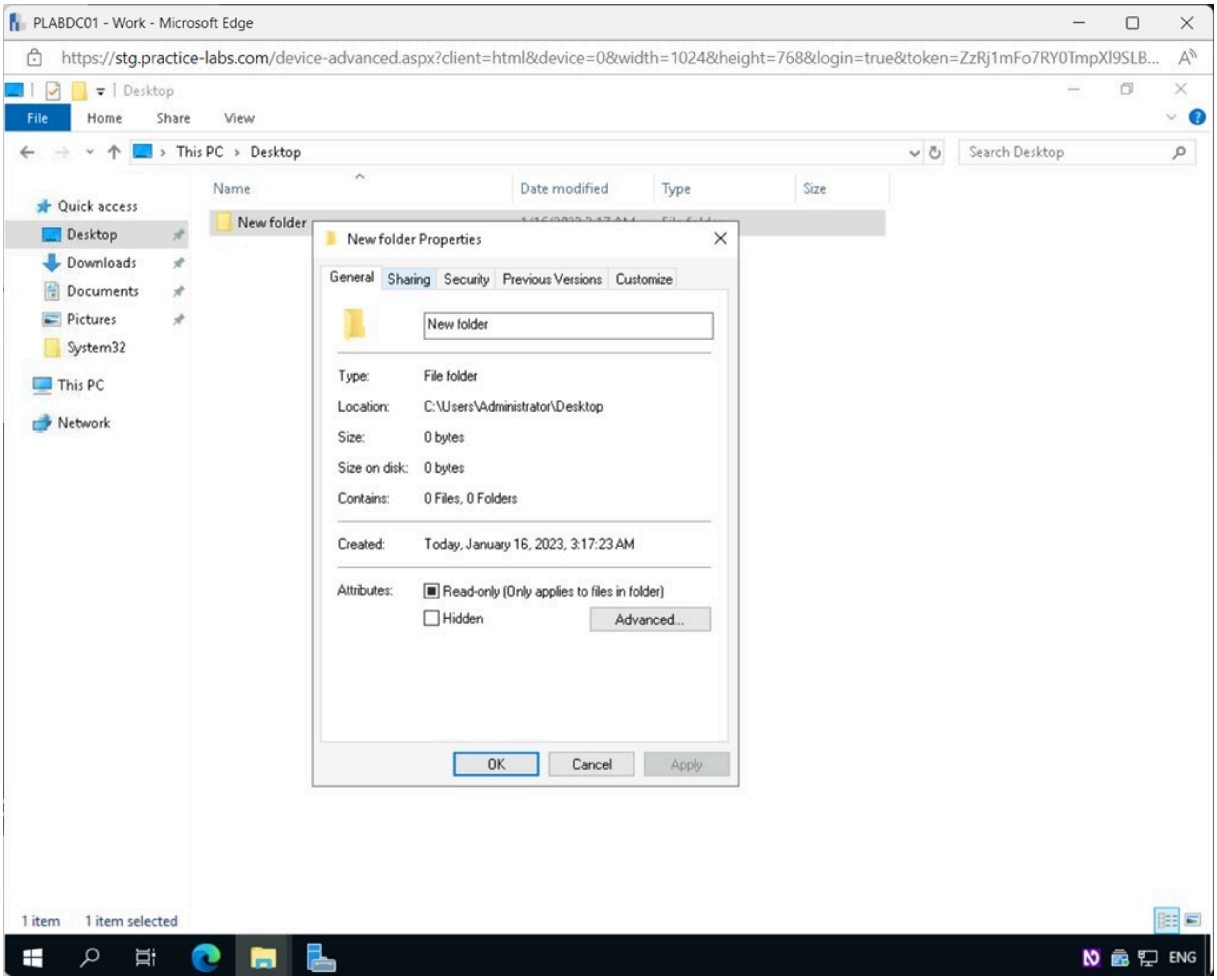

Figure 1.5 Screenshot of PLABDC01: Displaying selecting the Sharing tab in the New folder Properties dialog box.

Step 6

In the **New folder Properties - Sharing** tab, click the **Advanced Sharing** button.

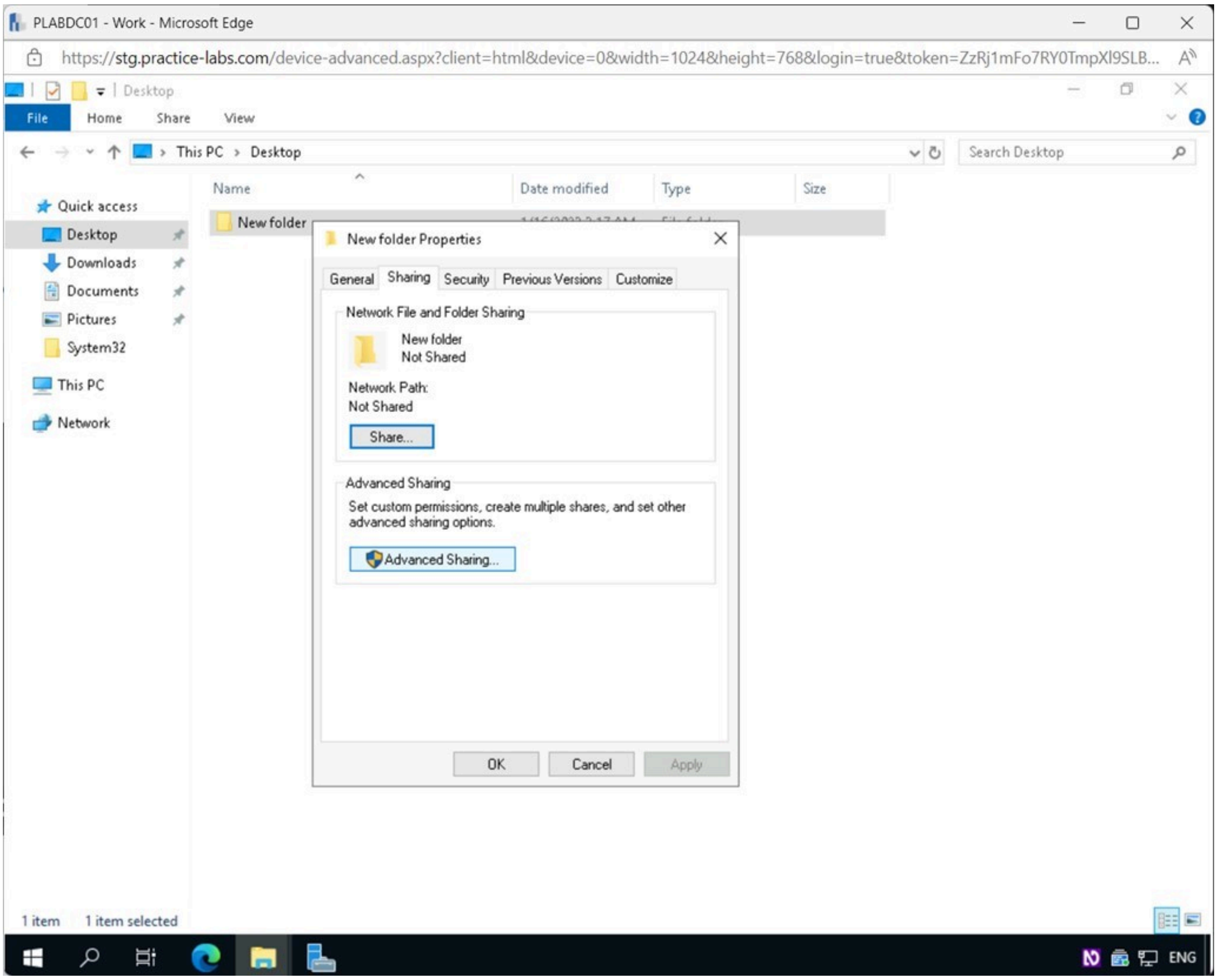

Figure 1.6 Screenshot of PLABDC01: Displaying selecting Advanced Sharing in the New folder Properties- Sharing tab.

Step 7

In the **Advanced Sharing** pop-up window, tick the **Share this folder** checkbox.

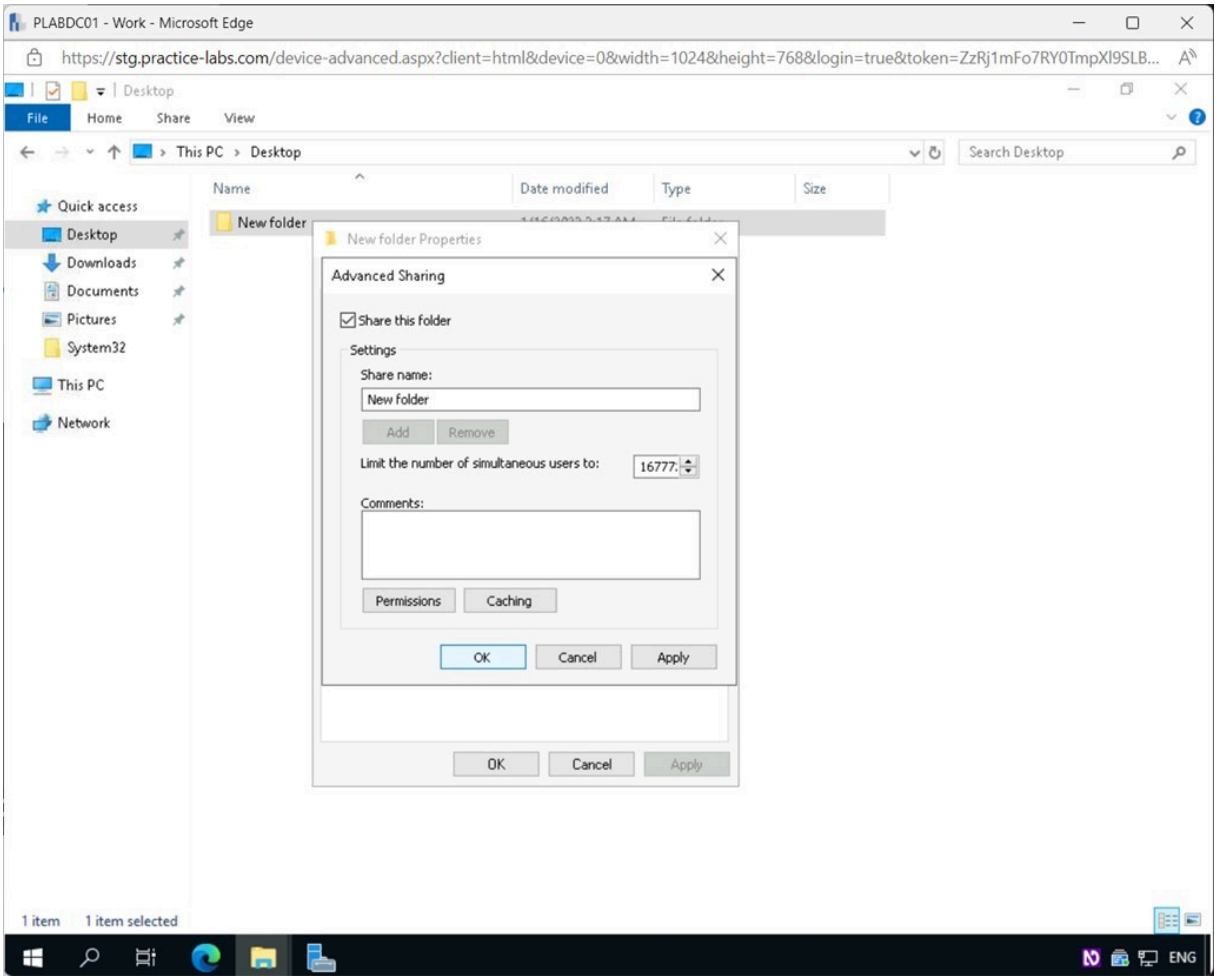

Figure 1.7 Screenshot of PLABDC01: Displaying the Advanced Sharing pop-up window showing the required settings performed and OK selected.

Step 8

Back on the **New folder Properties - Sharing** tab, select the **Security** tab.

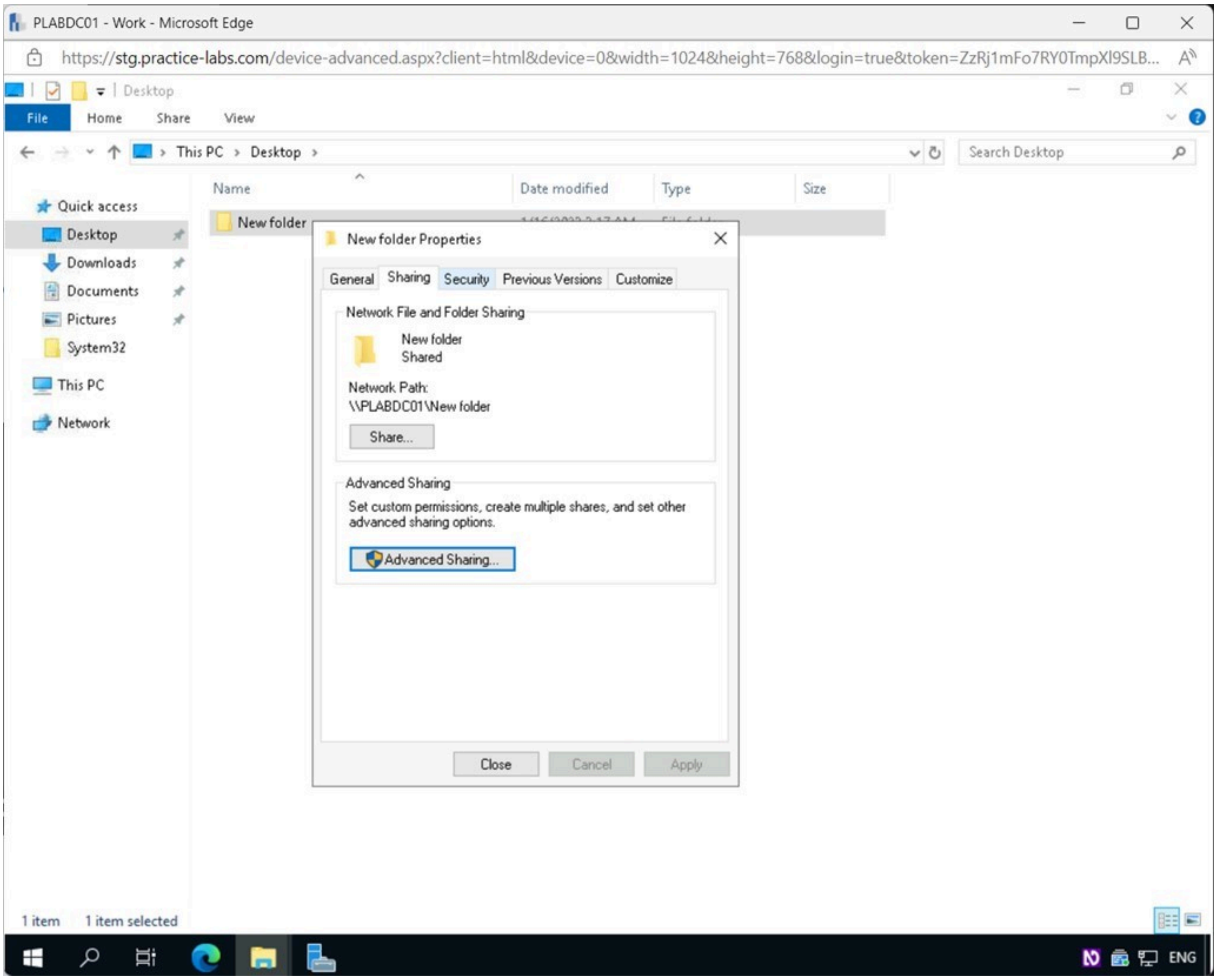

Figure 1.8 Screenshot of PLABDC01: Displaying selecting the Security tab in the New folder Properties dialog box.

Step 9

In the **New folder Properties - Security** tab, click **Edit**.

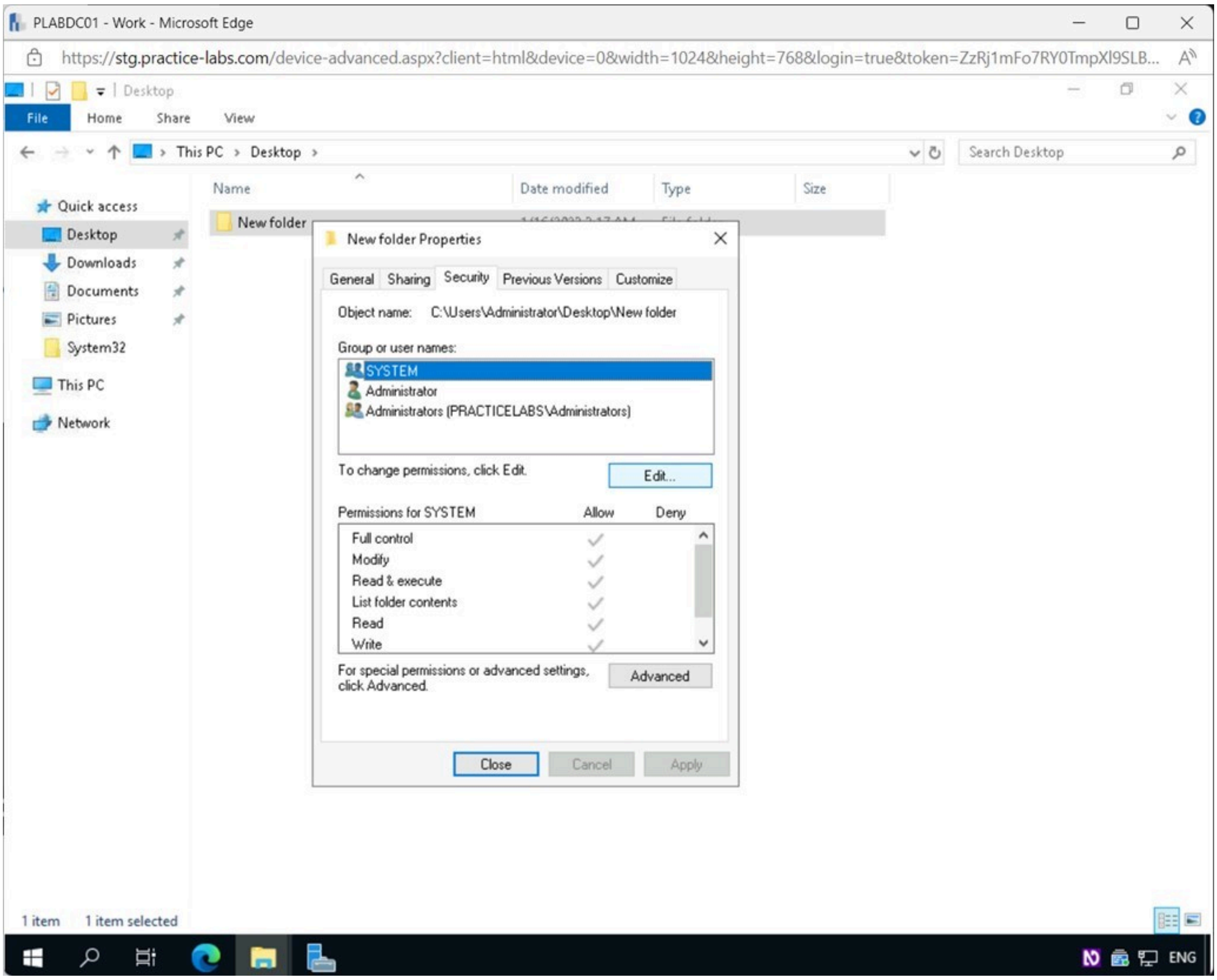

Figure 1.9 Screenshot of PLABDC01: Displaying selecting Edit in the New folder Properties- Security tab.

Step 10

In the **Permissions for New folder** pop-up window, click **Add**.

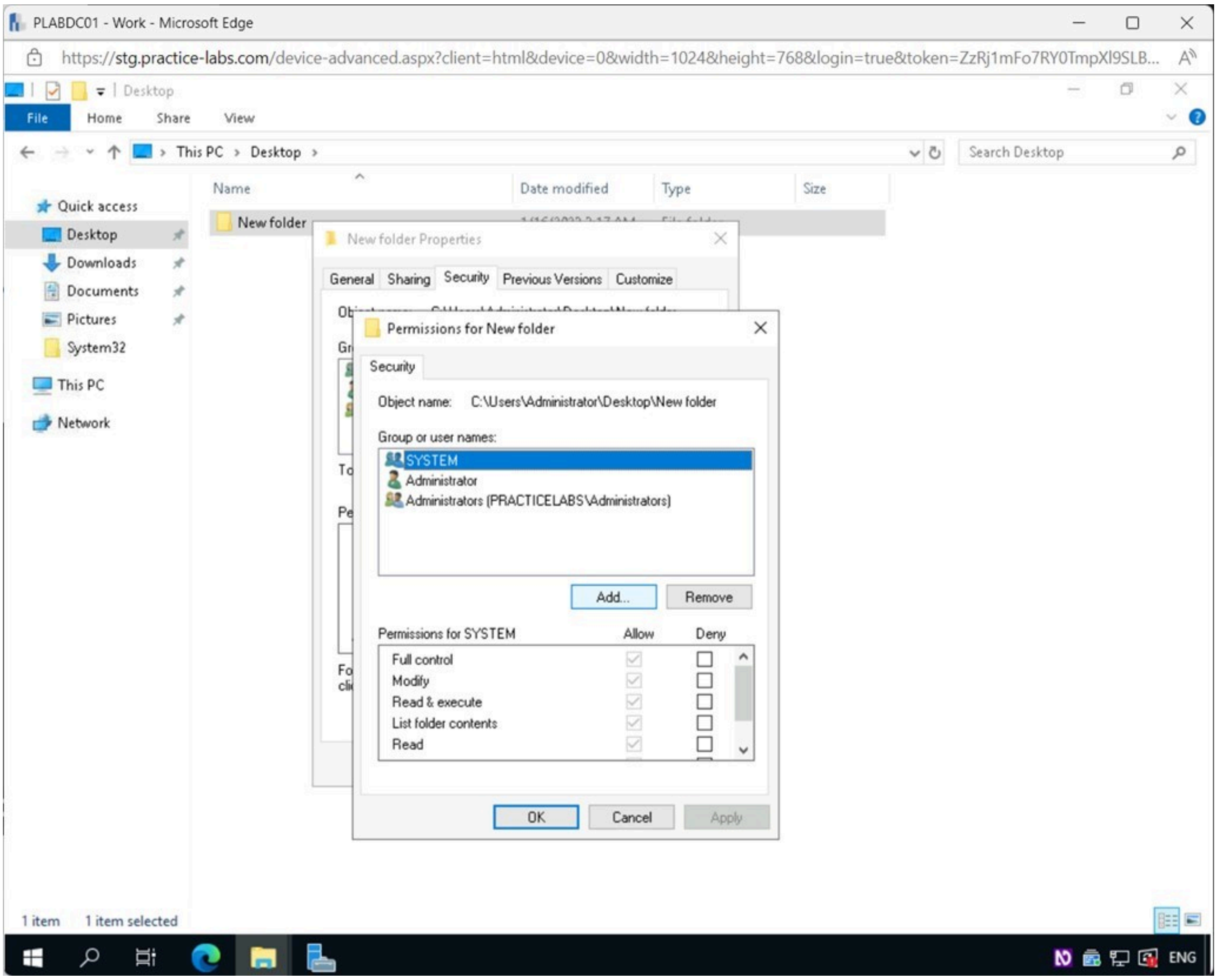

Figure 1.10 Screenshot of PLABDC01: Displaying selecting Add in the Permissions for New folder pop-up window.

Step 11

In the **Select Users, Computers, Service Accounts, or Groups** pop-up window, type the following under the **Enter the object names to select** field:

```
backup
```

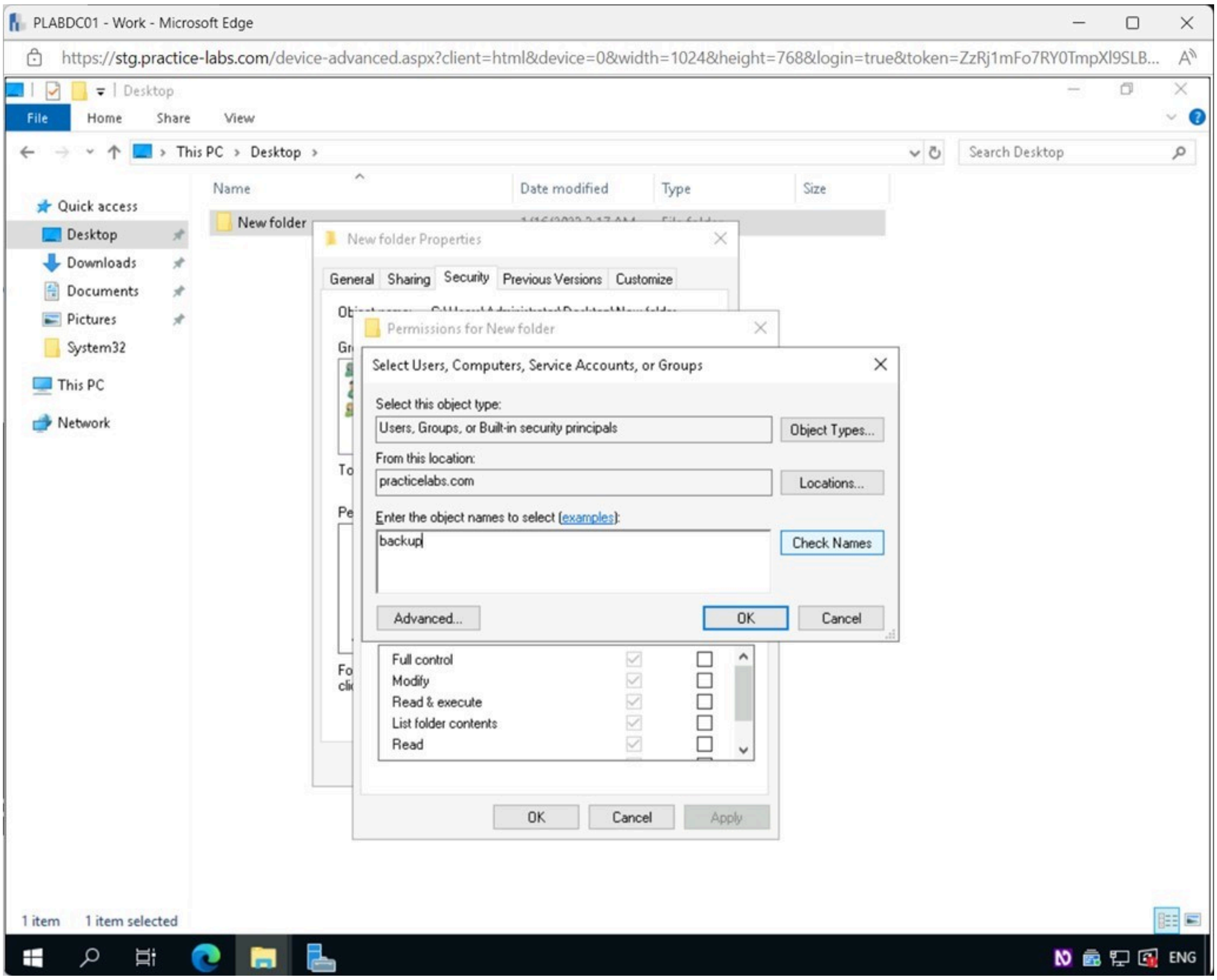

Figure 1.11 Screenshot of PLABDC01: Displaying the Select Users, Computers, Service Accounts, or Groups pop-up window, showing the required object name typed in and Check Names selected.

Step 12

Notice the **Backup Operators** group has been found.

Click **OK.**

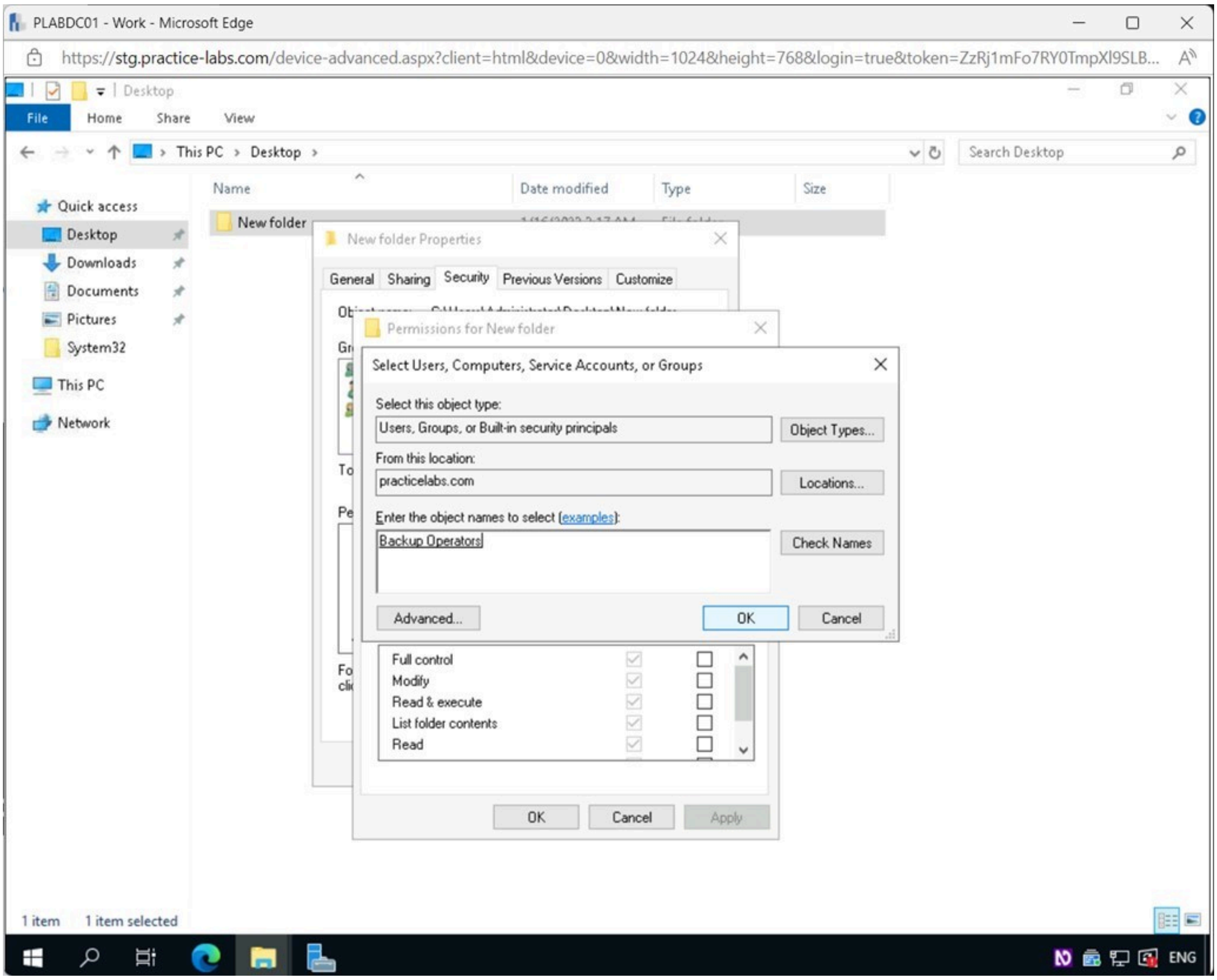

Figure 1.12 Screenshot of PLABDC01: Displaying selecting OK in the Select Users, Computers, Service Accounts, or Groups pop-up window.

Step 13

Back on the **Permissions for New folder** pop-up window, select **Backup Operators** and then select **Full Control** under the **Permissions for Backup Operations** section.

*The following **Permissions** are available:*

***Full control** - Directory contents and file contents can be viewed, edited, created, moved, and removed by users. Users with full control of a file or folder can also alter directory permissions.*
***Modify** - Users have the ability to read, write, delete, and add to files and directories, as well as alter file and directory characteristics.*
***Read & Execute** - Users are able to execute scripts and other executable files.*
***Read** - Users can view files and their attributes.*
***Write** - Users can write to and edit files.*

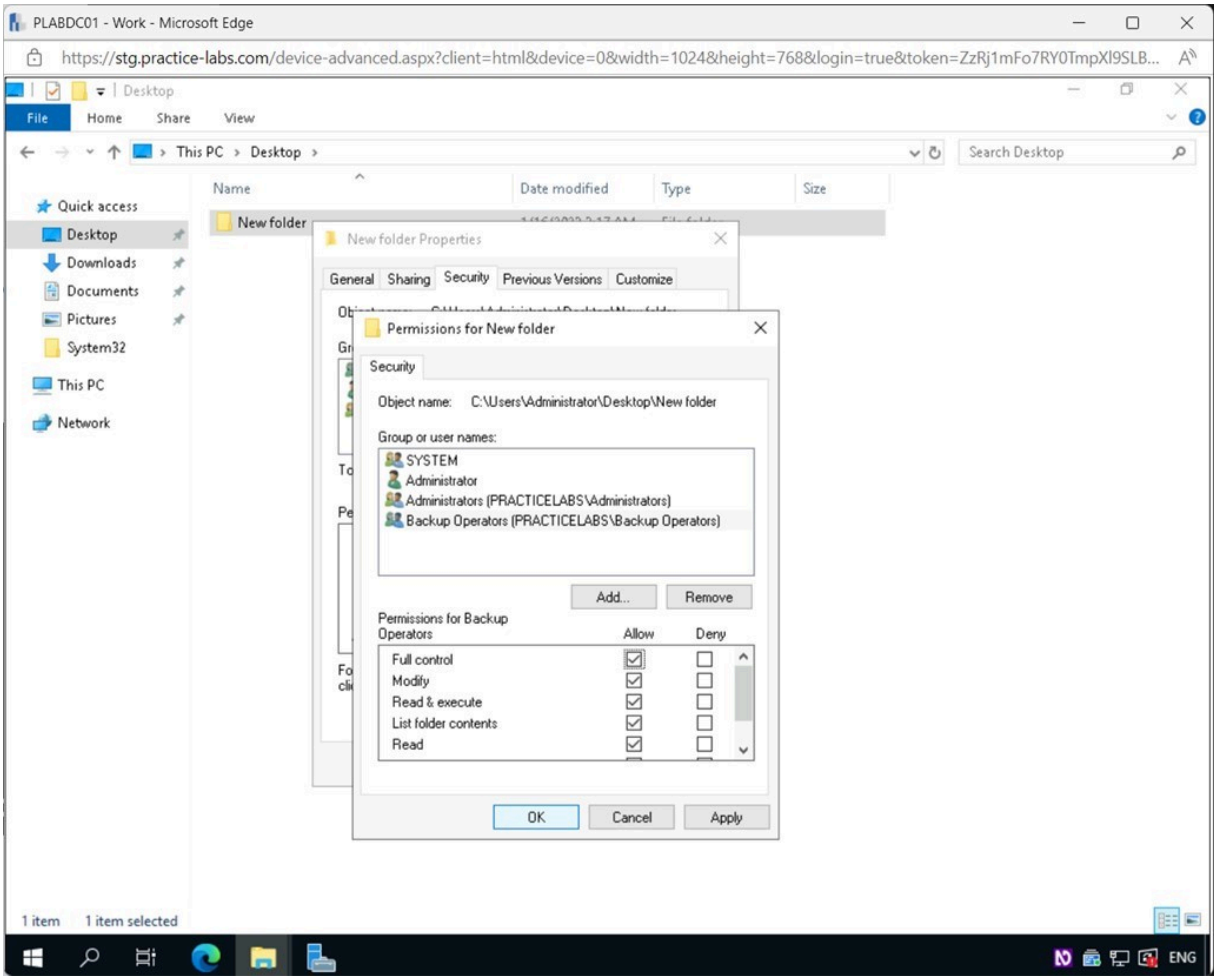

Figure 1.13 Screenshot of PLABDC01: Displaying the Permissions for New folder pop-up window, showing the required settings performed and the OK button selected.

Step 14

Back on the **New folder Properties - Security** tab, click **Advanced.**

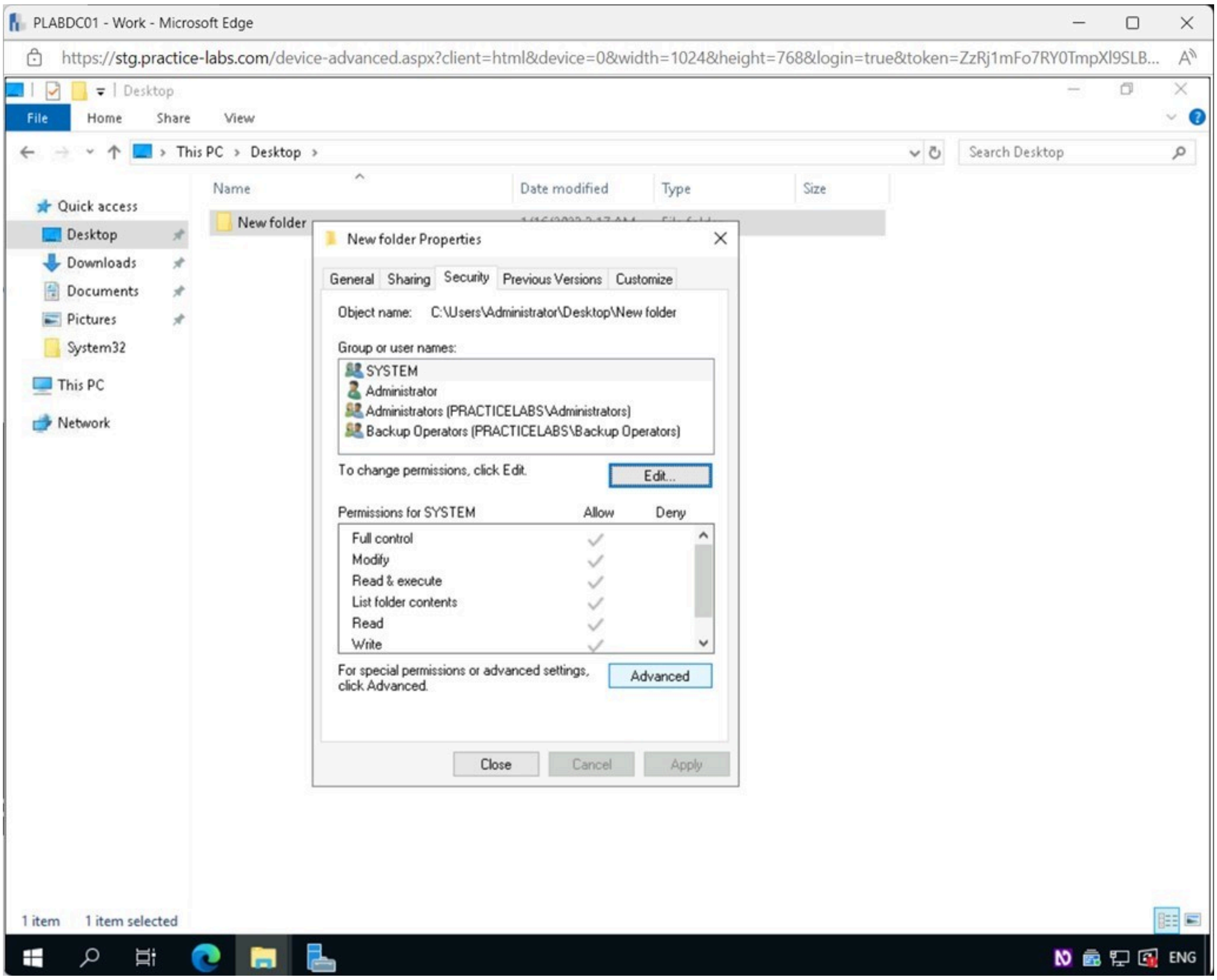

Figure 1.14 Screenshot of PLABDC01: Displaying selecting Advanced in the New folder Properties- Security tab.

Step 15

In the **Advanced Security Settings for New folder** window, select the **Effective Access** tab.

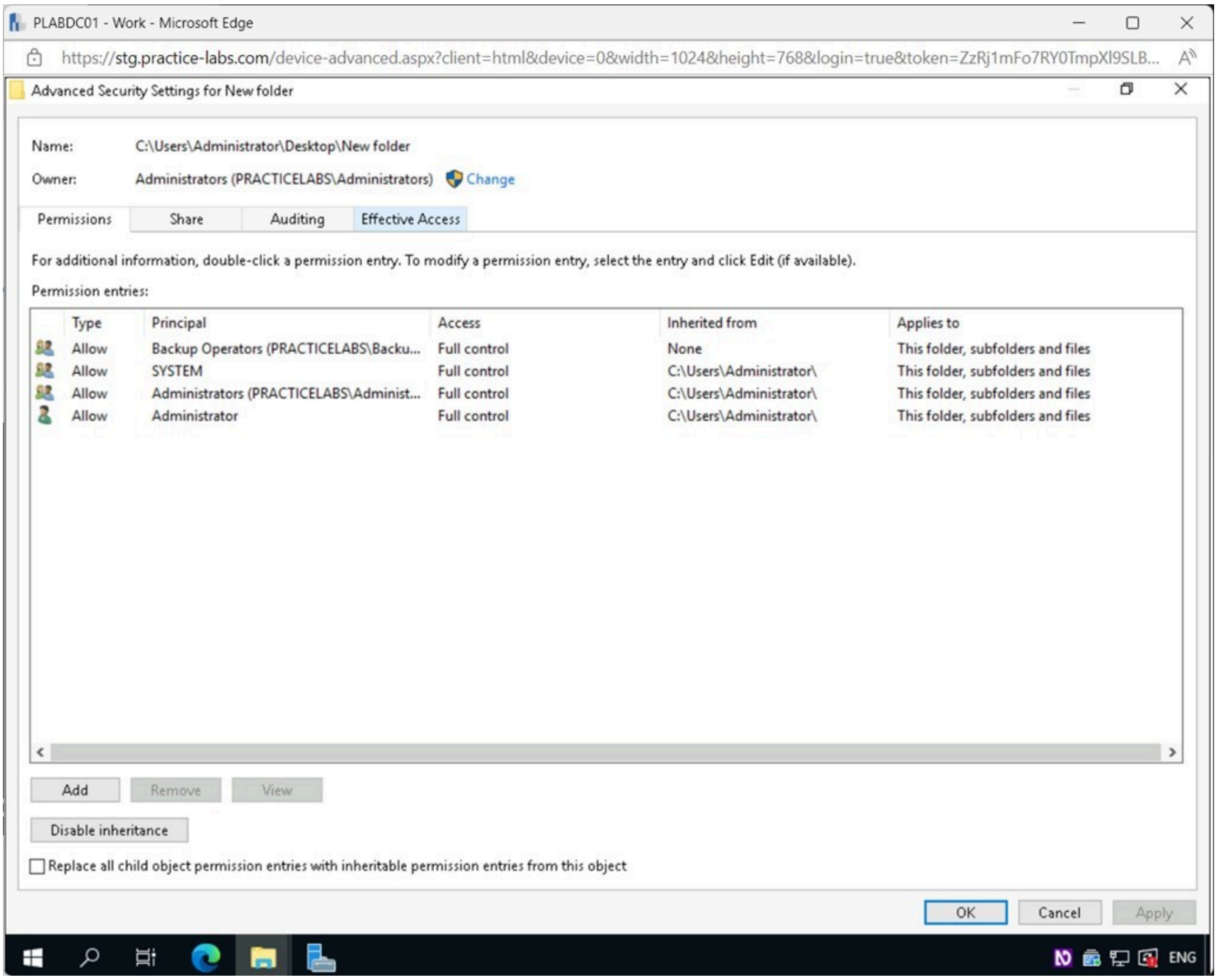

Figure 1.15 Screenshot of PLABDC01: Displaying selecting the Effective Access tab in the Advanced Security Settings for New folder window.

Step 16

From the **Advanced Security Settings for New folder - Effective Access** tab, click the **Select a user** link.

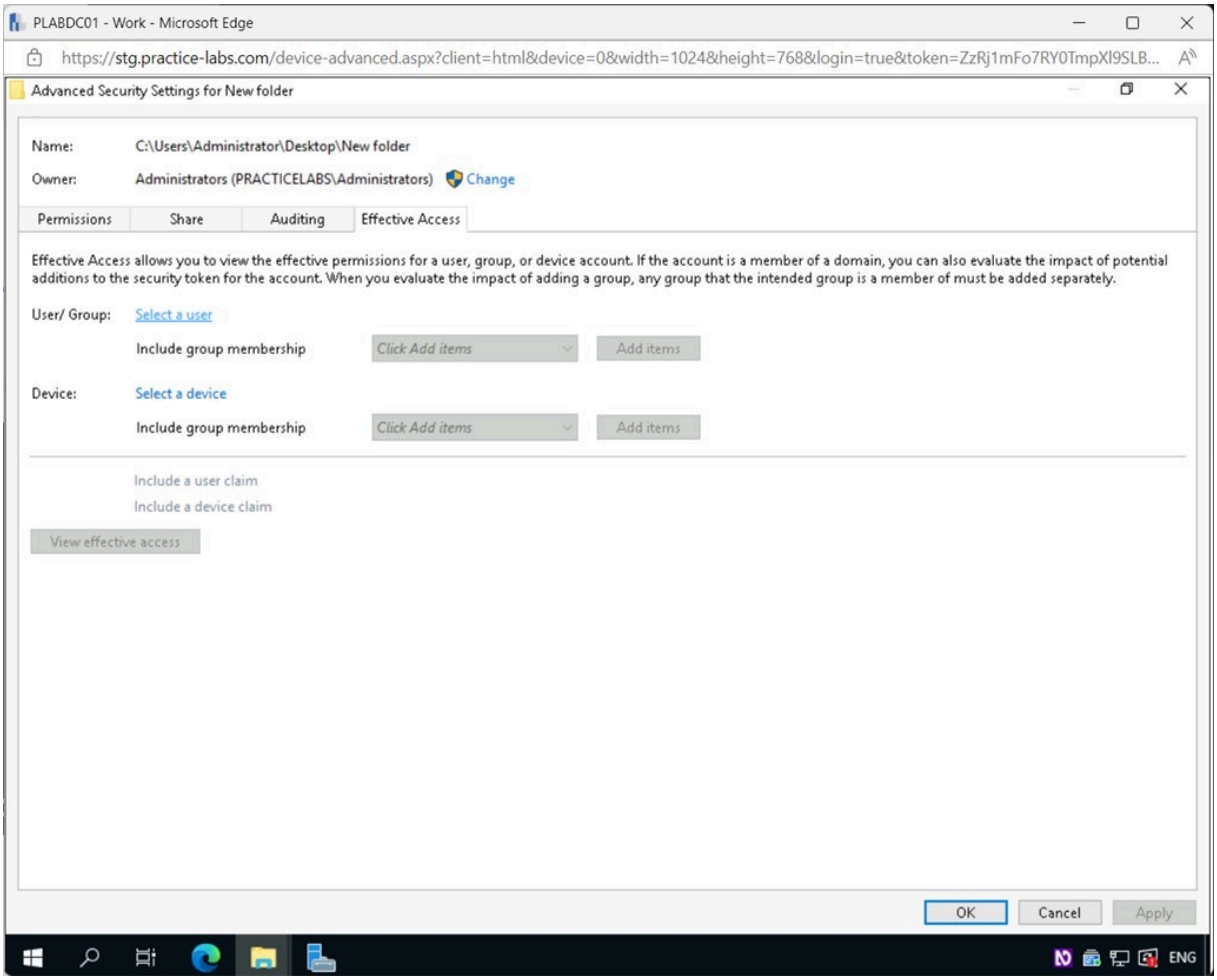

Figure 1.16 Screenshot of PLABDC01: Displaying clicking the Select a user link in the Advanced Security Settings for New folder - Effective Access tab.

Step 17

In the **Select User, Computer, Service Account, or Group** pop-up window, type the following for the **Enter the object name to select** field:

```
backup
```

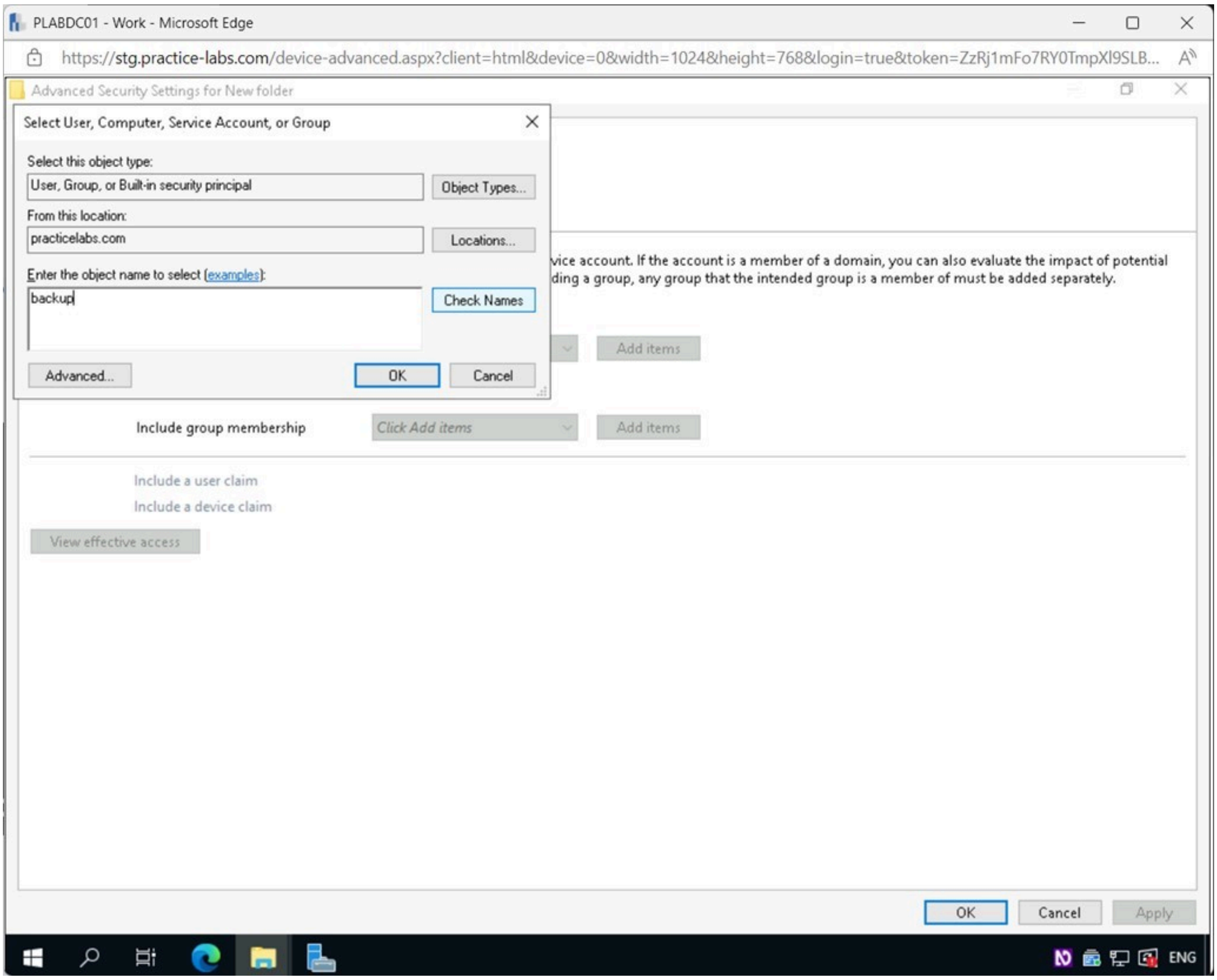

Figure 1.17 Screenshot of PLABDC01: Displaying the Select User, Computers, Service Account, or Group pop-up window, showing the required object name entered and Check Names selected.

Step 18

Notice the **Backup Operators** group has been found.

Click **OK.**

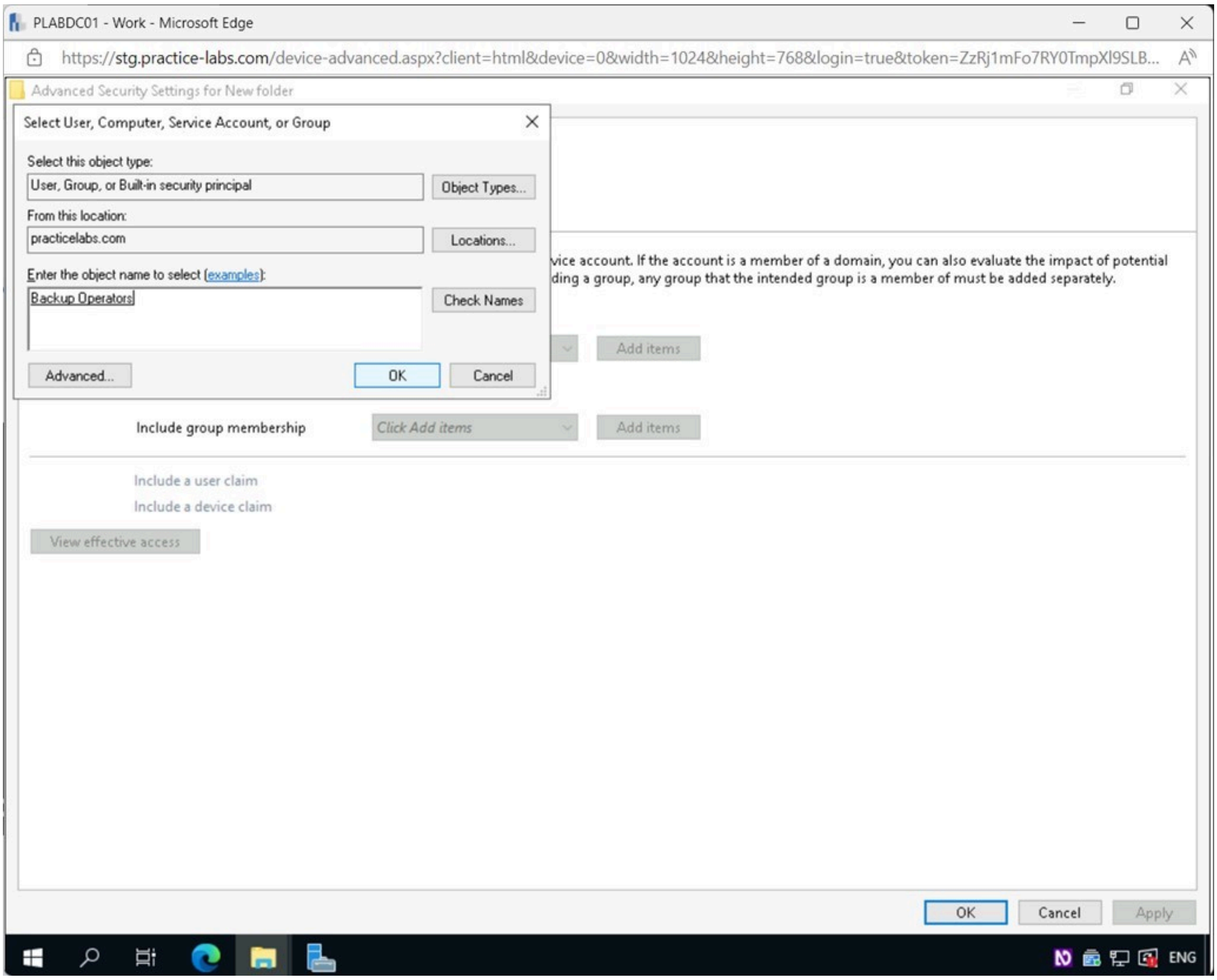

Figure 1.18 Screenshot of PLABDC01: Displaying selecting OK in the Select User, Computers, Service Accounts, or Group pop-up window.

Step 19

The **Backup Operators** group has now been added to the **Effective Access** window.

Click the **View effective access** button.

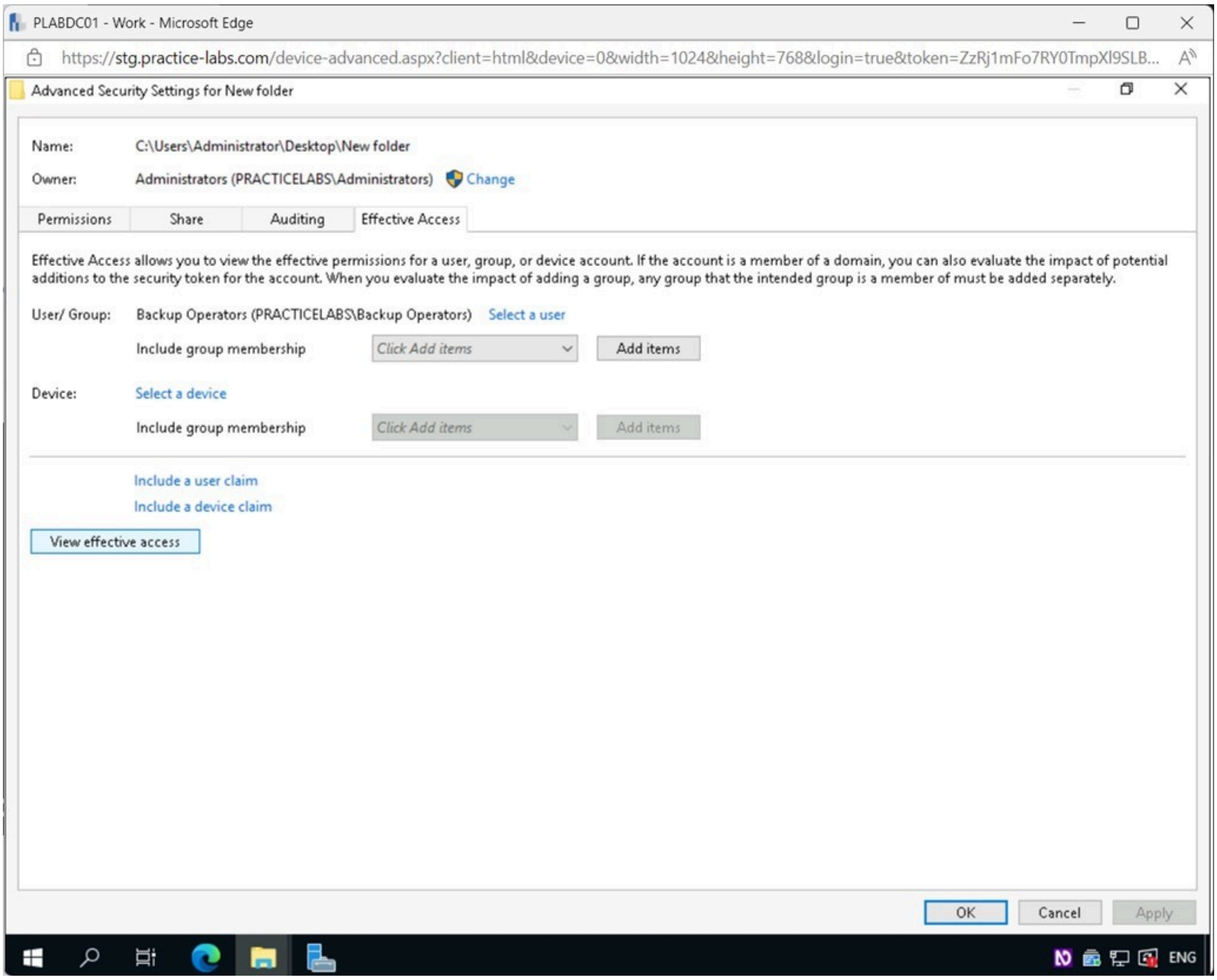

Figure 1.19 Screenshot of PLABDC01: Displaying selecting the View effective access button in the Effective Access window.

Step 20

Here, you can view the type of **Effective Access** the **Backup Operators** have for the **New folder**.

Click **OK.**

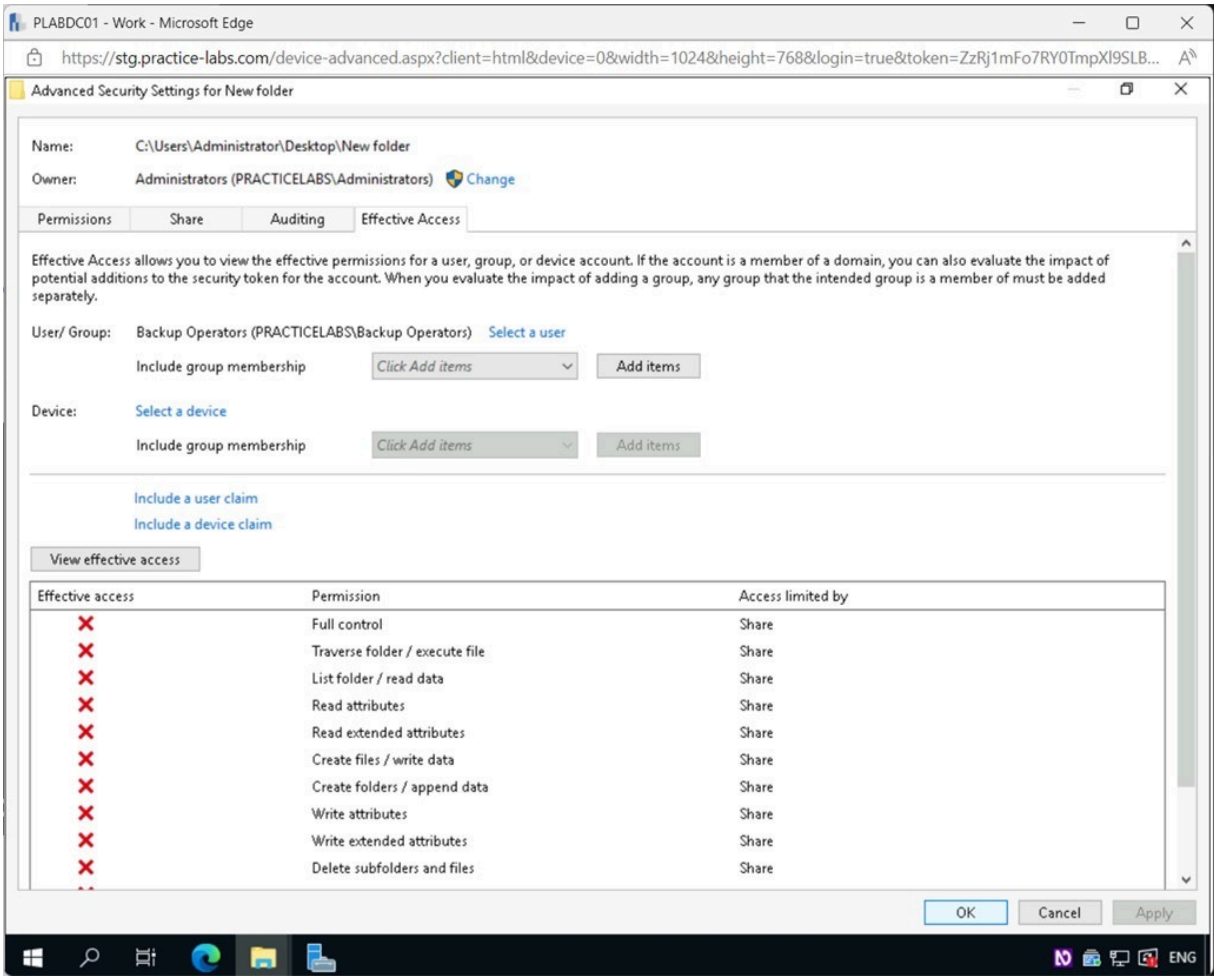

Figure 1.20 Screenshot of PLABDC01: Displaying selecting OK in the Advanced Security Settings for New folder - Effective Access tab.

Step 21

Click **Close** in the **New folder Properties** pop-up window.

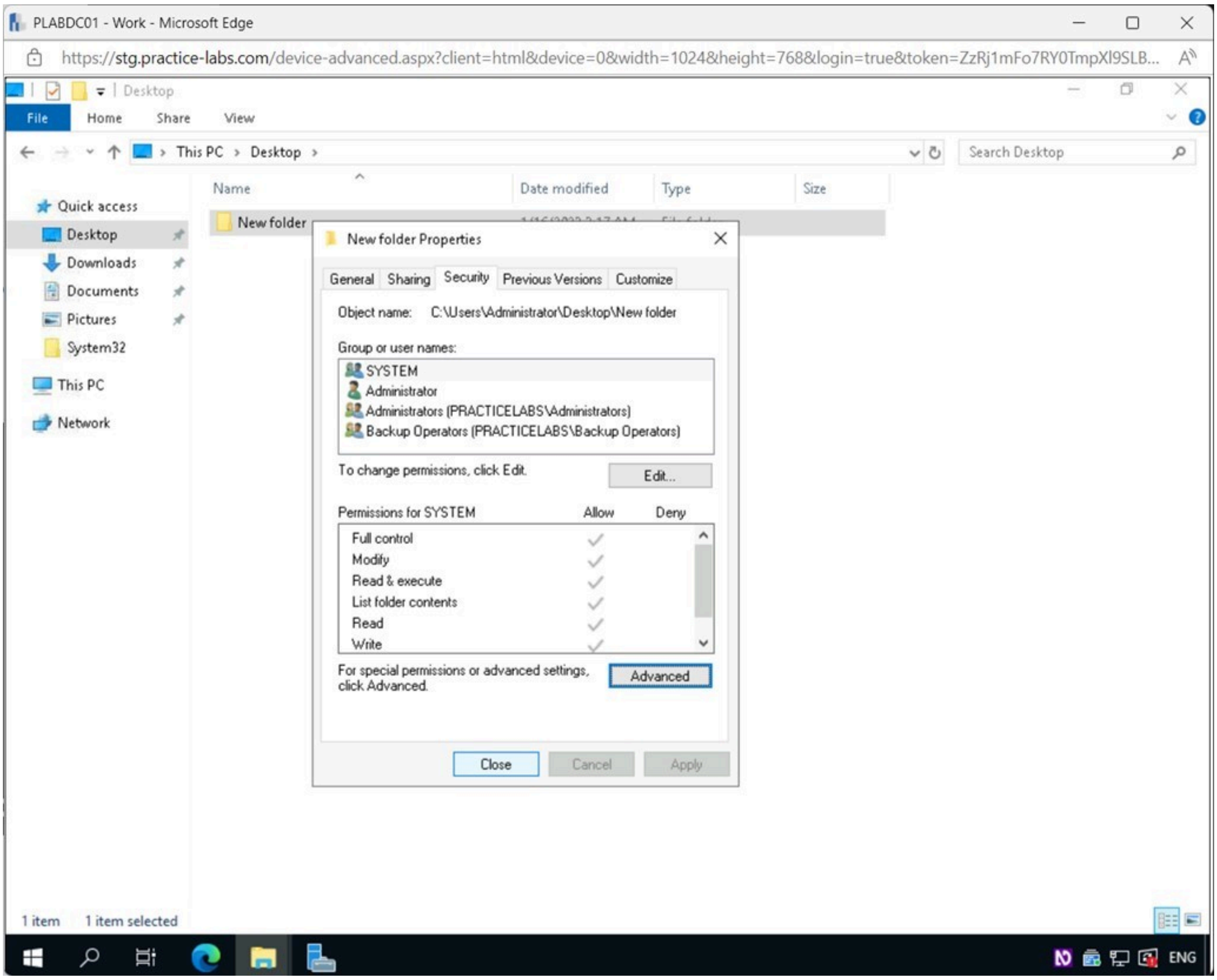

Figure 1.21 Screenshot of PLABDC01: Displaying selecting Close in the New folder Properties pop-up window.

Step 22

Close the **File Explorer - Desktop** window.

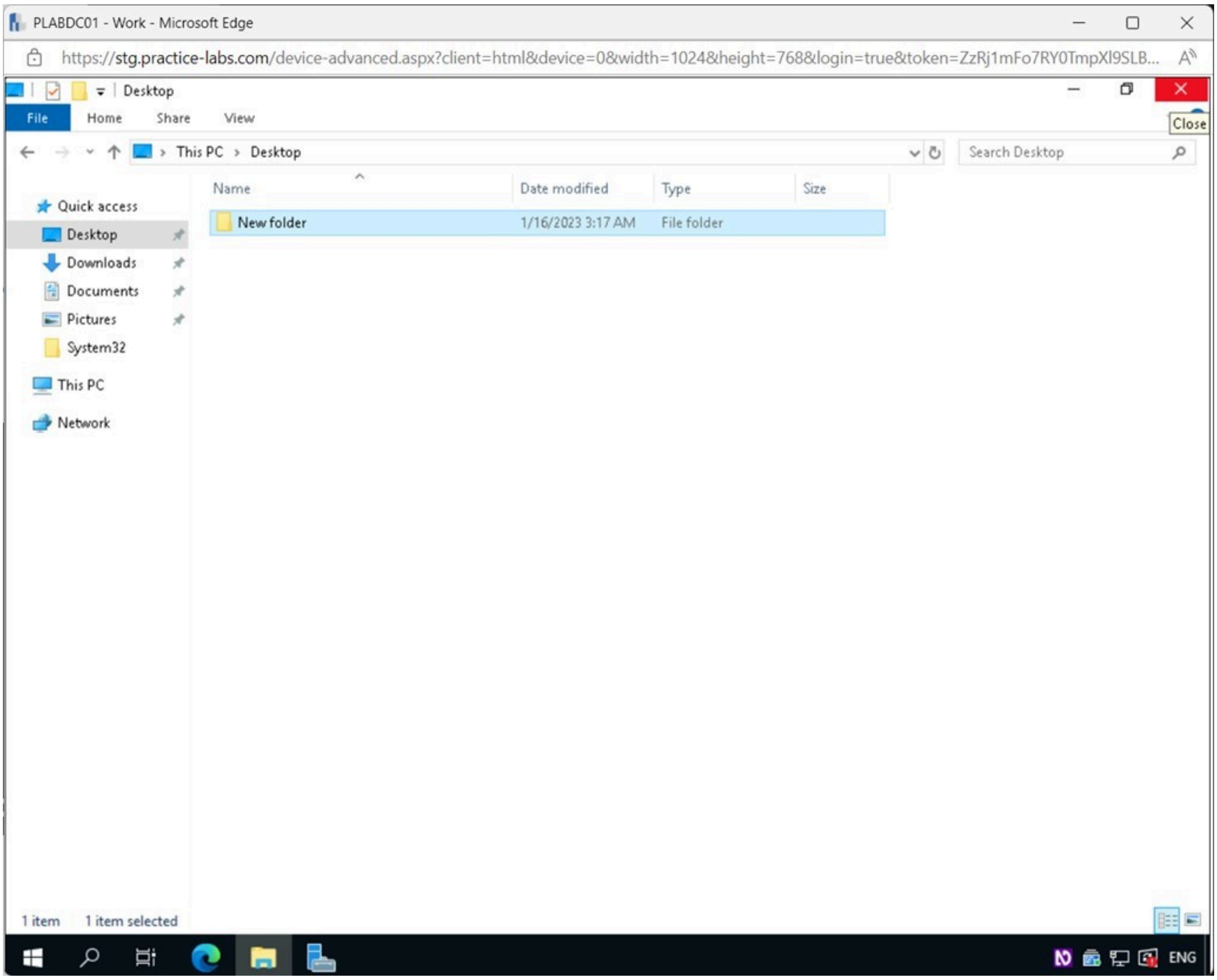

Figure 1.22 Screenshot of PLABDC01: Displaying closing the File Explorer - Desktop window.

Task 2 - Insufficient Disk Space Errors

There are some common causes for insufficient disk space errors in Windows. Inadequate free space on the drive, file corruption, or device driver issues. In any case, the solutions to such problems are typically straightforward. At times, it may be sufficient to close unnecessary windows or applications. It's important to remember that disk space is a scarce resource that may quickly add up to cost.

In this task, you will use the Disk Cleanup tool to free up disk space.

Step 1

Ensure you are connected to **PLABDC01**.

Click the Windows **Start** charm, and expand the **Windows Administrative Tools** menu.

Scroll down and select **Disk Cleanup**.

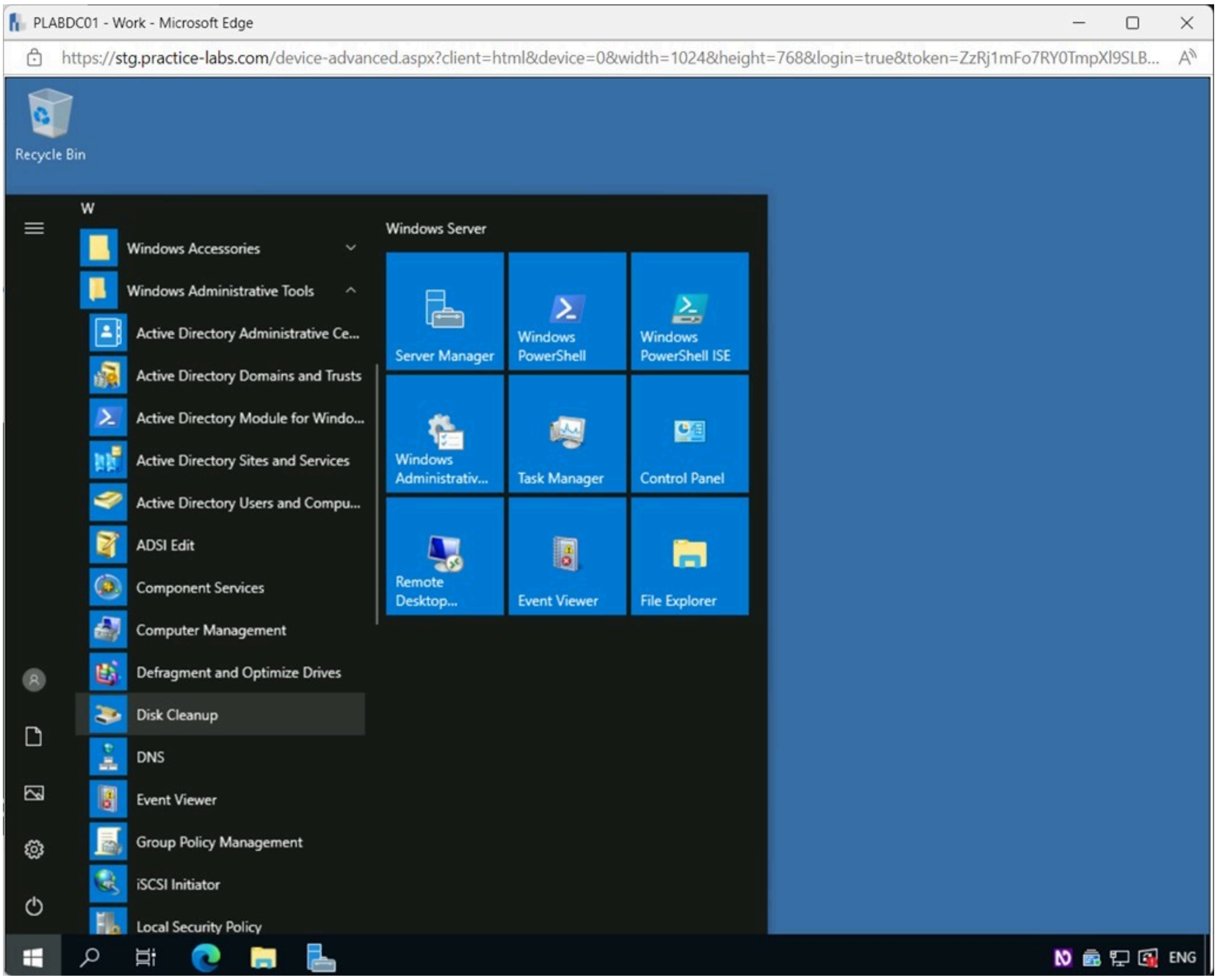

Figure 1.23 Screenshot of PLABDC01: Displaying selecting Disk Cleanup from the Windows Start menu.

Step 2

The program takes about a minute to load while it scans the device.

In the **Disk Cleanup for (C:)** window, select the **More Options** tab.

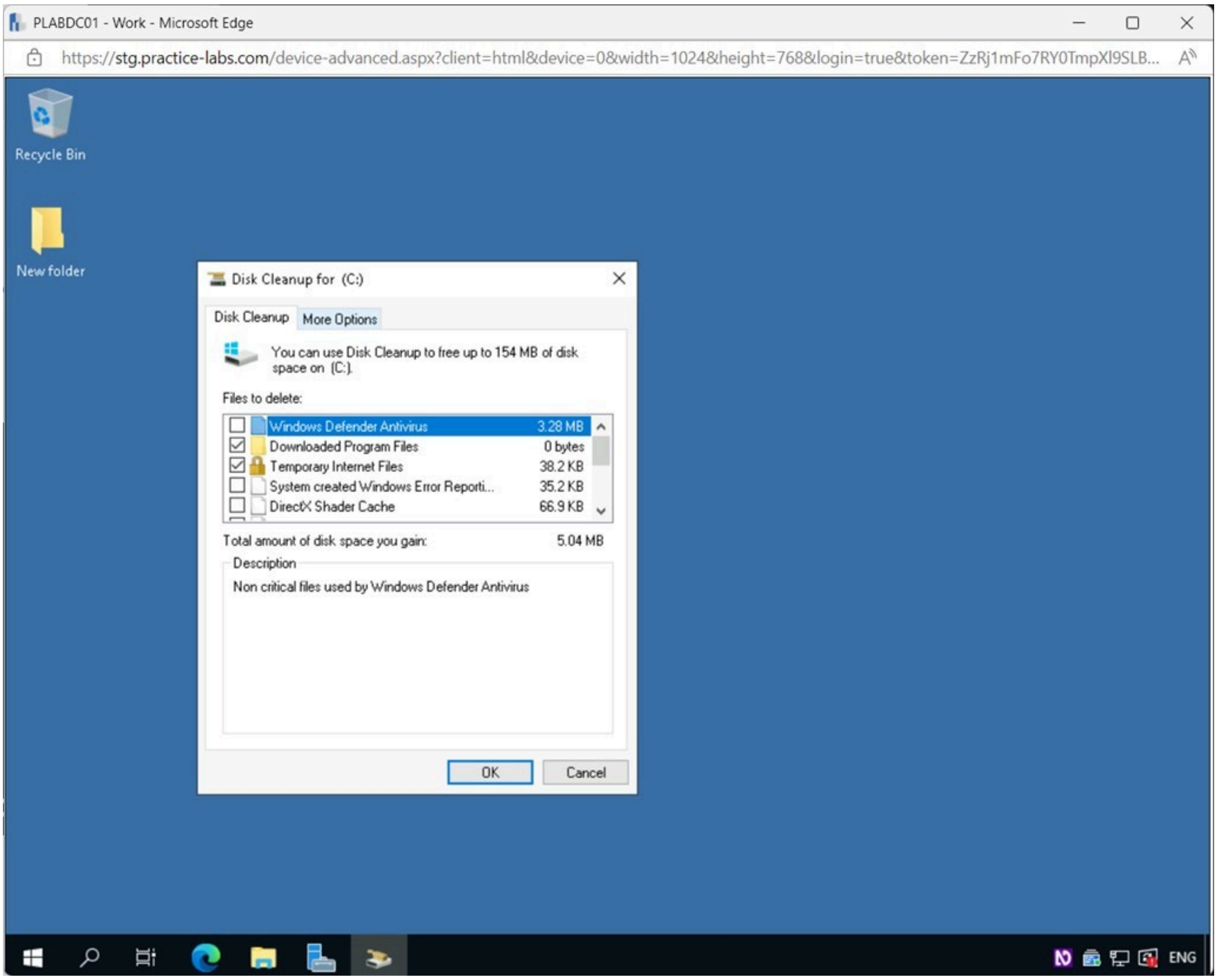

Figure 1.24 Screenshot of PLABDC01: Displaying selecting the More Options tab in the Disk Cleanup for (C:) window.

***Note: Disk Cleanup** scans the disk for temporary files, Internet cache data, and unneeded software files. It can erase any of these files quickly and easily, freeing up valuable space.*

Step 3

In the **Disk Cleanup - More Options** tab, click **Clean up**.

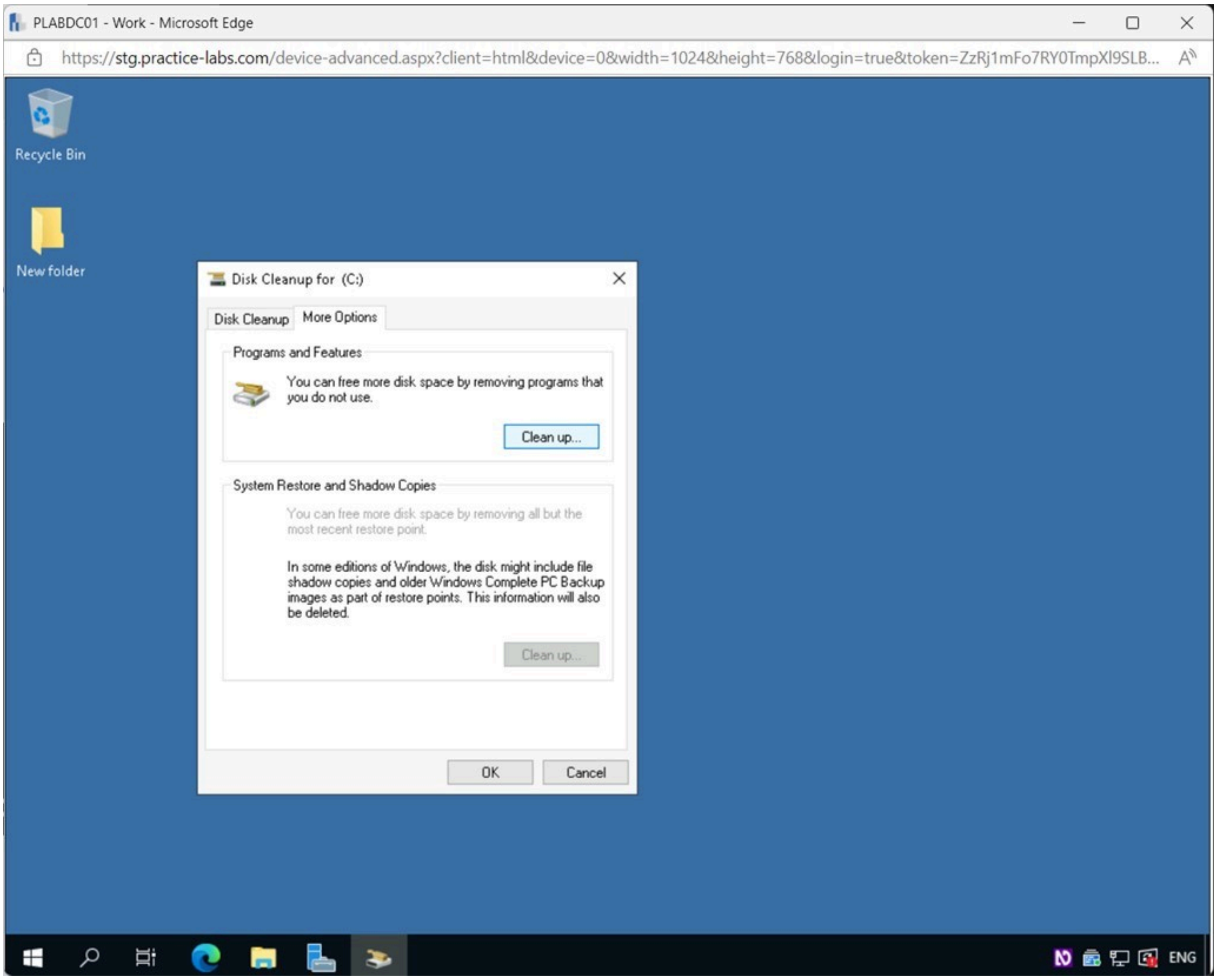

Figure 1.25 Screenshot of PLABDC01: Displaying selecting Clean up in the Disk Cleanup - More Options tab.

> ***Note:*** *In the* ***More Options*** *tab, you can clean up programs and applications. You can also erase shadow files, older backups and restore points.*

Step 4

In the **Programs and Features** window, users can remove programs that are not required anymore.

> ***Note:*** *You will not make any configuration changes in this window.*

Close the **Programs and Features** window.

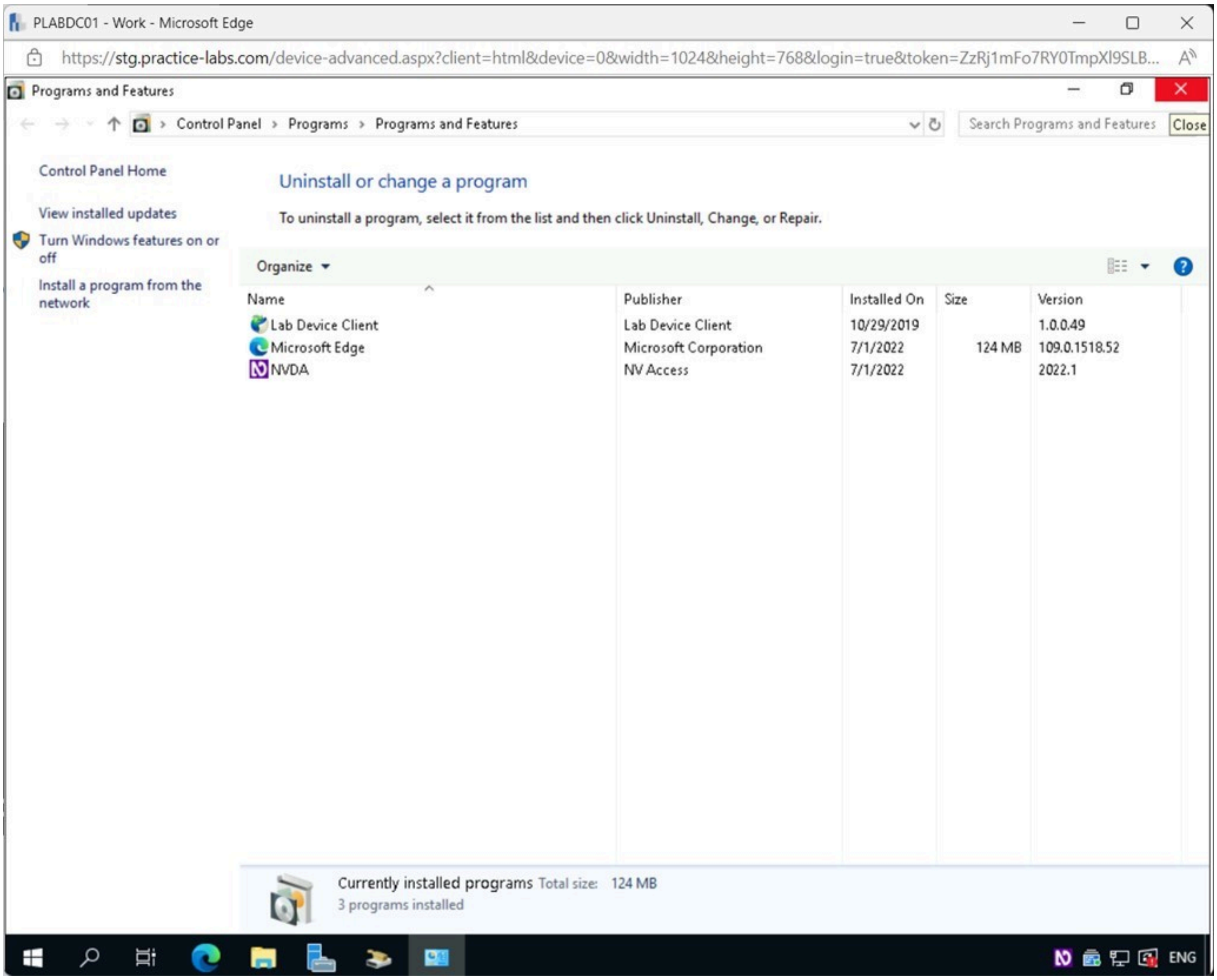

Figure 1.26 Screenshot of PLABDC01: Displaying closing the Programs and Features window.

Step 5

Back on the **Disk Cleanup for (C:)** window, select the **Disk Cleanup** tab.

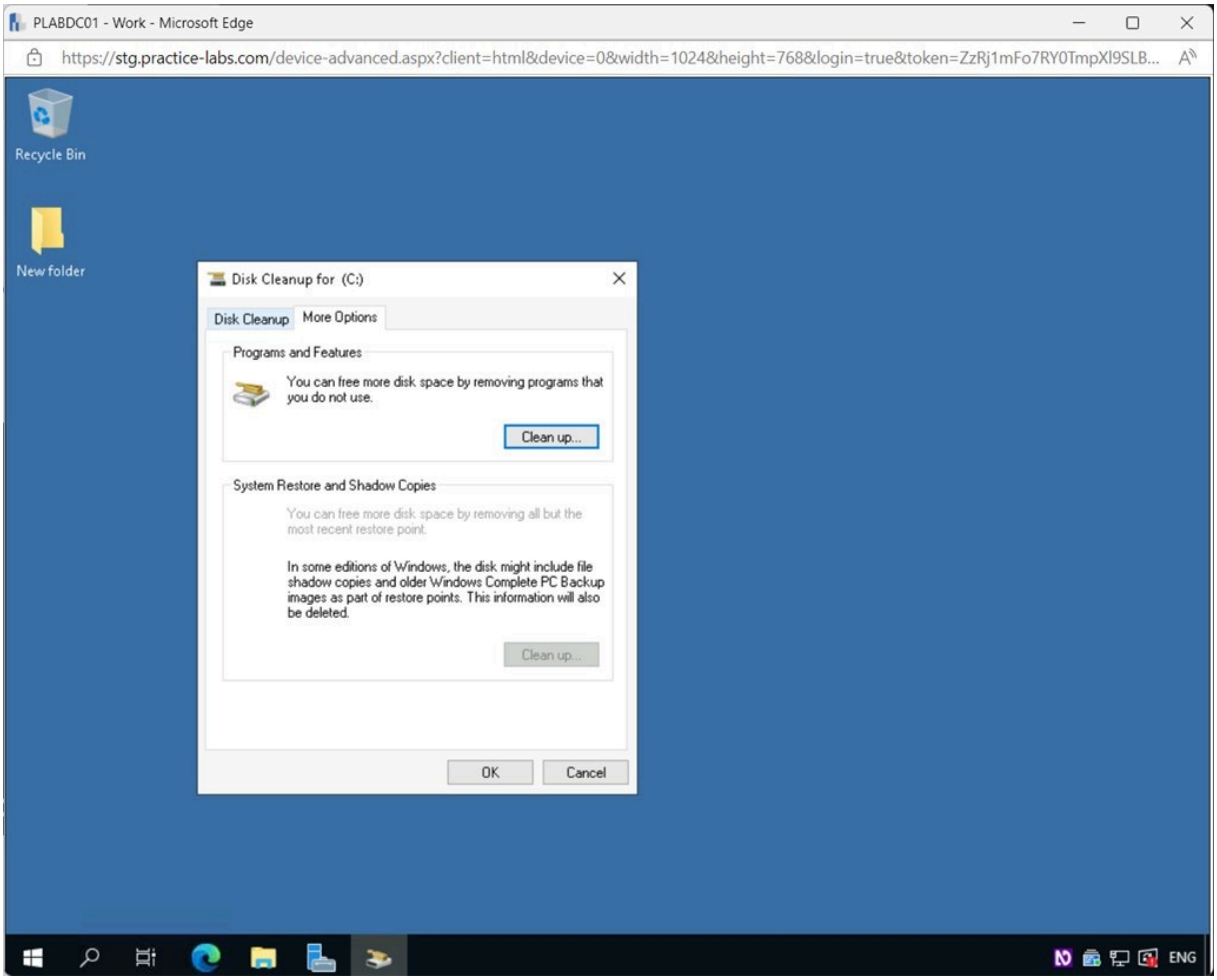

Figure 1.27 Screenshot of PLABDC01: Displaying selecting the Disk Cleanup tab in the Disk Cleanup for (C:) window.

Step 6

In the **Disk Cleanup for (C:) - Disk Cleanup** tab, tick all the checkboxes under the **Files to delete** section.

> ***Note:*** *As you can see, there are not a lot of files to be cleaned up in this lab device. Notice the* ***Total amount of disk space you gain*** *is* ***20.2 MB****. The value may differ. This process can take longer when being done on a production system due to the larger amount of data.*

Click **OK**.

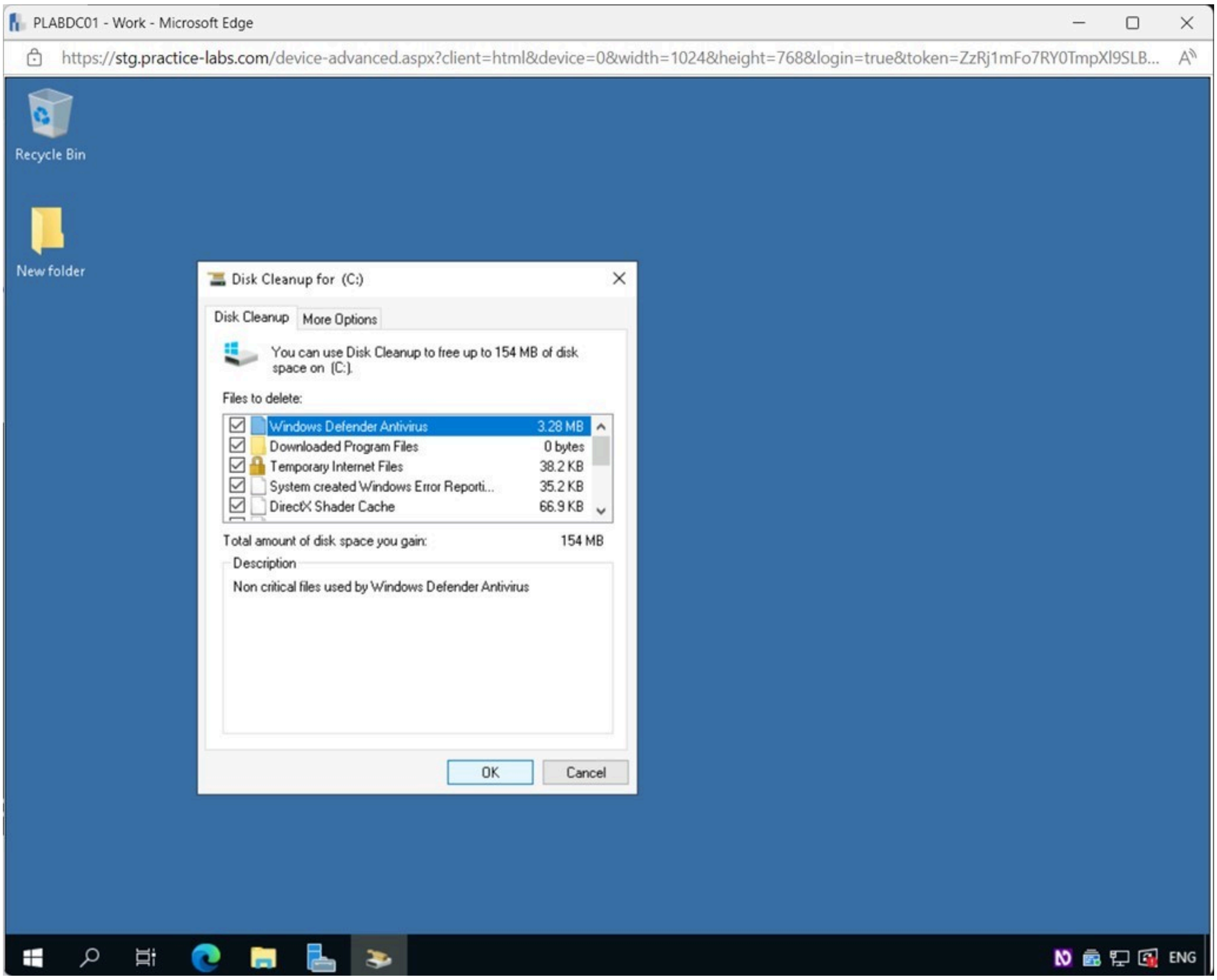

Figure 1.28 Screenshot of PLABDC01: Displaying the Disk Cleanup for (C:) - Disk Cleanup tab, showing the required settings performed and the OK button selected.

Step 7

In the **Disk Cleanup** pop-up window, click **Delete Files**.

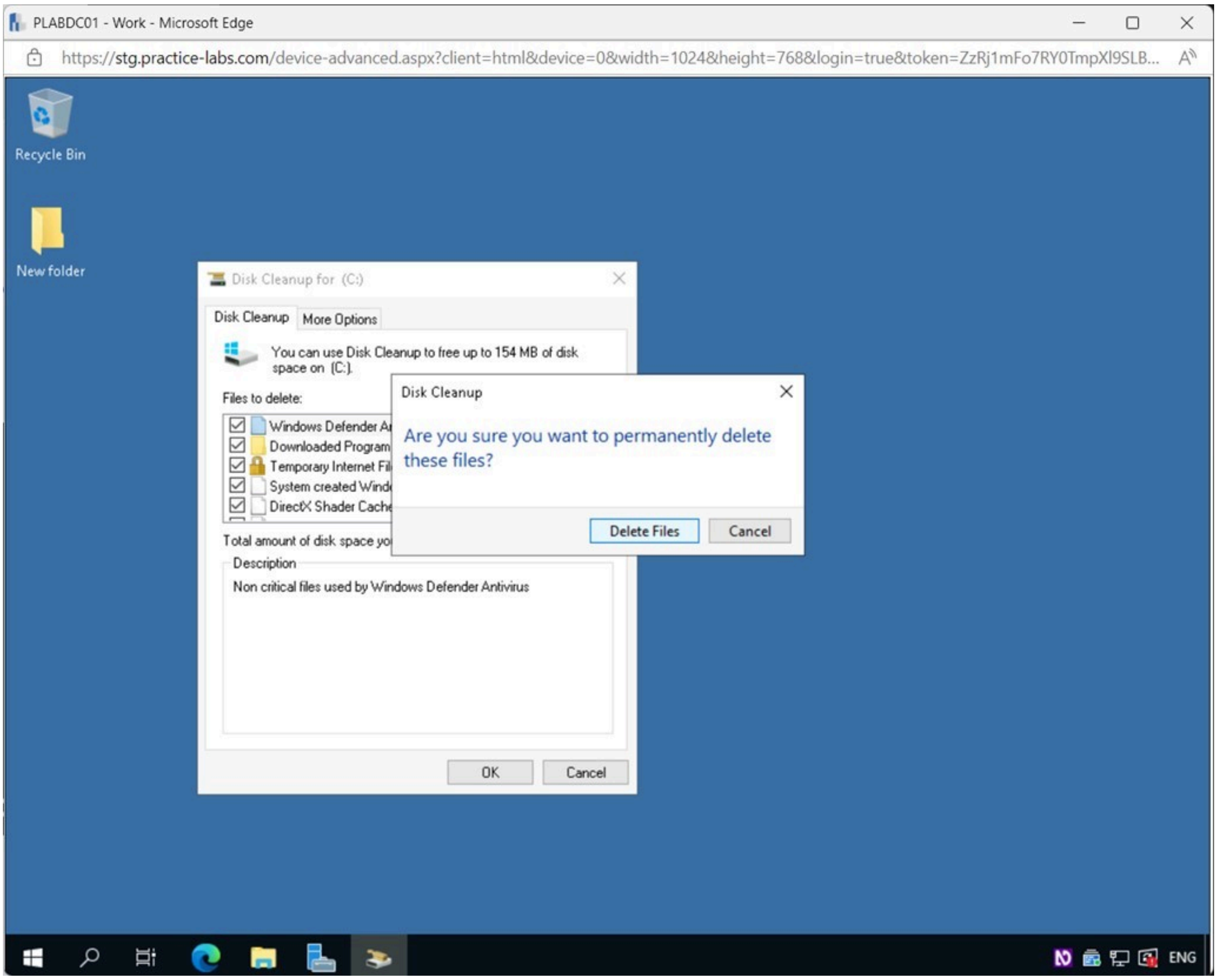

Figure 1.29 Screenshot of PLABDC01: Displaying selecting Delete Files in the Disk Cleanup pop-up window.

Step 8

The **Disk Cleanup** runs and will close automatically once finished.

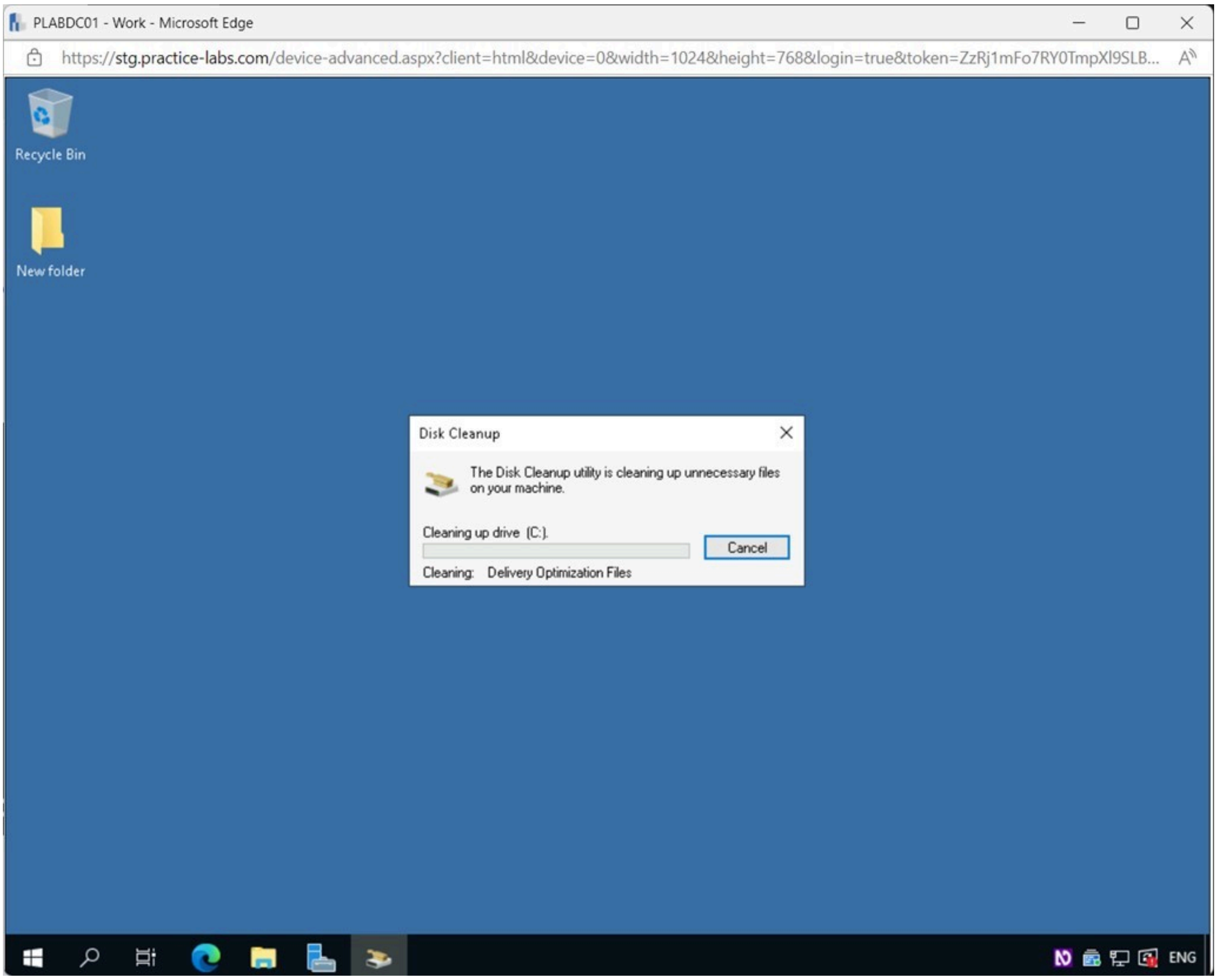

Figure 1.30 Screenshot of PLABDC01: Displaying the progress of the Disk utility in the Disk Cleanup pop-up window.

Task 3 - Slow Performance of Disks

Over time when creating and deleting files, the disk can become cluttered and fragmented, which will degrade its performance. Open spaces are created on the platter, and files that are being saved will be broken up into chunks and spread all over the disk to store them. Defragmenting the hard drive reorganizes these files and puts them continuously on the disk so it can find and load the necessary data quickly.

In this task, you will use the Defragment and Optimize Drives tool to increase the performance of the disk.

Step 1

Connect to **PLABDC01**.

Click the Windows **Start** charm, and expand the **Windows Administrative Tools** menu.

Scroll down and select **Defragment and Optimize Drives**.

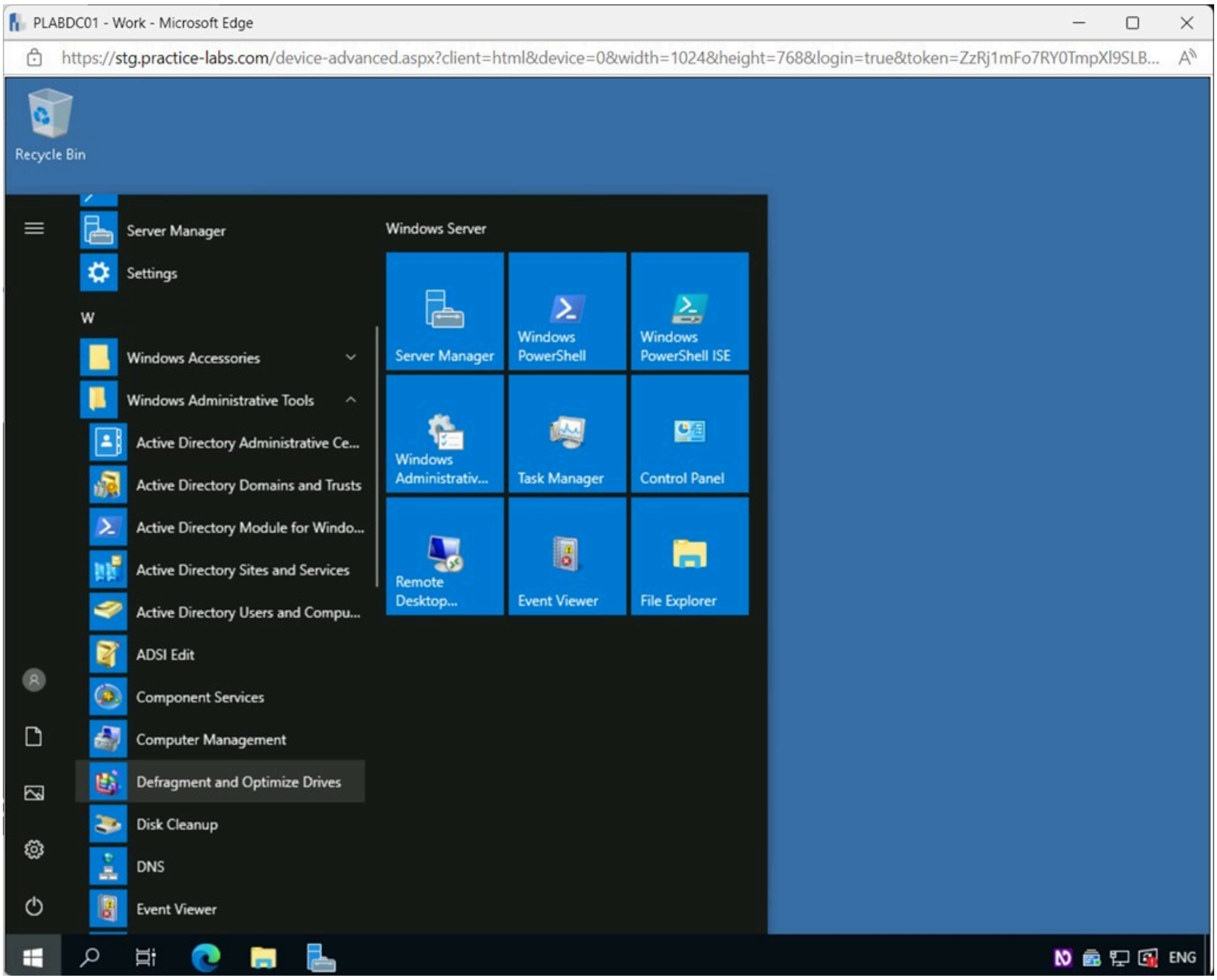

Figure 1.31 Screenshot of PLABDC01: Displaying selecting Defragment and Optimize Drives from the Windows Start menu.

Step 2

In the **Optimize Drives** window, notice that there is a **Thin provisioned drive**.

Click **Optimize.**

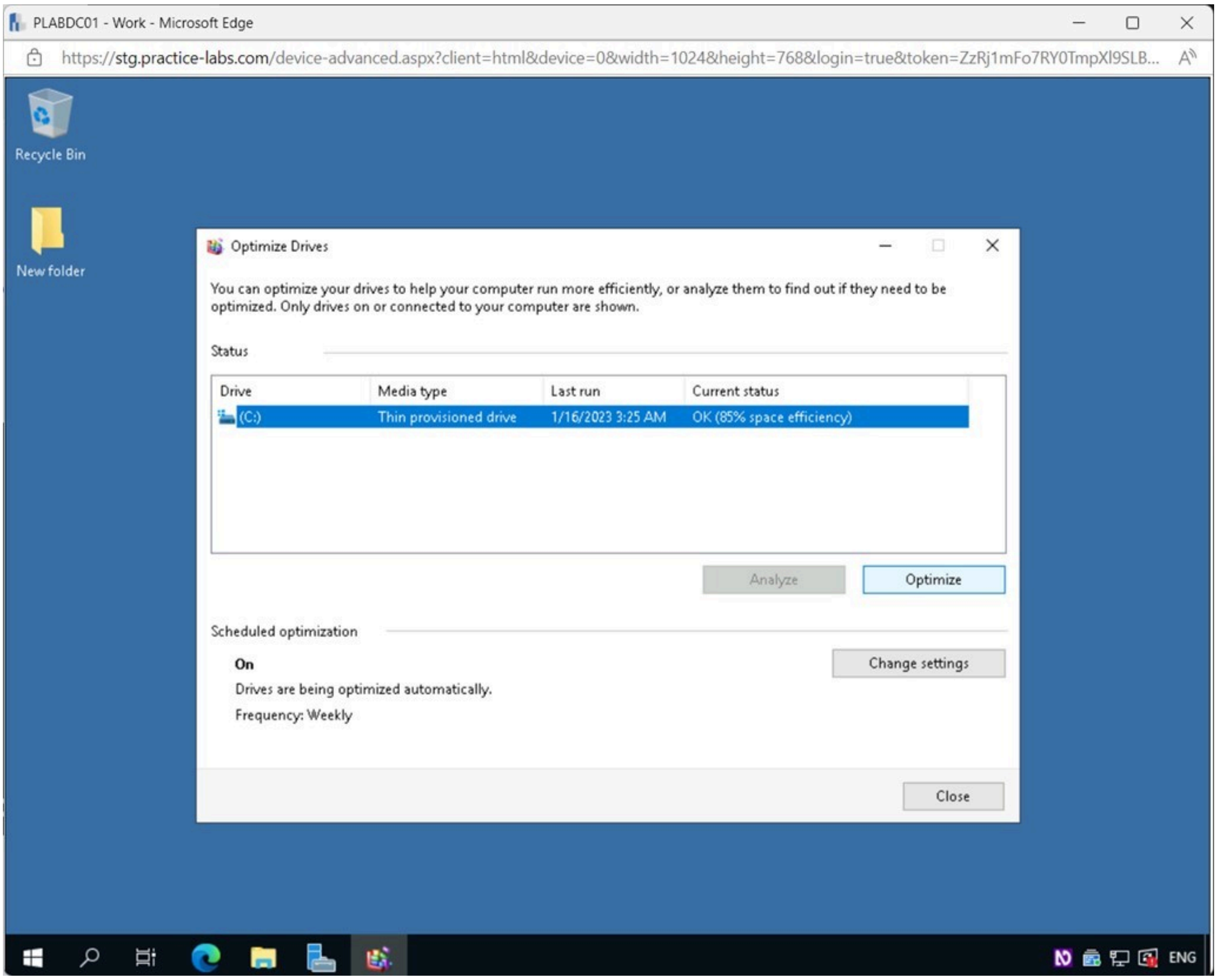

Figure 1.32 Screenshot of PLABDC01: Displaying selecting Optimiz in the Optimize Drives window.

> ***Note:*** *Defragmenting a standard hard drive is safe, simple, and healthy. Regular defragmentation and other maintenance chores help your computer work smoothly. Hard Disk Drive (HDD) will use the analyze feature. Solid-state Drives (SSDs) don't need defragmentation. If you have an SSD, clicking Optimize on the defrag screen will run a TRIM operation. TRIM instructs the SSD on which data blocks can be reused after a file is erased.*

Step 3

The **Optimize** function runs and reorganizes the drive.

Click the **Change settings** button.

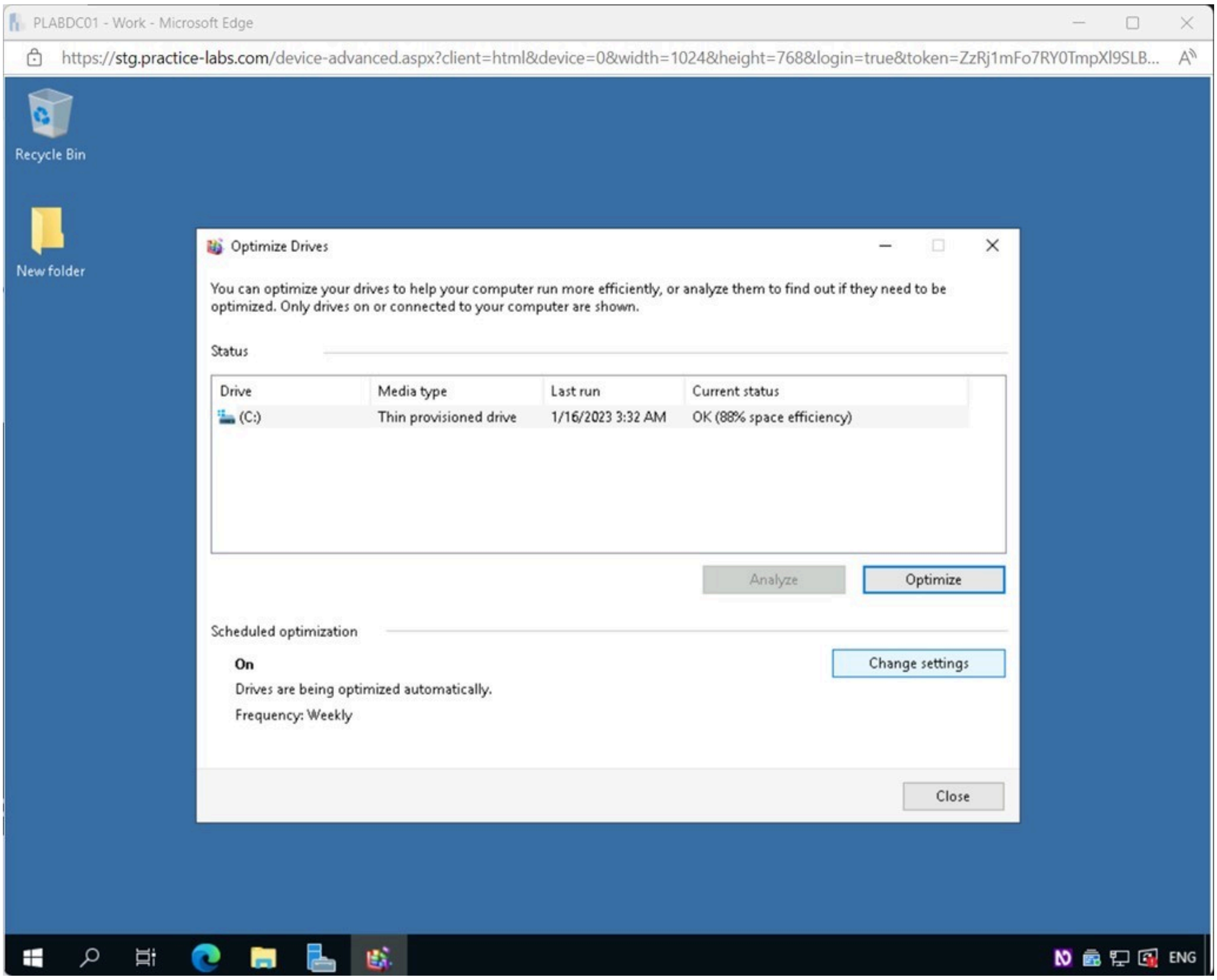

Figure 1.33 Screenshot of PLABDC01: Displaying selecting Change settings in the Optimize Drives window.

Step 4

In the **Optimize Drives** pop-up window, users can set a schedule for the computer to automatically optimize its drives and keep the device running smoothly.

You will not make any configuration changes here.

Click **Cancel.**

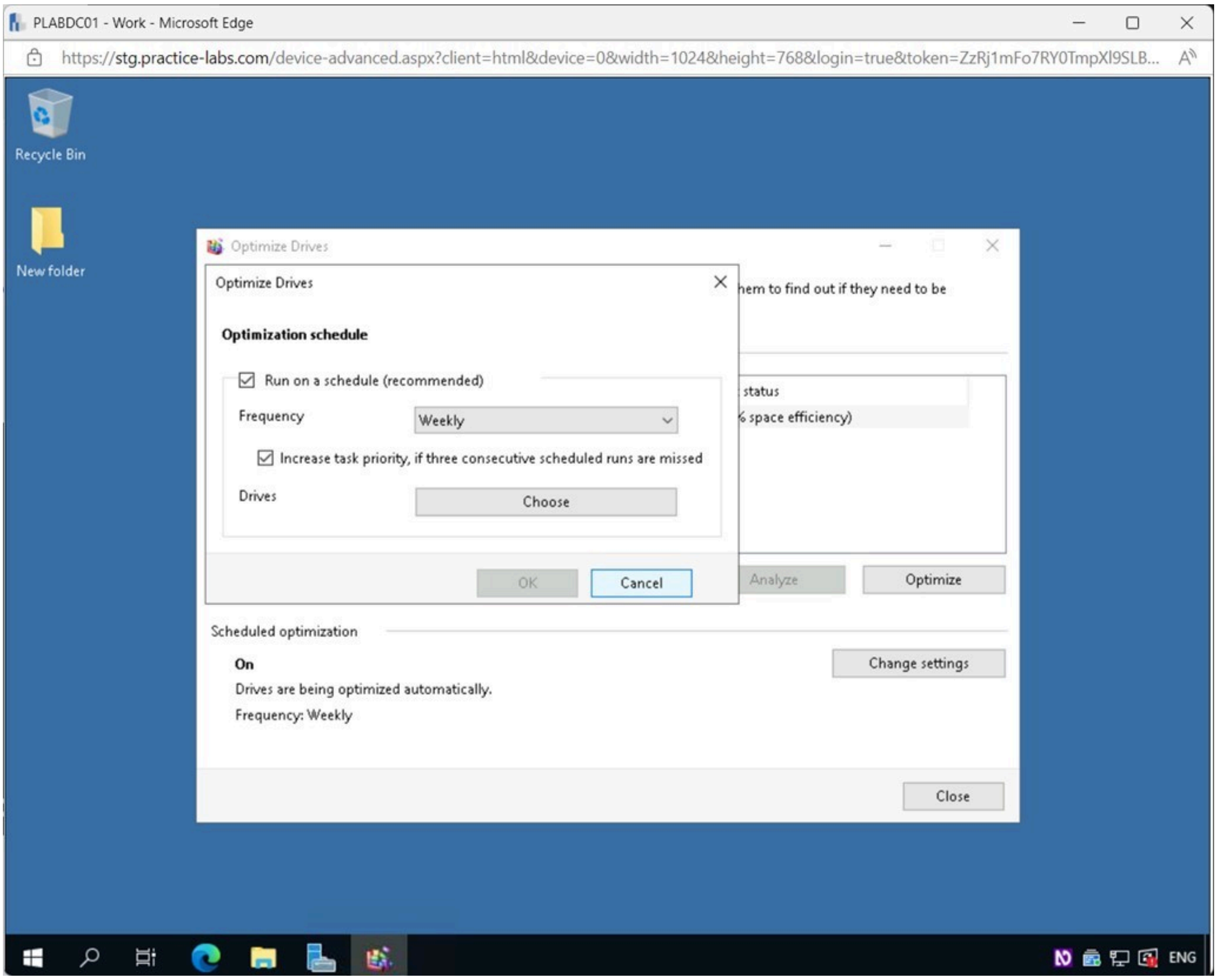

Figure 1.34 Screenshot of PLABDC01: Displaying selecting Cancel in the Optimize Drives pop-up window.

Step 5

Back on the **Optimize Drives** window, click **Close**.

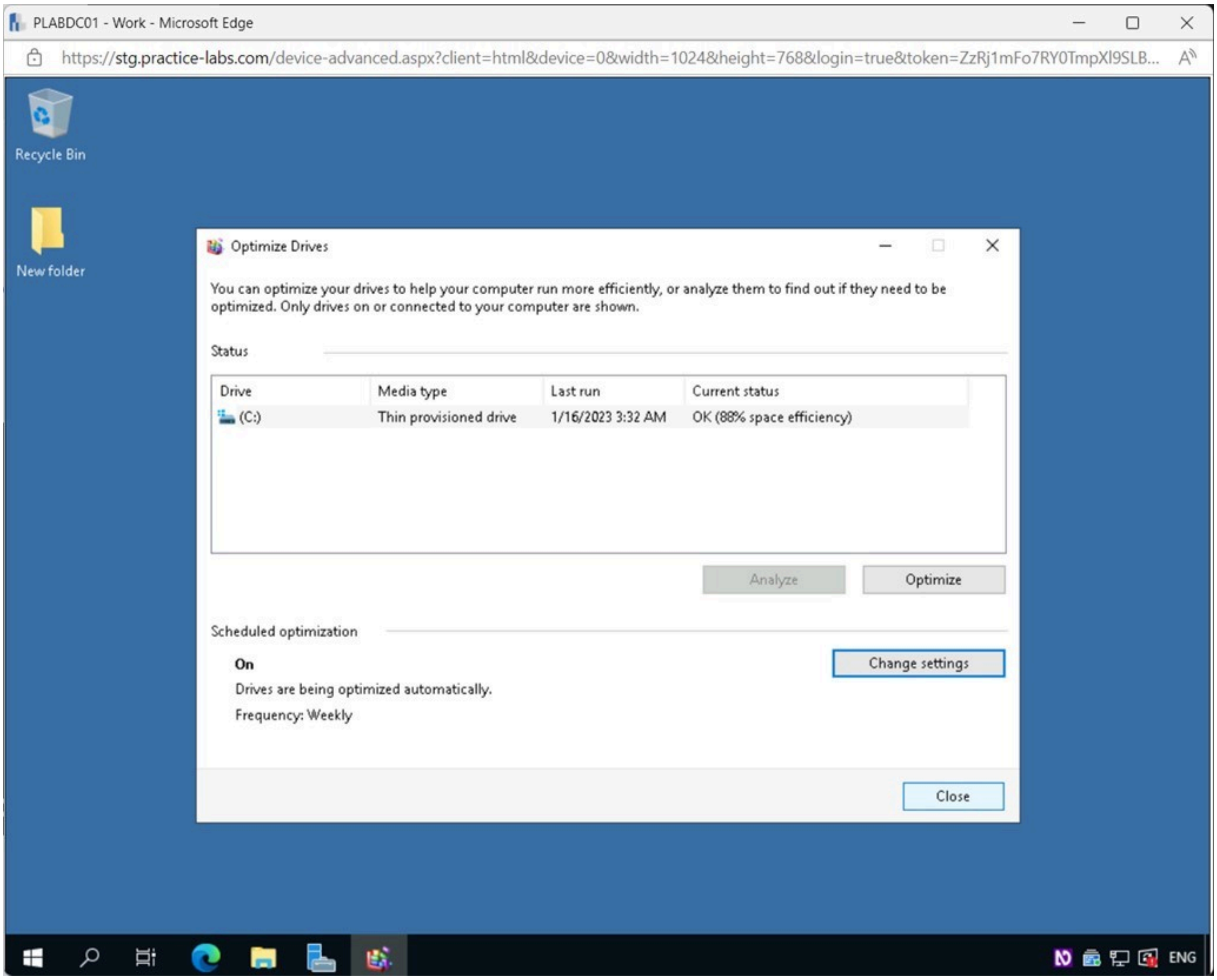

Figure 1.35 Screenshot of PLABDC01: Displaying selecting Close in the Optimize Drives window.

Task 4 - Unable to Access Logical Drive Errors

When attempting to access a flash drive or utilizing Computer Management on a remote server, you may see the **Not accessible** error. You may see this error when attempting to resize or move a disk that still contains certain Windows system files from a prior installation. If this happens frequently, you won't be able to use Windows 10's partition and disk management tools.

In this task, you will access the Component Services MMC and view the Access permissions for different users that can cause access issues when not properly configured.

Step 1

Ensure you are connected to **PLABDC01**.

Restore the **Server Manager** window from the **Taskbar**.

Click the **Tools** menu and then select **Component Services**.

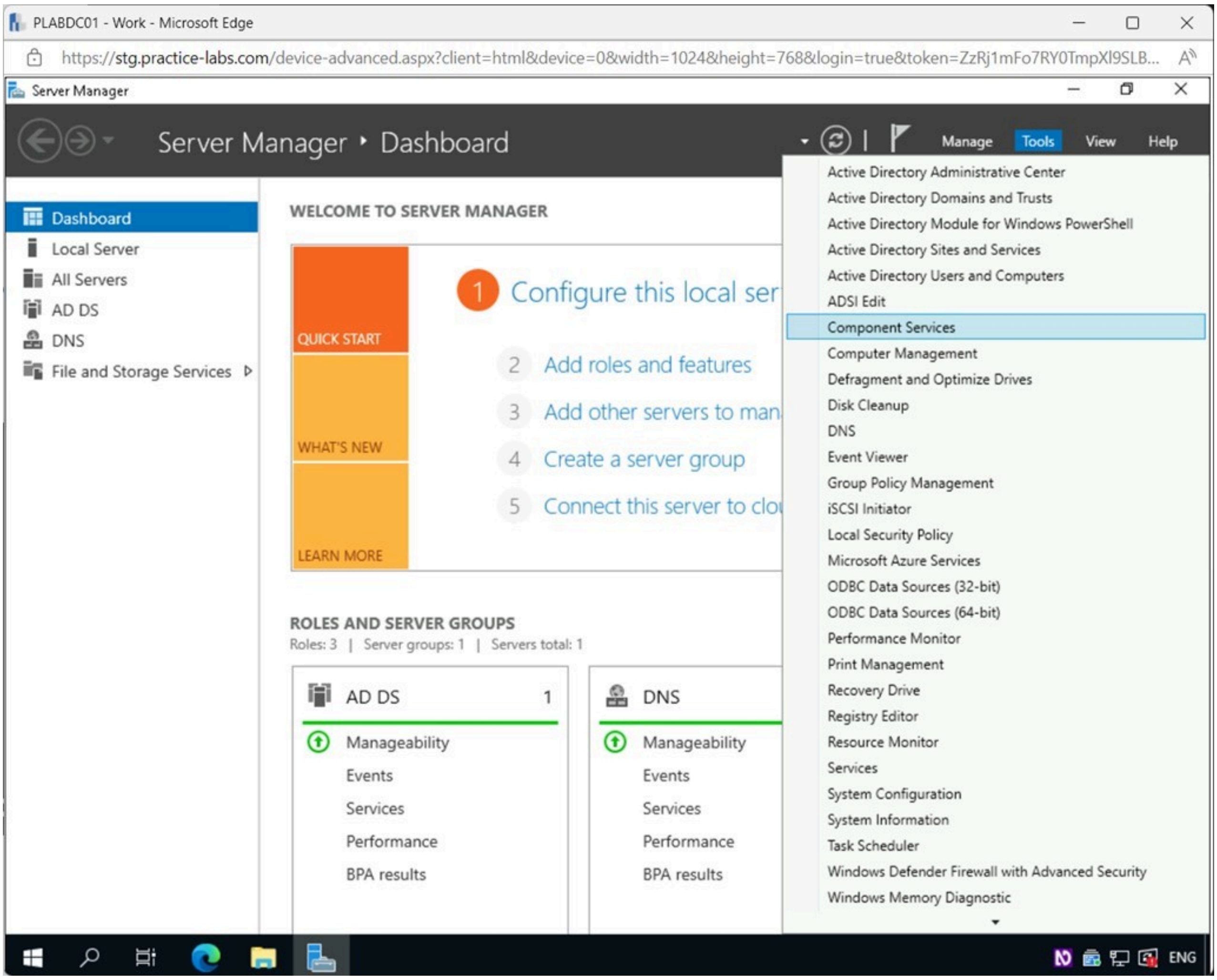

Figure 1.36 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Component Services is selected.

Step 2

In the **Component Services** window, expand **Component Services > Computers** on the left pane.

Right-click on **My Computer** and select **Properties**.

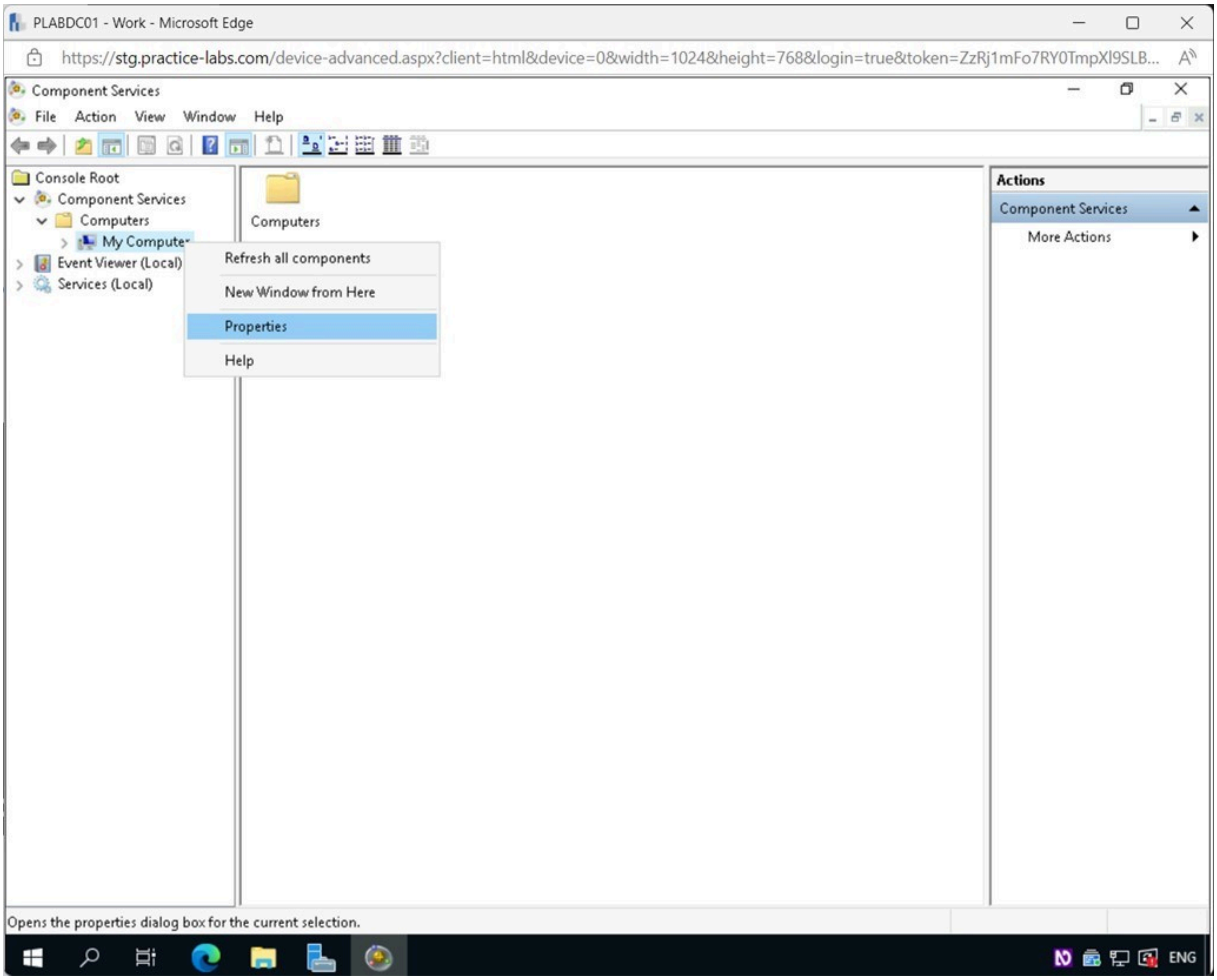

Figure 1.37 Screenshot of PLABDC01: Displaying the Component Services window. My Computer is right-clicked and Properties selected.

Step 3

In the **My Computer Properties** dialog box, select the **COM Security** tab.

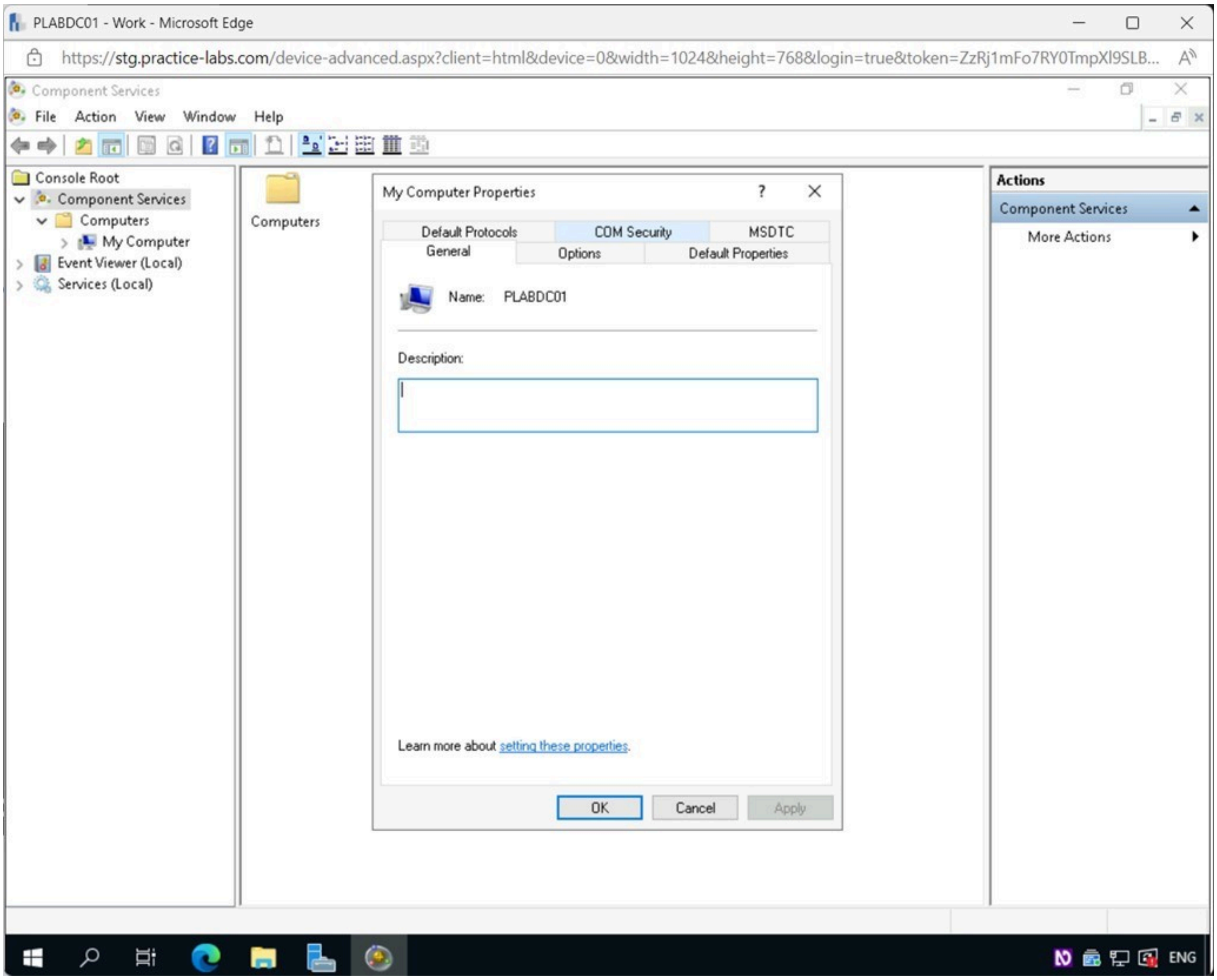

Figure 1.38 Screenshot of PLABDC01: Displaying selecting the COM Security tab in the My Computer Properties dialog box.

Step 4

In the **My Computer Properties - COM Security** tab, read the **Access Permissions** section to get an understanding of what can be done.

Click **Edit Limits** in the **Access Permissions** section.

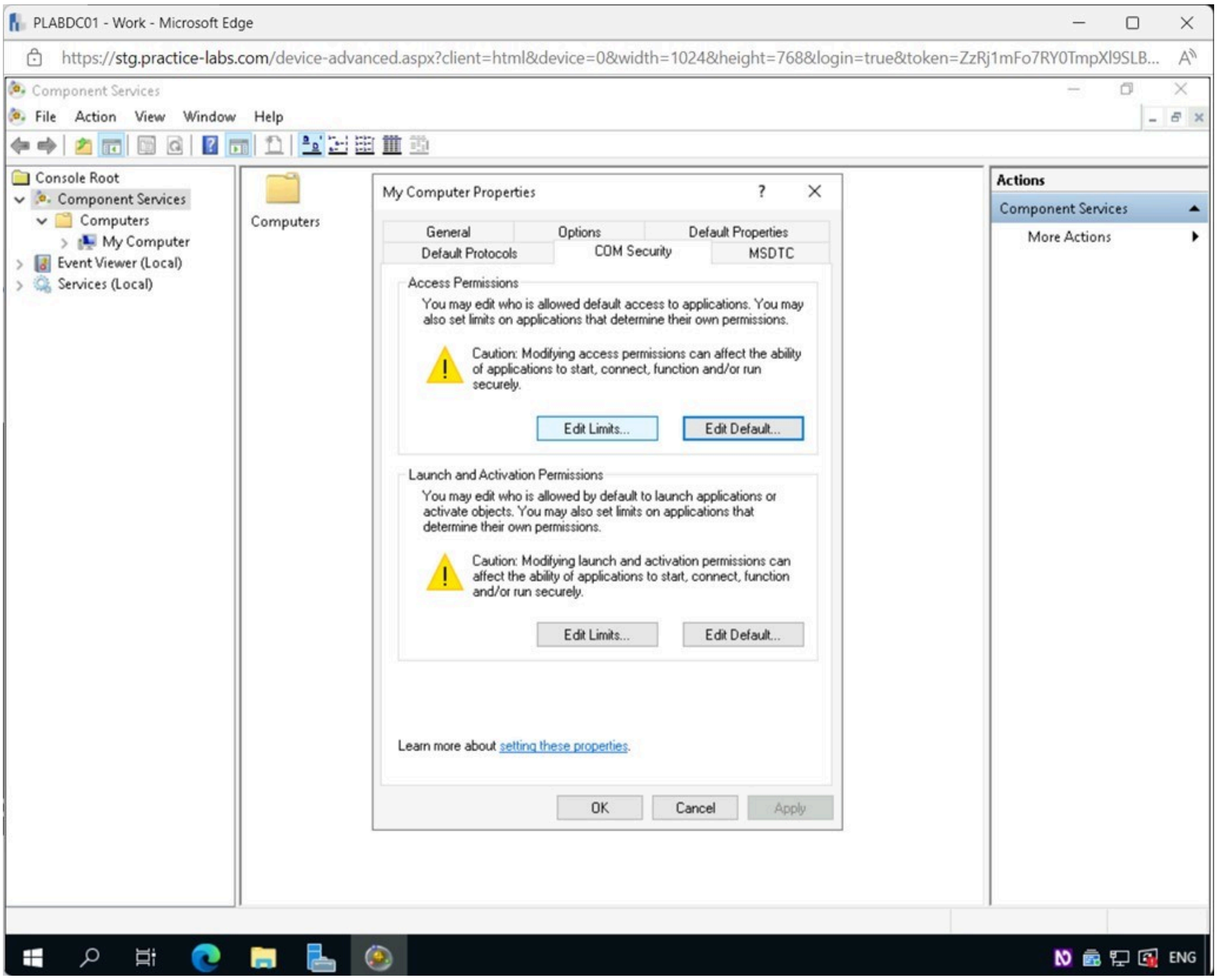

Figure 1.39 Screenshot of PLABDC01: Displaying the My Computer Properties - COM Security tab, showing Edit Limits selected.

Step 5

In the **Access Permission** pop-up window, review the access information for different users.

Scroll down and select **ANONYMOUS LOGON**. Notice that both **Local** and **Remote** access is selected. If this is not properly configured, it could be a common cause of access issues and should be checked.

Click **OK.**

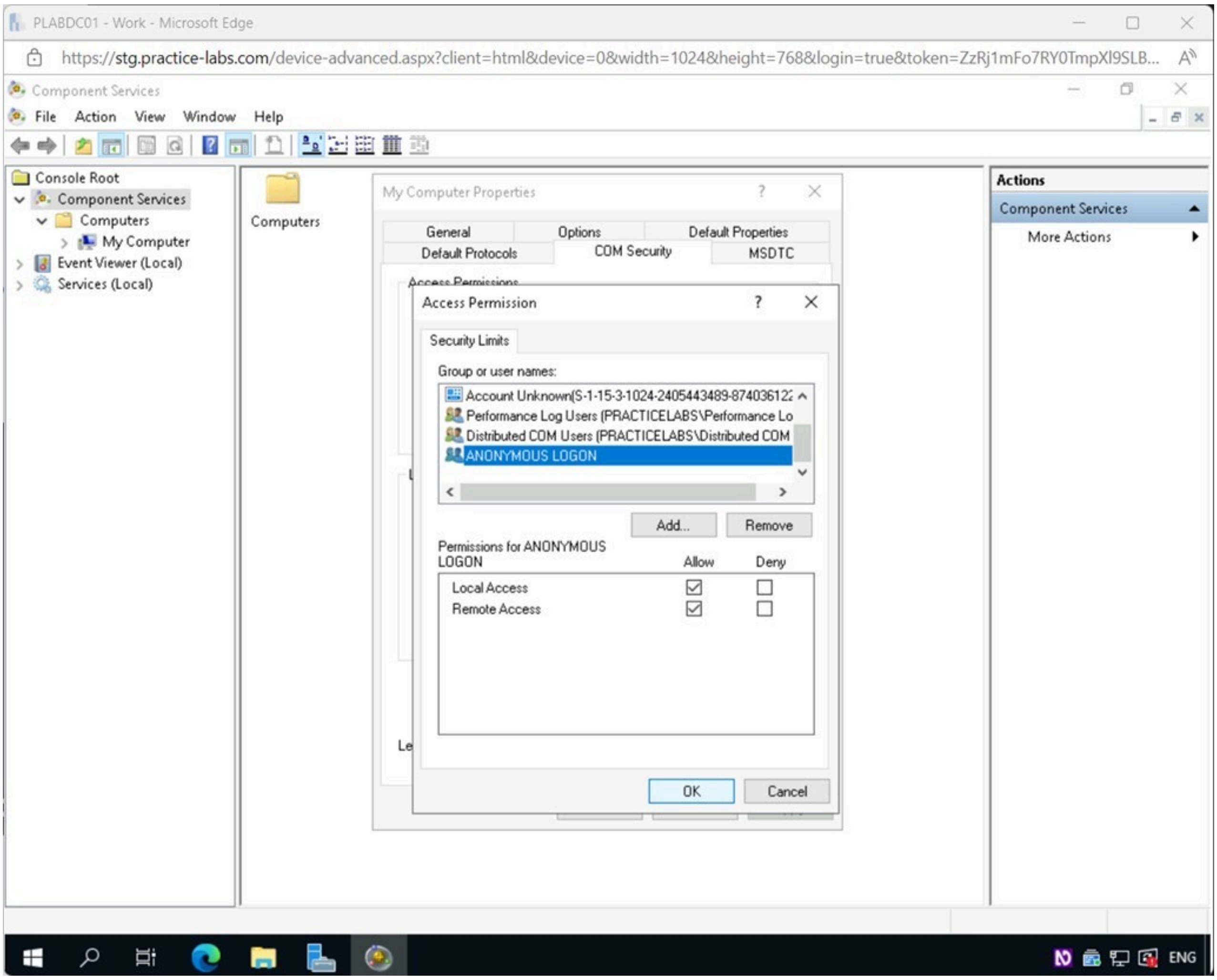

Figure 1.40 Screenshot of PLABDC01: Displaying selecting OK in the Access Permissions pop-up window.

Step 6

Back on the **My Computer Properties - COM Security** tab, read the **Launch and Activation Permissions** section to get an understanding of what can be done.

Click **Edit Limits** under the **Launch and Activation Permissions** section.

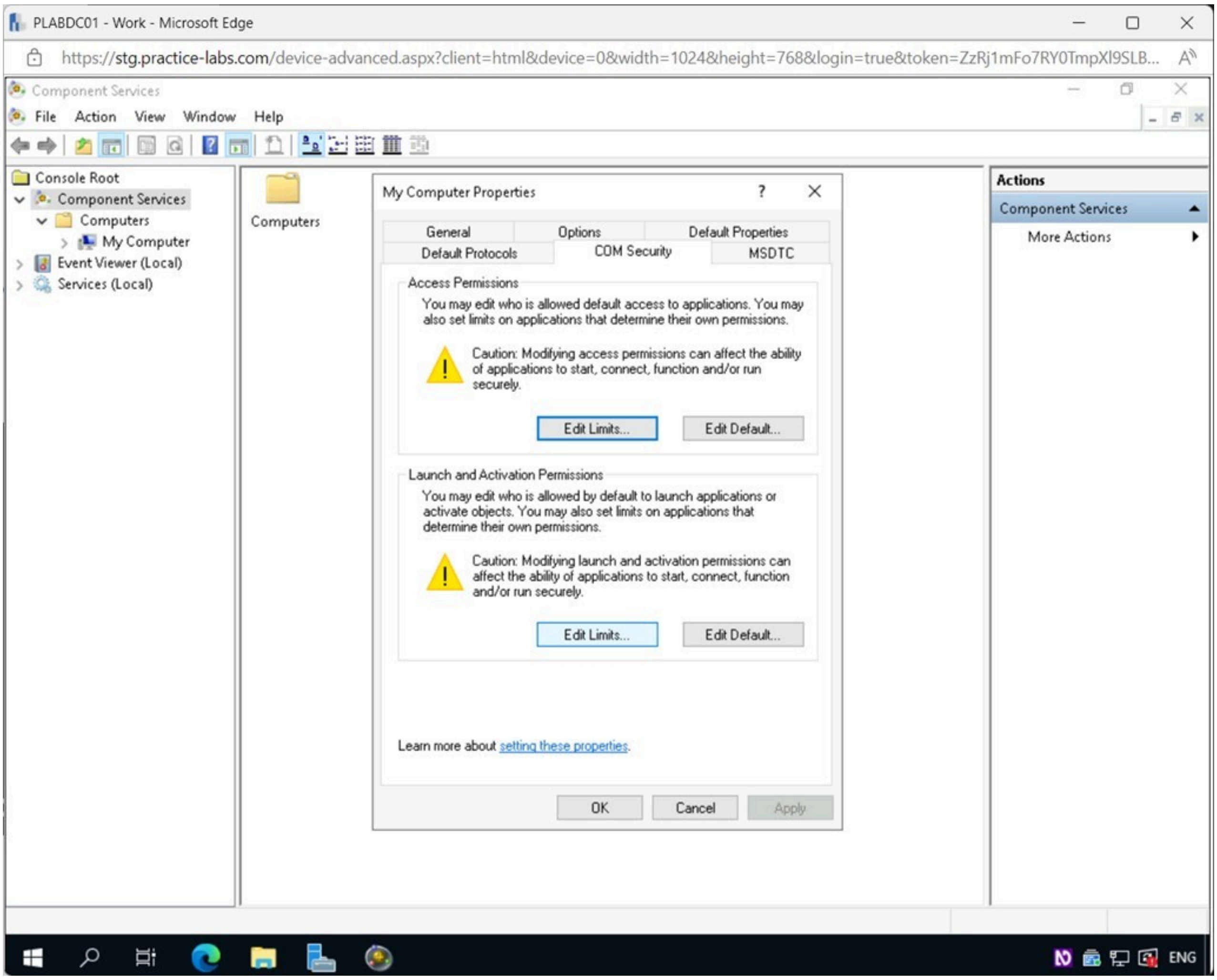

Figure 1.41 Screenshot of PLABDC01: Displaying the My Computer Properties - COM Security tab. Edit Limits button in the Launch and Activation Permissions section is selected.

Step 7

In the **Launch and Activation Permissions** pop-up window, notice the permission for the group **Everyone** only allows **Local** access.

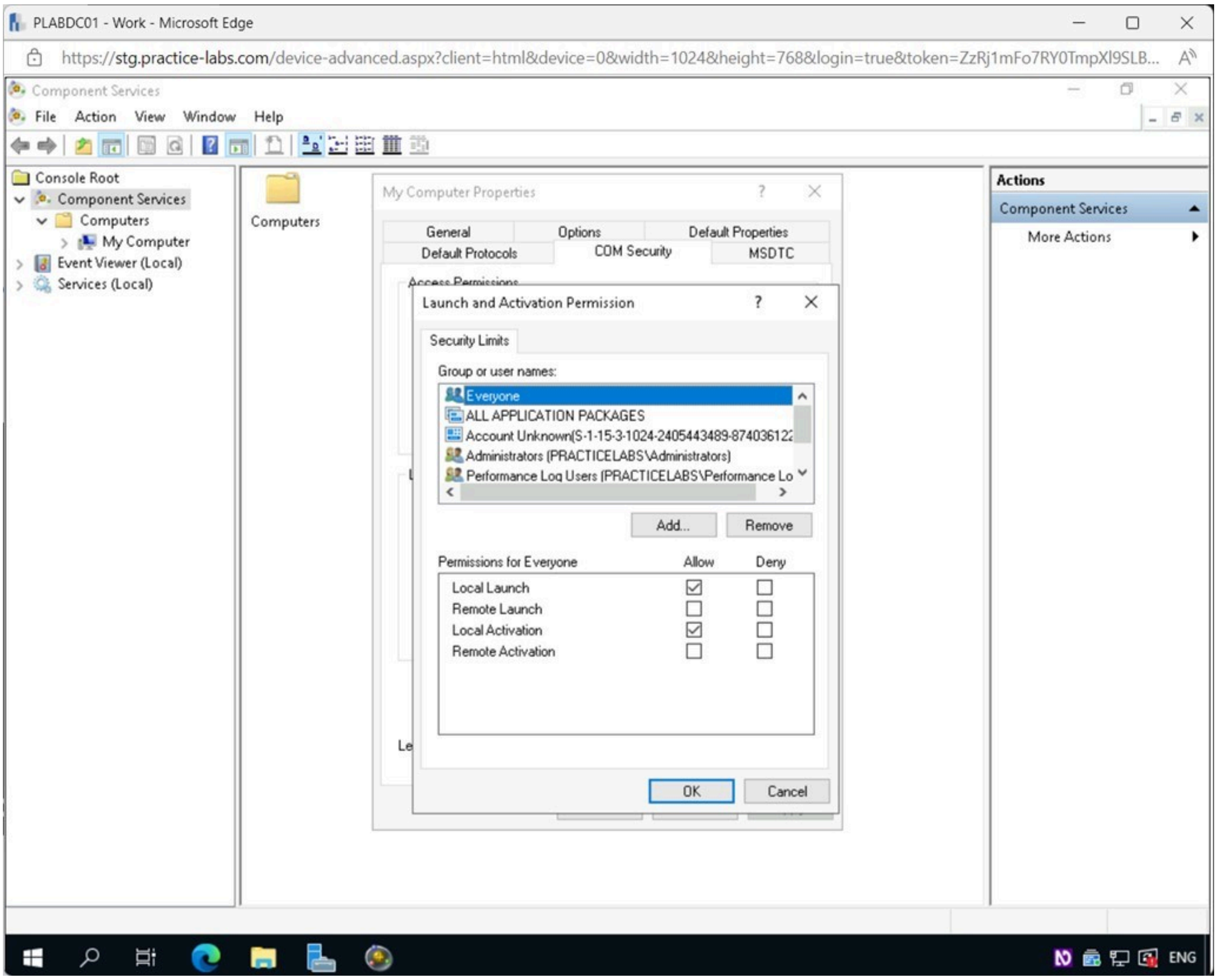

Figure 1.42 Screenshot of PLABDC01: Displaying the Launch and Activation Permissions pop-up window, showing the group Everyone selected.

Step 8

Select the **Administrators** group.

Notice the **Administrators** group has both **Local** and **Remote** access.

Click **OK**.

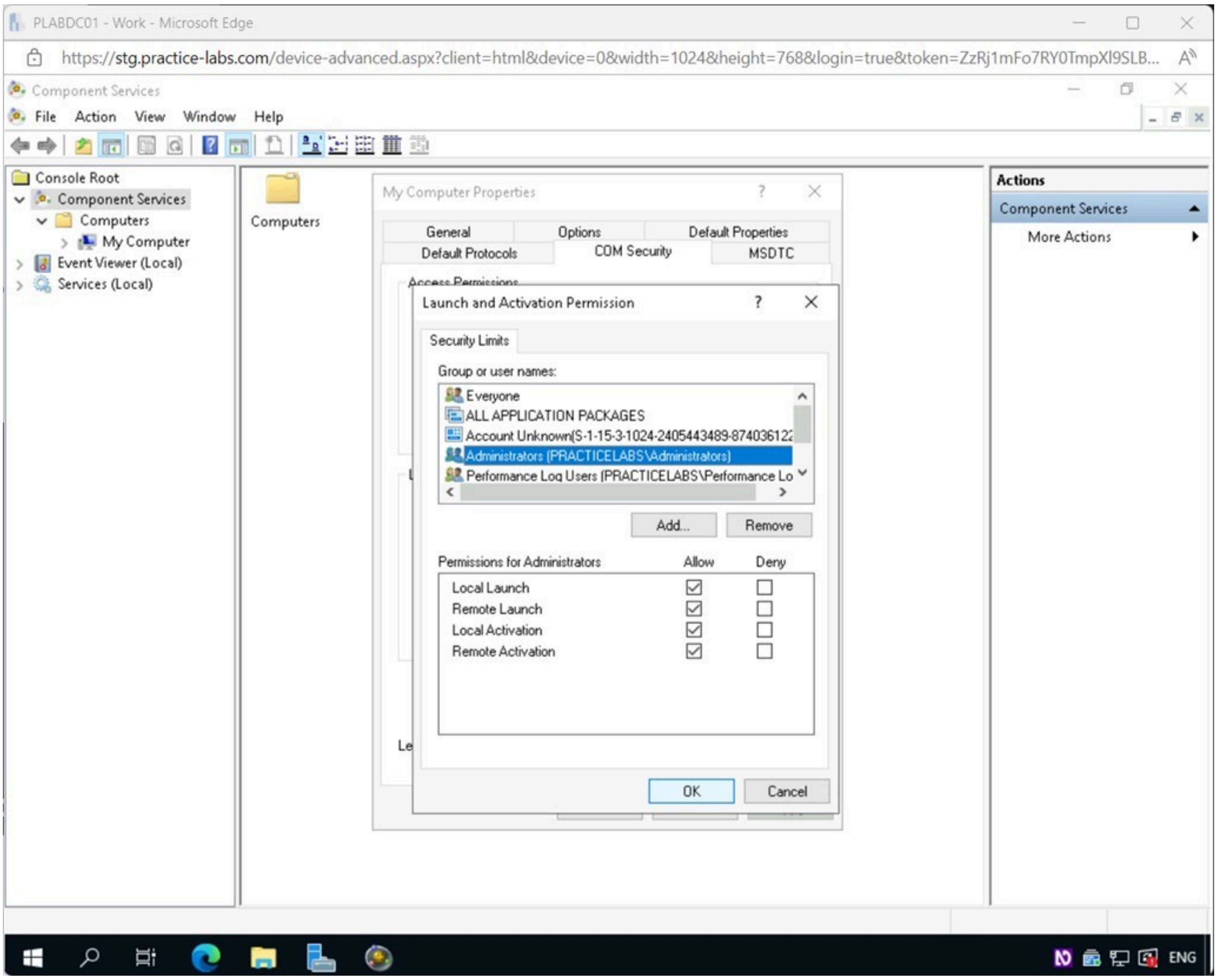

Figure 1.43 Screenshot of PLABDC01: Displaying the Launch and Activation Permissions pop-up window, showing the Administrator group selected and the OK button highlighted.

Step 9

Back on the **My Computer Properties - COM Security** tab, click the **Learn more about setting these properties** link.

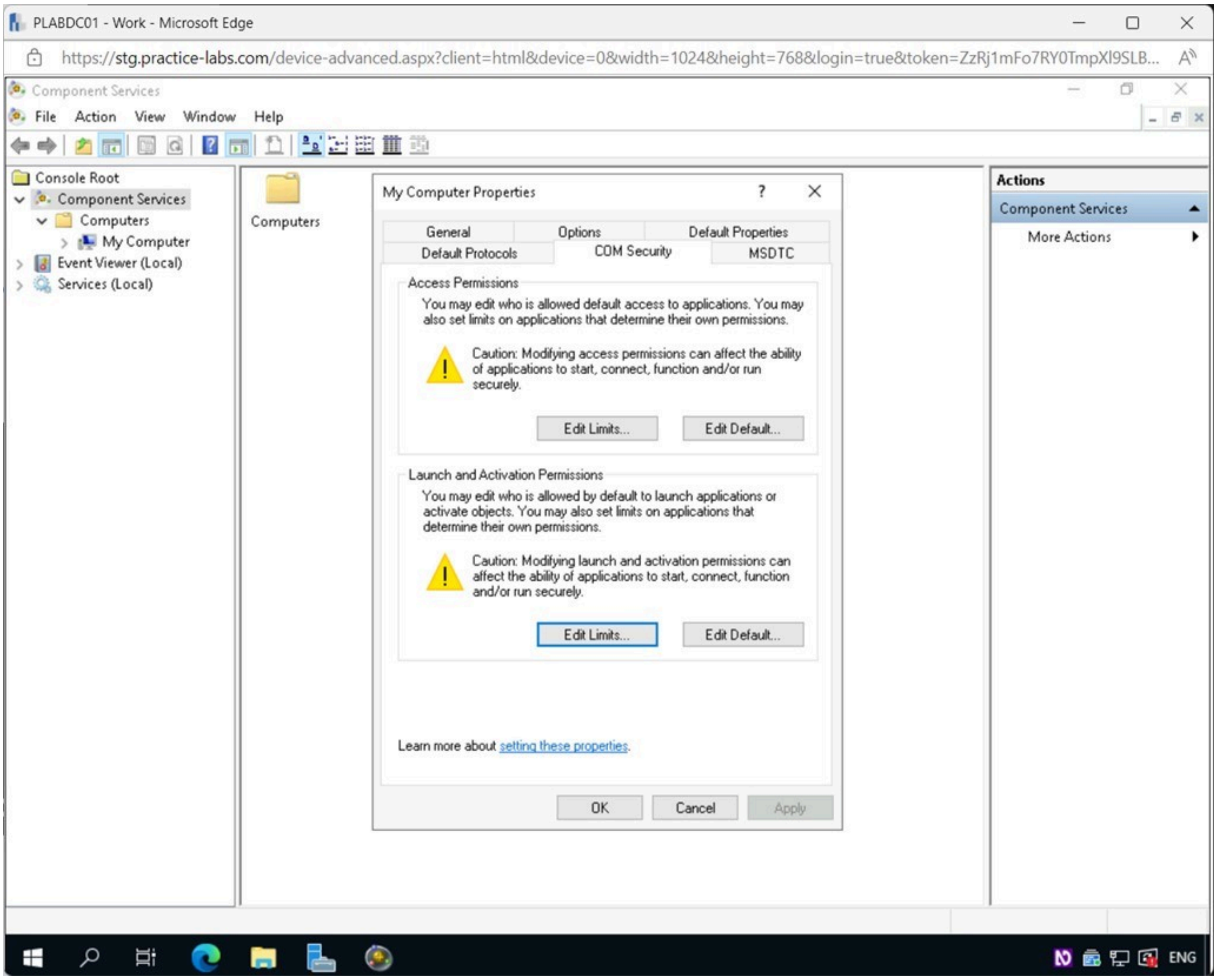

Figure 1.44 Screenshot of PLABDC01: Displaying the My Computer Properties - COM Security tab, showing the setting these properties link selected.

Step 10

The page will open on the **Microsoft Edge** browser window.

Review the information available on **Component Services Administration.**

Once done, close **Microsoft Edge**.

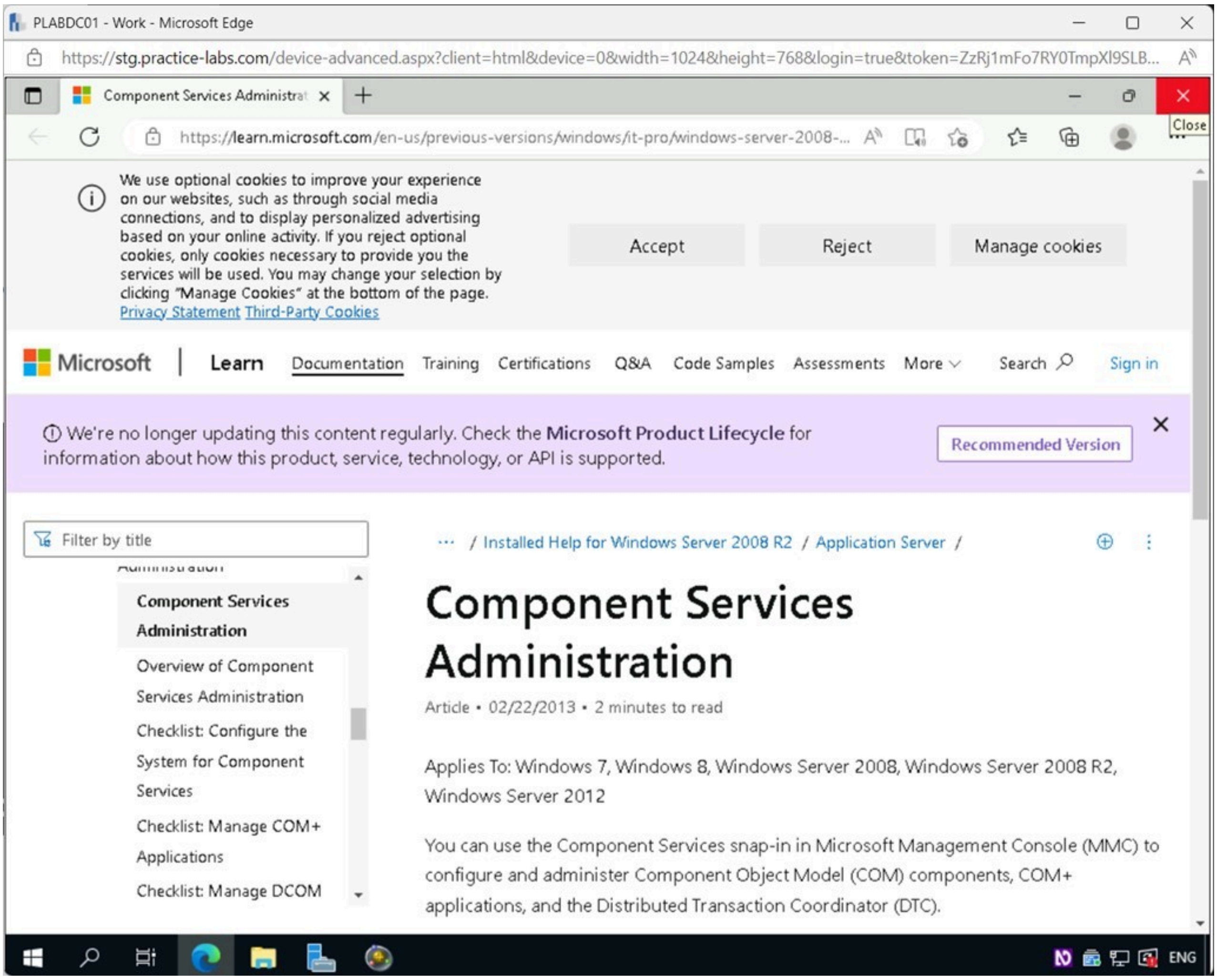

Figure 1.45 Screenshot of PLABDC01: Displaying the Microsoft Edge browser window, showing the Component Services Administration webpage. Close icon on the top-right corner is selected.

Step 11

Close the **My Computer Properties** dialog box.

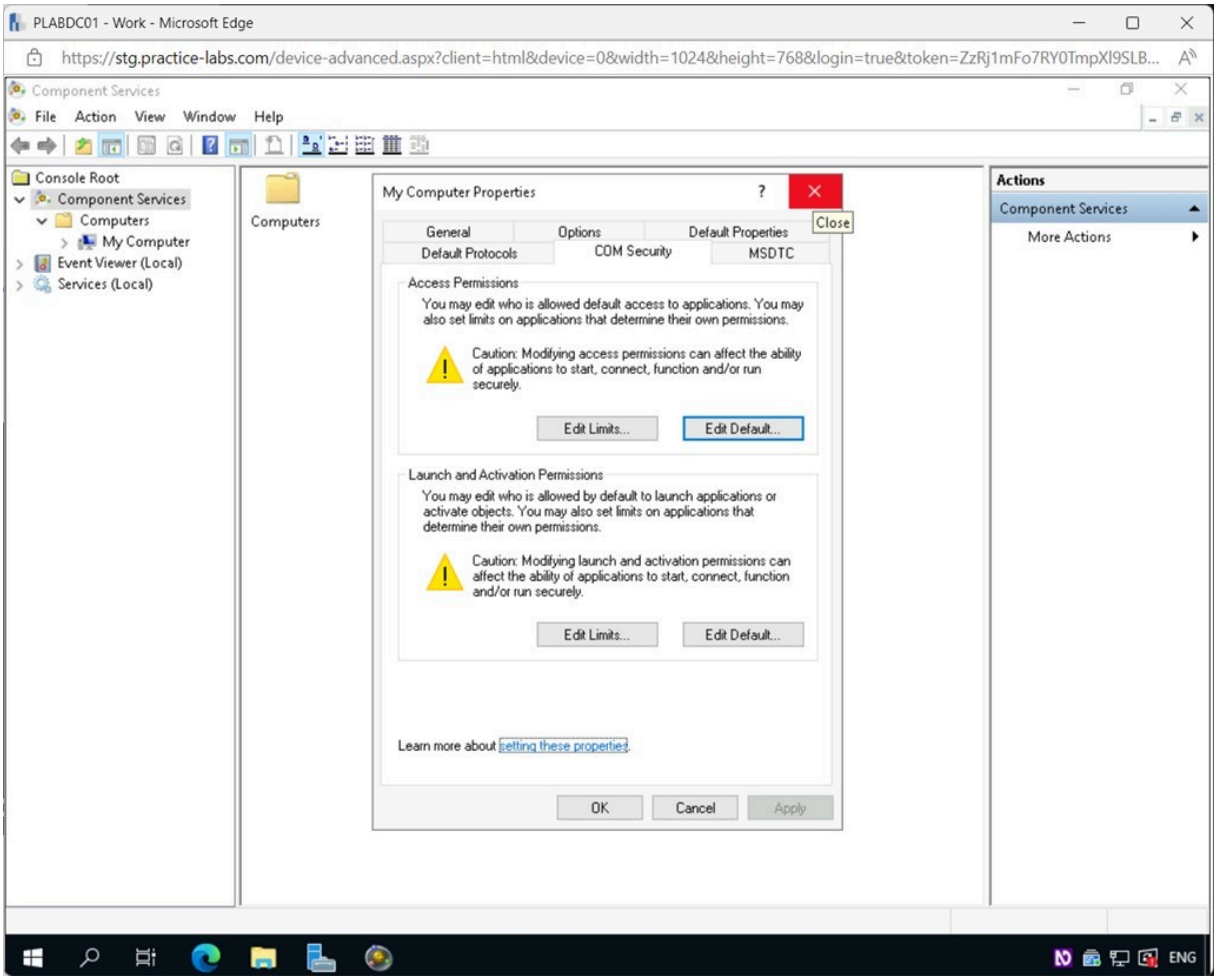

Figure 1.46 Screenshot of PLABDC01: Displaying closing the My Computer Properties dialog box.

Step 12

Close the **Component Services** window.

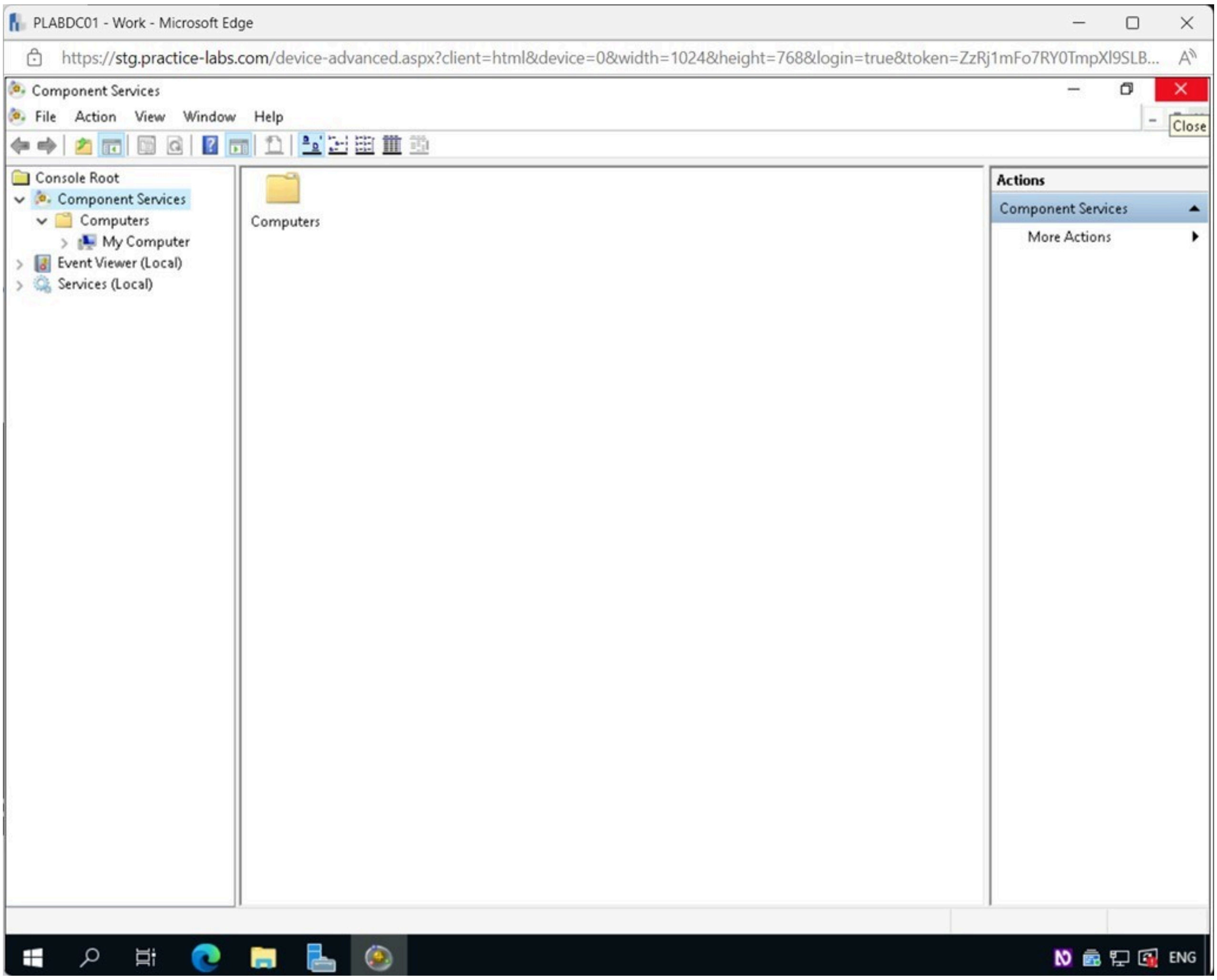

Figure 1.47 Screenshot of PLABDC01: Displaying closing the Component Services window.

Minimize the **Server Manager** window.

Task 5 - Page / Swap / Scratch File or Partition Issues

The hard drive has a file called the page file. It is also known as a swap file, swap partition, and scratch file. For data in RAM that has not been used recently or data that can't be kept because the RAM has filled up, this file is a reserved hard drive area that acts as an extension of RAM. By default, Windows maintains the page file. But administrators or users can change the default size number to match their needs.

In this task, you will learn where to make changes to the page file on a Windows machine. You will also view information about the Swap Partition on a Linux machine.

Step 1

Connect to **PLABDC01**.

Click the Windows **Start** charm and type the following:

```
control panel
```

Select **Control Panel** from the **Best Match** pop-up menu.

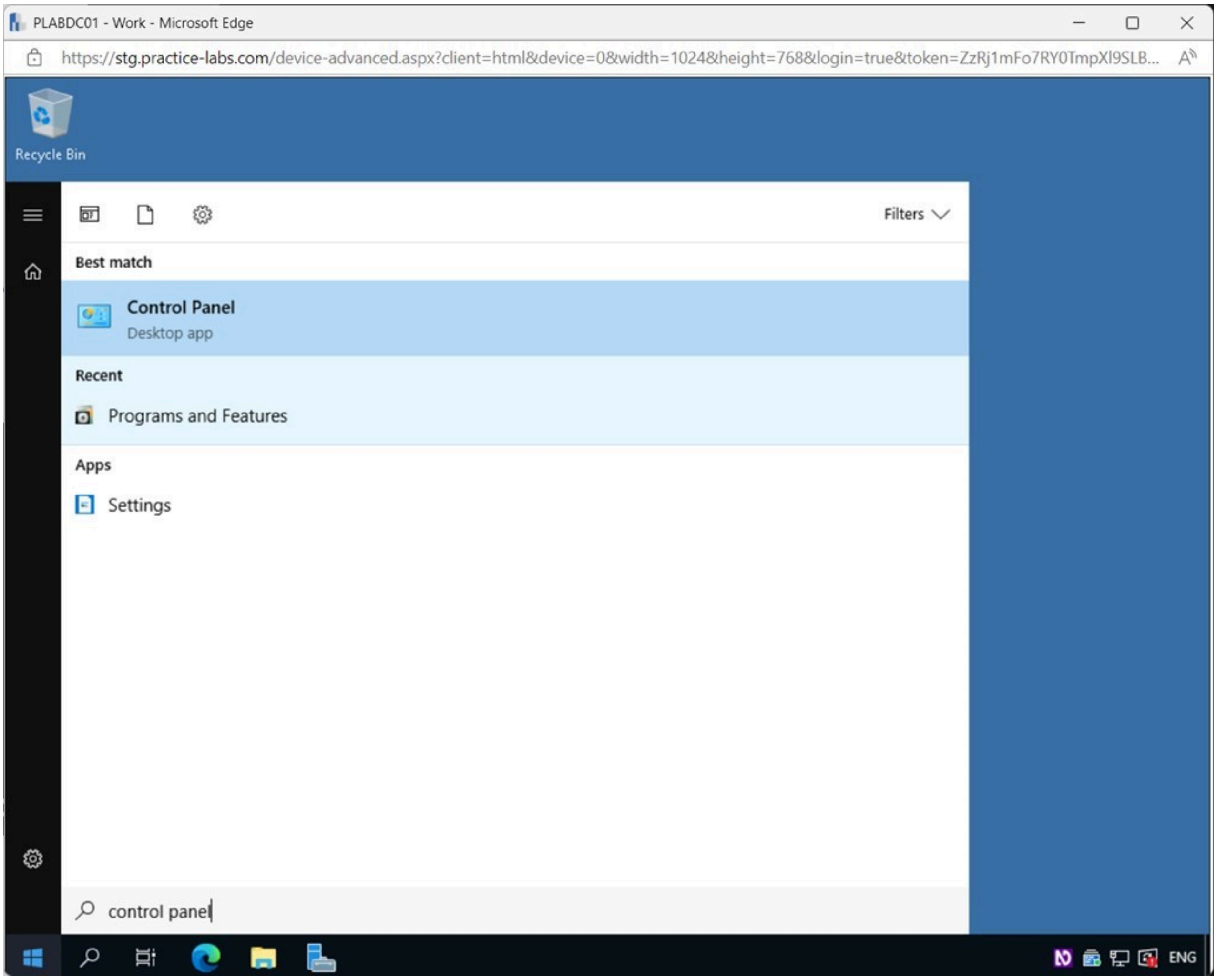

Figure 1.48 Screenshot of PLABDC01: Displaying selecting Control Panel from the Best Match pop-up menu.

Step 2

The **Control Panel** is displayed in **Category** view.

Click the **View by** drop-down menu and select **Large icons**.

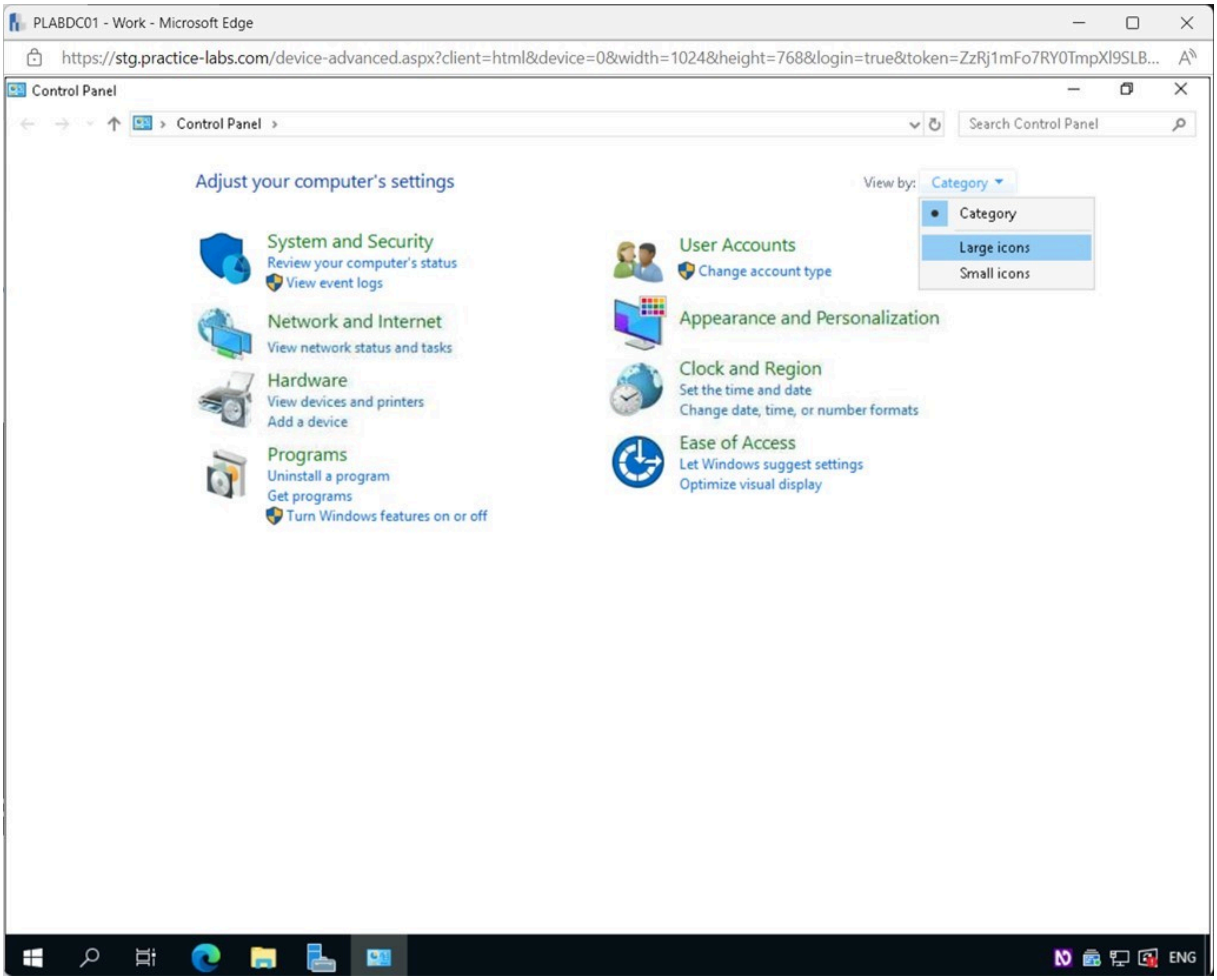

Figure 1.49 Screenshot of PLABDC01: Displaying the Control Panel window. The View by drop-down menu is accessed, and Large icons is selected.

Step 3

In the **All Control Panel Items** window, select **System**.

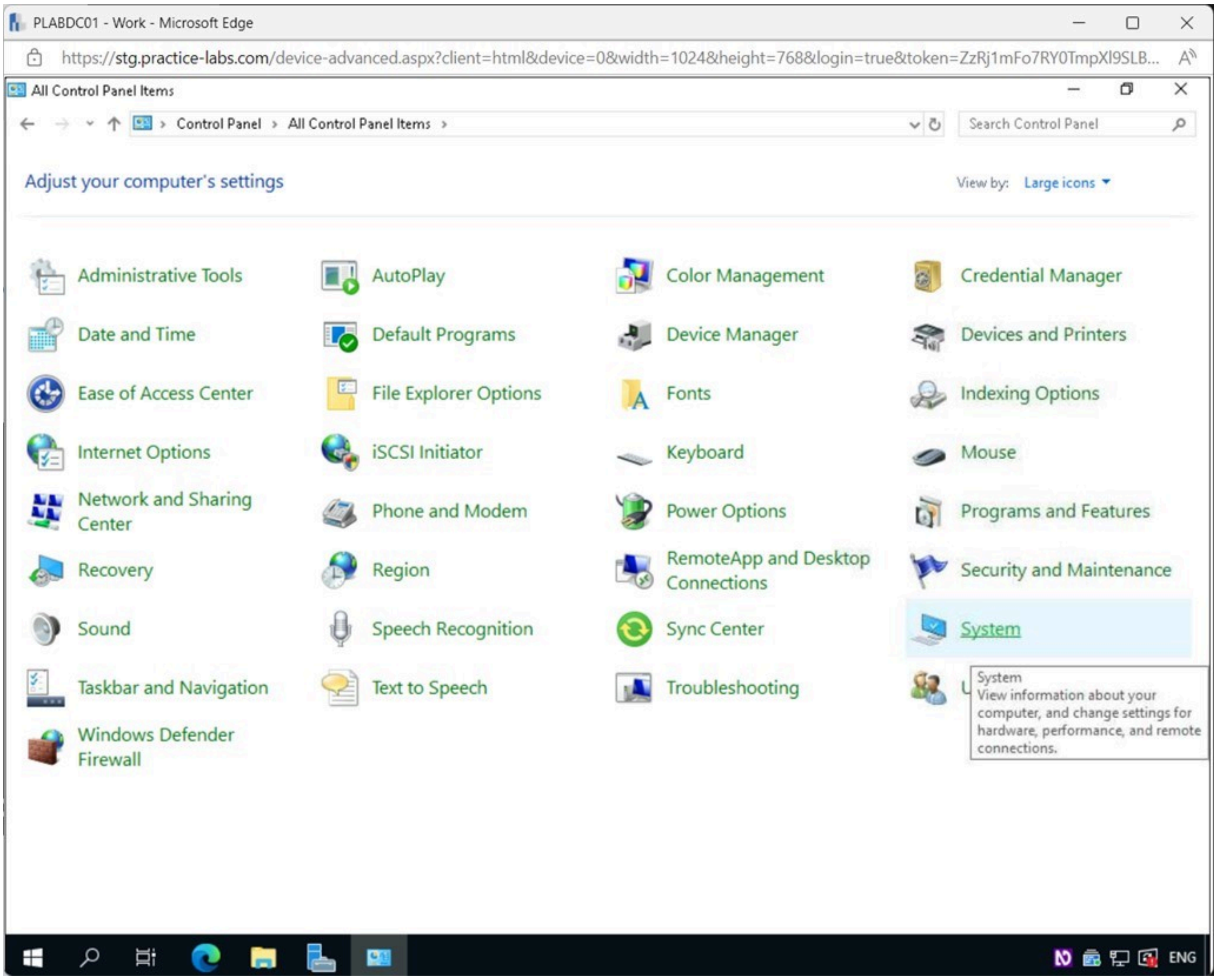

Figure 1.50 Screenshot of PLABDC01: Displaying selecting System in the All Control Panel Items window.

Step 4

In the **System** window, click the **Advanced system settings** link on the left pane.

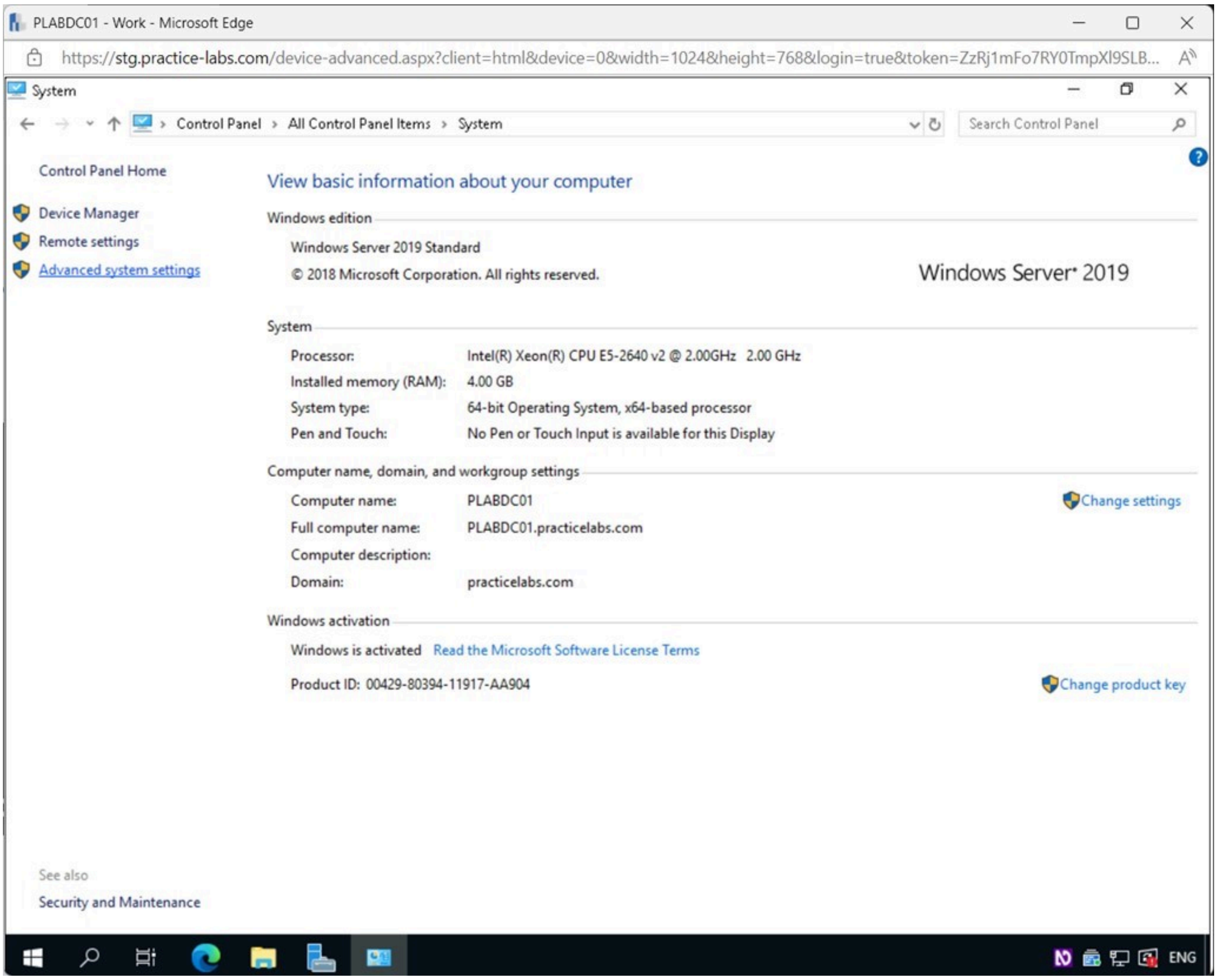

Figure 1.51 Screenshot of PLABDC01: Displaying selecting the Advanced system settings link in the System window.

Step 5

In the **System Properties** window, click **Settings**.

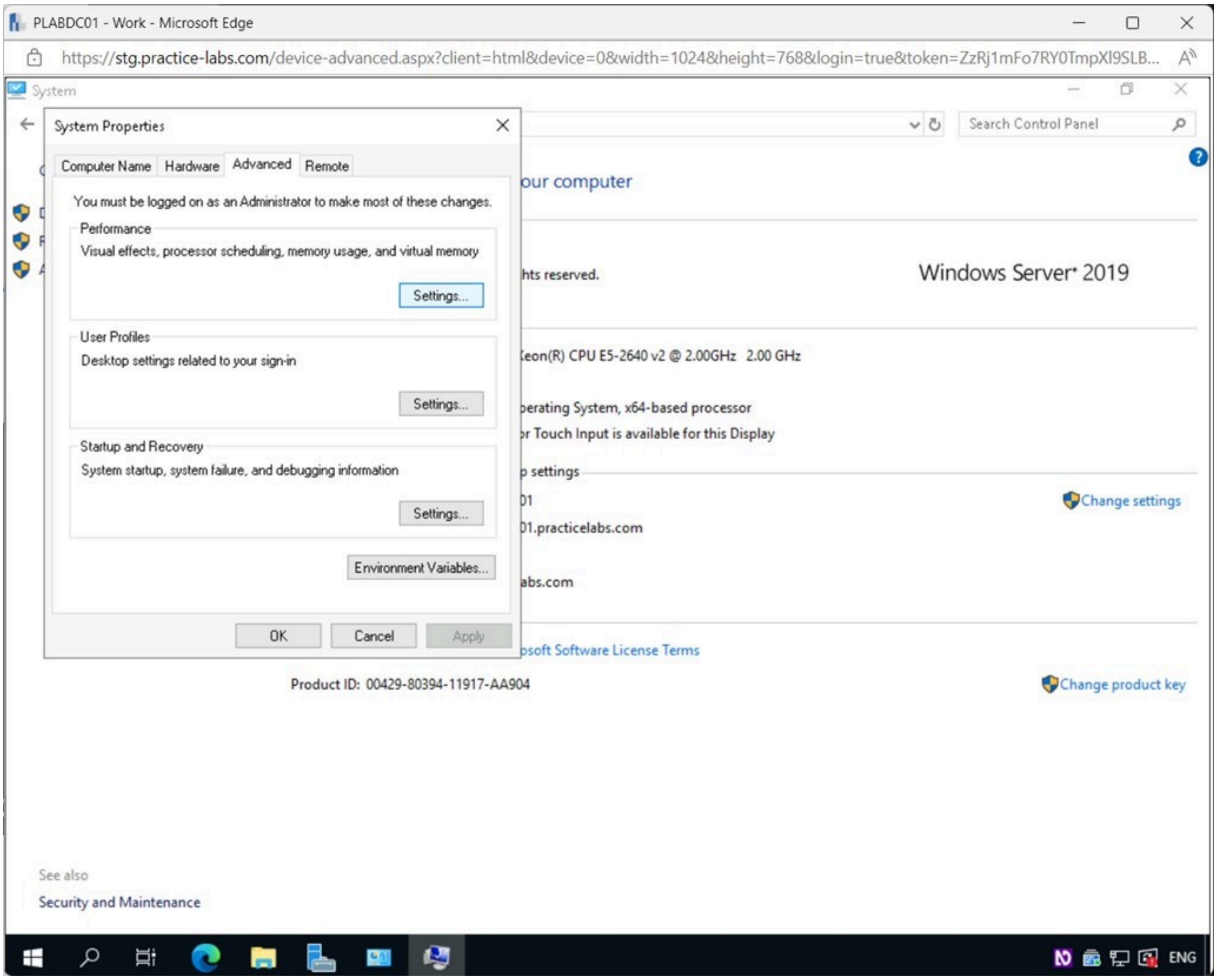

Figure 1.52 Screenshot of PLABDC01: Displaying selecting Settings in the System Properties - Advanced tab.

Step 6

In the **Performance Options** pop-up window, select the **Advanced** tab.

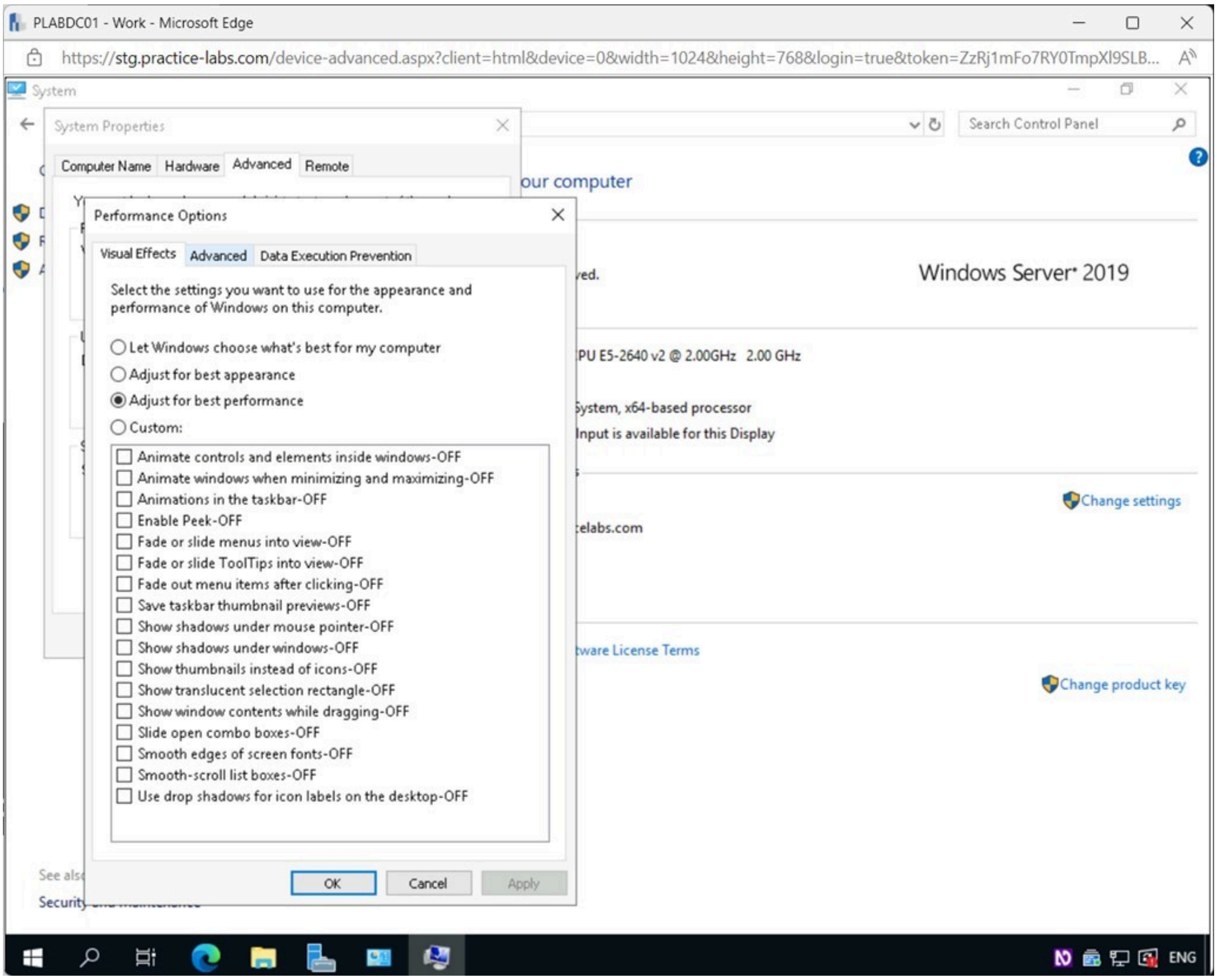

Figure 1.53 Screenshot of PLABDC01: Displaying selecting the Advanced tab in the Performance Options pop-up window.

Step 7

In the **Performance Options - Advanced** tab, click **Change** under the **Virtual memory** section.

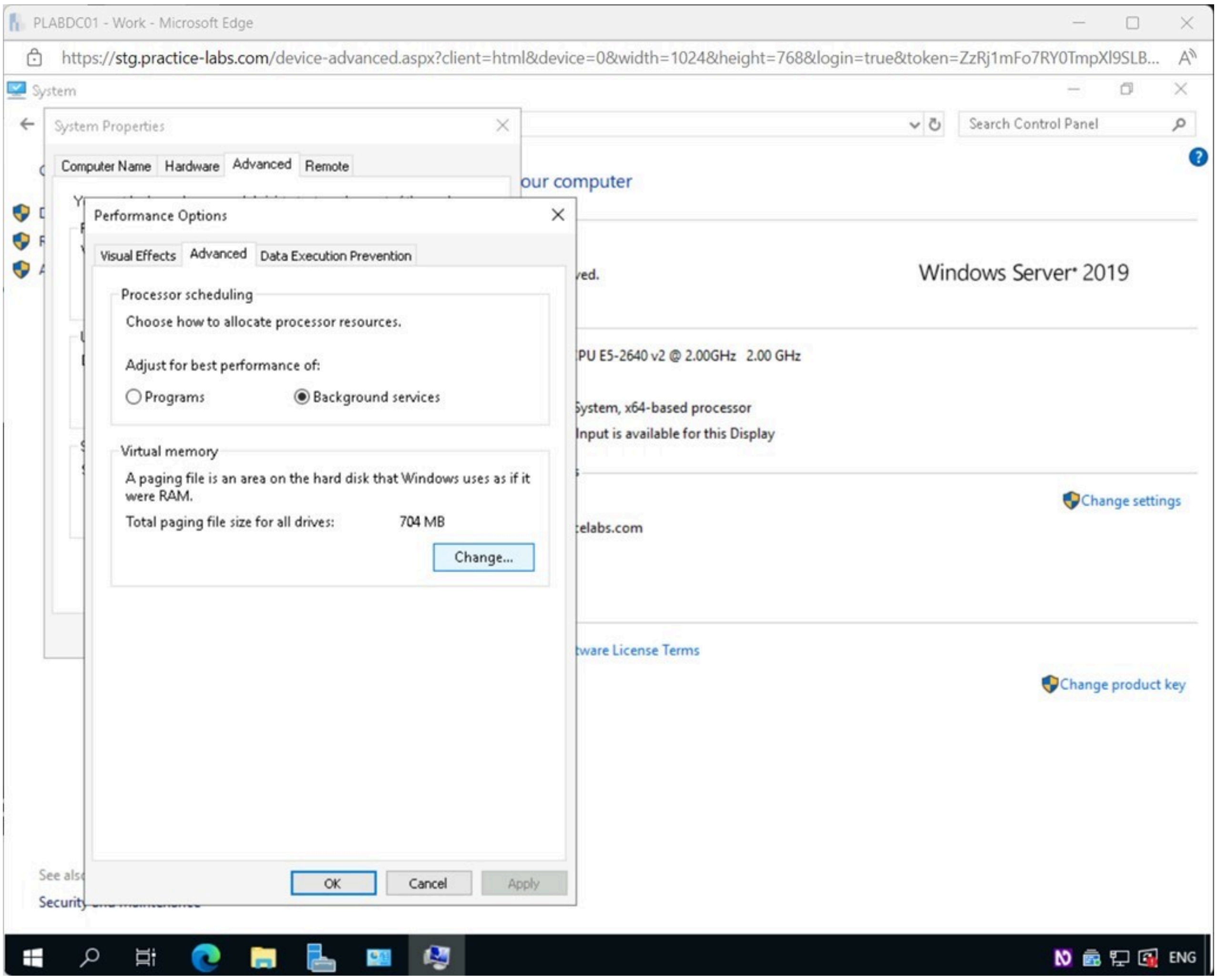

Figure 1.54 Screenshot of PLABDC01: Displaying selecting Change in the Performance Options - Advanced tab.

Step 8

In the **Virtual Memory** pop-up window, untick the **Automatically manage paging file size for all drives** checkbox.

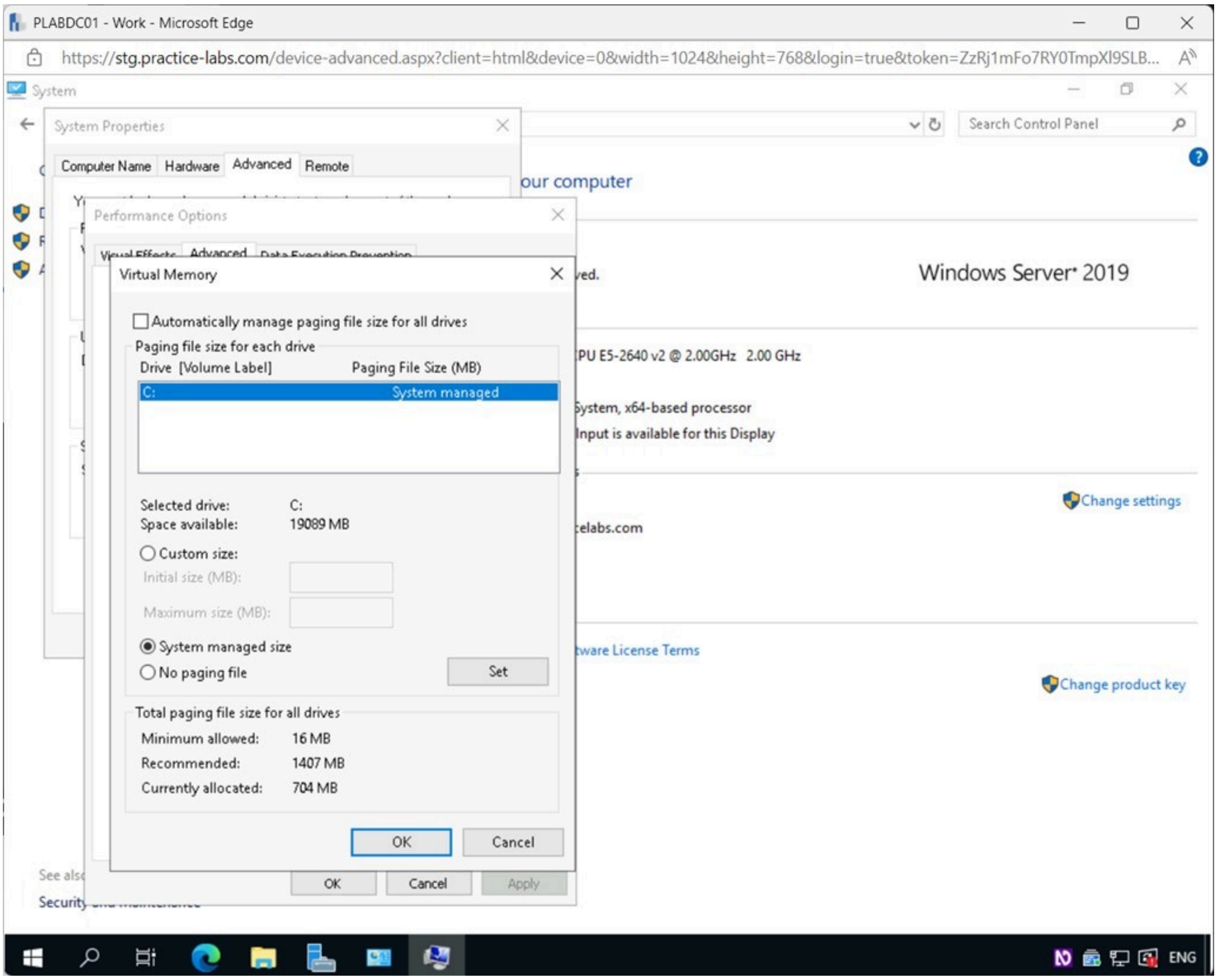

Figure 1.55 Screenshot of PLABDC01: Displaying the Virtual Memory pop-up window, showing the required settings performed.

Step 9

Choose the **Custom size** option and type the following:

```
Initial size(MB):  200
Maximum size(MB):  1407
```

Click **Set**.

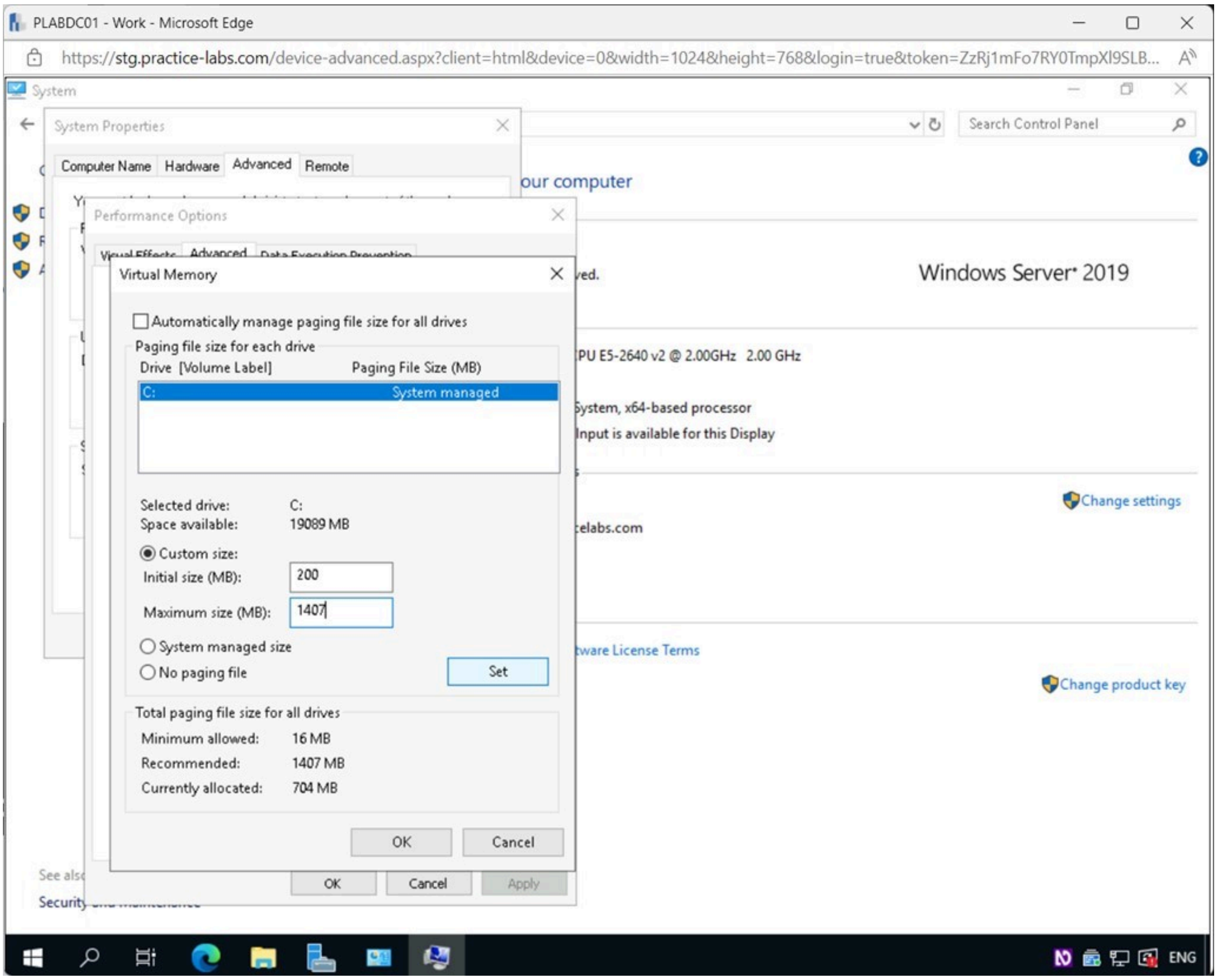

Figure 1.56 Screenshot of PLABDC01: Displaying the Virtual Memory pop-up window, showing the required settings performed and the Set button selected.

> ***Note:*** *Notice the recommended sizes under the* ***Total paging file size for all drives*** *section are 16MB and 1407MB. We chose 200MB as the initial size because anything smaller can cause system errors and affect logging.*

Step 10

Click **OK**.

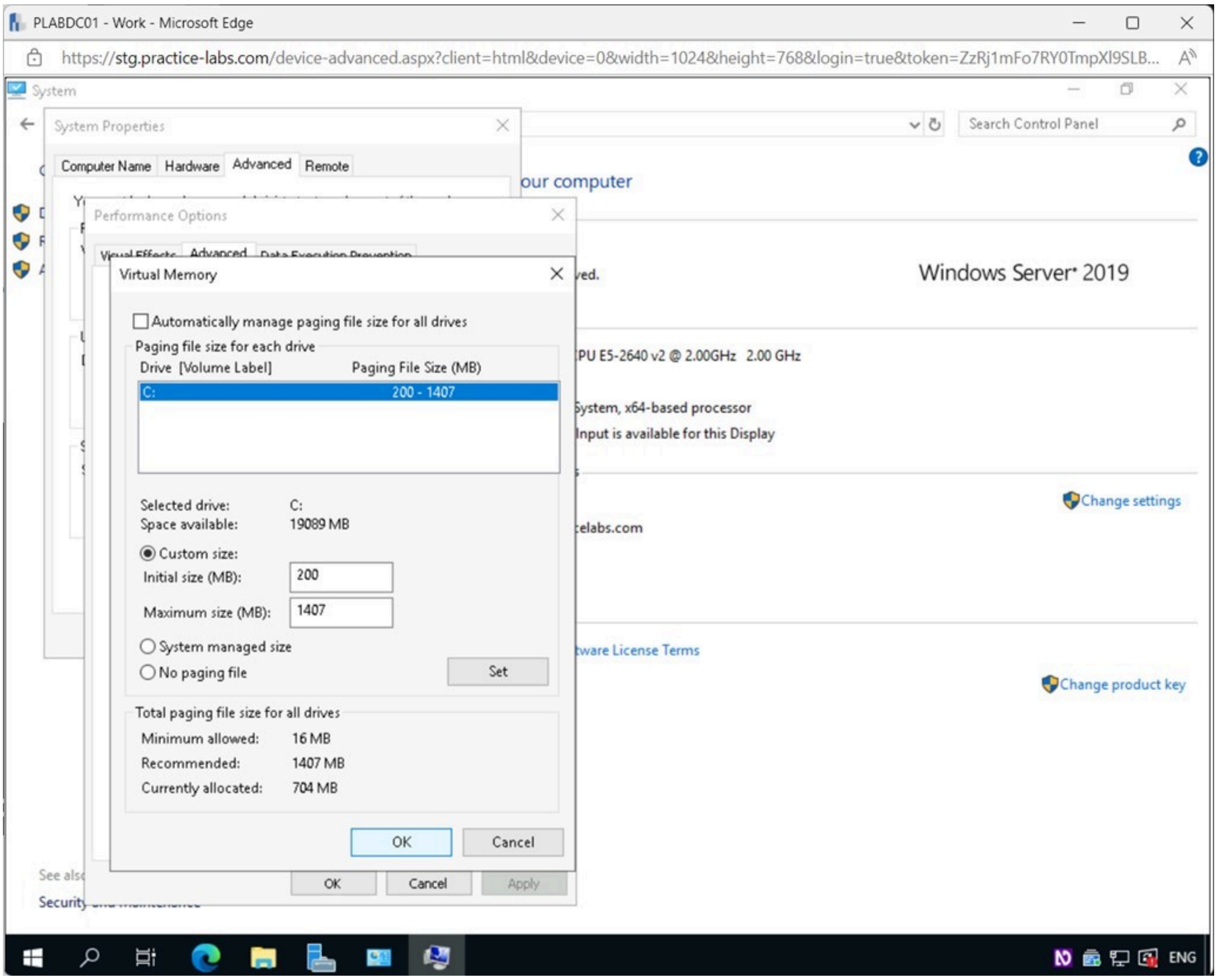

Figure 1.57 Screenshot of PLABDC01: Displaying selecting OK in the Virtual Memory pop-up window.

Step 11

Click **OK** on the **System Properties** pop-up window.

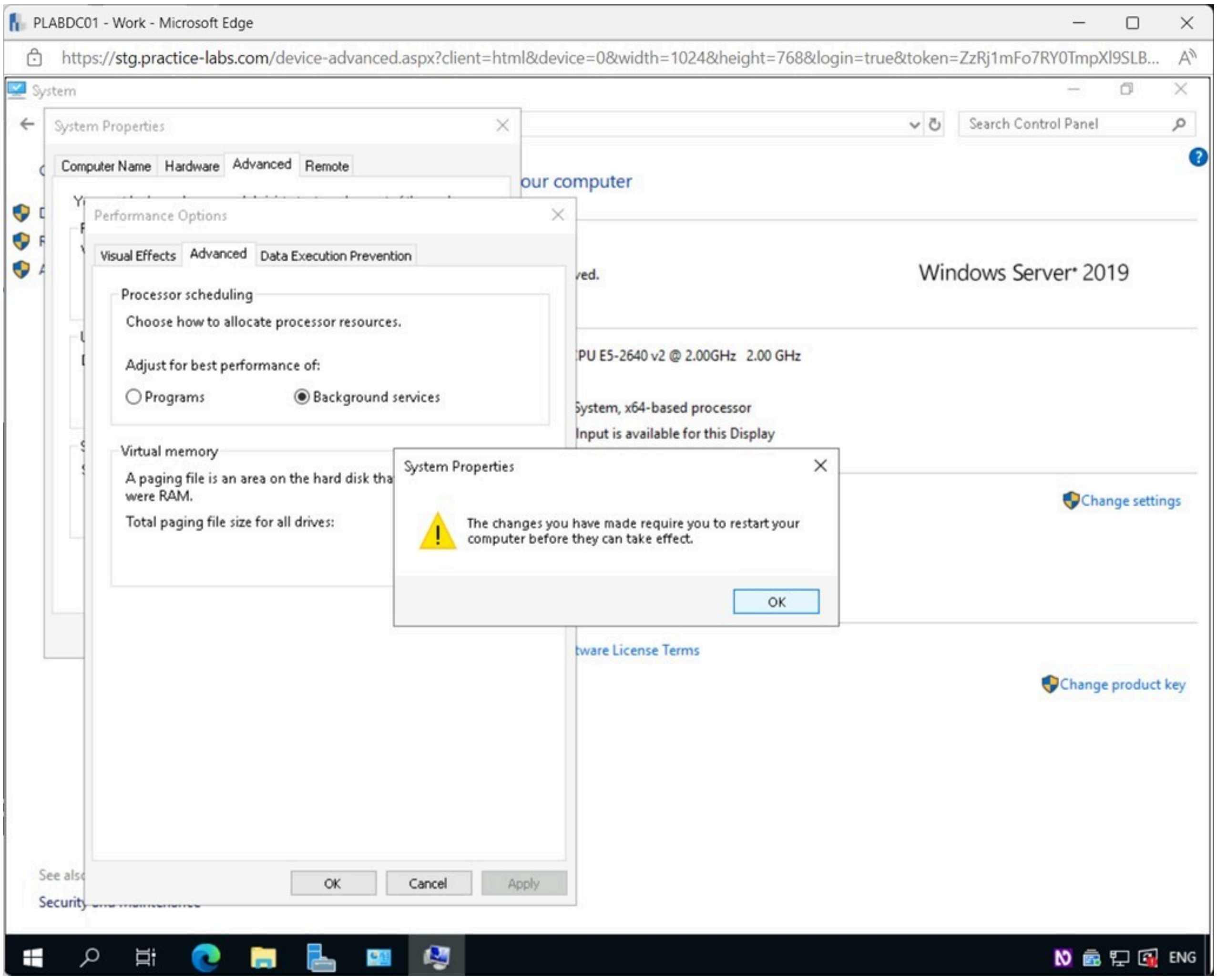

Figure 1.58 Screenshot of PLABDC01: Displaying selecting OK in the System Properties pop-up window.

Step 12

Click **OK** on the **Performance Options** window.

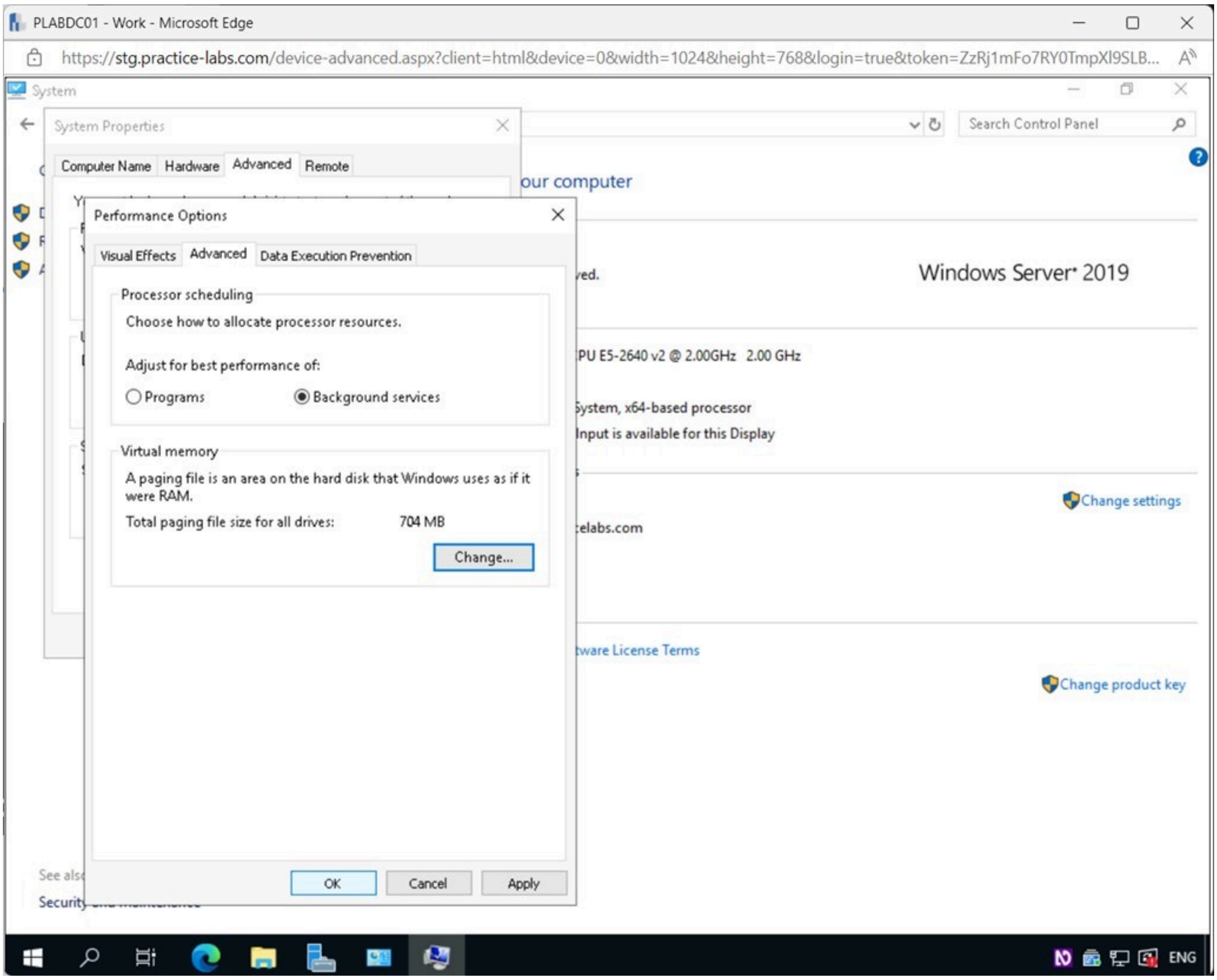

Figure 1.59 Screenshot of PLABDC01: Displaying selecting OK in the Performance Options window.

Step 13

Click **OK** on the **System Properties** window.

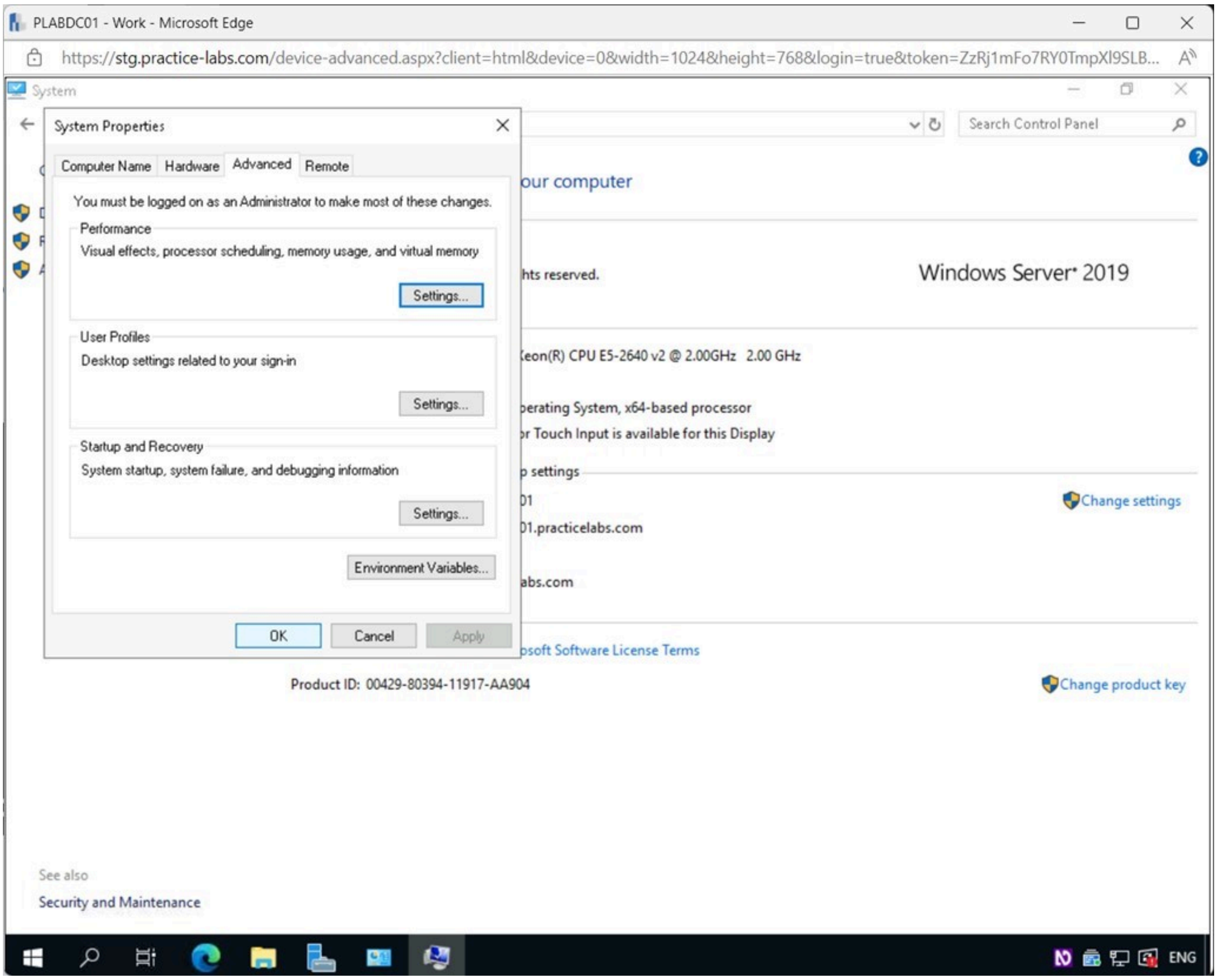

Figure 1.60 Screenshot of PLABDC01: Displaying selecting OK in the System Properties window.

Step 14

In the **Microsoft Windows** pop-up window, click **Restart Later**.

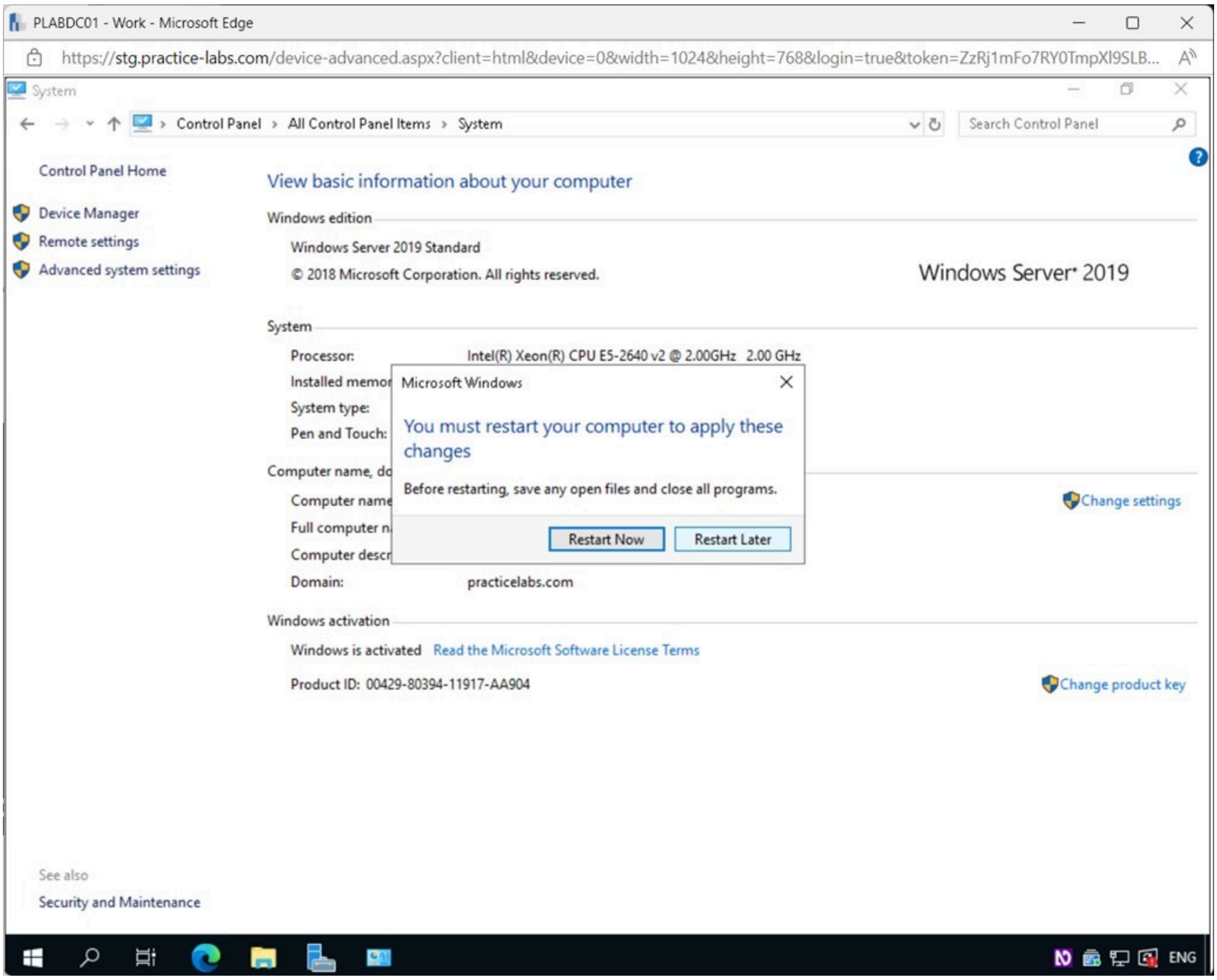

Figure 1.61 Screenshot of PLABDC01: Displaying selecting Restart Later on the Microsoft Windows pop-up window.

Step 15

Close the **System** window.

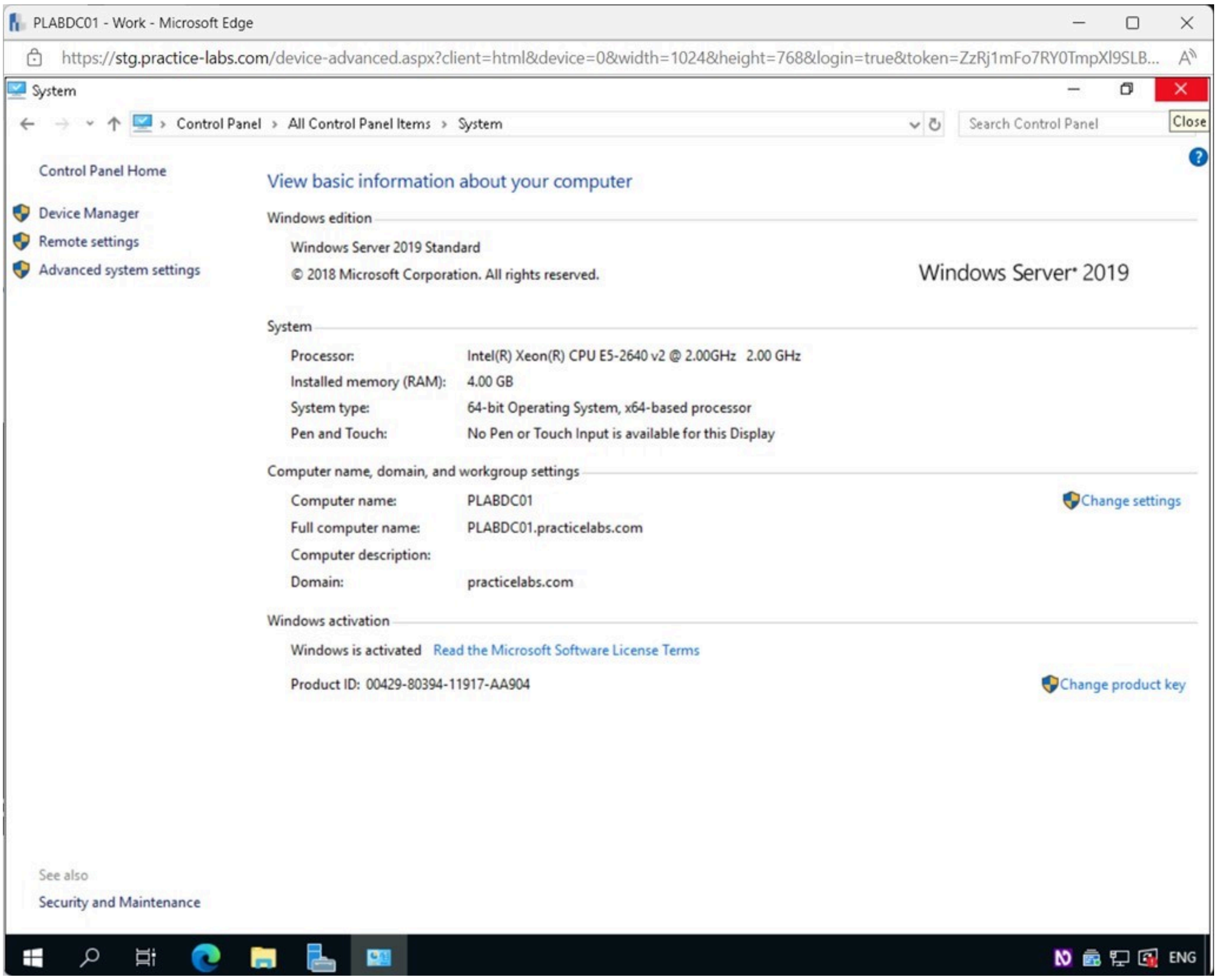

Figure 1.62 Screenshot of PLABDC01: Displaying closing the System window.

Step 16

Connect to **PLABUBUNTU**.

Click the **Terminal** icon on the **favorites** bar on the left side of the screen.

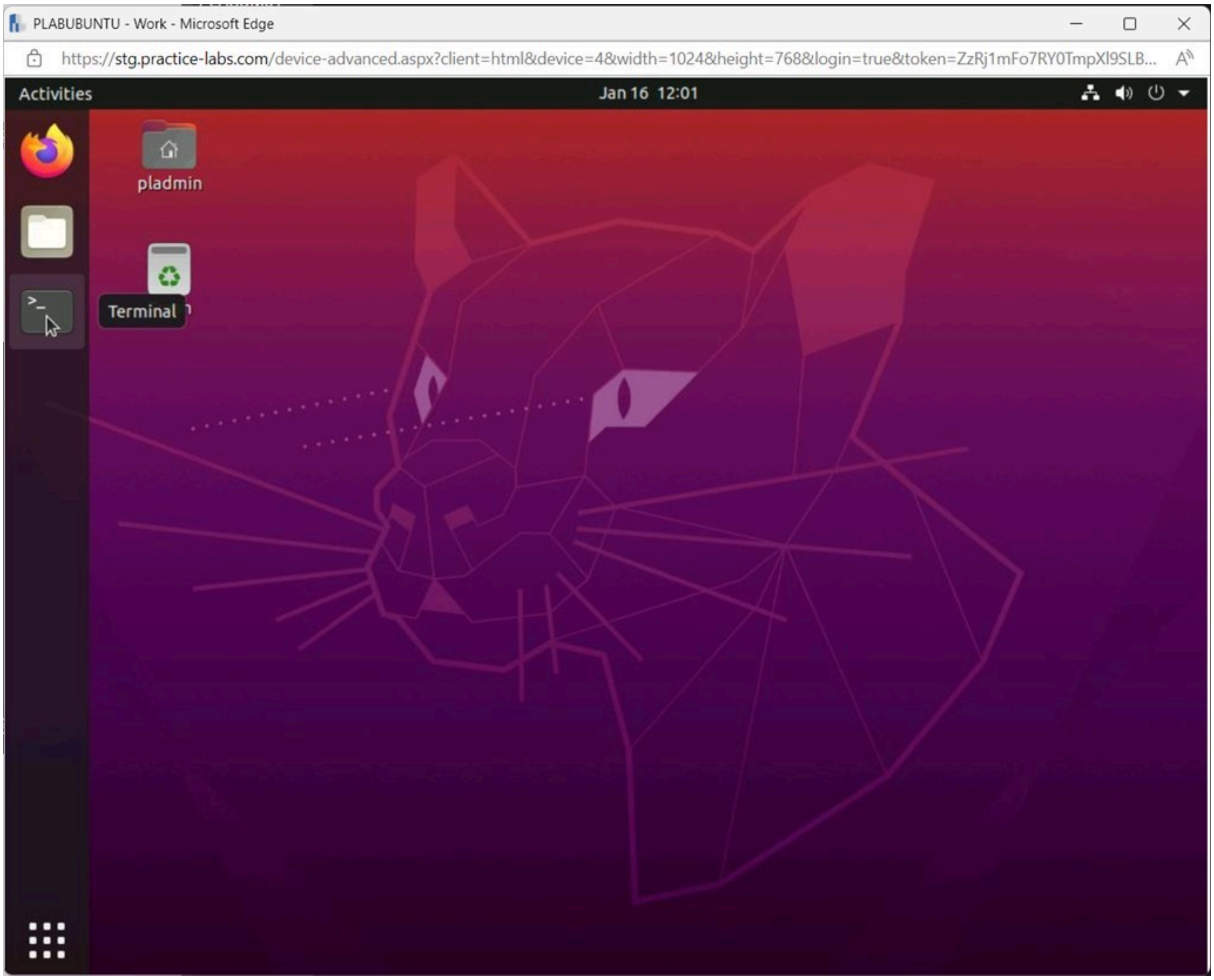

Figure 1.63 Screenshot of PLABUBUNTU: Displaying selecting Terminal in the PLABUBUNTU desktop.

Step 17

In the **Terminal** window, type the following command and press **Enter**:

```
swapon -s
```

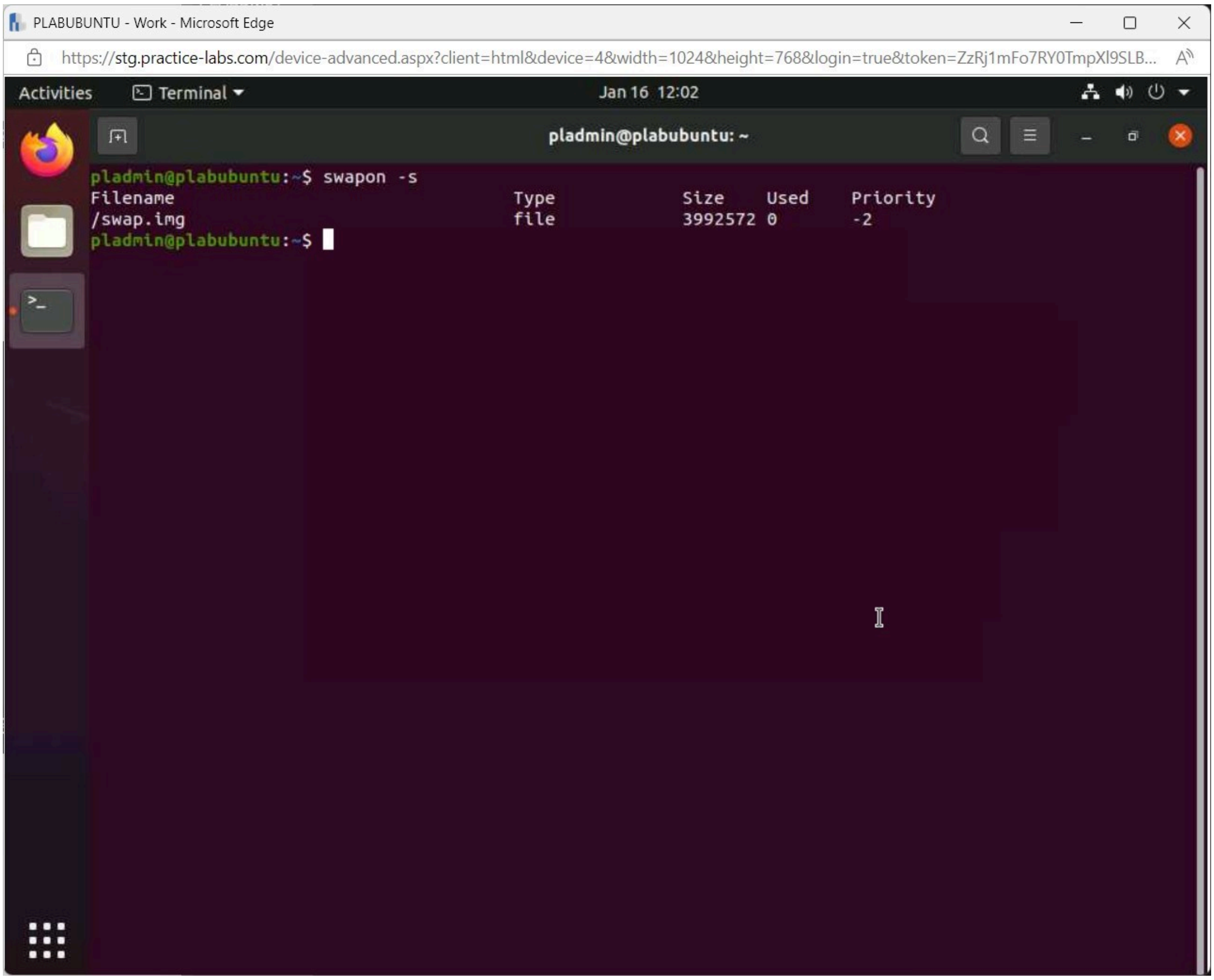

Figure 1.64 Screenshot of PLABUBUNTU: Displaying the Terminal window, showing the required command entered and executed.

Note: *The command executes, and we can see some basic information about the swap file.*

Step 18

Type the following and press **Enter**:

```
free -m
```

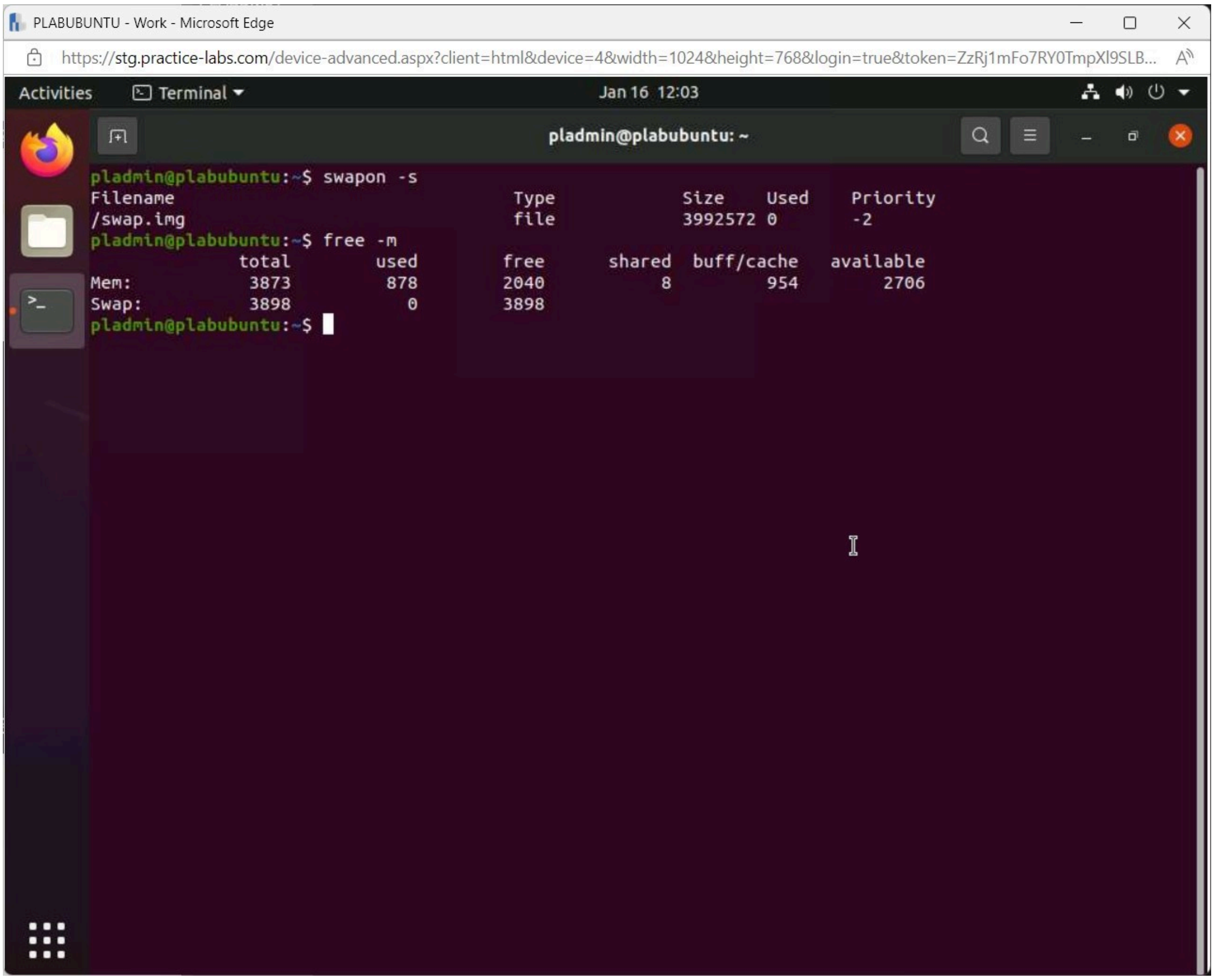

Figure 1.65 Screenshot of PLABUBUNTU: Displaying the Terminal window, showing the required command entered and executed.

> ***Note:*** *The command executes, and we can see some more basic information about the swap file.*

Step 19

Close the **Terminal** window.

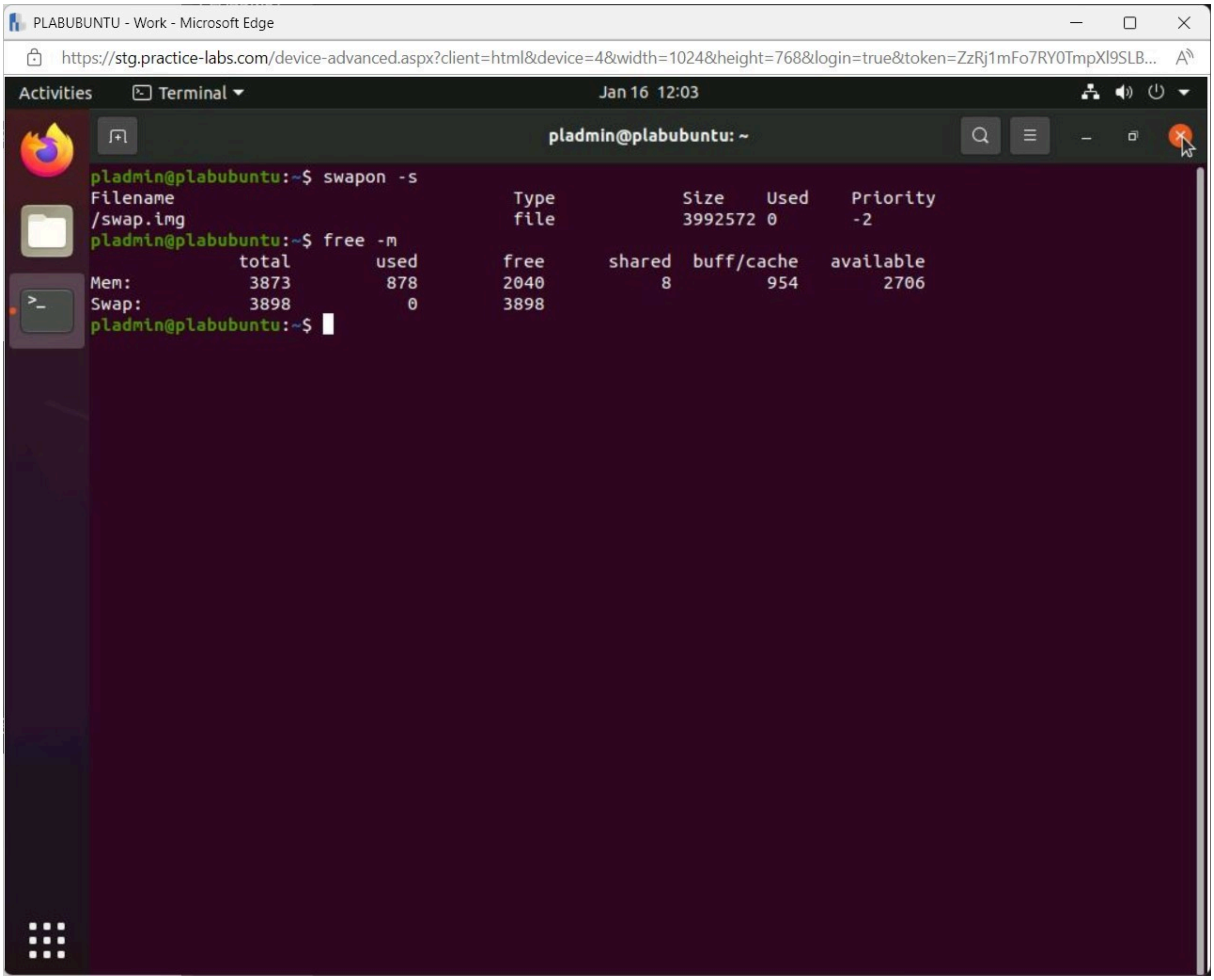

Figure 1.66 Screenshot of PLABUBUNTU: Displaying closing the Terminal window.

Task 6 - Caching Issues

A cache is a quick memory buffer that stores recently accessed data for later use. This helps deal with data requests more quickly than if they were to retrieve the data from its primary storage location. Data that has already been accessed can be cached for later use. The data in a cache can be utilized using software or fast-access hardware, such as RAM. A cache is a temporary storage area that is used to speed up data access before it is transferred to a more permanent location. A cache's quickness comes at the cost of only storing a subset of data for a limited amount of time.

In this task, you will view where the cache settings can be configured.

Step 1

Connect to **PLABDC01**.

Right-click the Windows **Start** charm and select **Device Manager**.

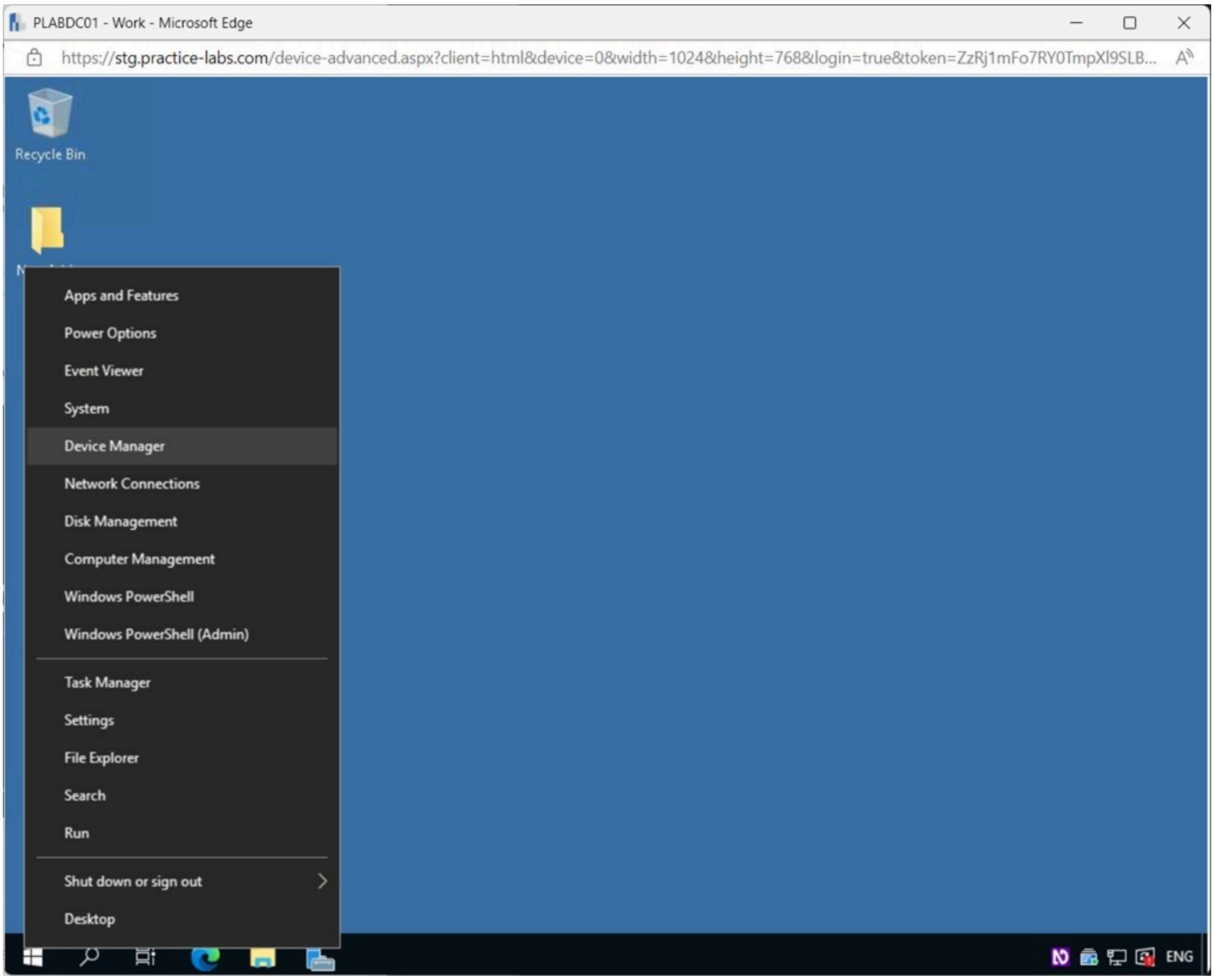

Figure 1.67 Screenshot of PLABDC01: Displaying right-clicking the Windows Start Charm and selecting Device Manager.

Step 2

In the **Device Manager** window, expand **Disk drives**.

Right-click on **Microsoft Virtual Disk** and select **Properties**.

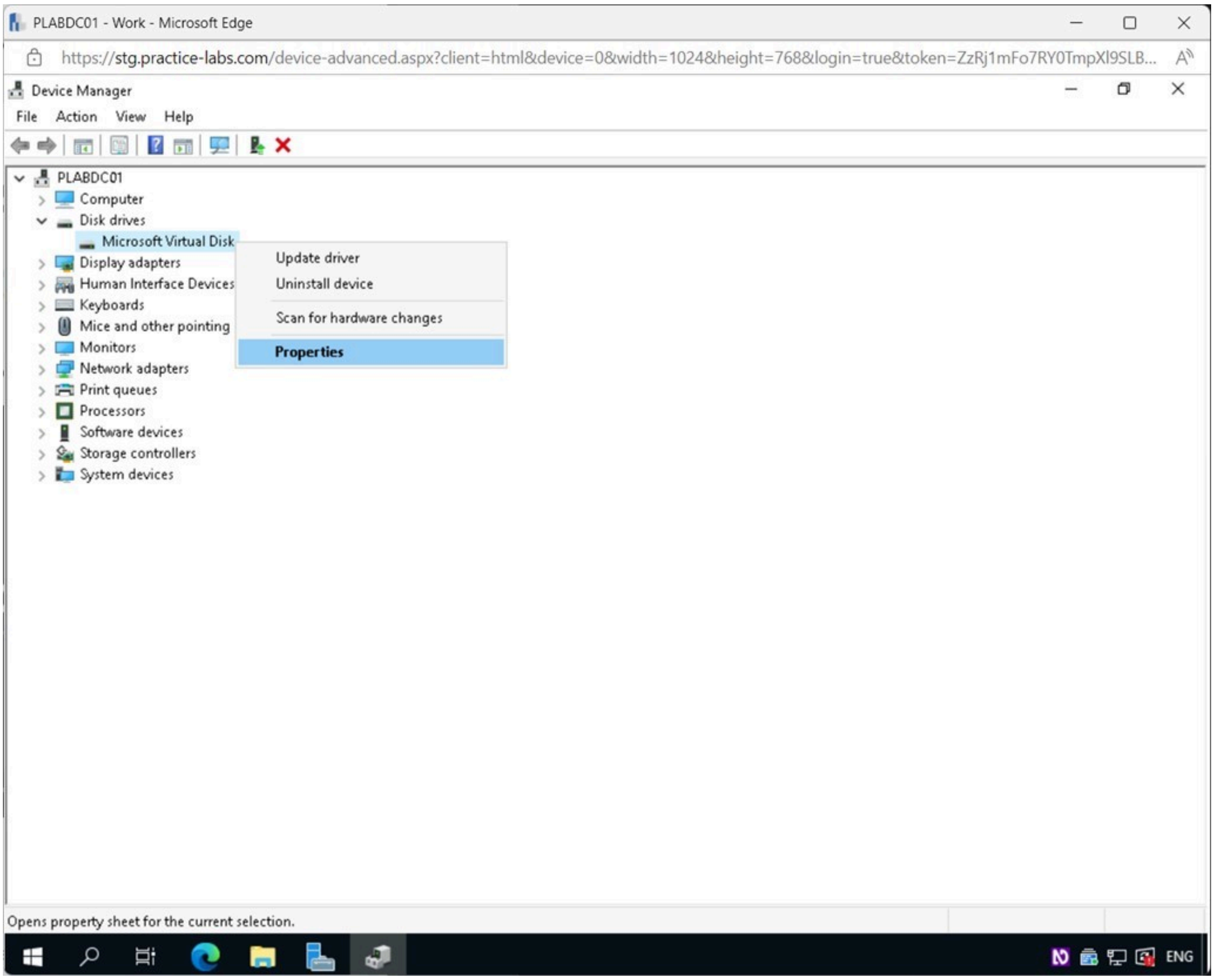

Figure 1.68 Screenshot of PLABDC01: Displaying the Device Manager window. Microsoft Virtual Disk is right-clicked, and Properties selected.

Step 3

In the **Microsoft Virtual Disk Properties** window, select the **Policies** tab.

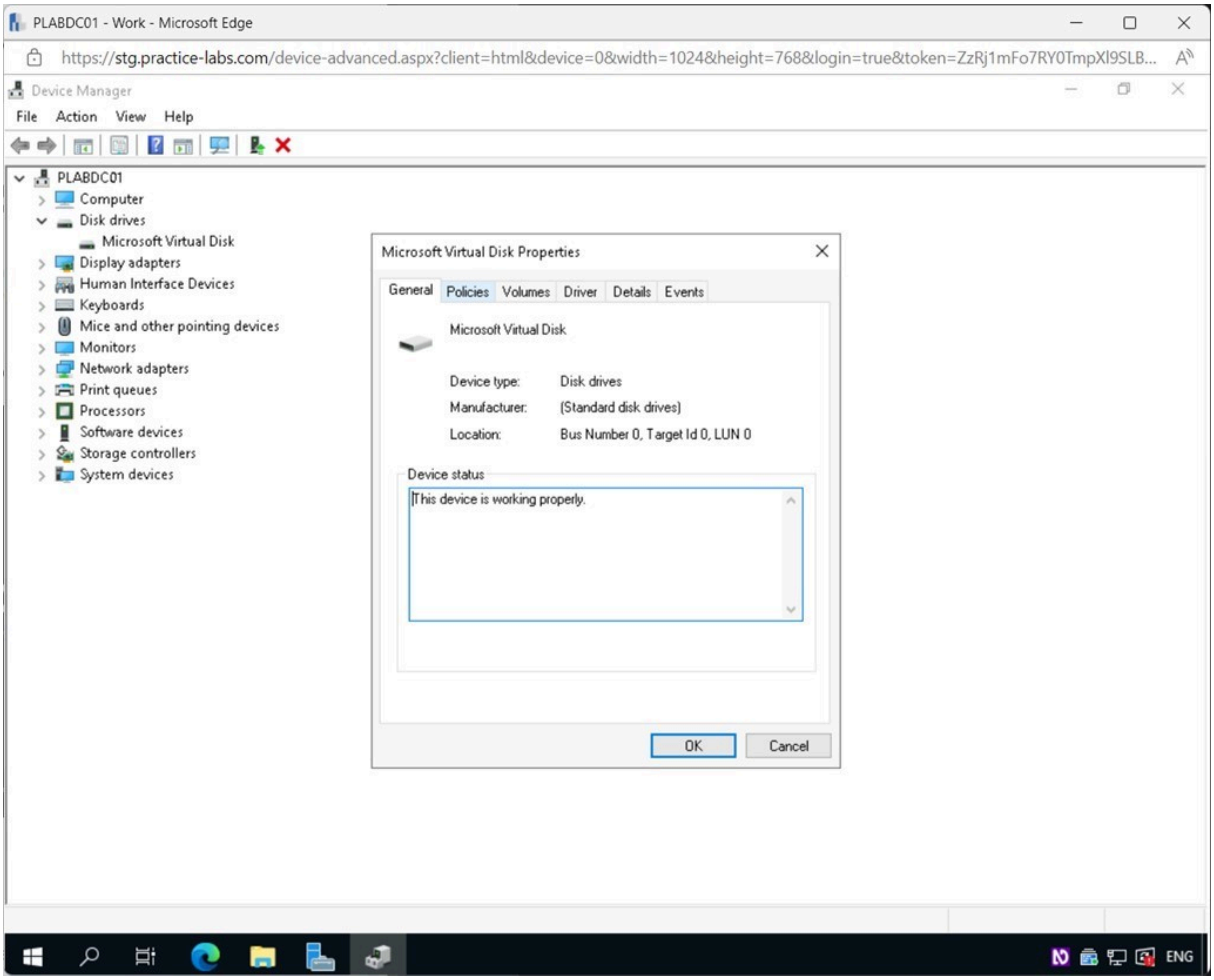

Figure 1.69 Screenshot of PLABDC01: Displaying selecting the Policies tab in the Microsoft Virtual Disk Properties window.

Step 4

In the **Microsoft Virtual Disk Properties - Policies** tab, caching can be turned on and off.

> ***Note:*** *Everything is correctly configured here for caching to be used. Notice that an additional feature to help save data can be enabled but is not recommended unless separate power supplies are used.*

Click **OK**.

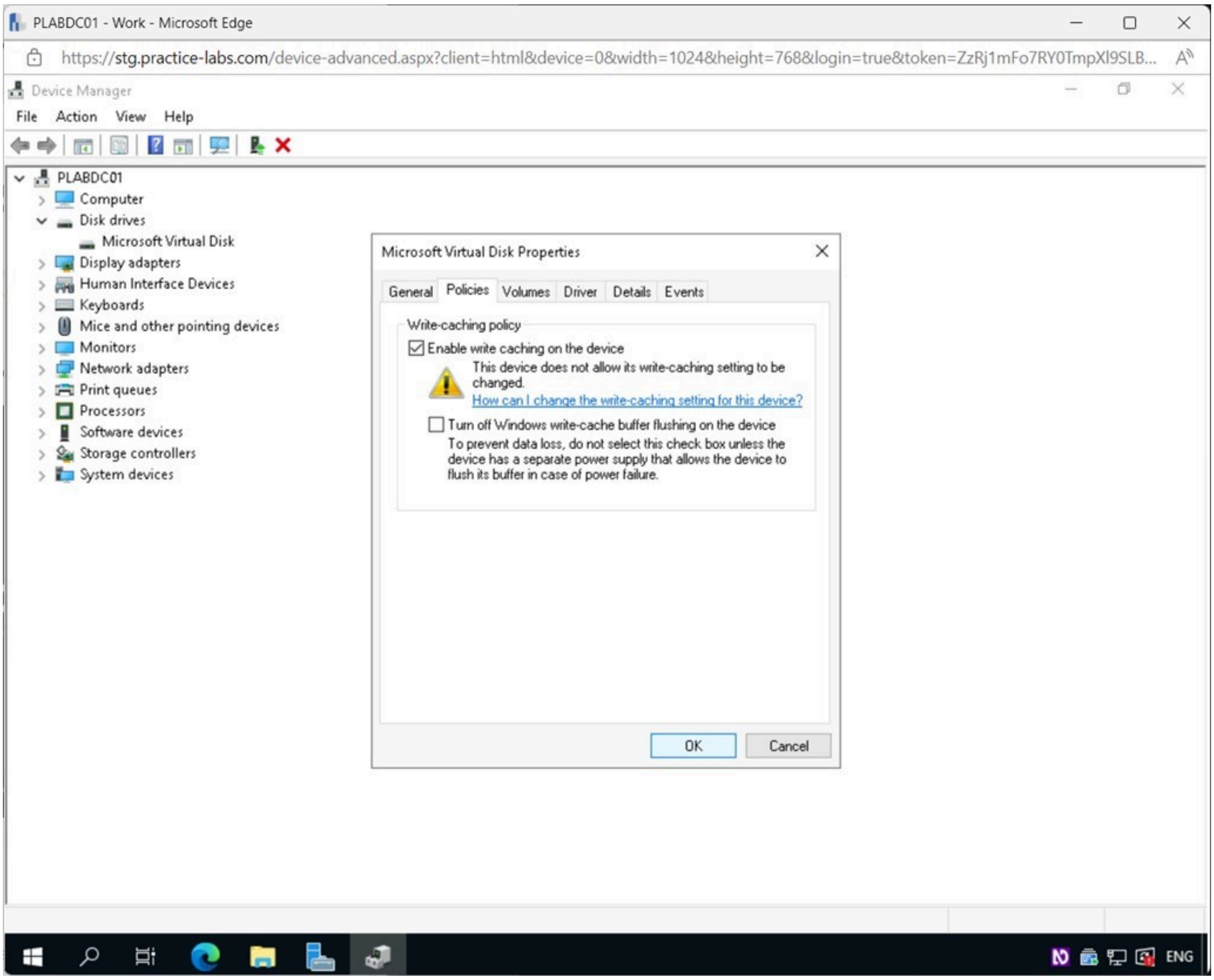

Figure 1.70 Screenshot of PLABDC01: Displaying selecting OK in the Microsoft Virtual Disk Properties - Policies tab.

Step 5

Close the **Device Manager** window.

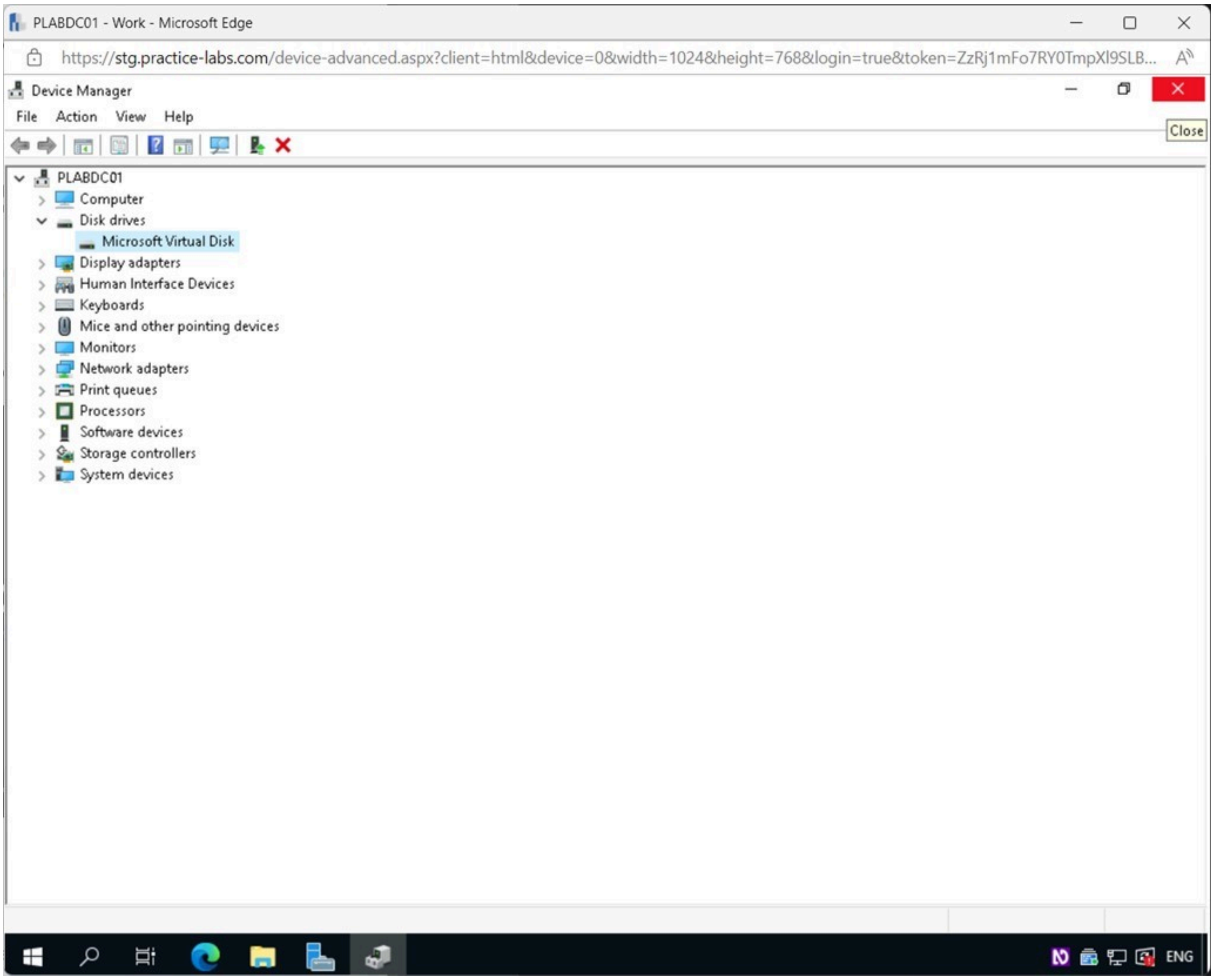

Figure 1.71 Screenshot of PLABDC01: Displaying closing the Device Manager window.

Exercise 2 - Tools and Techniques for Troubleshooting

Operating systems usually have built-in capabilities that can be used to help diagnose issues and get to resolutions faster.

In this exercise, common tools & techniques for troubleshooting storage issues will be discussed.

Learning Outcomes

After completing this exercise, you should be able to:

- Use the Check Disk (chkdsk) Tool
- Explore the System File Checker (SFC) Tool
- Use the Event Viewer

- Know about Monitoring Tools

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01 -** (Windows Server 2019 - Domain Controller)

Task 1 - Check Disk (chkdsk)

To ensure the logical integrity of the file system, you can use the CHKDSK command, which is accessible via the Command Prompt. Check Disk (or CHKDSK) is the program's full name and does what its name implies: it checks the disk for problems like bad sectors and attempts to rectify them. CHKDSK creates and displays the disk status based on the partition file system.

In this task, you will run chkdsk and look at the options that can be used with it.

Step 1

Connect to **PLABDC01**.

Click the Windows **Start** charm and type the following:

```
cmd
```

Select **Command Prompt** from the **Best match** pop-up menu.

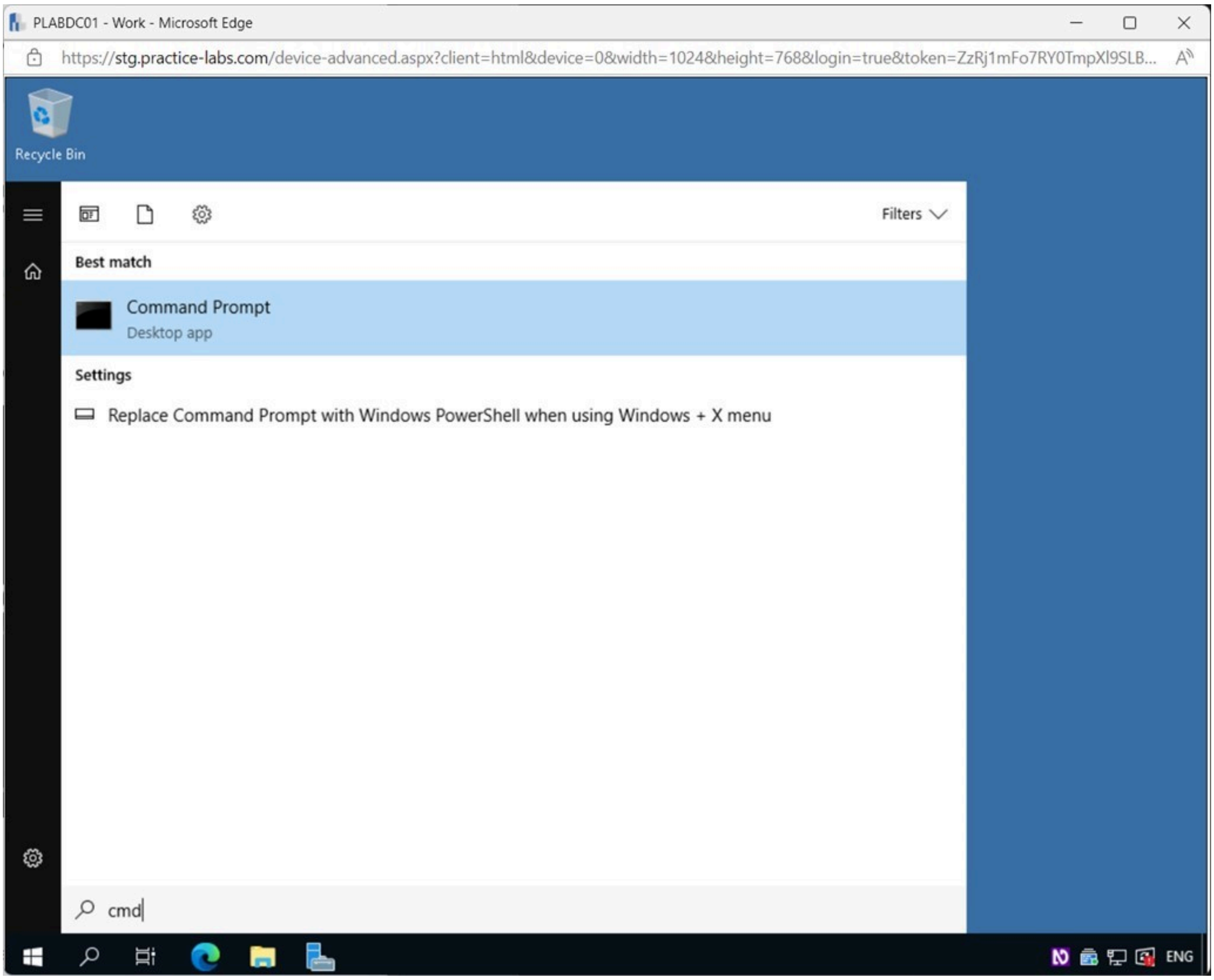

Figure 2.1 Screenshot of PLABDC01: Displaying selecting Command Prompt from the Best match pop-up menu.

Step 2

In the **Command Prompt** window, type the following command and press **Enter**:

```
chkdsk /?
```

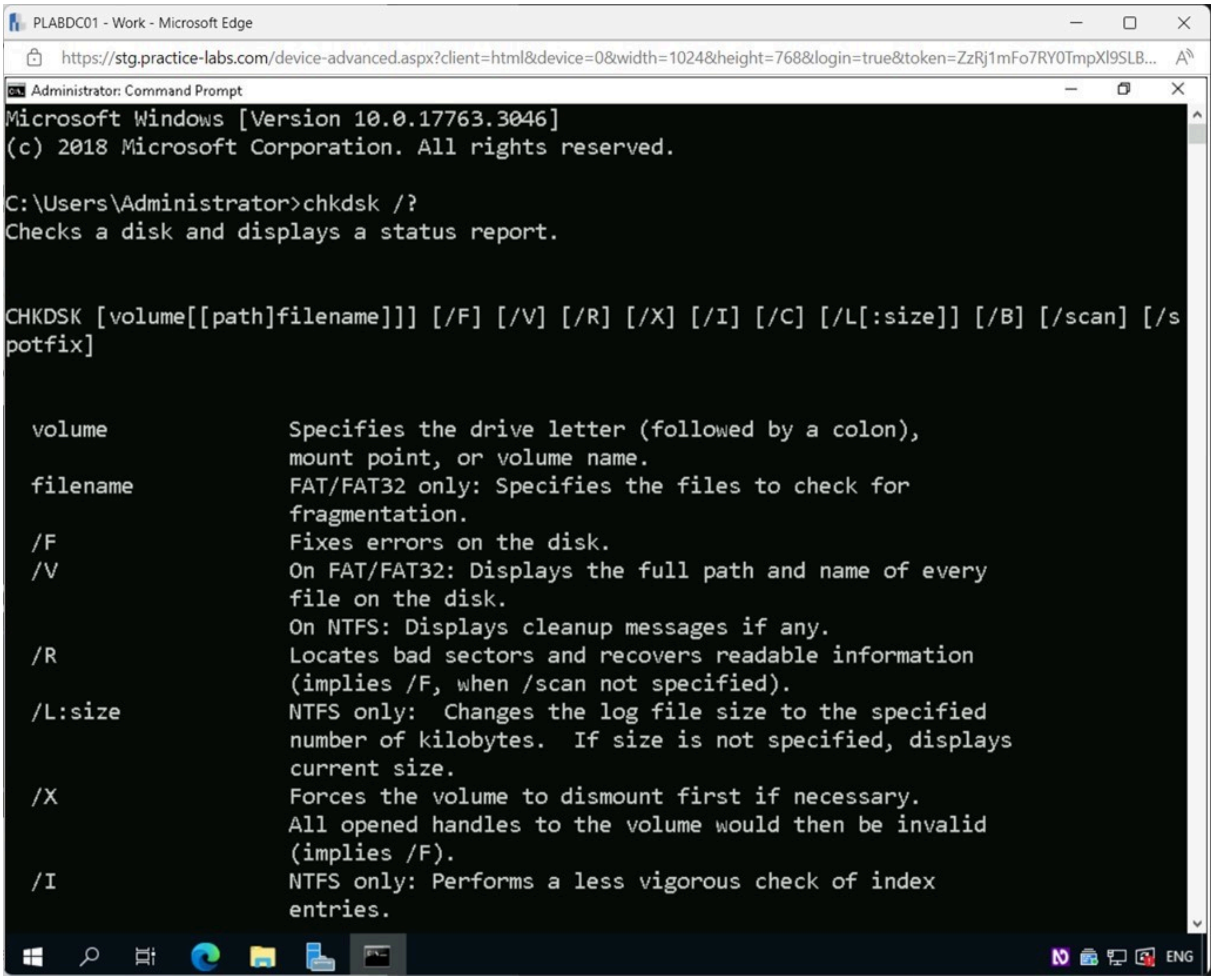

Figure 2.2 Screenshot of PLABDC01: Displaying the Command Prompt window, showing the required command entered and executed.

> ***Note:*** *The help switch parameter (/?) used with the chkdsk command shows help information on how to use the command and the different parameters that can be used with it.*

Step 3

Type the following command and press **Enter**:

```
chkdsk
```

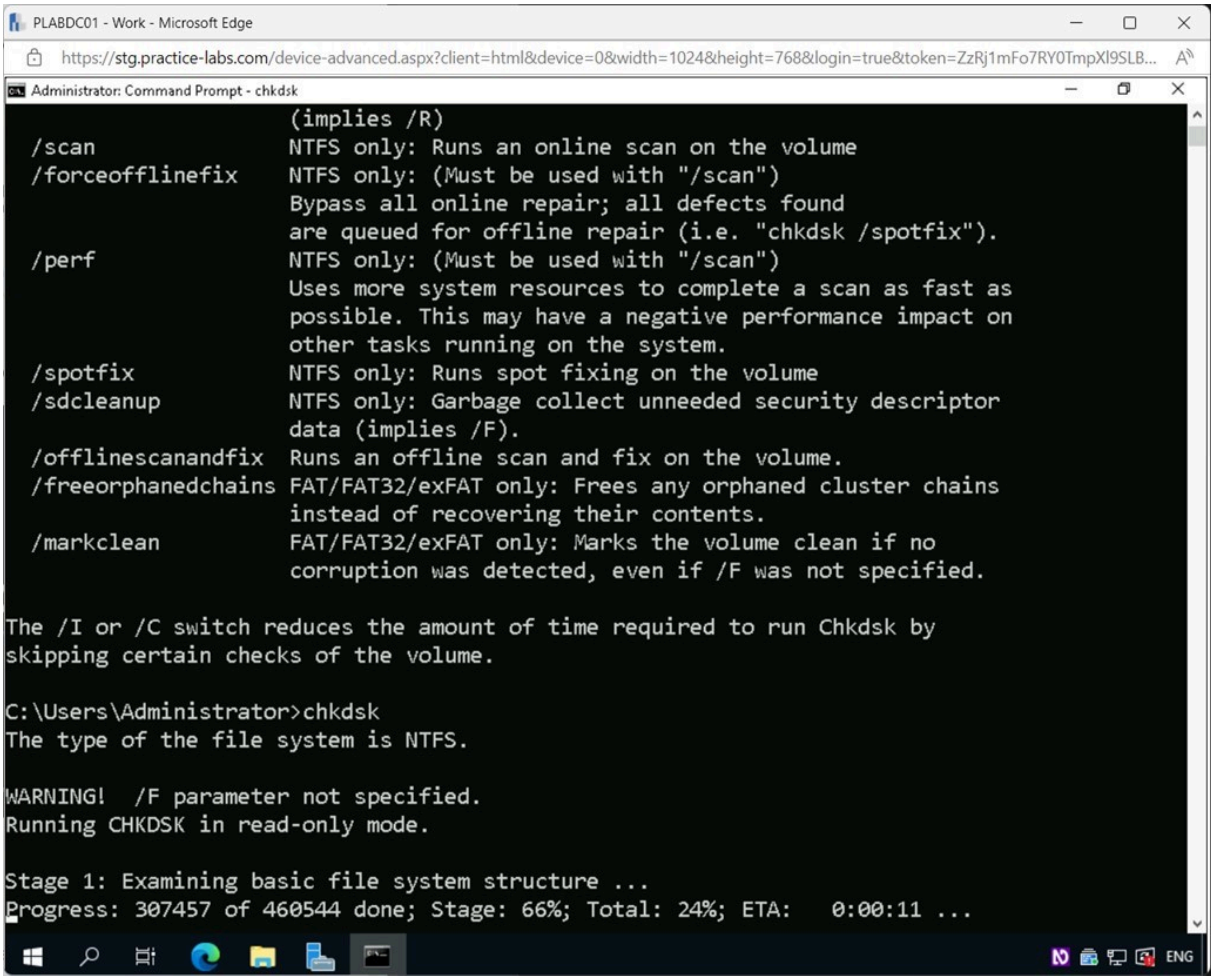

Figure 2.3 Screenshot of PLABDC01: Displaying the Command Prompt window, showing the required command entered and executed.

> ***Note:*** *The command attempts to find any bad sectors on the disk and repair them.*

Step 4

Next, type the following command to clear the **Command Prompt** window:

```
cls
```

Press **Enter.**

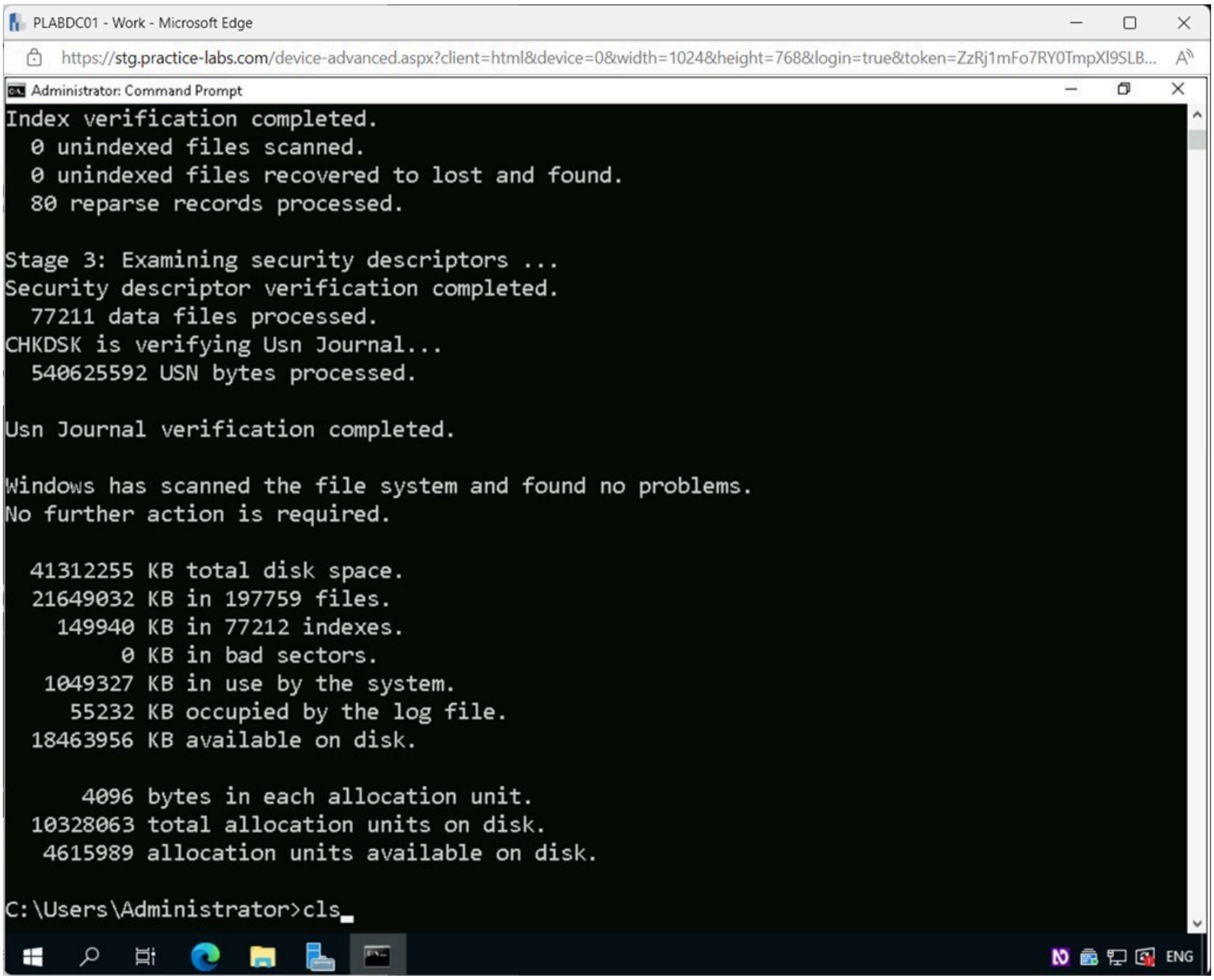

Figure 2.4 Screenshot of PLABDC01: Displaying the Command Prompt window, showing the required command entered.

Keep the **Command Prompt** window open.

Task 2 - System File Checker (SFC)

All recent versions of Windows have the System File Checker (SFC) utility. This program can be used to fix damaged Windows system files. SFC may be launched with administrative capabilities either from within Windows itself or through the Windows recovery environment.

In this task, you will look at the different options that can be run with the sfc command. Due to restrictions in the lab environment, you will not actually execute the commands.

Step 1

Ensure you are connected to **PLABDC01** and the **Command Prompt** window is open.

Type the following command and press **Enter**:

```
sfc /?
```

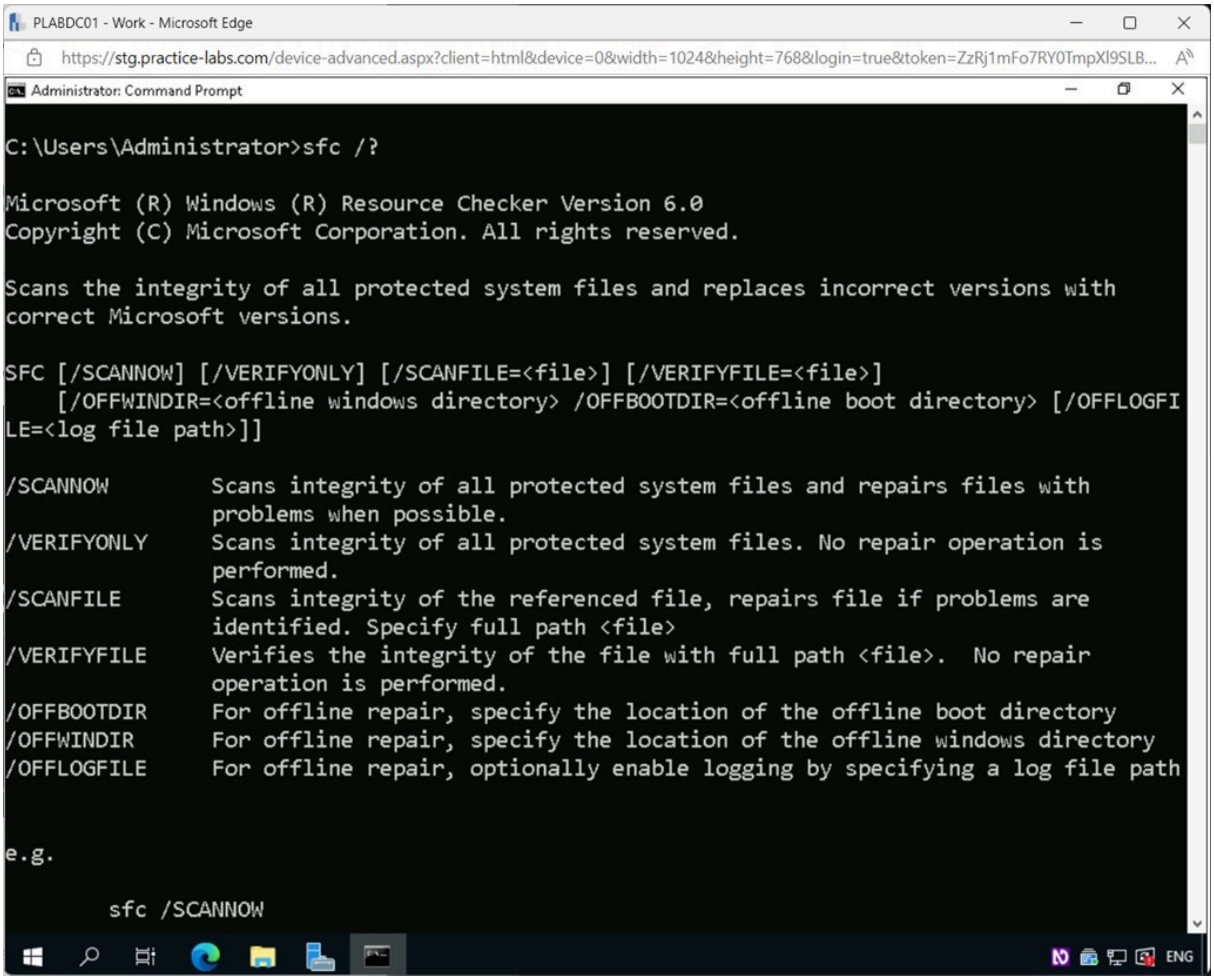

Figure 2.5 Screenshot of PLABDC01: Displaying the Command Prompt window, showing the required command entered and executed.

Note: *The help switch parameter (/?) used with the sfc command shows help information on how to use the command and the different parameters that can be used with it.*

Step 2

Close the **Command Prompt** window.

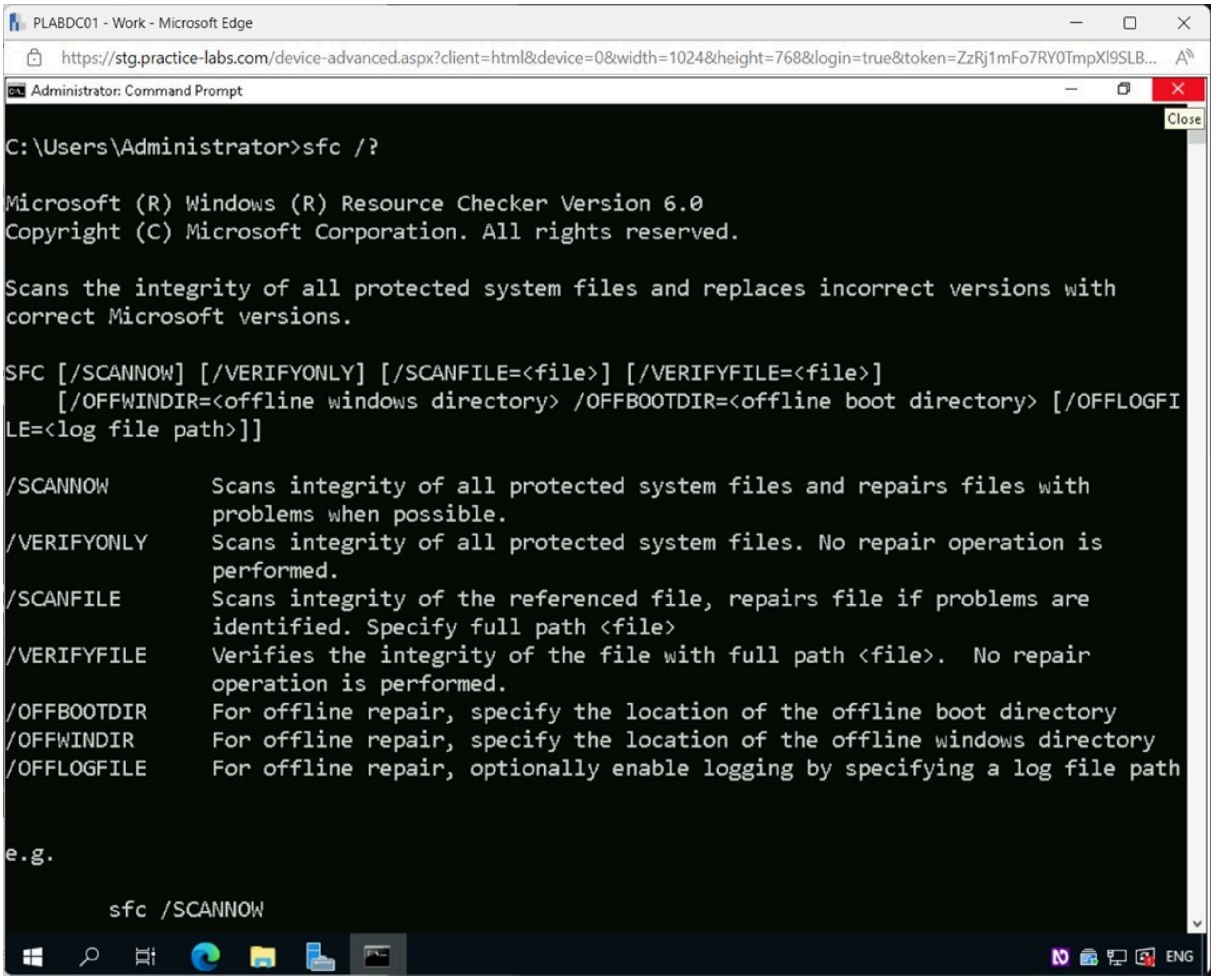

Figure 2.6 Screenshot of PLABDC01: Displaying closing the Command Prompt window.

Task 3 - Event Viewer

Information concerning application, security, system, and setup events can be seen in the Event Viewer. A user logging in, an issue occurring in the program, or an application not starting properly are examples of events. Information such as the event ID, the level of severity, and the program or service that created the log event are all displayed in the event log.

In this task, you will view the information available on the different types of Events.

Step 1

Ensure you are connected to **PLABDC01.**

Right-click the Windows **Start** charm and select **Event Viewer**.

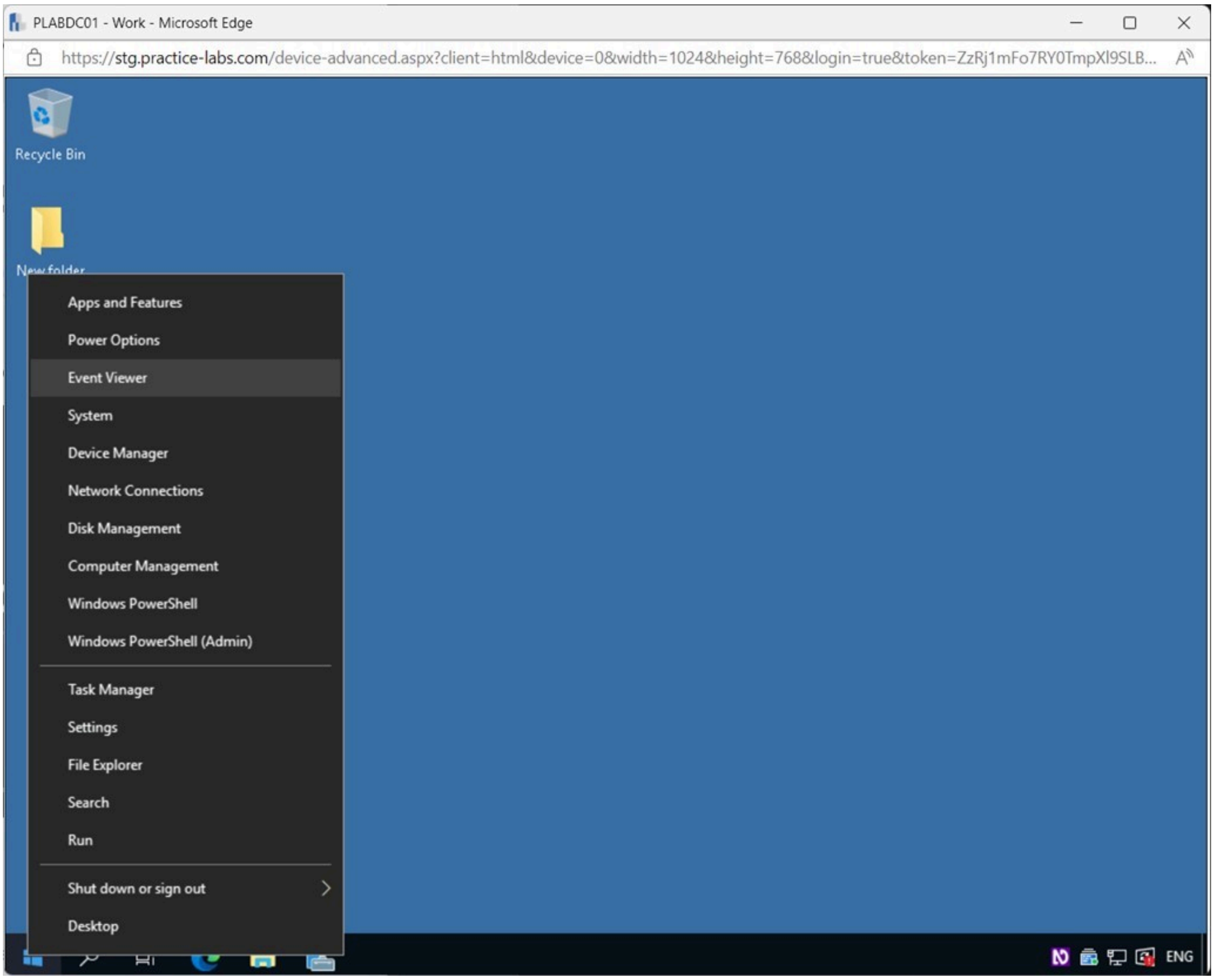

Figure 2.7 Screenshot of PLABDC01: Displaying right-clicking the Windows Start Charm and selecting Event Viewer.

Step 2

In the **Event Viewer** window, expand **Windows Logs** on the left pane.

Select **System**.

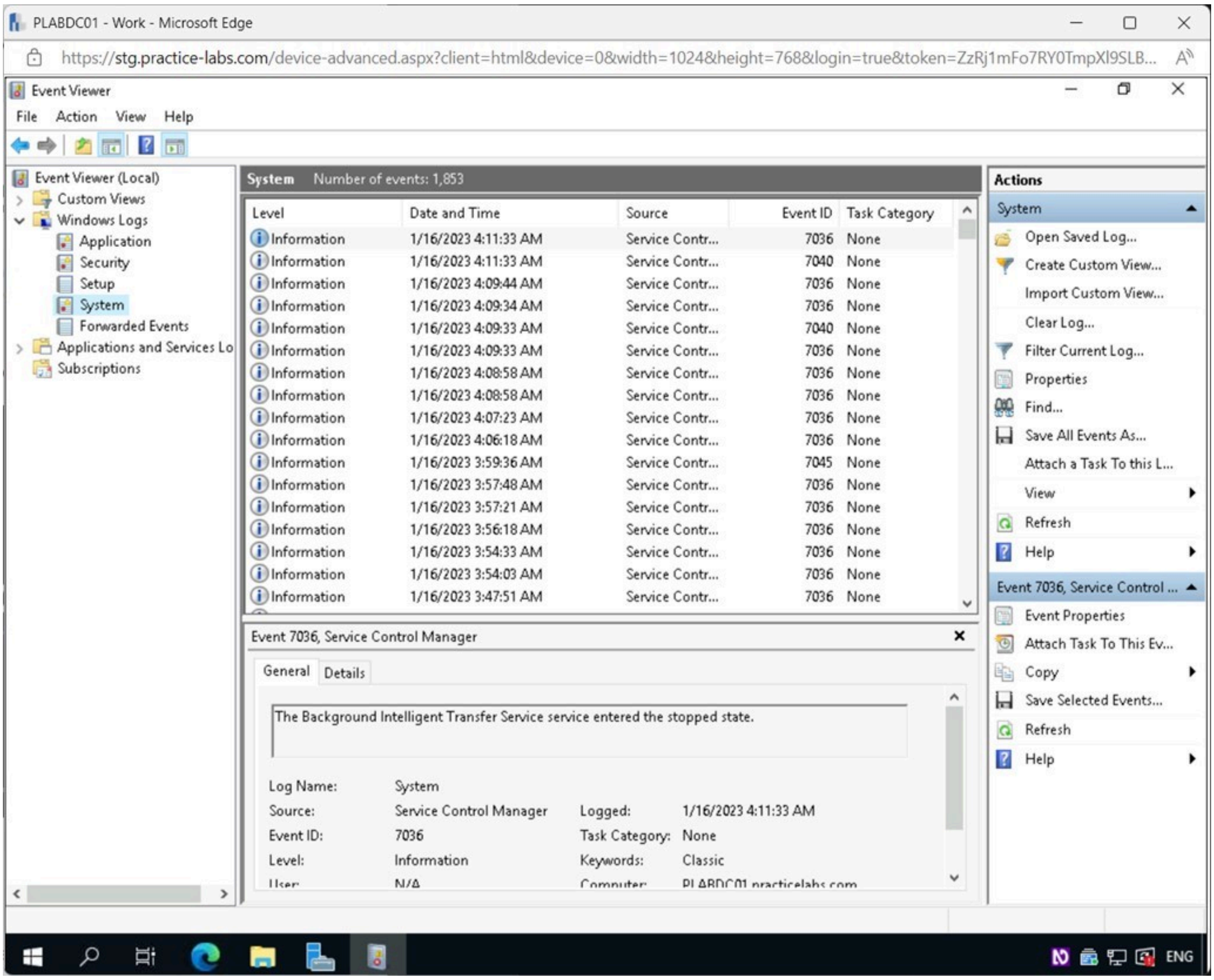

Figure 2.8 Screenshot of PLABDC01: Displaying the Event Viewer window, showing System selected.

Step 3

In the **System** pane, right-click on the first **Information** event available and select **Event Properties**.

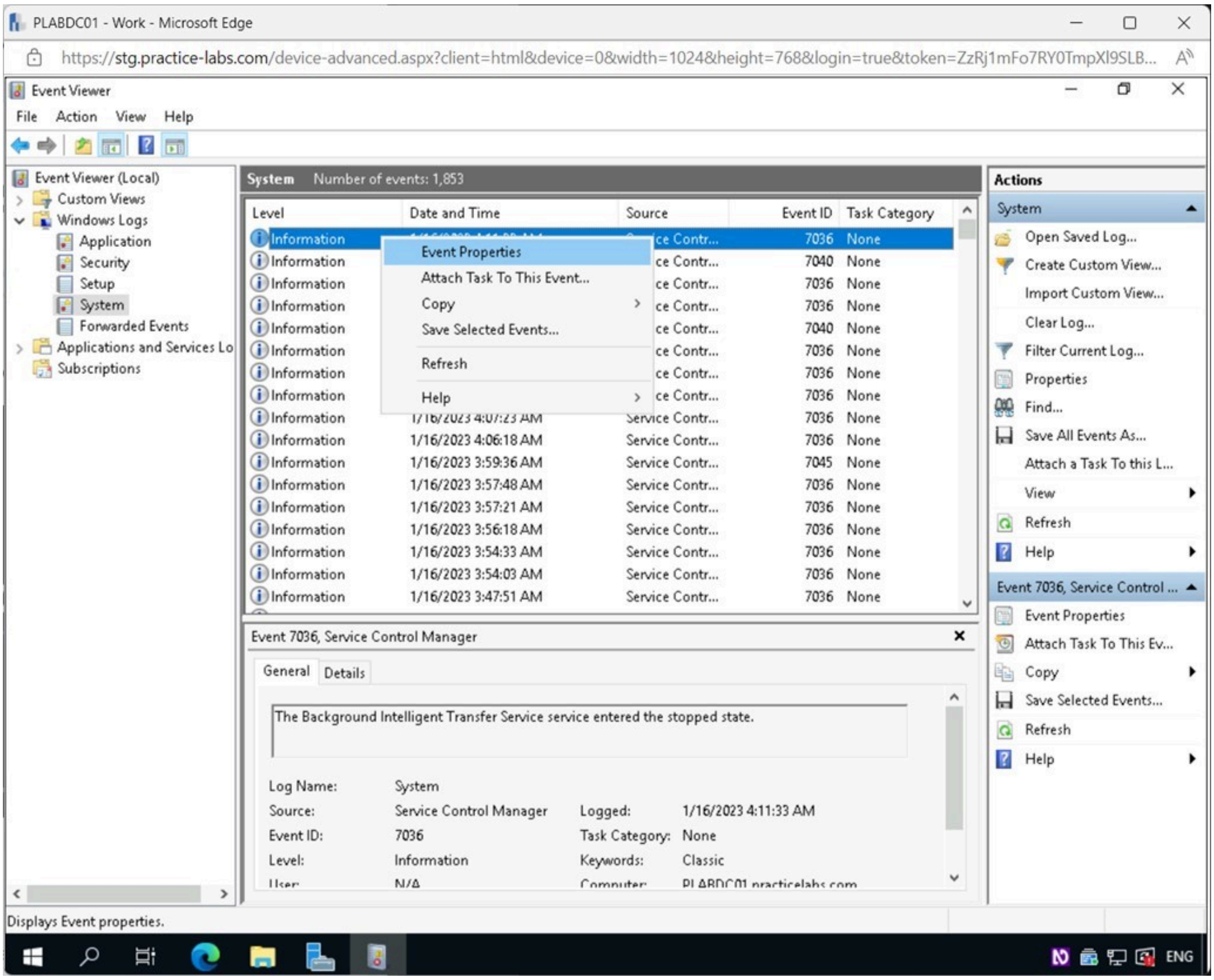

Figure 2.9 Screenshot of PLABDC01: Displaying the Event Viewer window, showing an Information event right-clicked and Event Properties selected.

> ***Note: Information*** *will contain details about the successful completion of an application, driver, or service.*

Step 4

In the **Event Properties - Event 7036, Service Control Manager** window, take note of the **Event ID**, which will be used later to research the event.

> ***Note:*** *Your* ***Event ID*** *might differ from what's shown in the screenshot.*

Click the **Details** tab.

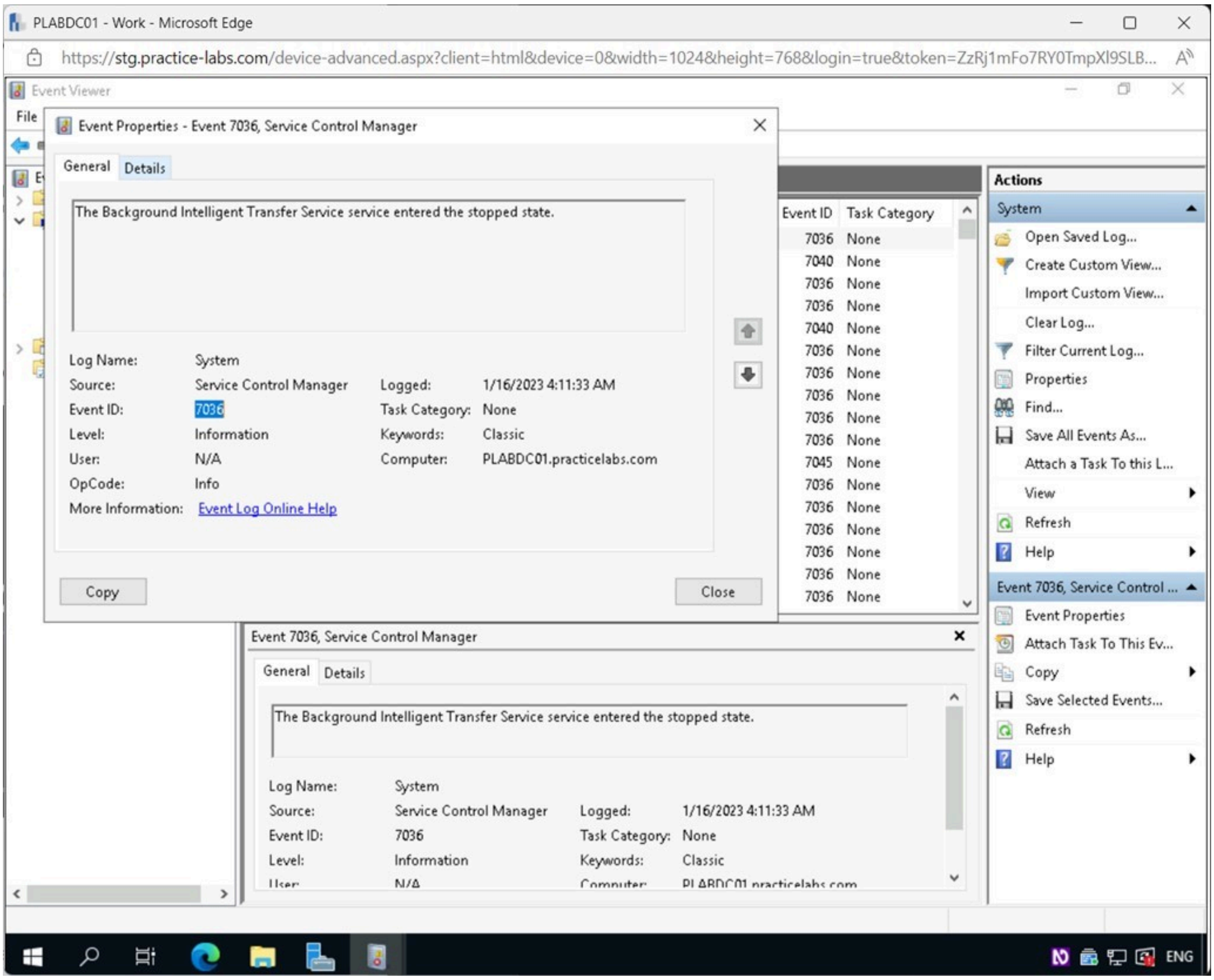

Figure 2.10 Screenshot of PLABDC01: Displaying the Event Properties - Event 7036, Service Control Manager window, showing the Details tab selected.

Step 5

In the **Event Properties - Event 7036, Service Control Manager - Detail** tab, expand **System** by clicking on the **+** sign next to it.

View the information available on the **System** event.

Click the **Microsoft Edge** icon on the **Taskbar**.

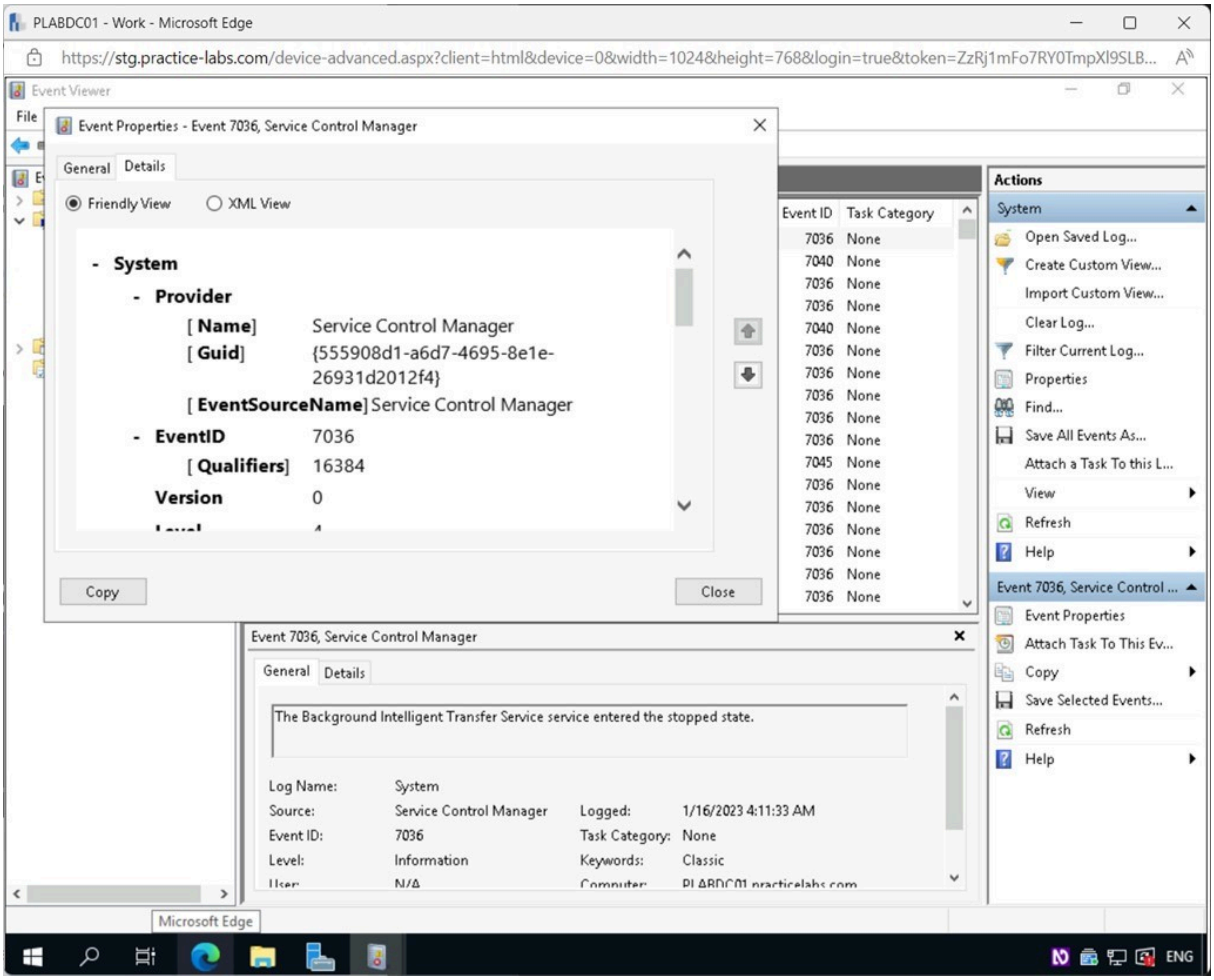

Figure 2.11 Screenshot of PLABDC01: Displaying the Event Properties - Event 7036, Service Control Manager - Detail tab. Microsoft Edge icon on the Taskbar is selected.

Step 6

In the **Microsoft Edge** browser window, type the following in the address bar and press **Enter**:

```
Windows event id (Your Event ID)
```

Note: *Enter the* ***Event ID*** *you took note of in* ***Step 4****.*

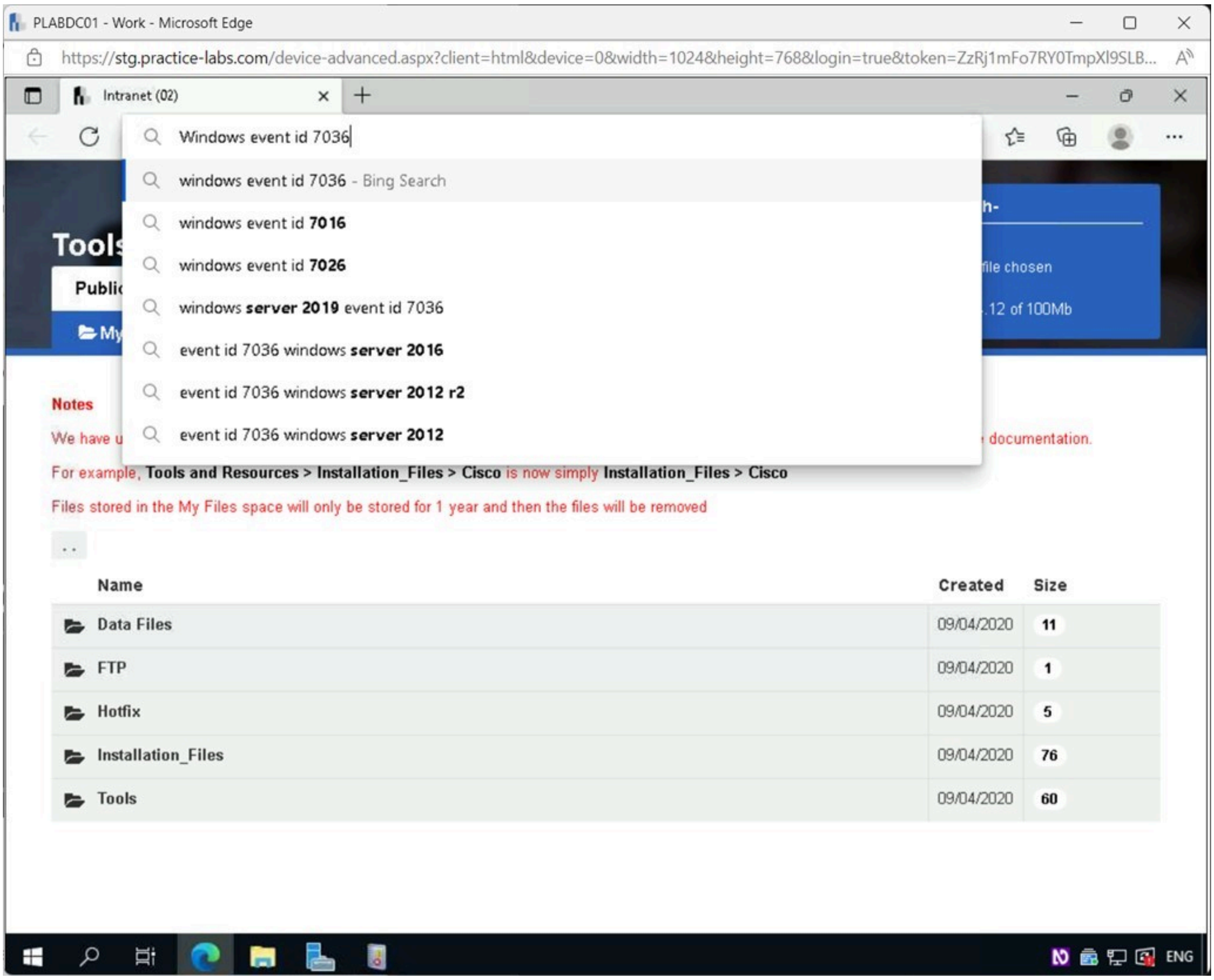

Figure 2.12 Screenshot of PLABDC01: Displaying the Microsoft Edge browser window, showing the required search text entered in the address bar.

Step 7

Select the first search result.

> ***Note:*** *Your first search result can vary from what's shown on the screenshot.*

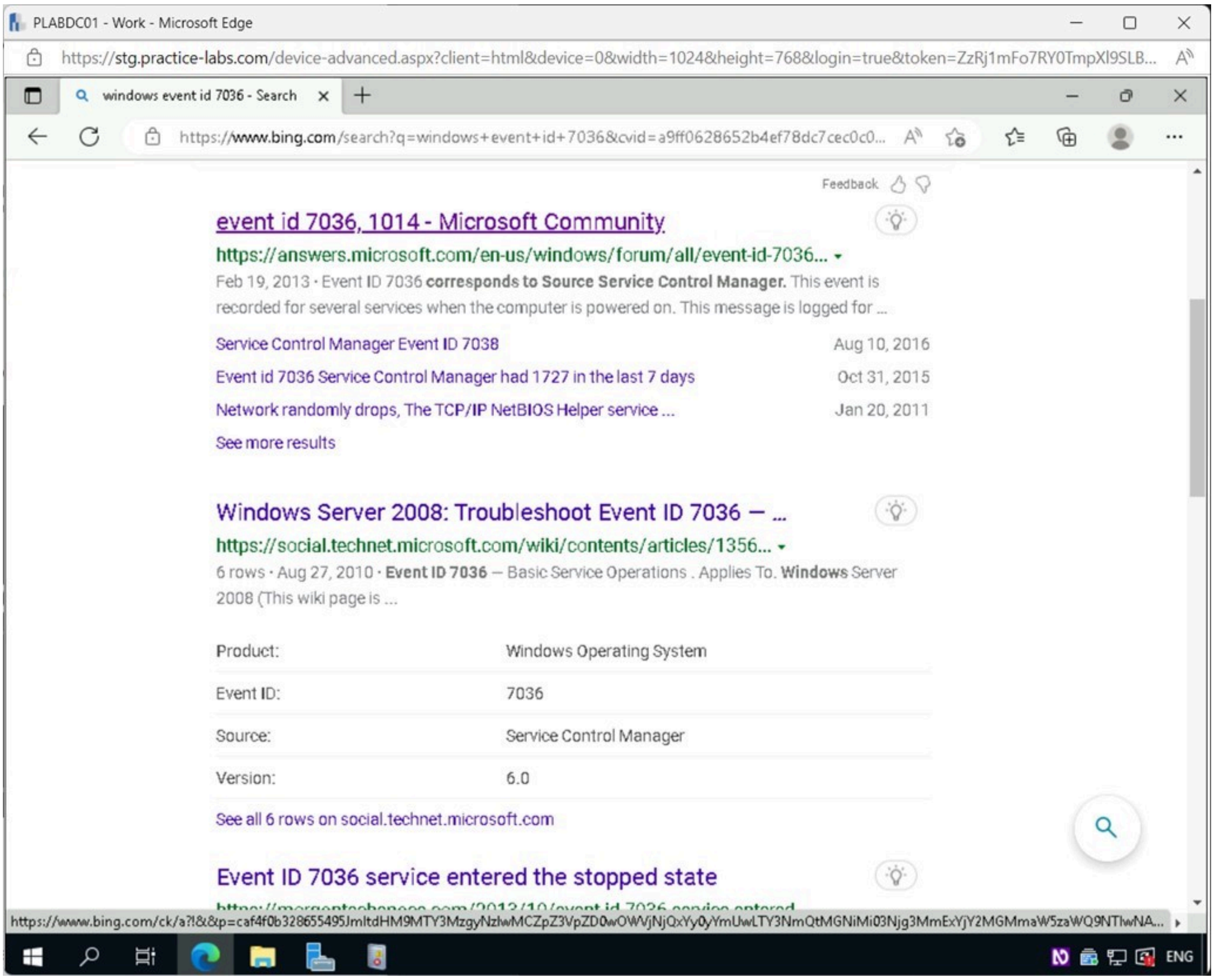

Figure 2.13 Screenshot of PLABDC01: Displaying selecting the required search result in the Microsoft Edge window.

Step 8

On the webpage, review the information available about the **Event ID**.

Minimize the **Microsoft Edge** browser window.

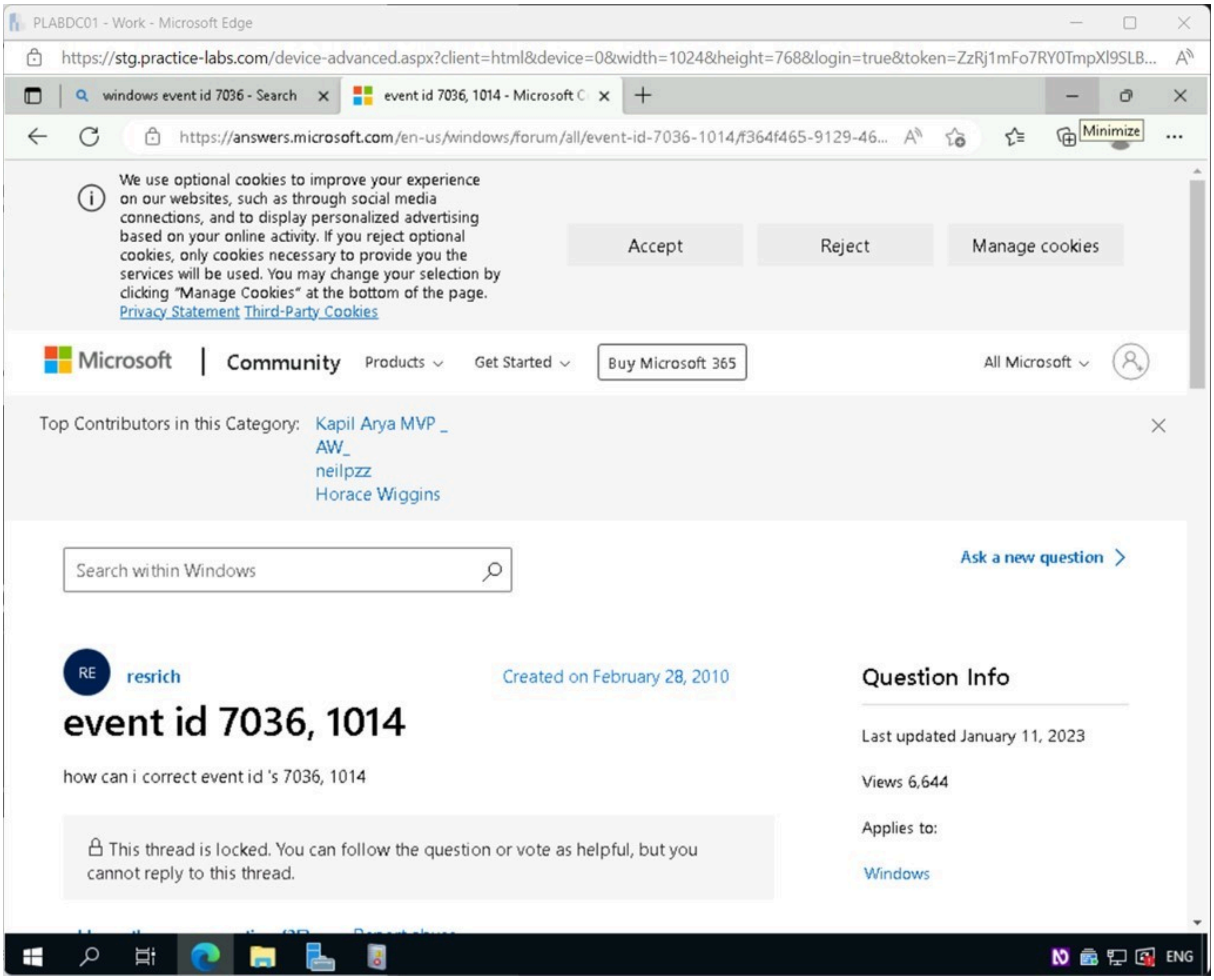

Figure 2.14 Screenshot of PLABDC01: Displaying information about the required Event ID in Microsoft Edge. Minimize icon on the top-right corner is selected.

Step 9

Close the **Event Properties - Event 7036, Service Control Manager** window.

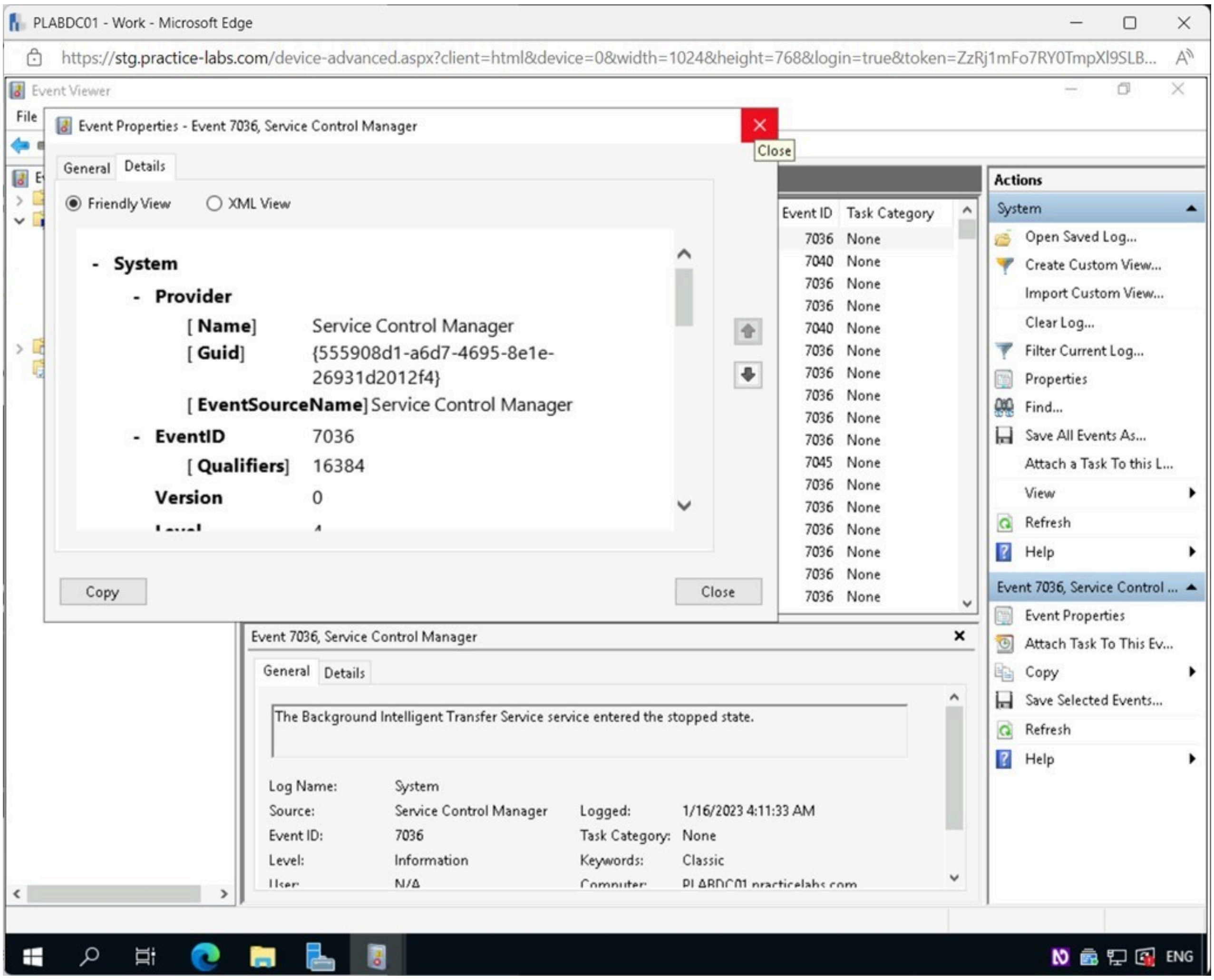

Figure 2.15 Screenshot of PLABDC01: Displaying closing the Event Properties - Event 7036, Service Control Manager window.

Step 10

Back on the **Event Viewer** window, scroll down the **System** pane and select the first **Warning** event that appears.

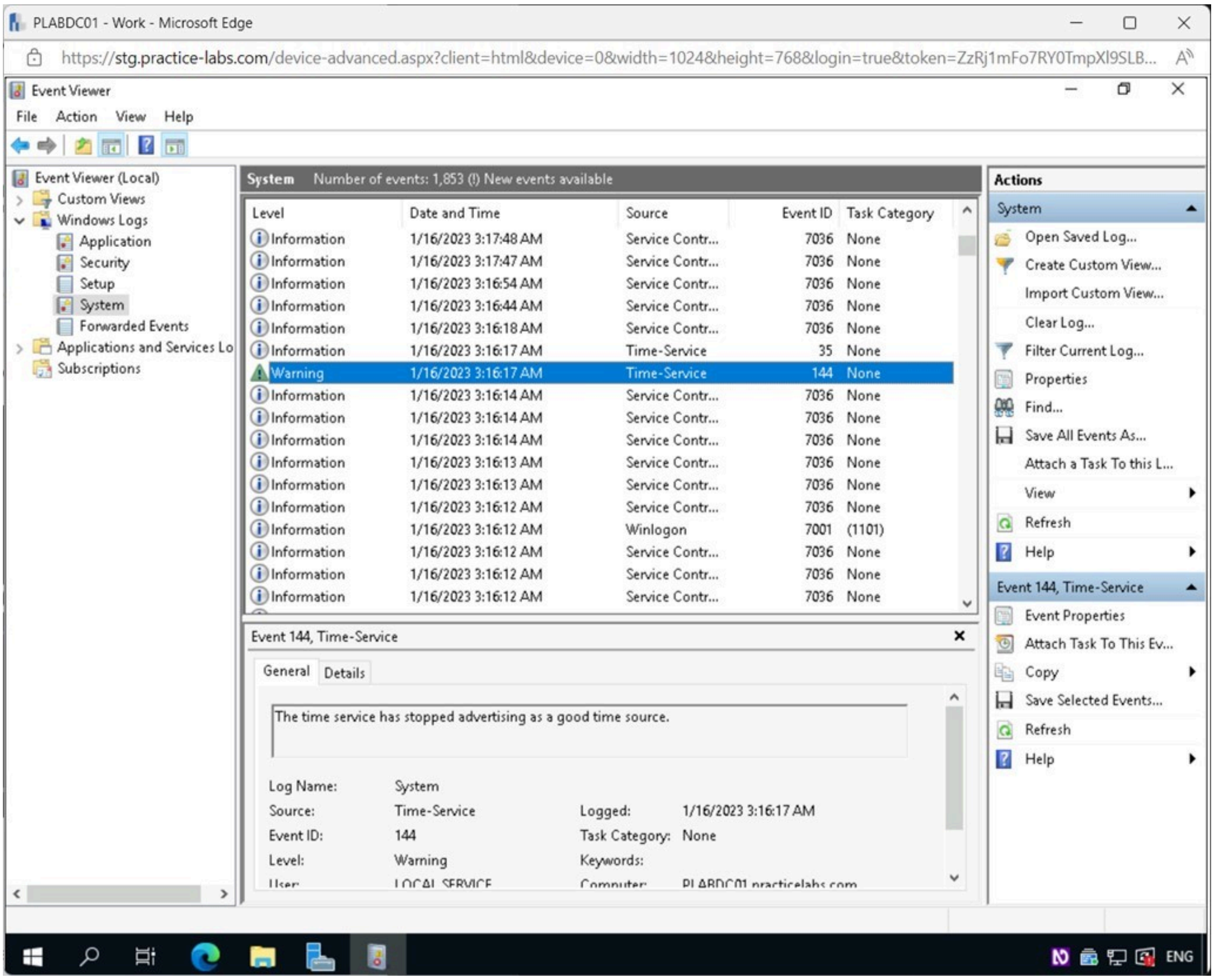

Figure 2.16 Screenshot of PLABDC01: Displaying the Event Viewer window, showing a Warning event in the System pane selected.

> ***Note: Warning*** *is an occurrence that, while not catastrophic in and of itself, might foreshadow a future difficulty. For instance, a warning will be recorded if disk space drops below a certain threshold.*

Step 11

Take note of the **Event ID** in the bottom pane.

Restore **Microsoft Edge** from the **Taskbar**.

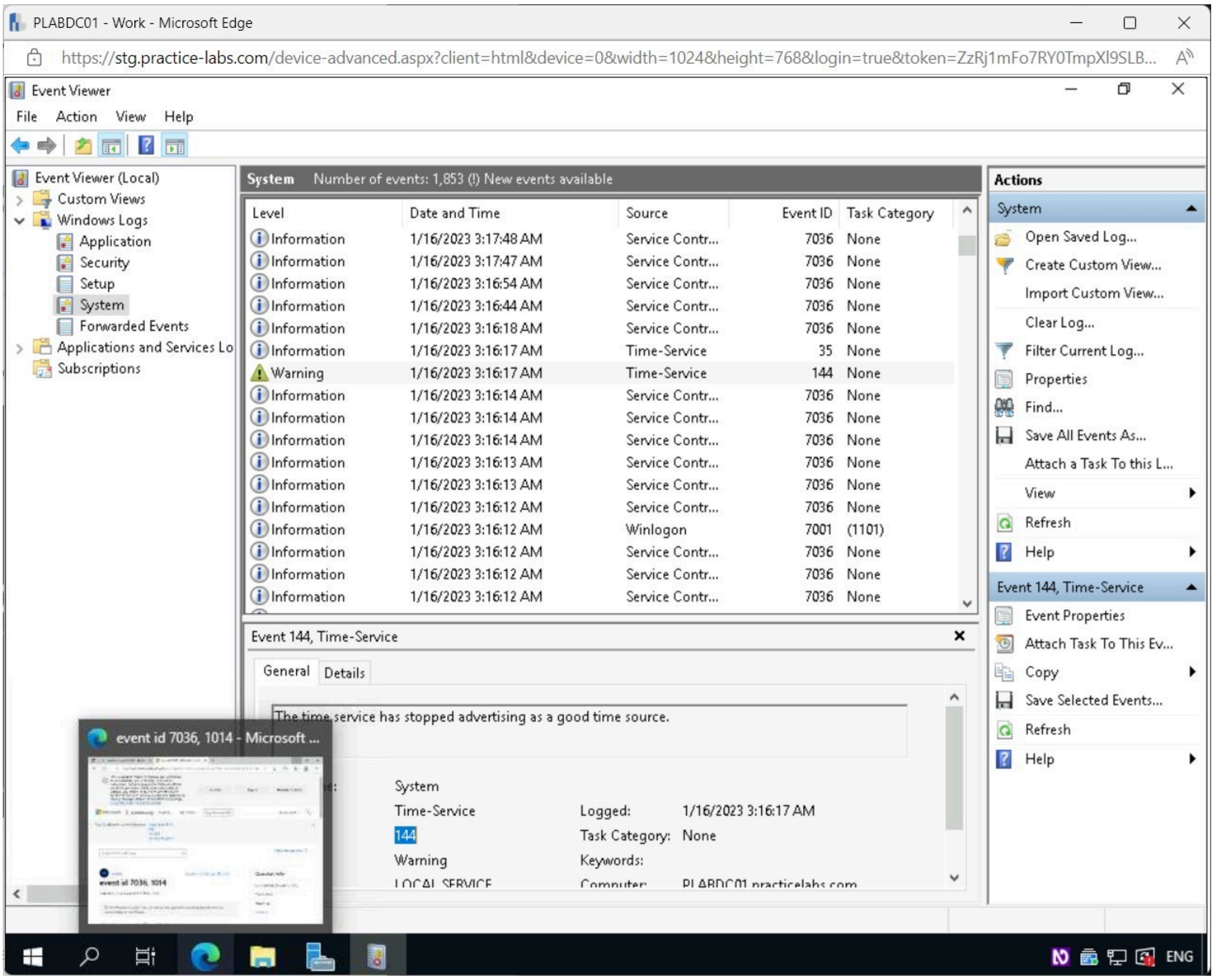

Figure 2.17 Screenshot of PLABDC01: Displaying the Event Viewer window, showing the required Event ID selected.

Step 12

In the **Microsoft Edge** browser window, **Close** the tab from the previous search.

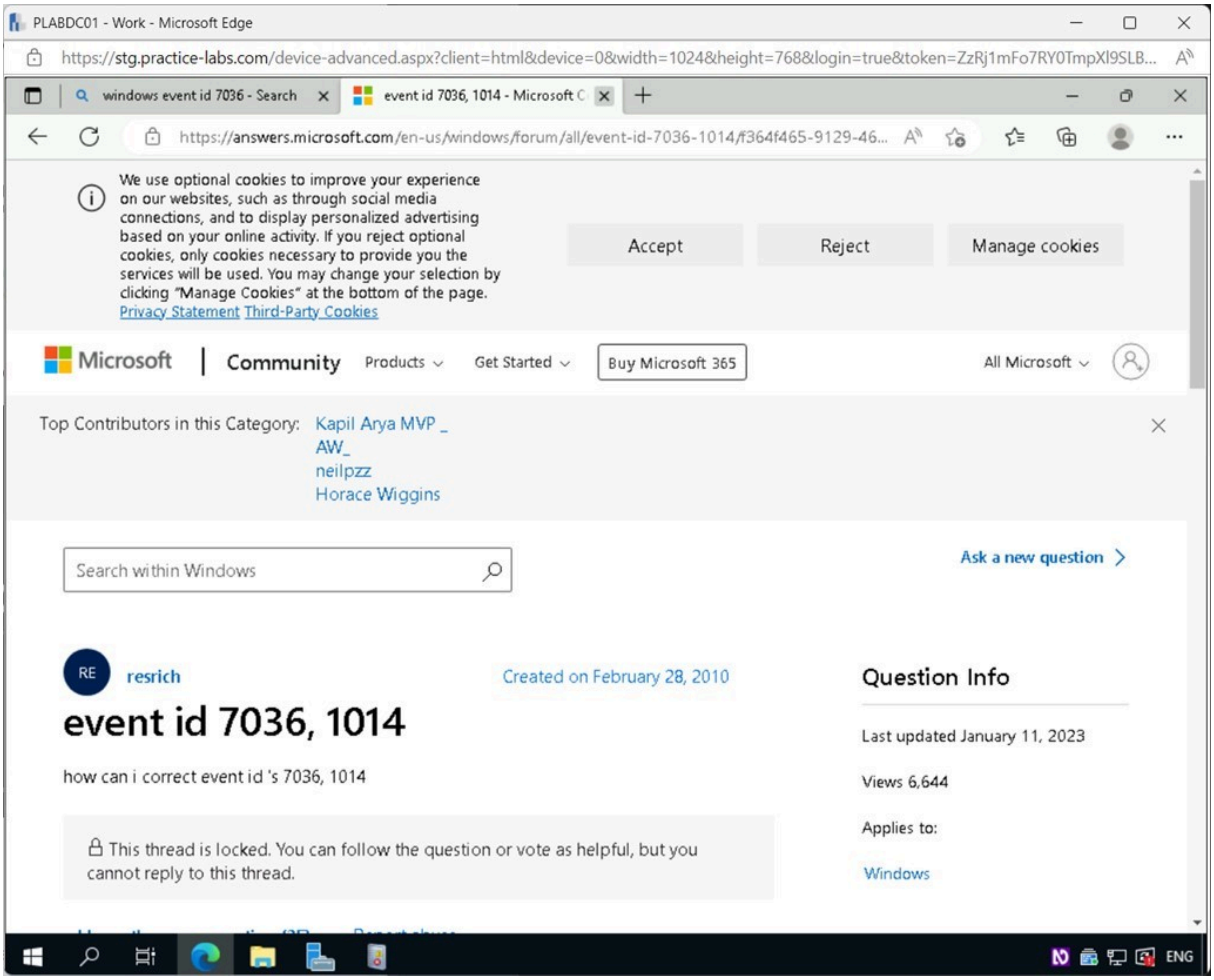

Figure 2.18 Screenshot of PLABDC01: Displaying closing the required tab in the Microsoft Edge browser window.

Step 13

Type the following into the search textbox and press **Enter**:

```
Windows event id (Your Event Id)
```

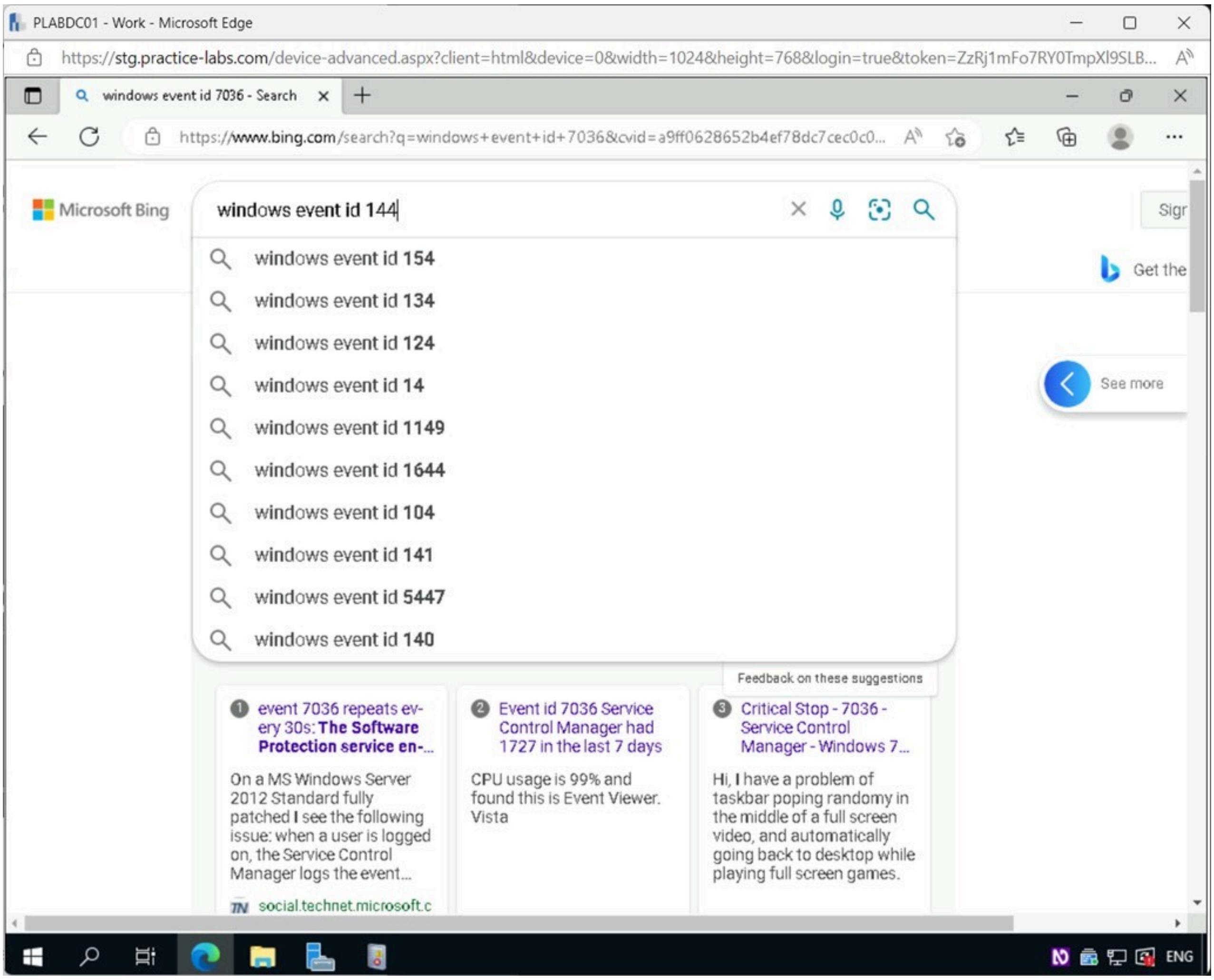

Figure 2.19 Screenshot of PLABDC01: Displaying entering the required search text in the Microsoft Edge browser window.

Step 14

Select the first search result.

> ***Note:*** *Your first search result can vary from what's shown on the screenshot.*

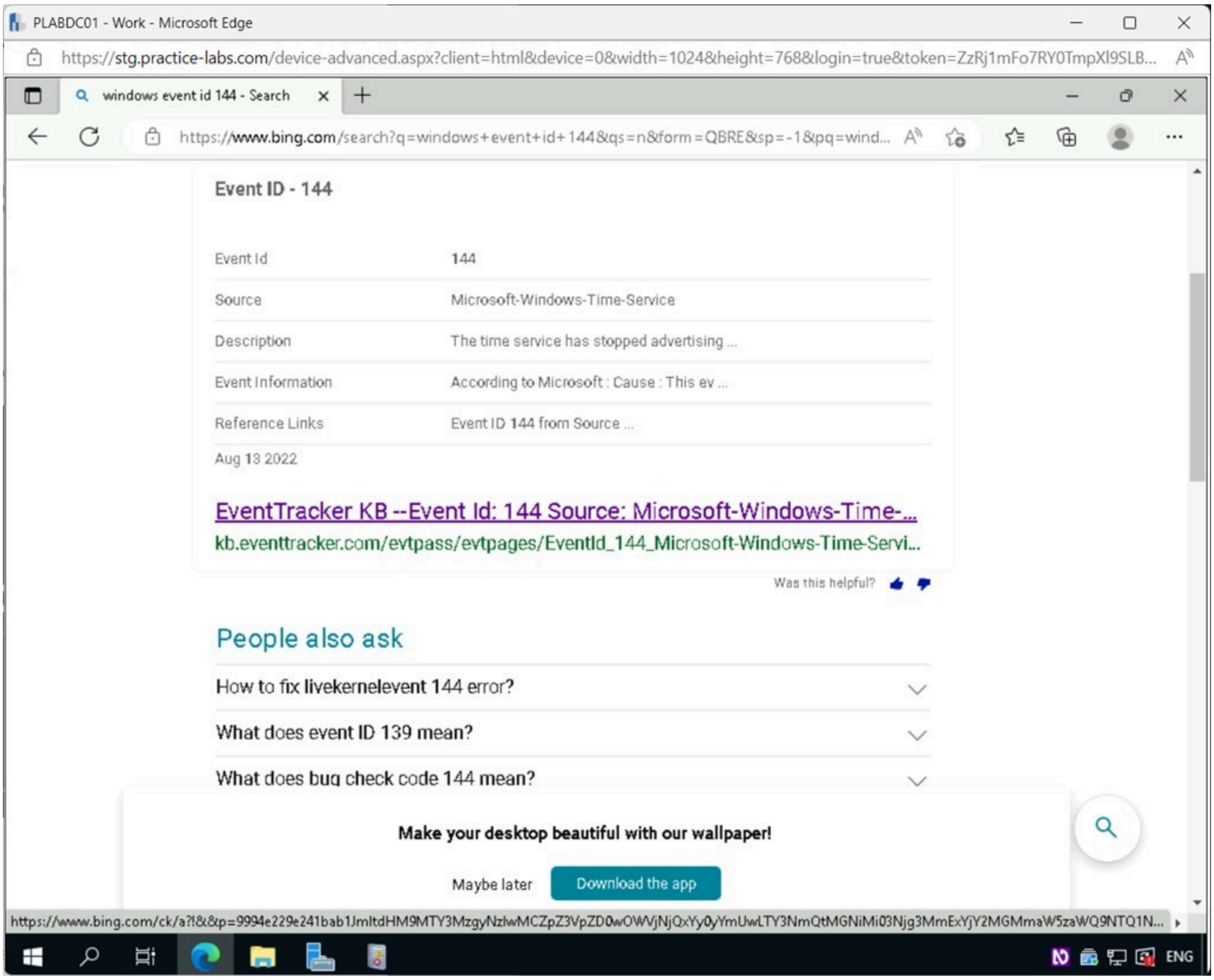

Figure 2.20 Screenshot of PLABDC01: Displaying selecting the required search result in the Microsoft Edge window.

Step 15

On the webpage, review the information available about the **Event ID**.

Minimize the **Microsoft Edge** browser window.

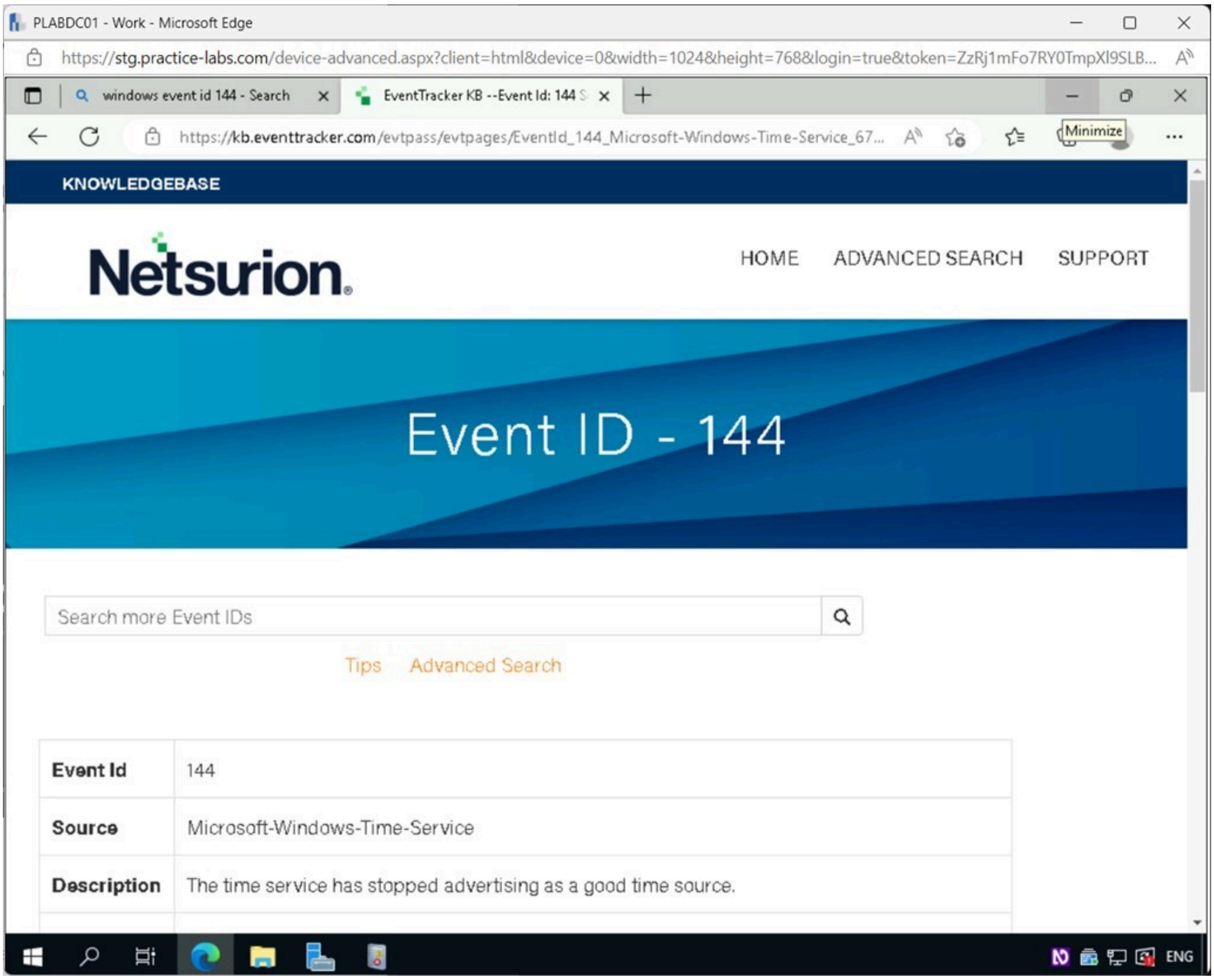

Figure 2.21 Screenshot of PLABDC01: Displaying information about the required Event ID in Microsoft Edge. Minimize icon on the top-right corner is selected.

Step 16

Back on the **Event Viewer** window, scroll down the **System** pane and select the first **Error** event that appears.

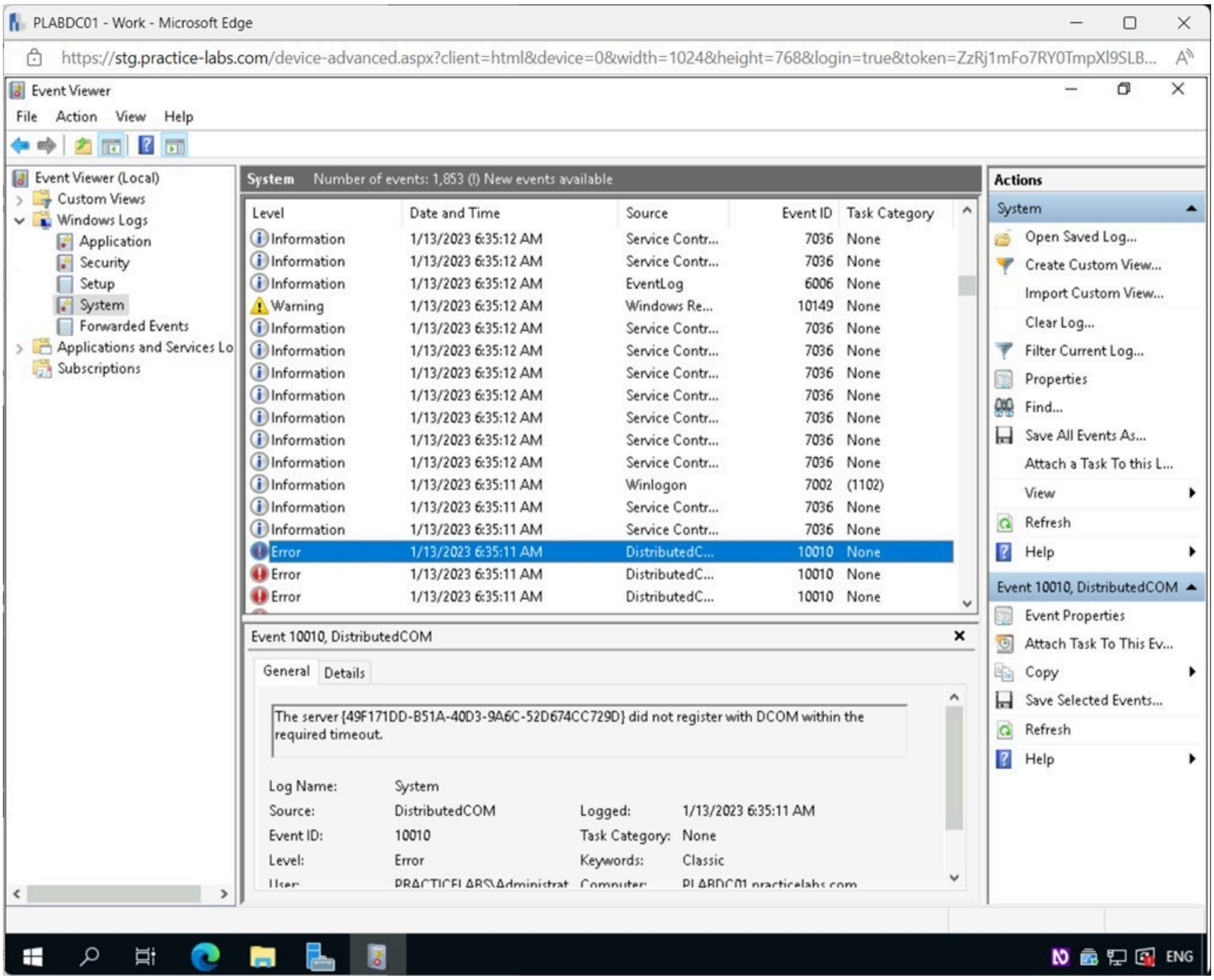

Figure 2.22 Screenshot of PLABDC01: Displaying the Event Viewer window, showing an Error event in the System pane selected.

> ***Note:*** *An* ***Error*** *is a major issue, such as the loss of data or a feature. For instance, an error will be recorded if a service is not successfully loaded.*

Step 17

Take note of the **Event ID** in the bottom pane.

Restore **Microsoft Edge** from the **Taskbar**.

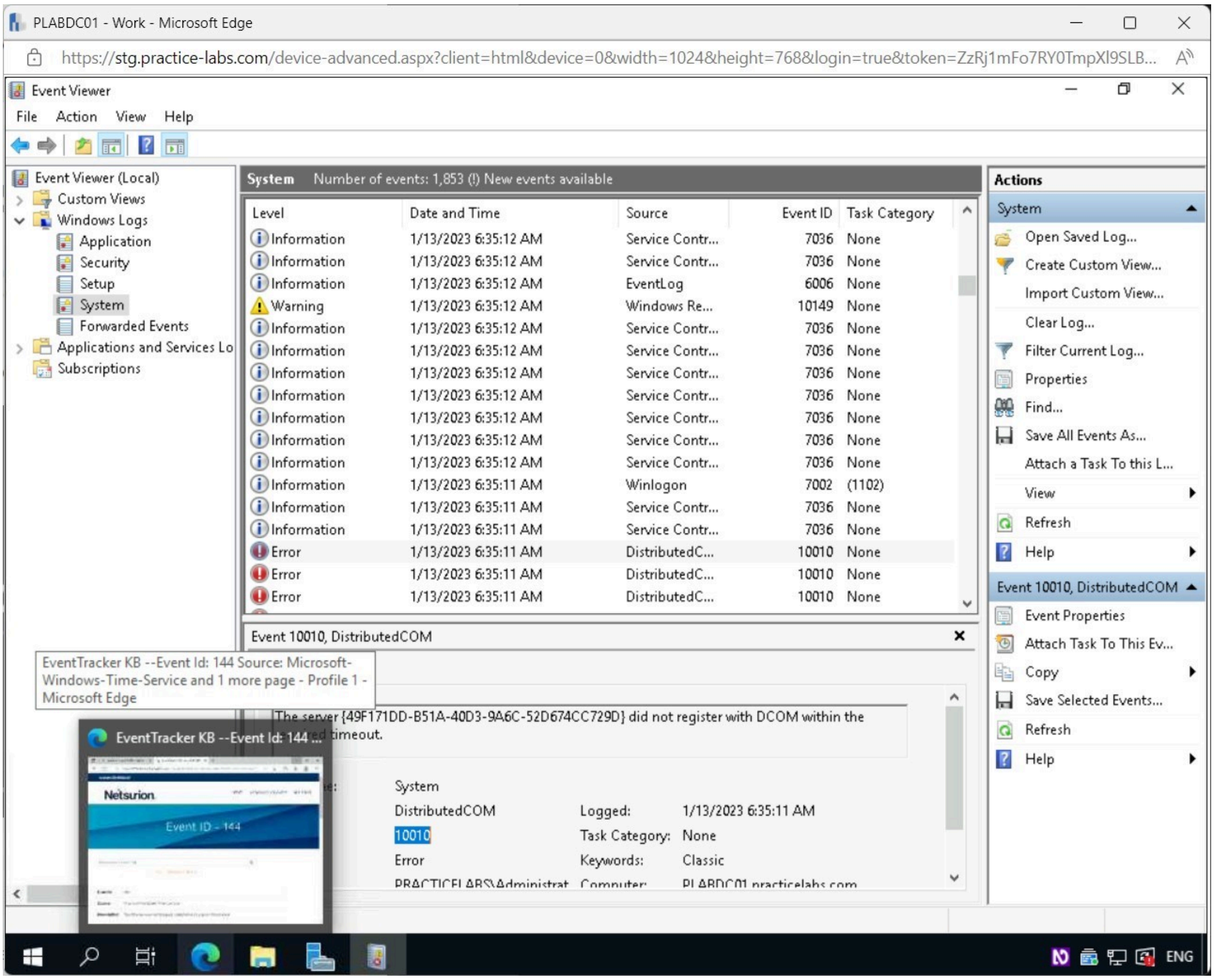

Figure 2.23 Screenshot of PLABDC01: Displaying the Event Viewer window, showing the required Event ID selected.

Step 18

In the **Microsoft Edge** browser window, **Close** the tab from the previous search.

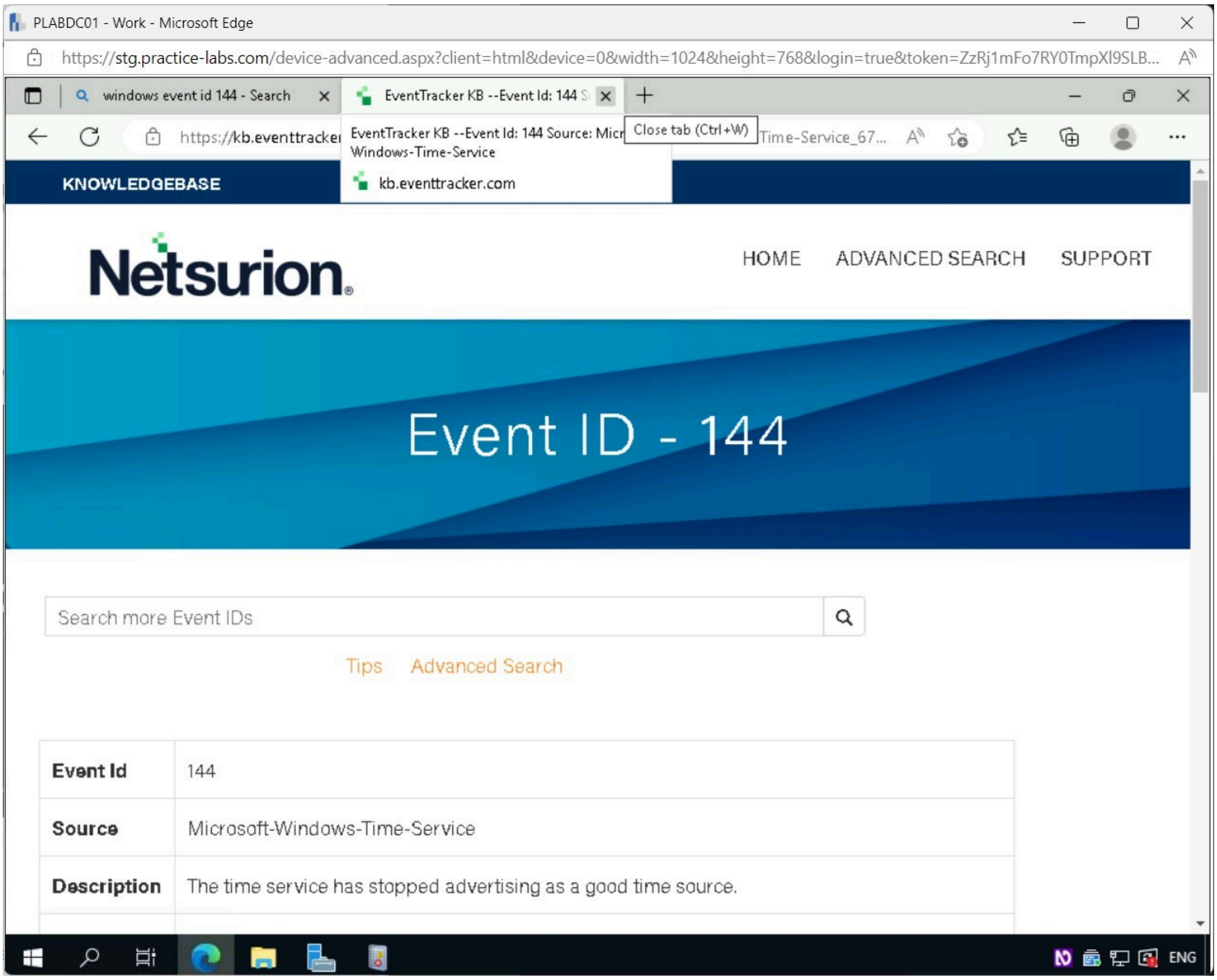

Figure 2.24 Screenshot of PLABDC01: Displaying closing the required tab in the Microsoft Edge browser window.

Step 19

Type the following into the search textbox and press **Enter**:

```
Windows event id (Your Event Id)
```

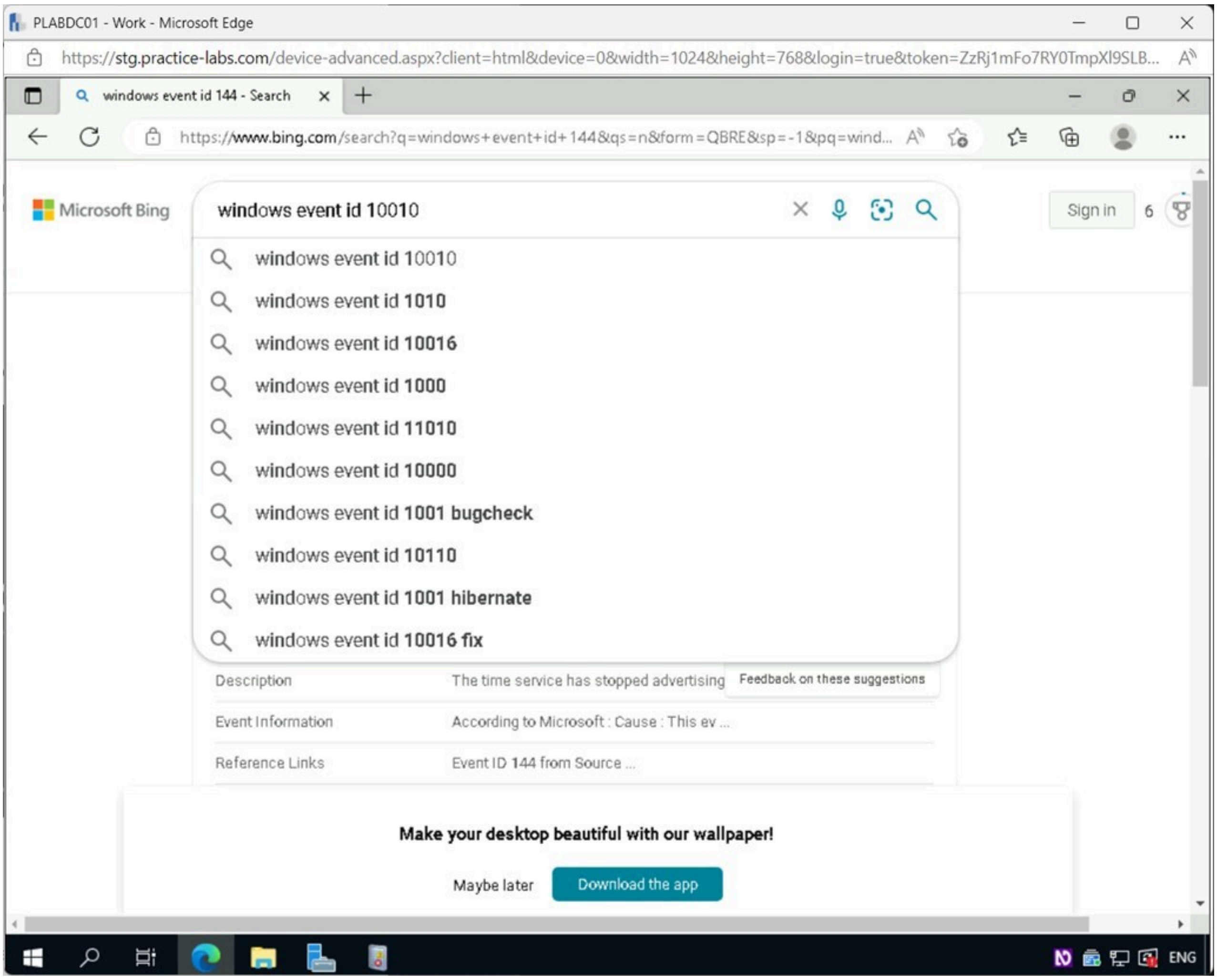

Figure 2.25 Screenshot of PLABDC01: Displaying entering the required search text in the Microsoft Edge browser window.

Step 20

Select the first search result.

> ***Note:*** *Your first search result can vary from what's shown on the screenshot.*

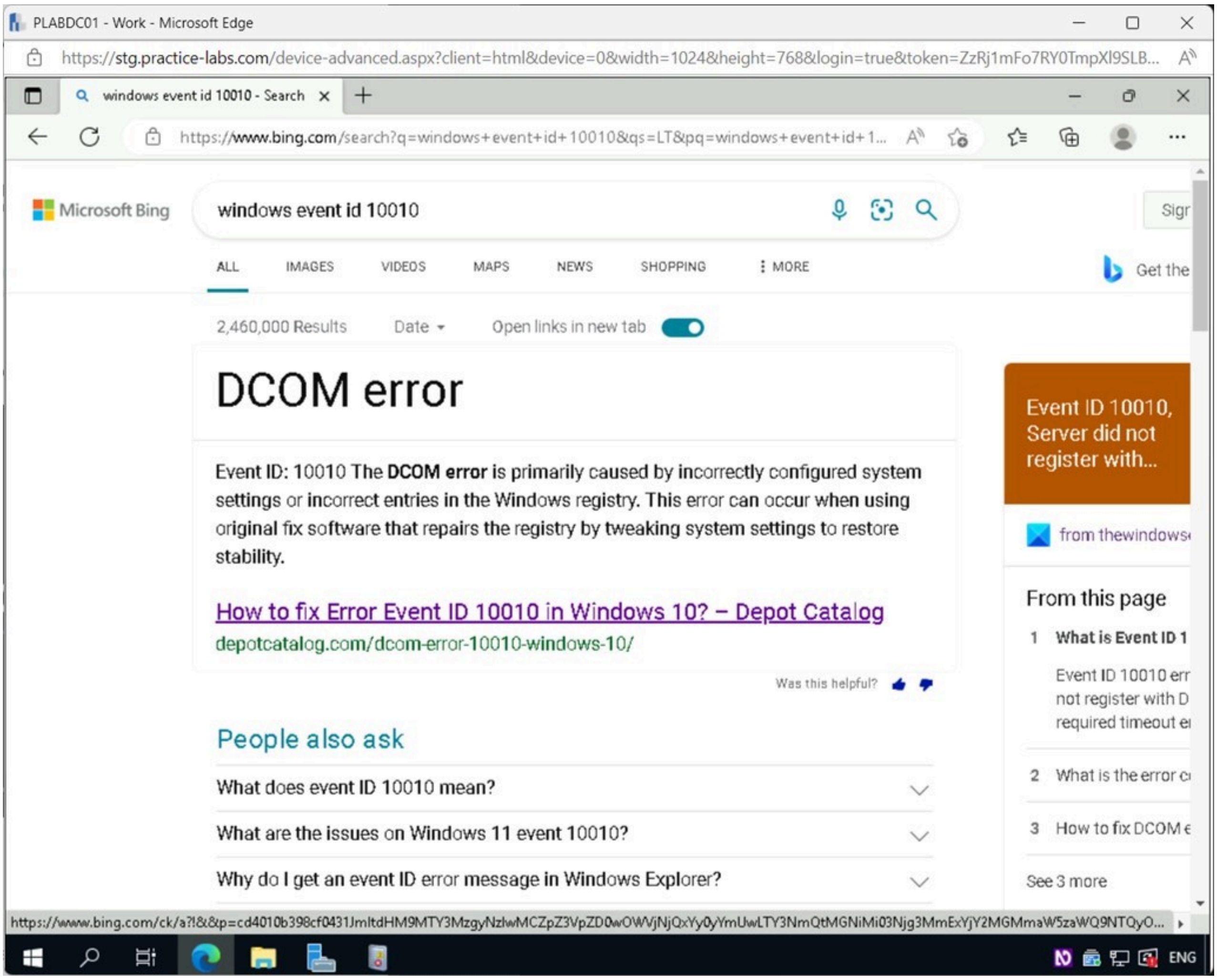

Figure 2.26 Screenshot of PLABDC01: Displaying selecting the required search result in the Microsoft Edge window.

Step 21

On the webpage, review the information available about the **Event ID**.

Close **Microsoft Edge**.

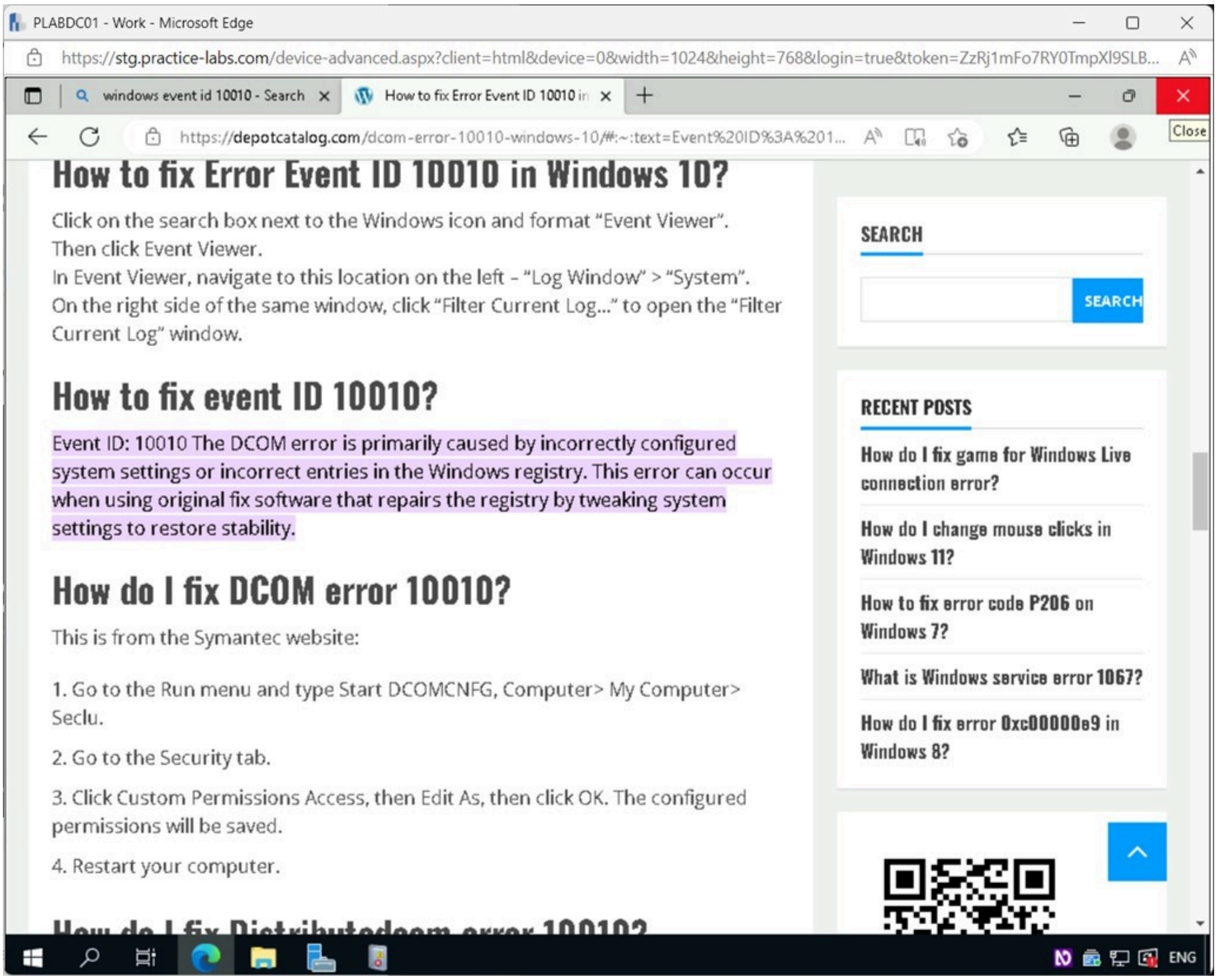

Figure 2.27 Screenshot of PLABDC01: Displaying closing the Microsoft Edge browser window.

Step 22

Close the **Event Viewer** window.

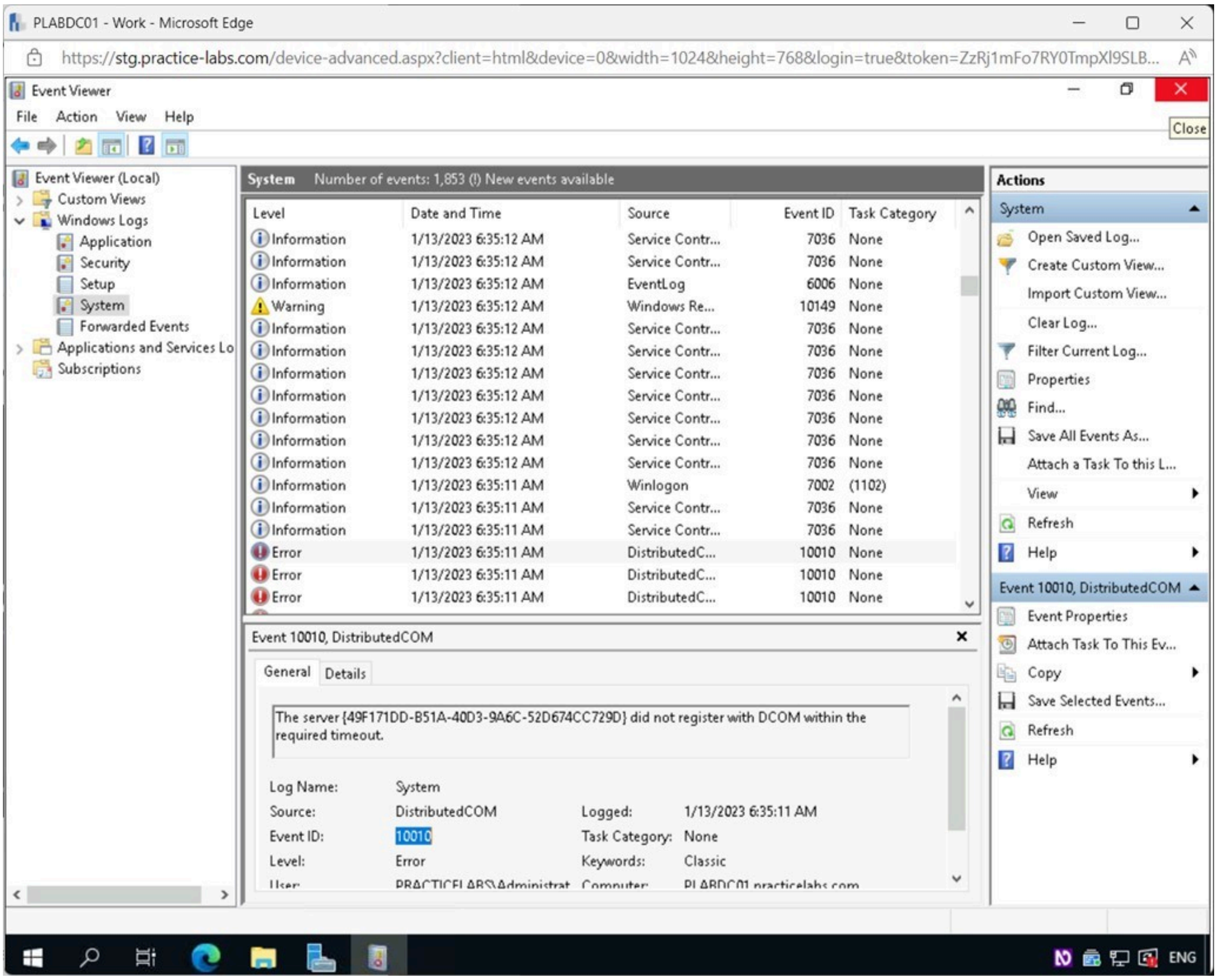

Figure 2.28 Screenshot of PLABDC01: Displaying closing the Event Viewer window.

Task 4 - Monitoring Tools

Microsoft Windows Performance Monitor lets administrators see how running programs affect system responsiveness. The program logs data in real-time and for later review. Windows Performance Monitor examines settings, performance counters, and event traces to measure system performance. Collecting data creates a Data Collector Set. Resource Monitor gives a similar look as what can be seen in performance monitor, but it does not offer the ability to log the data.

In this task, you will look at the performance monitor and see how to set up multiple types of data collector sets.

Step 1

Ensure you are connected to **PLABDC01.**

Restore the **Server Manager** window from the **Taskbar.**

Click the **Tools** menu and then select **Performance Monitor**.

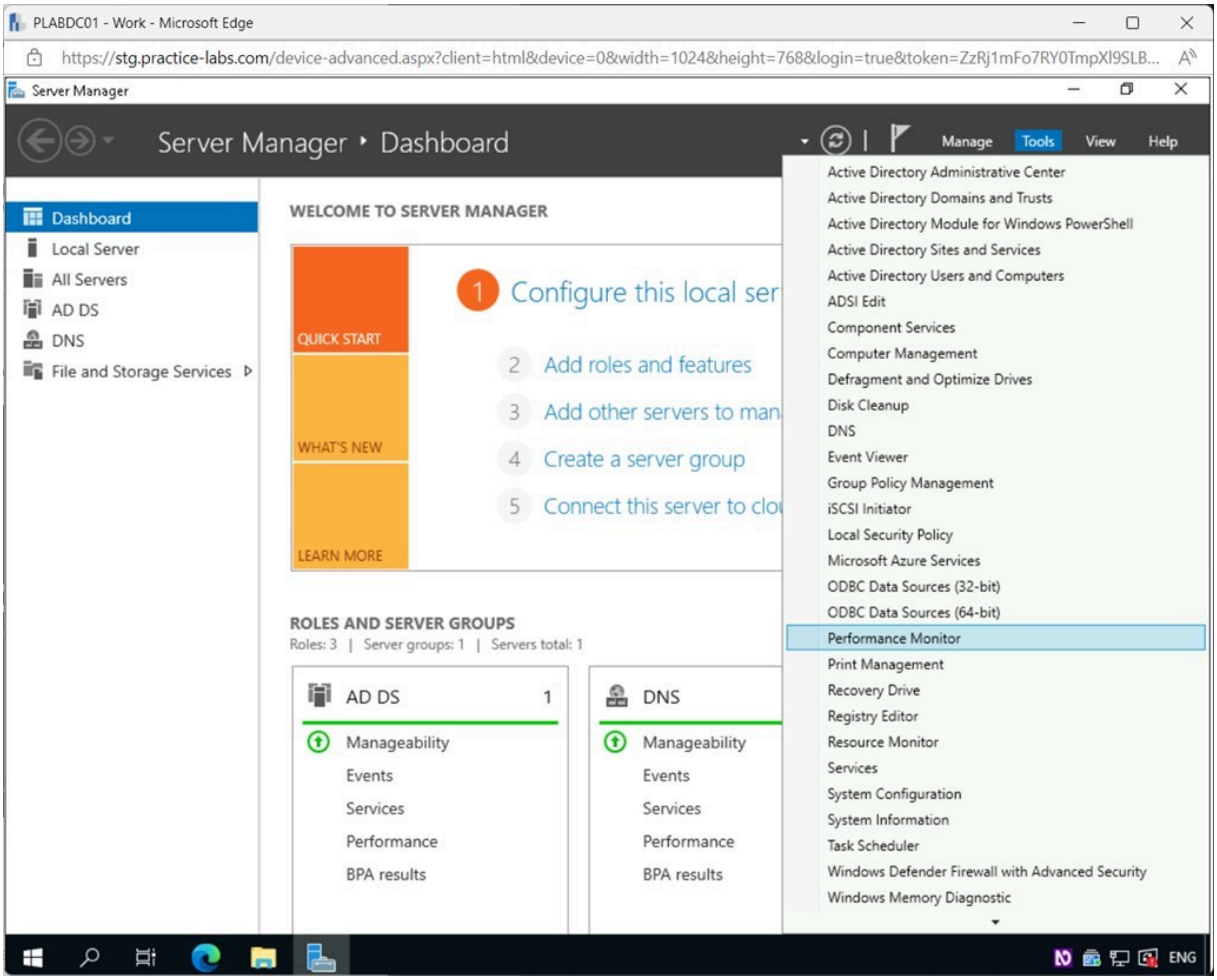

Figure 2.29 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Performance Monitor is selected.

Step 2

In the **Performance Monitor** window, expand **Data Collector Sets** on the left pane.

Right-click on **User Defined**, and select **New** > **Data Collector Set**.

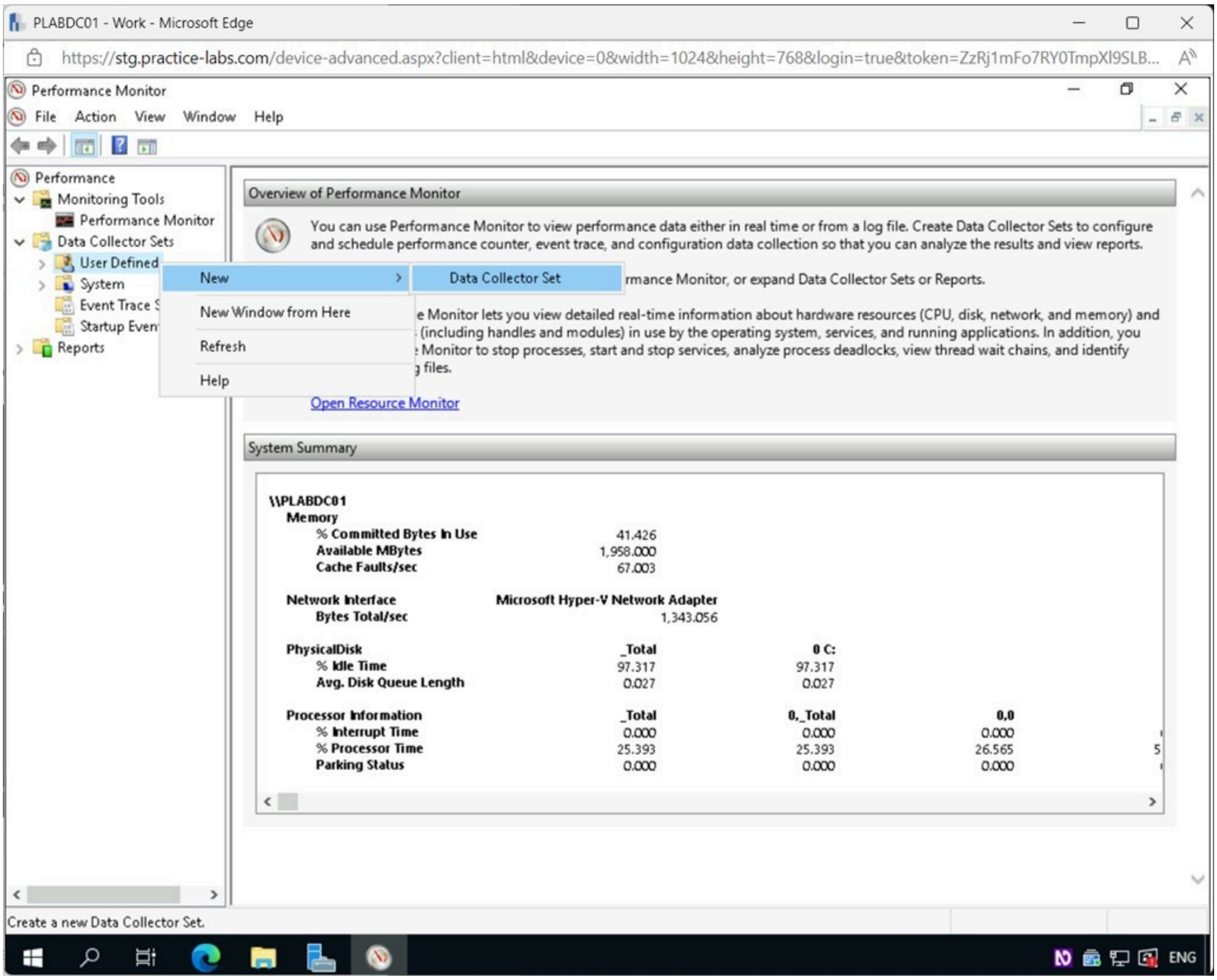

Figure 2.30 Screenshot of PLABDC01: Displaying the Performance Monitor window. User Defined is right-clicked, and New > Data Collector Set menu options are selected.

Step 3

In the **Create new Data Collector Set - How would you like to create this new data collector set?** window, type the following for the **Name** field:

```
Baseline Server
```

Click **Next**.

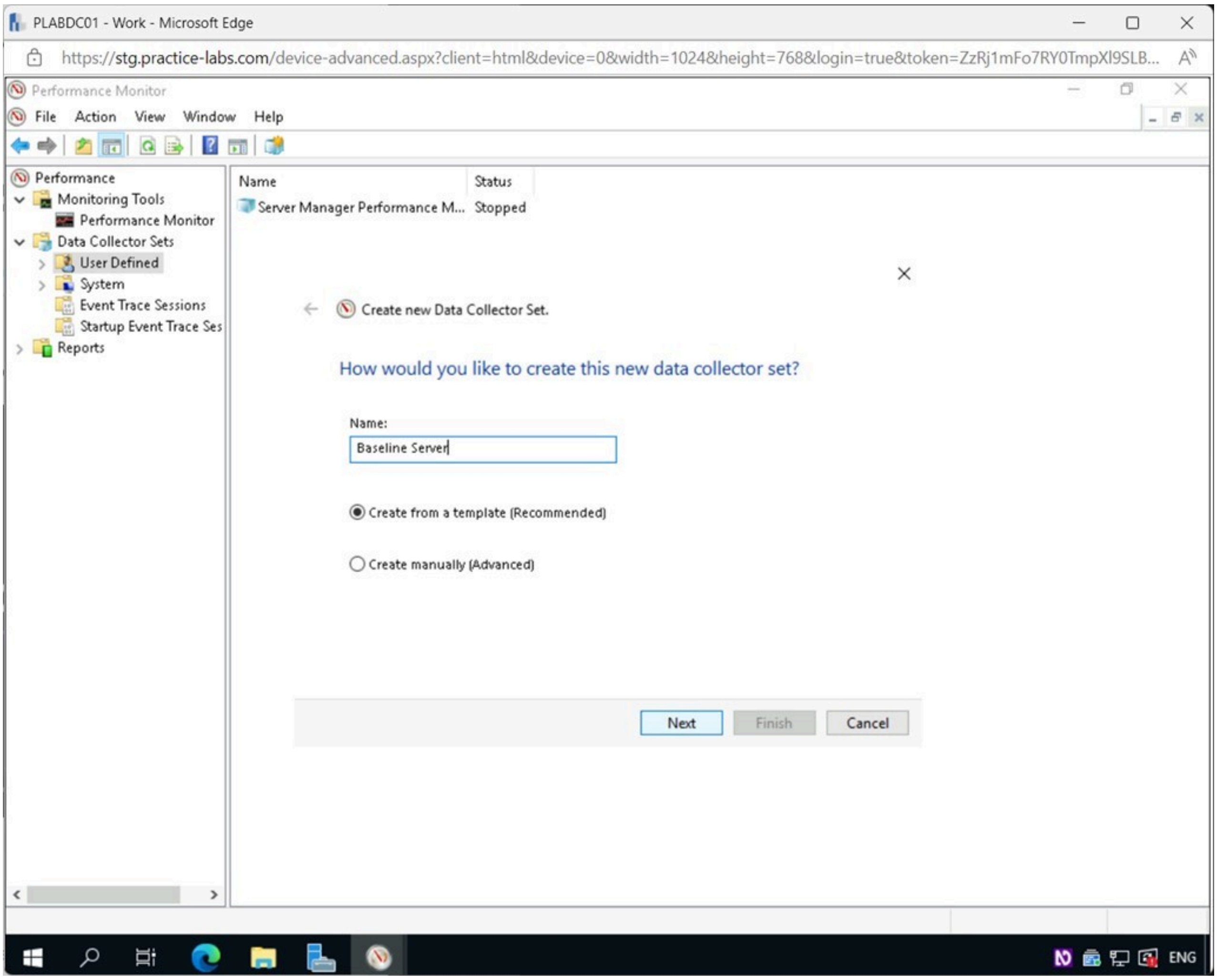

Figure 2.31 Screenshot of PLABDC01: Displaying the Create new Data Collector Set - How would you like to create this new data collector set? Window, showing the required Name typed in and the Next button selected.

Step 4

In the **Which template would you like to use?** page, select **System Diagnostics** and click **Next**.

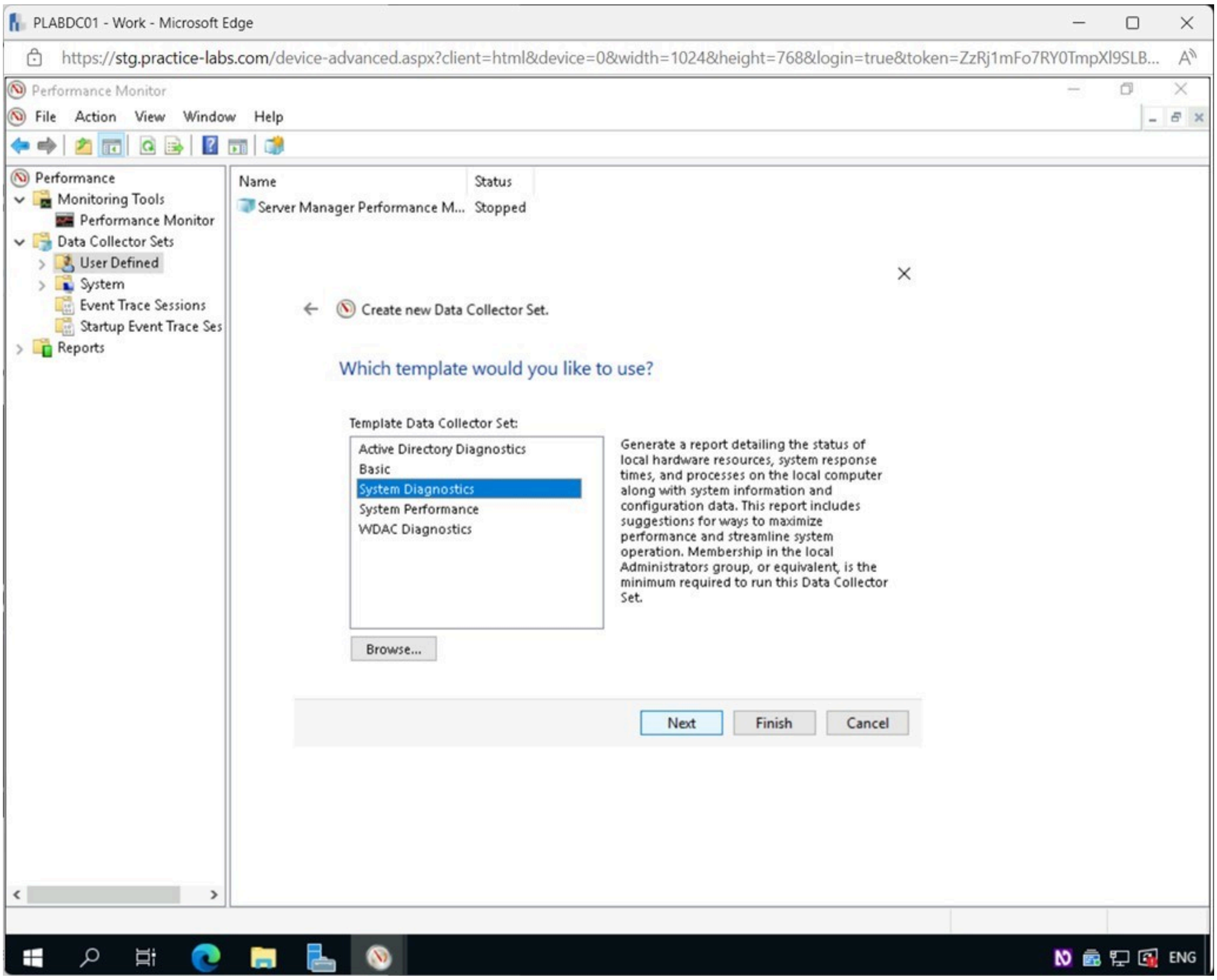

Figure 2.32 Screenshot of PLABDC01: Displaying the Which template would you like to use? page, showing System Diagnostics selected and the Next button highlighted.

Step 5

In the **Where would you like the data to be saved?** page, click **Next.**

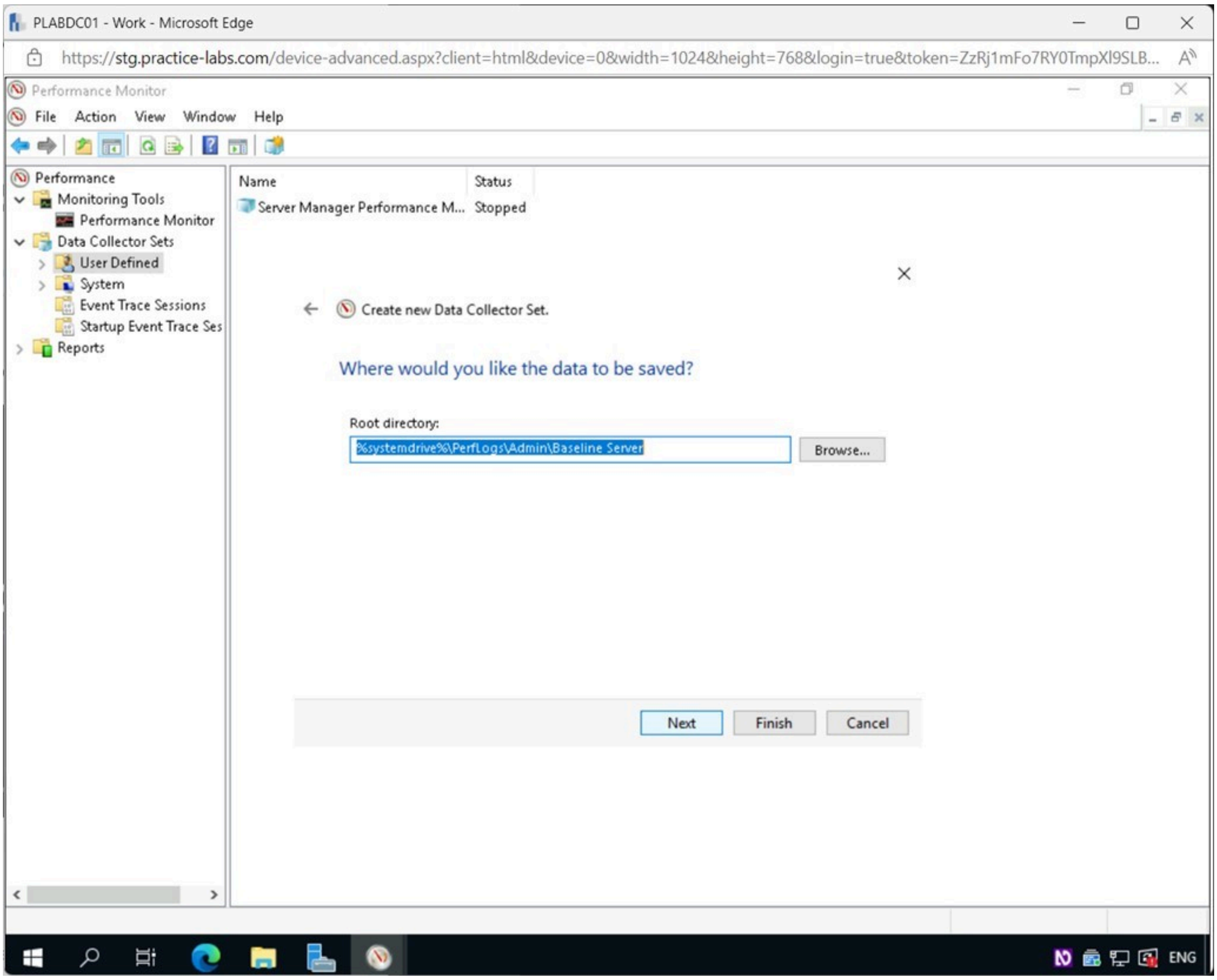

Figure 2.33 Screenshot of PLABDC01: Displaying clicking Next in the Where would you like the data to be saved? page.

Step 6

In the **Create the data collector set?** page, click **Finish.**

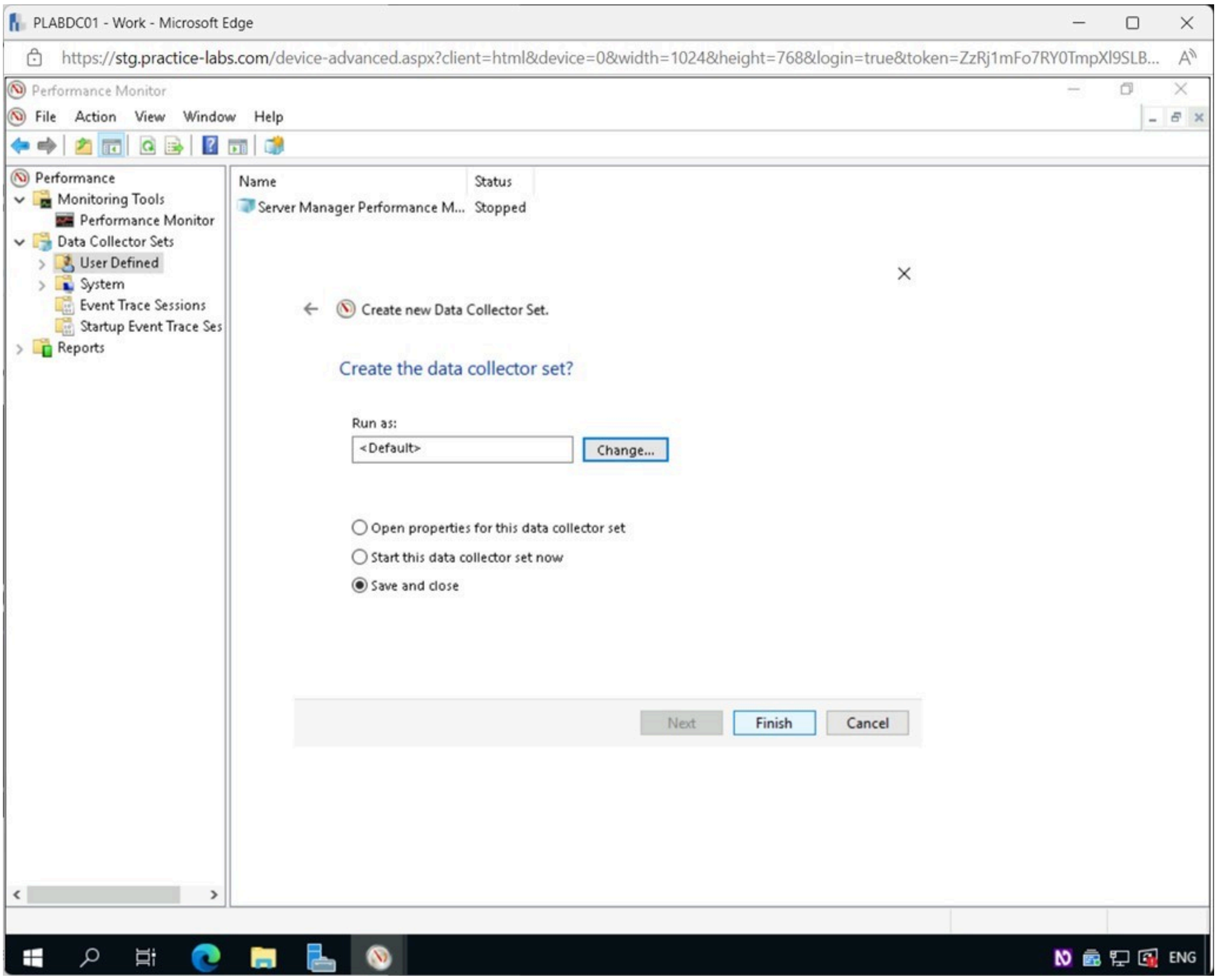

Figure 2.34 Screenshot of PLABDC01: Displaying clicking Finish in the Create the data collector set? page.

Step 7

Back on the **Performance Monitor** window, select **Baseline Server** in the right details pane, and then click the **Start the Data Collector Set** icon in the **Toolbar**.

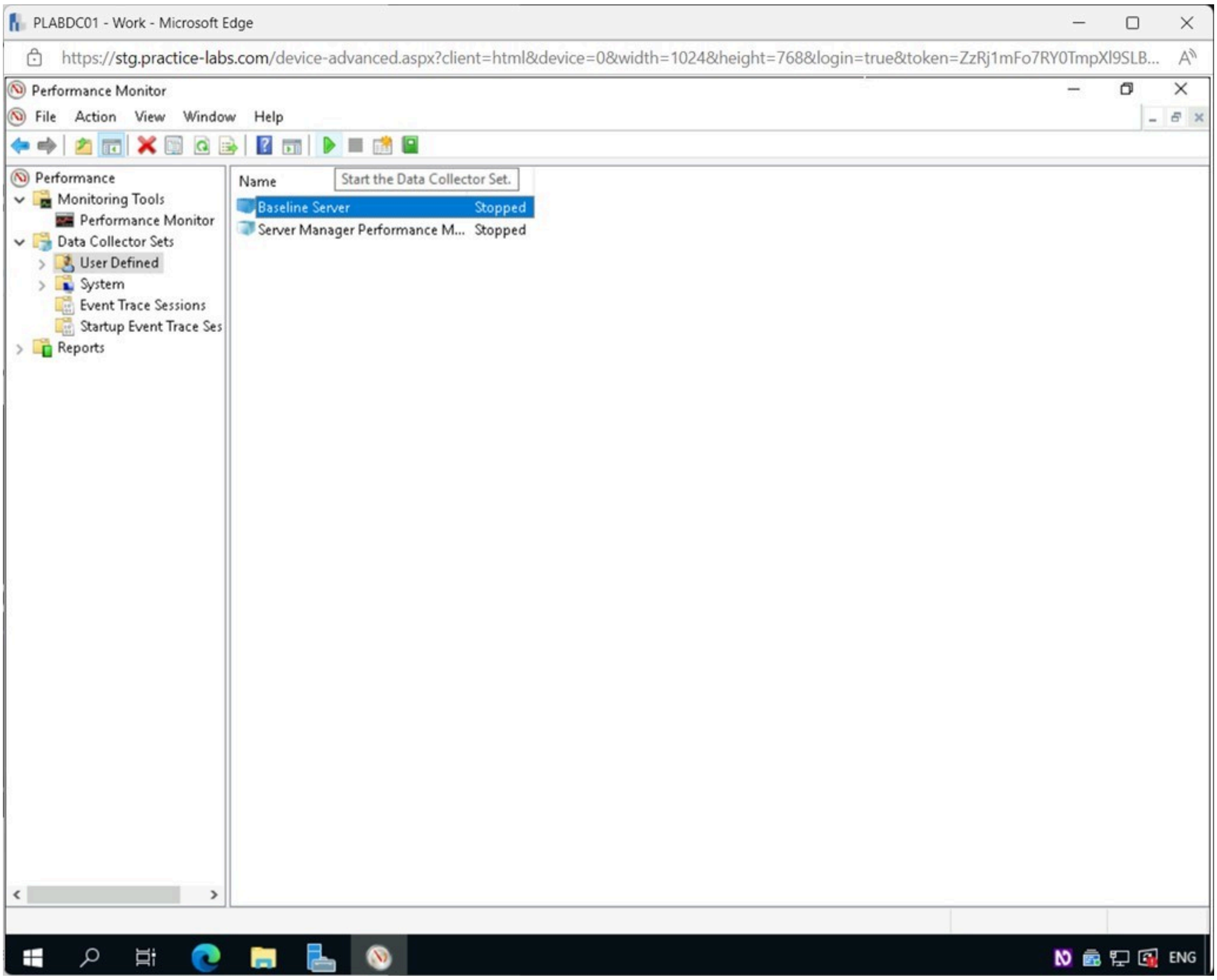

Figure 2.35 Screenshot of PLABDC01: Displaying selecting the Start the Data Collector Set icon in the Performance Monitor window.

Step 8

Let the process run for around one minute, and then press the **Stop the Data Collector Set** icon on the **Toolbar**.

> ***Note:*** *The process might stop automatically before you click the* ***Stop the Data Collector Set*** *icon.*

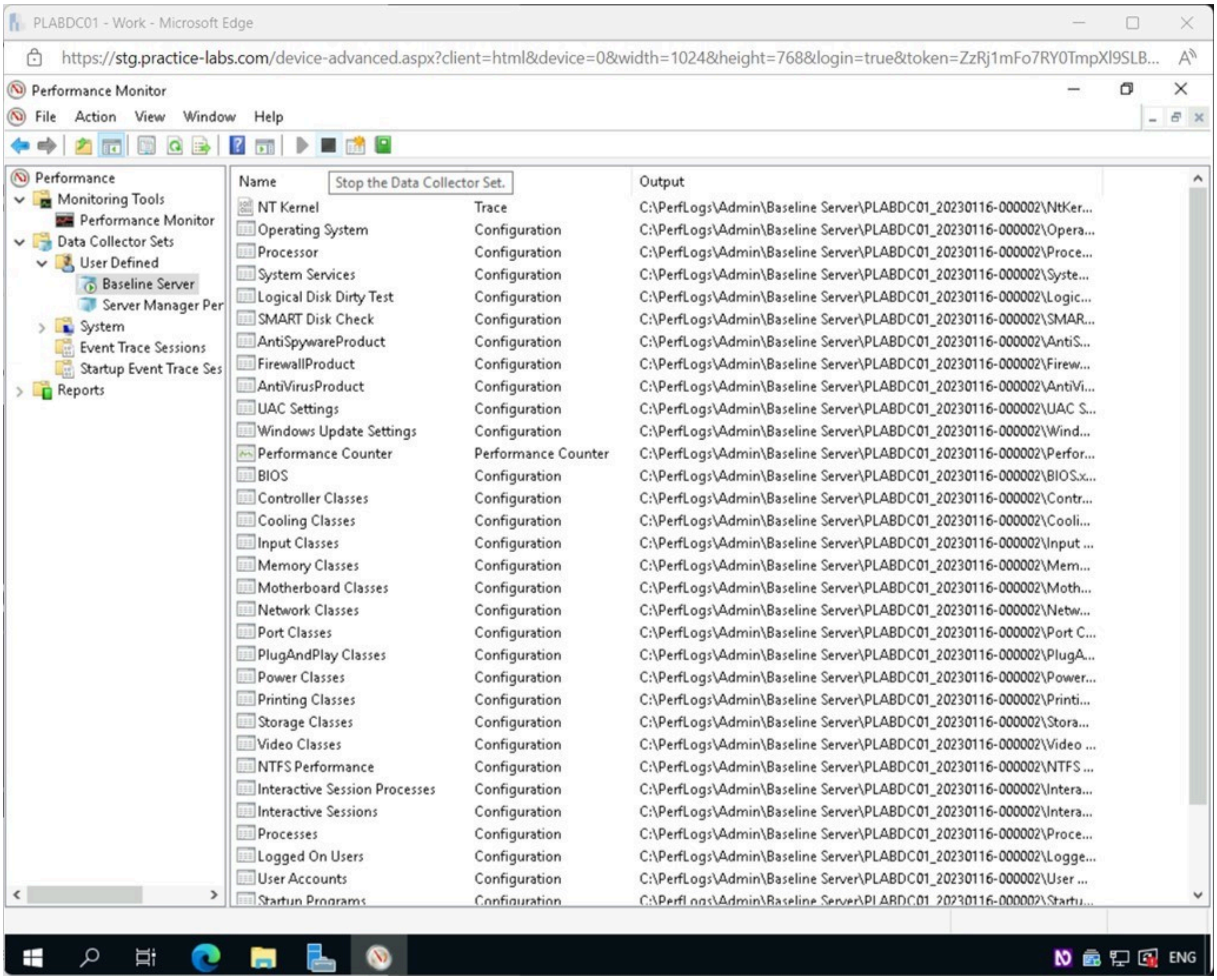

Figure 2.36 Screenshot of PLABDC01: Displaying selecting the Stop the Data Collector Set icon in the Performance Monitor window.

Step 9

In the **Performance Monitor** window, expand **Reports > User Defined > Baseline Server**

Select **PLABDC01_2023…**

The **data set** will run for roughly one minute and collect the information.

Notice the output from the data set. There is a lot of information and setting used in this predefined template.

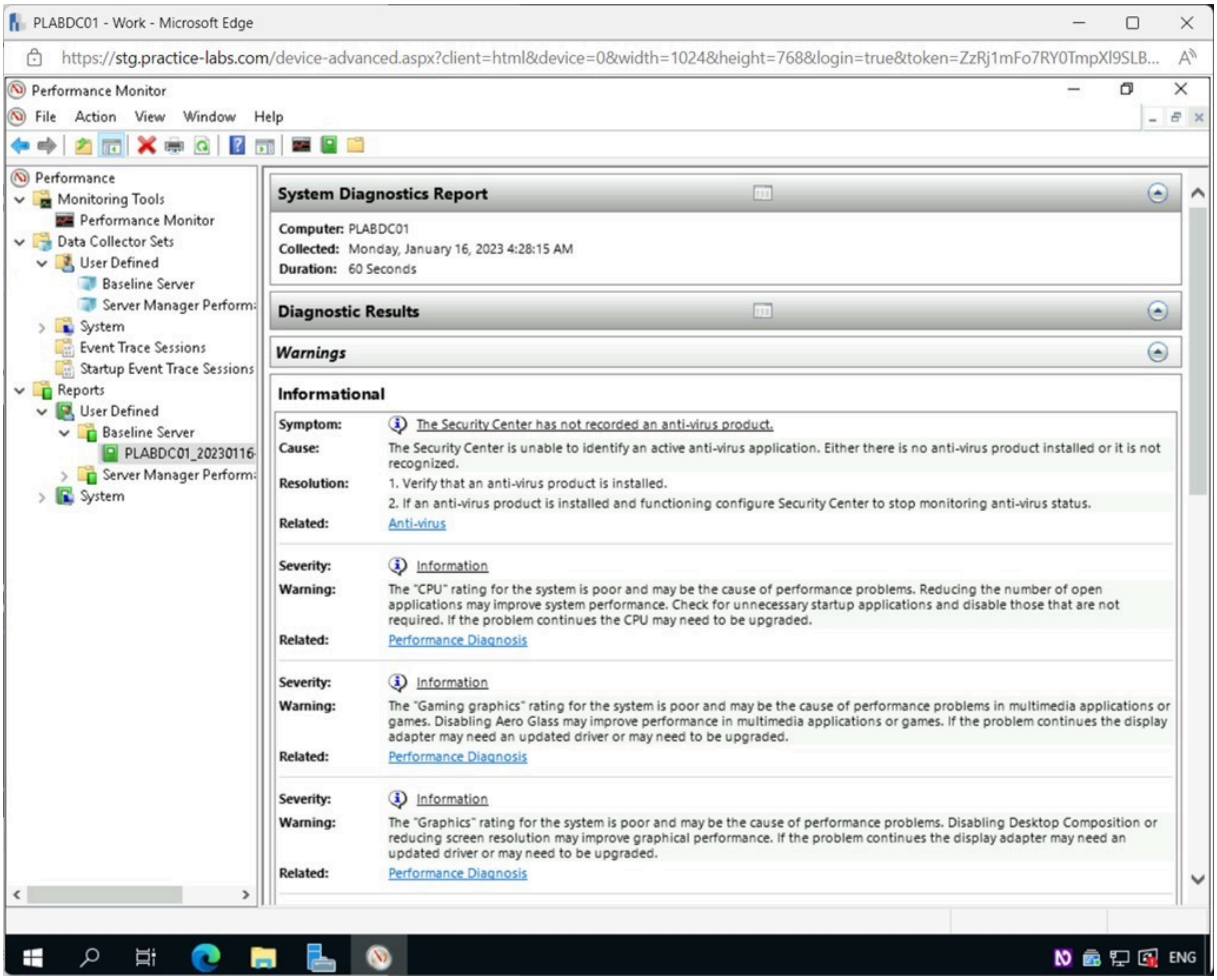

Figure 2.37 Screenshot of PLABDC01: Displaying the Performance Monitor window, showing the Reports for PLABDC01_2022 on the right pane.

Step 10

Right-click **User Defined** under the **Data Collector Sets** node and select **New** > **Data Collector Set**.

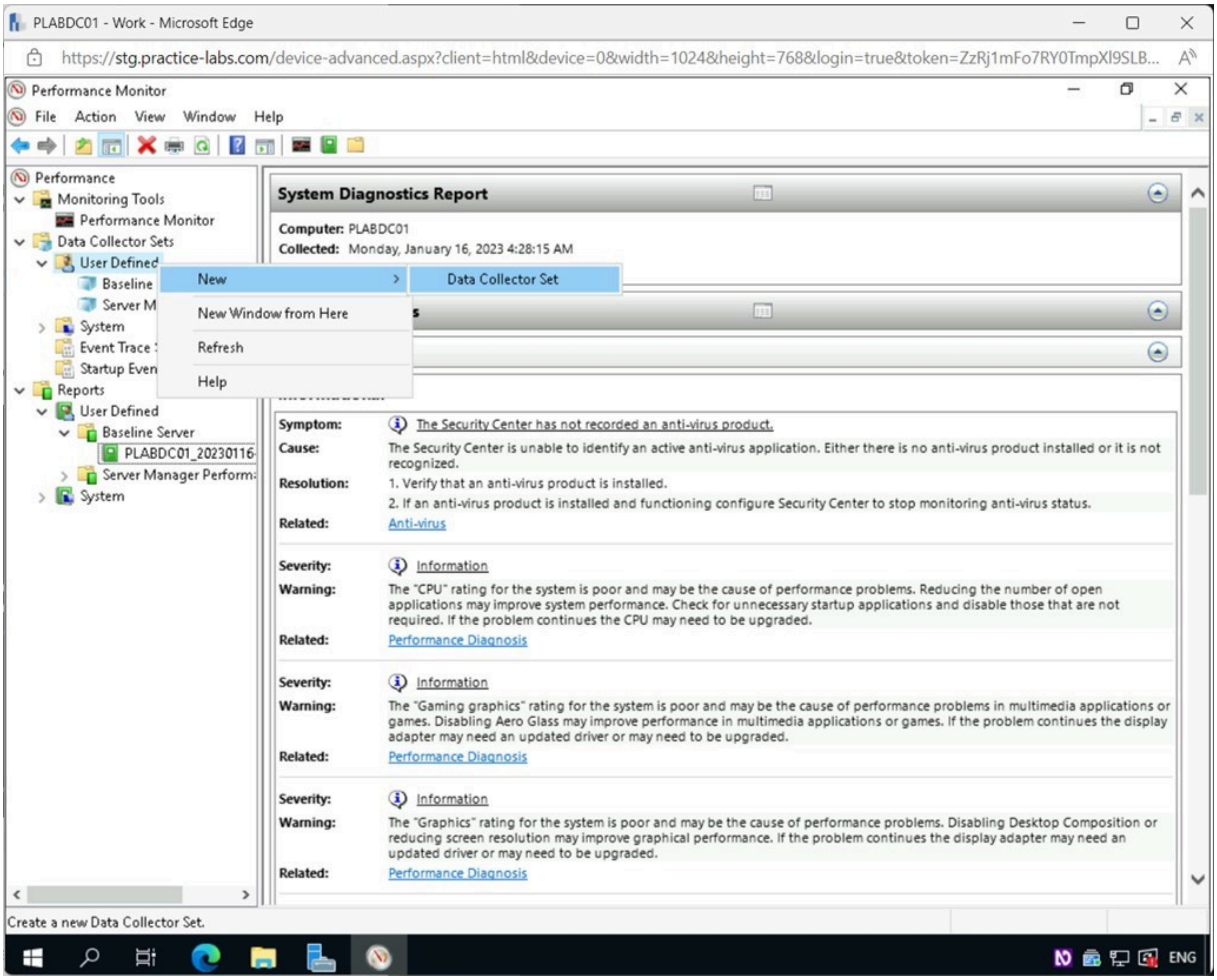

Figure 2.38 Screenshot of PLABDC01: Displaying the Performance Monitor window. User Defined is right-clicked, and New > Data Collector Set menu options are selected.

Step 11

In the **Create new Data Collector Set - How would you like to create this new data collector set?** window, type the following for the **Name** field:

```
General Setup
```

Choose the **Create manually(Advanced)** option.

Click **Next**.

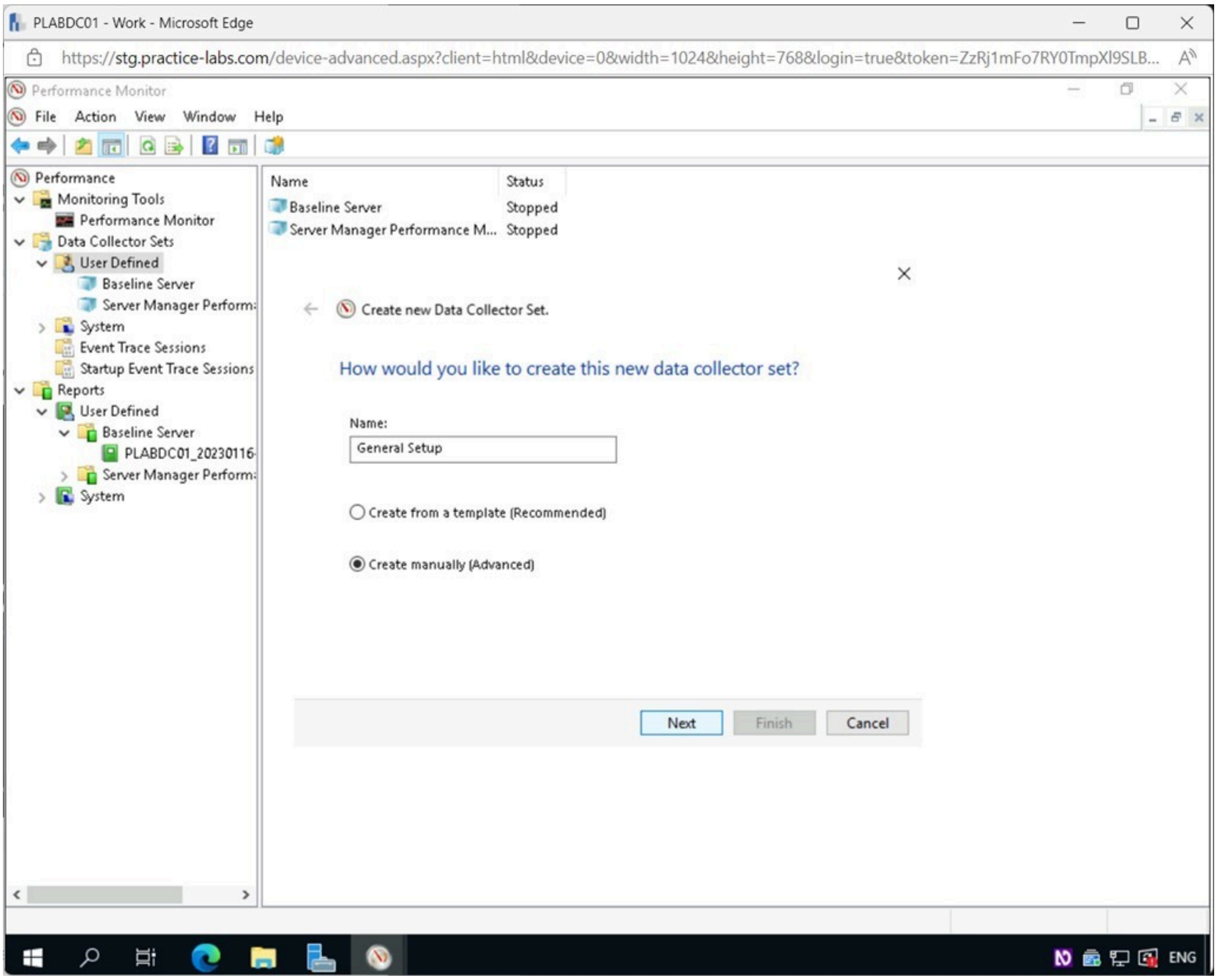

Figure 2.39 Screenshot of PLABDC01: Displaying the Create new Data Collector Set - How would you like to create this new data collector set? Window, showing the required settings performed and the Next button selected.

Step 12

In the **What type of data do you want to include?** page, tick the **Performance counter** checkbox.

Click **Next**.

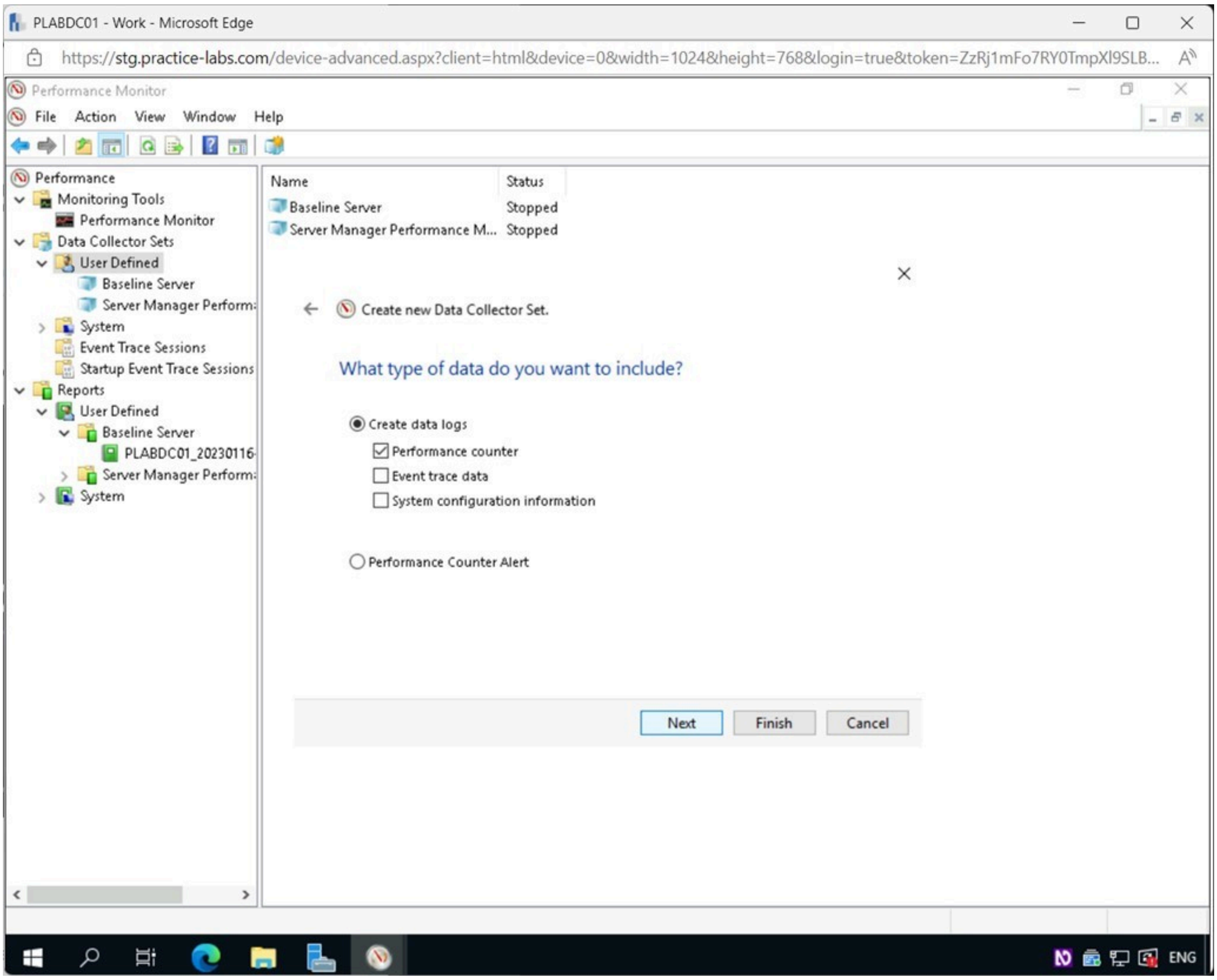

Figure 2.40 Screenshot of PLABDC01: Displaying the What type of data do you want to include? page, showing the required settings performed and the Next button selected.

Step 13

From the **Which performance counters would you like to log?** page, click **Add**.

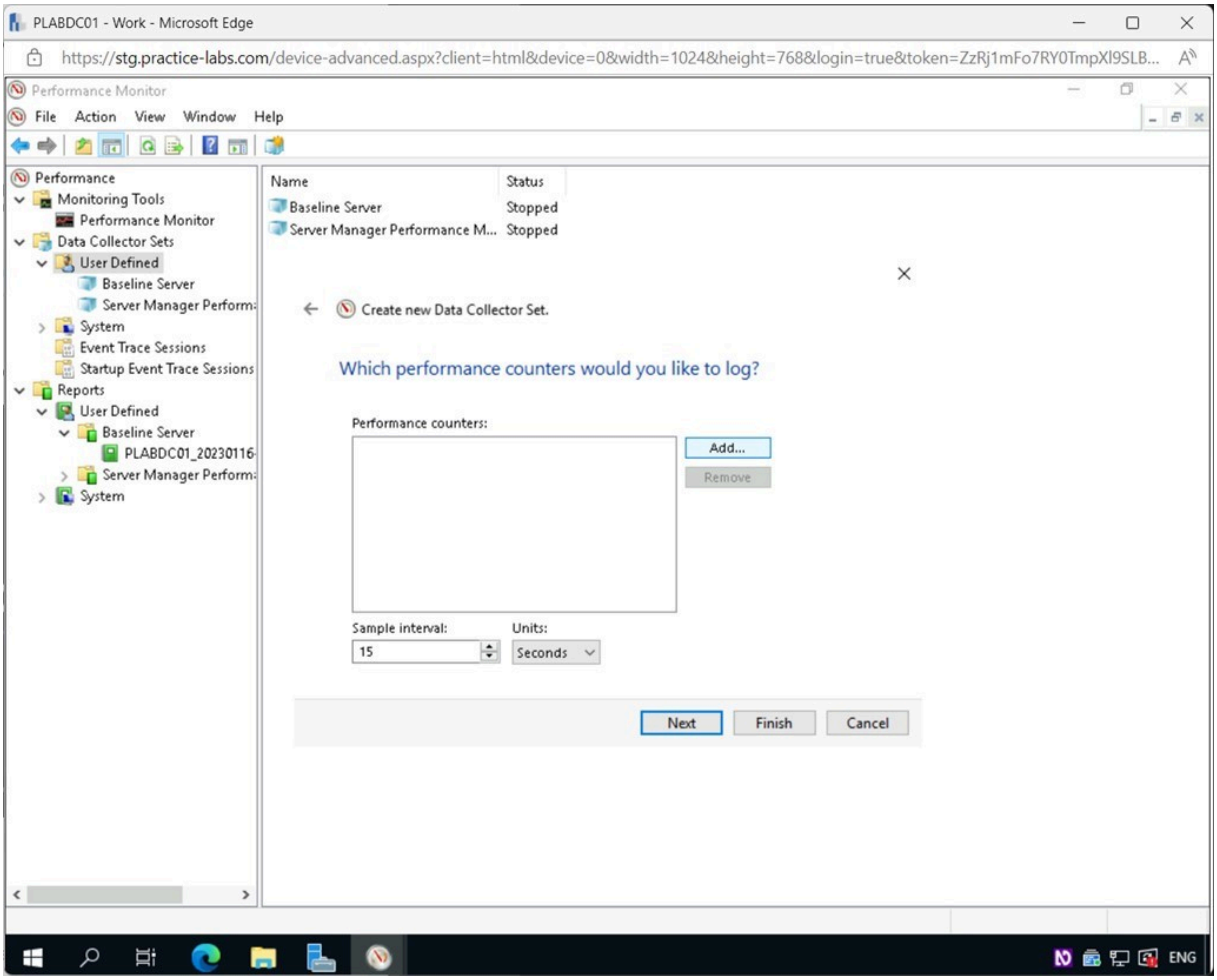

Figure 2.41 Screenshot of PLABDC01: Displaying the Which performance counters would you like to log? page, showing the Add button selected.

Step 14

In the **Available counters** window, scroll down and select **System**.

Click **Add>>.**

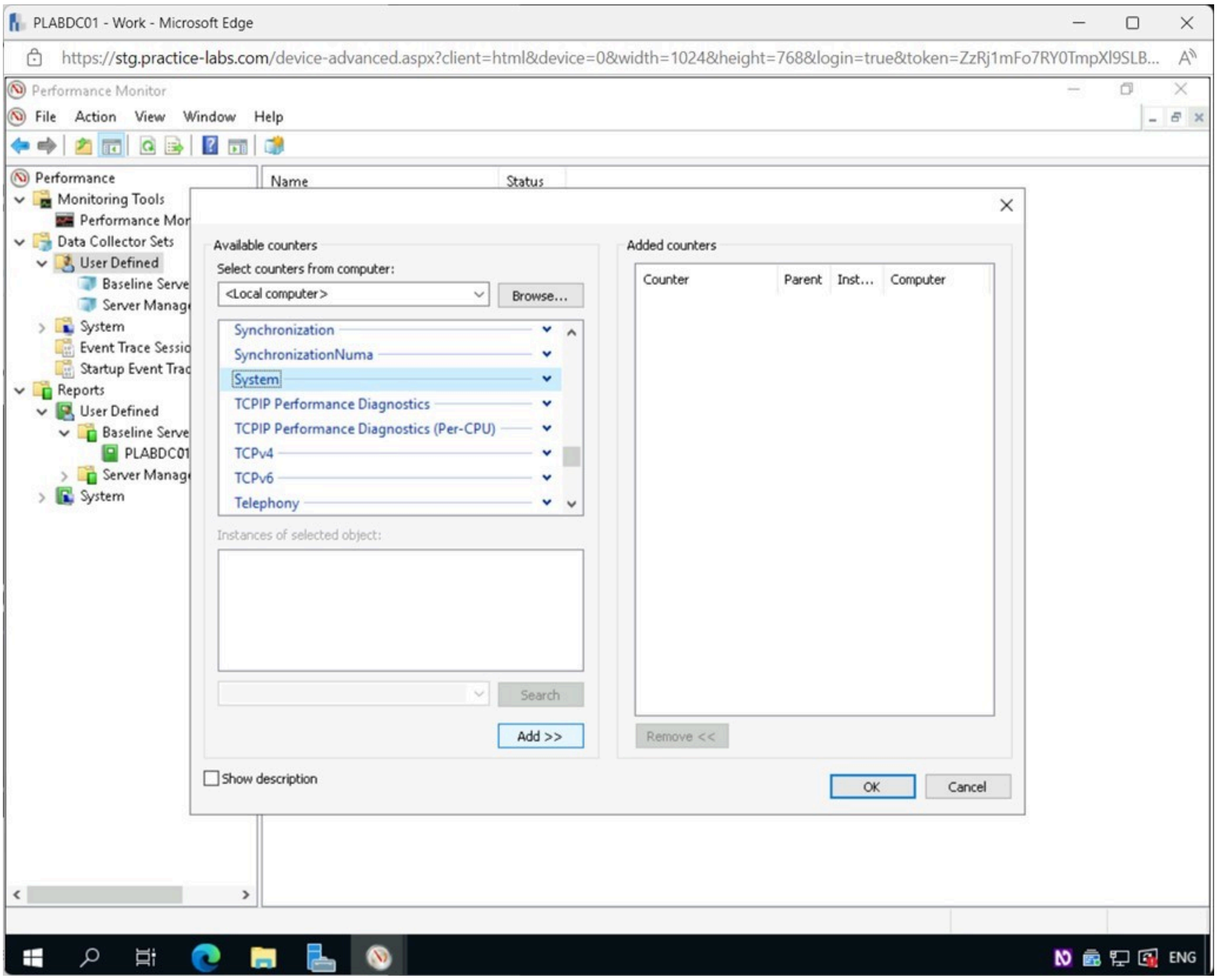

Figure 2.42 Screenshot of PLABDC01: Displaying the Available counters window, showing the Add>> button selected.

Step 15

System has now been added to the **Added counters** section.

Similarly, add the following categories:

- **Processor**
- **PhysicalDisk**
- **Network Interface**
- **Memory**

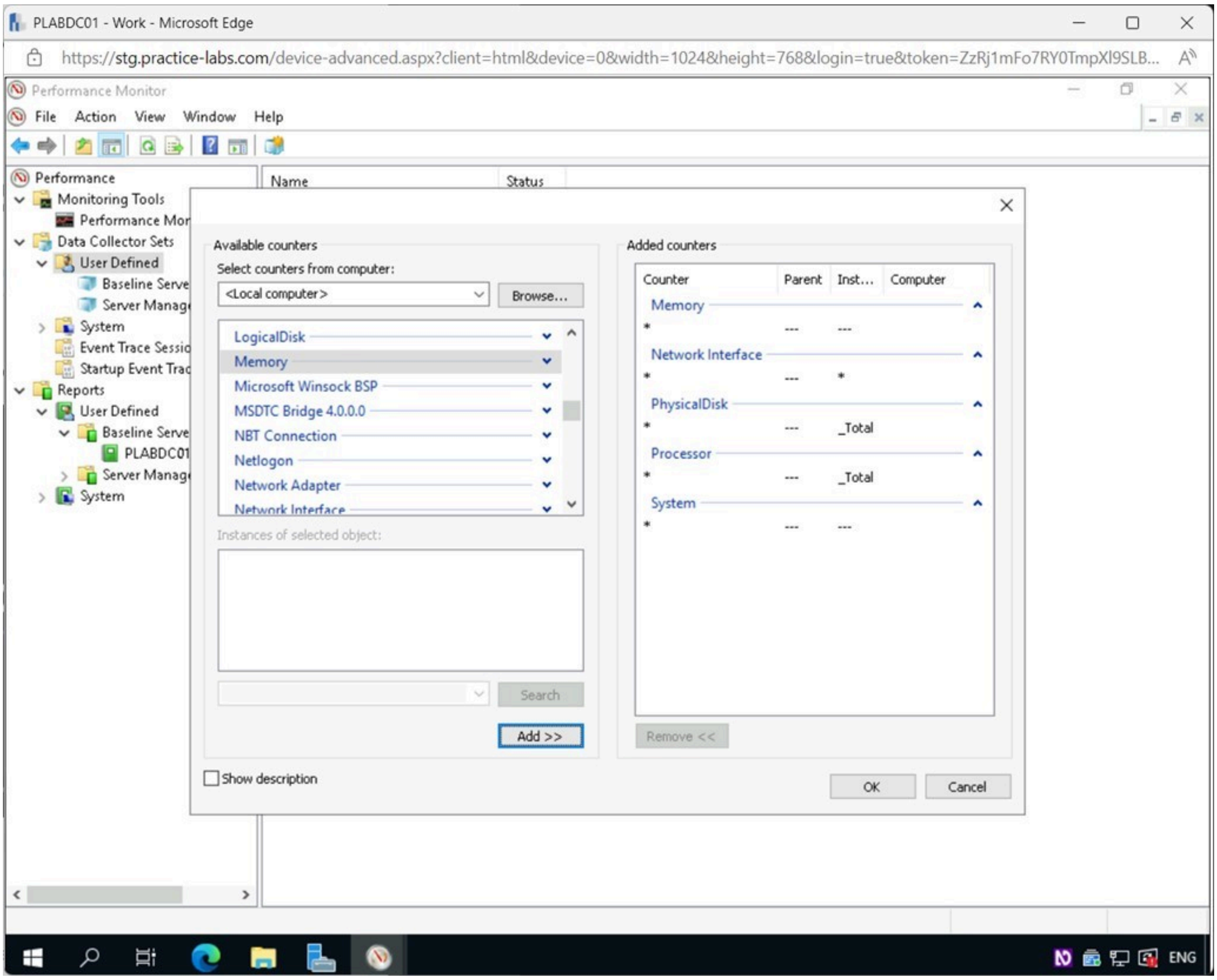

Figure 2.43 Screenshot of PLABDC01: Displaying the Available counters window, showing the required settings performed.

Step 16

Once all the required categories have been added to the **Added counters** section, click **OK**.

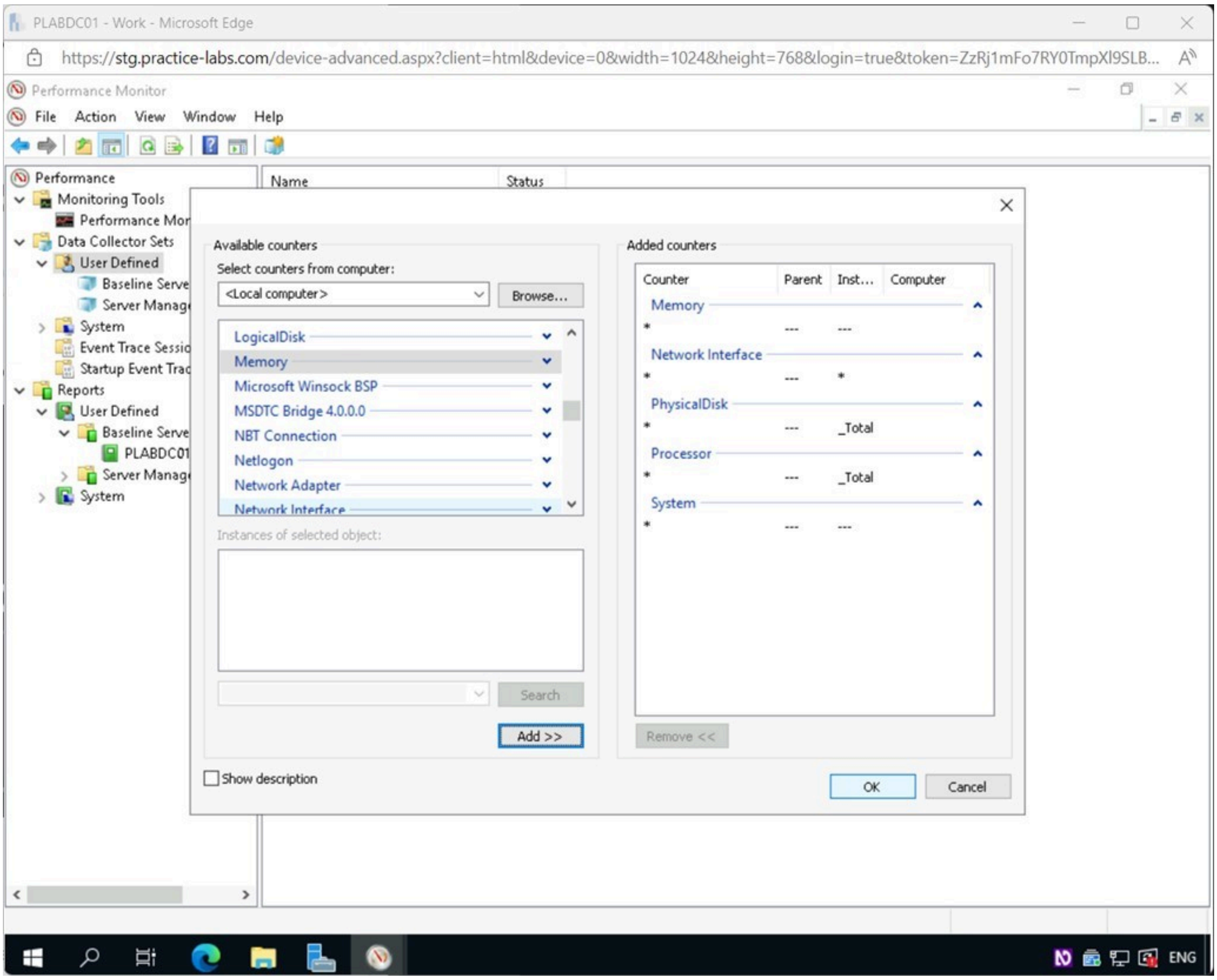

Figure 2.44 Screenshot of PLABDC01: Displaying selecting OK in the Available counters window.

Step 17

Back on the **Which performance counters would you like to log?** page, click **Finish**.

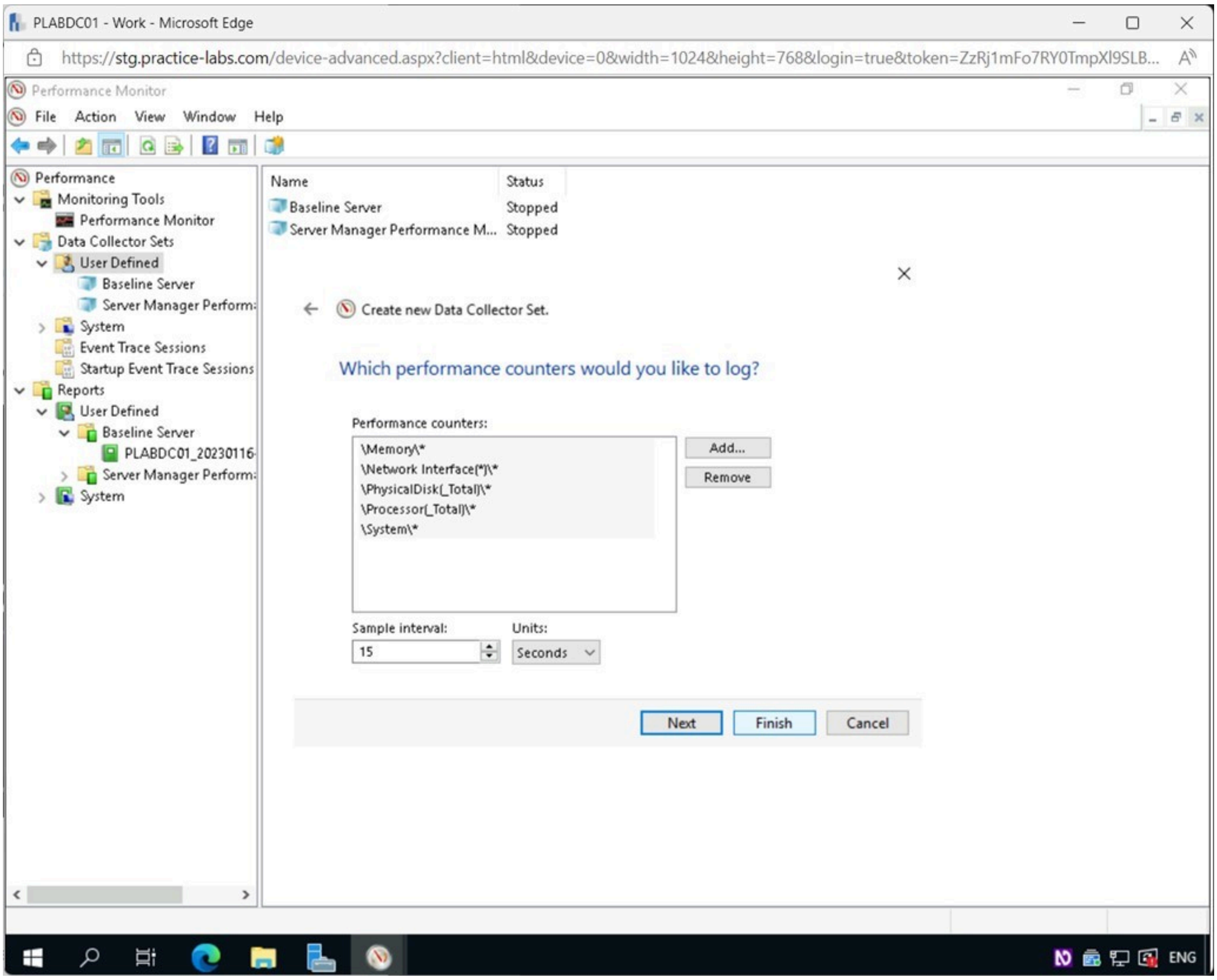

Figure 2.45 Screenshot of PLABDC01: Displaying selecting Finish in the Which performance counters would you like to log? page.

Step 18

Back on the **Performance Monitor** window, notice **General Setup** appears on the right details pane.

Select **General Setup** and then click the **Start the Data Collector Set** icon on the **Toolbar**.

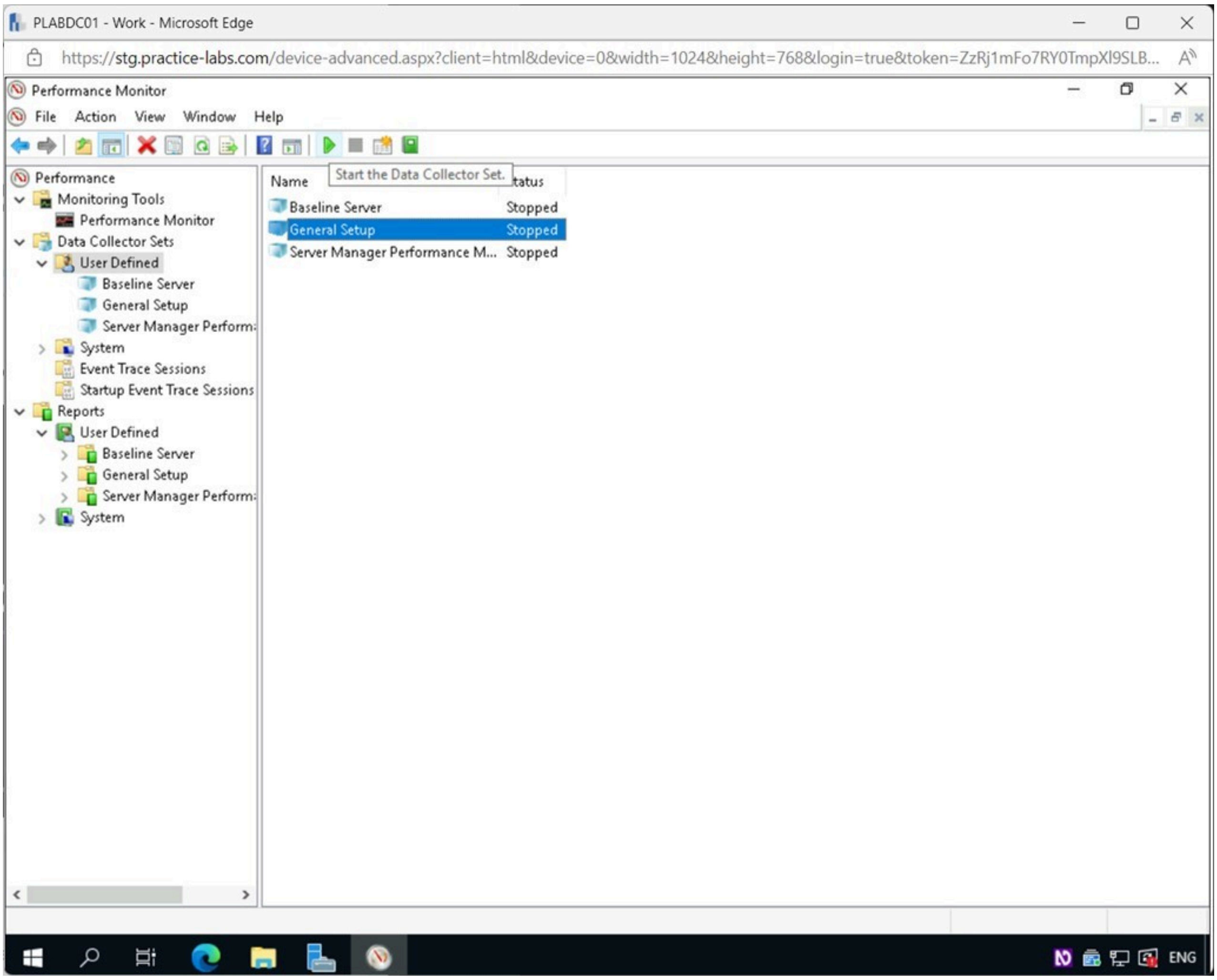

Figure 2.46 Screenshot of PLABDC01: Displaying the Performance Monitor window. Start the Data Collector Set icon on the Toolbar is selected.

Step 19

Let the process run for around one minute, and then press the **Stop the Data Collector Set** icon on the **Toolbar**.

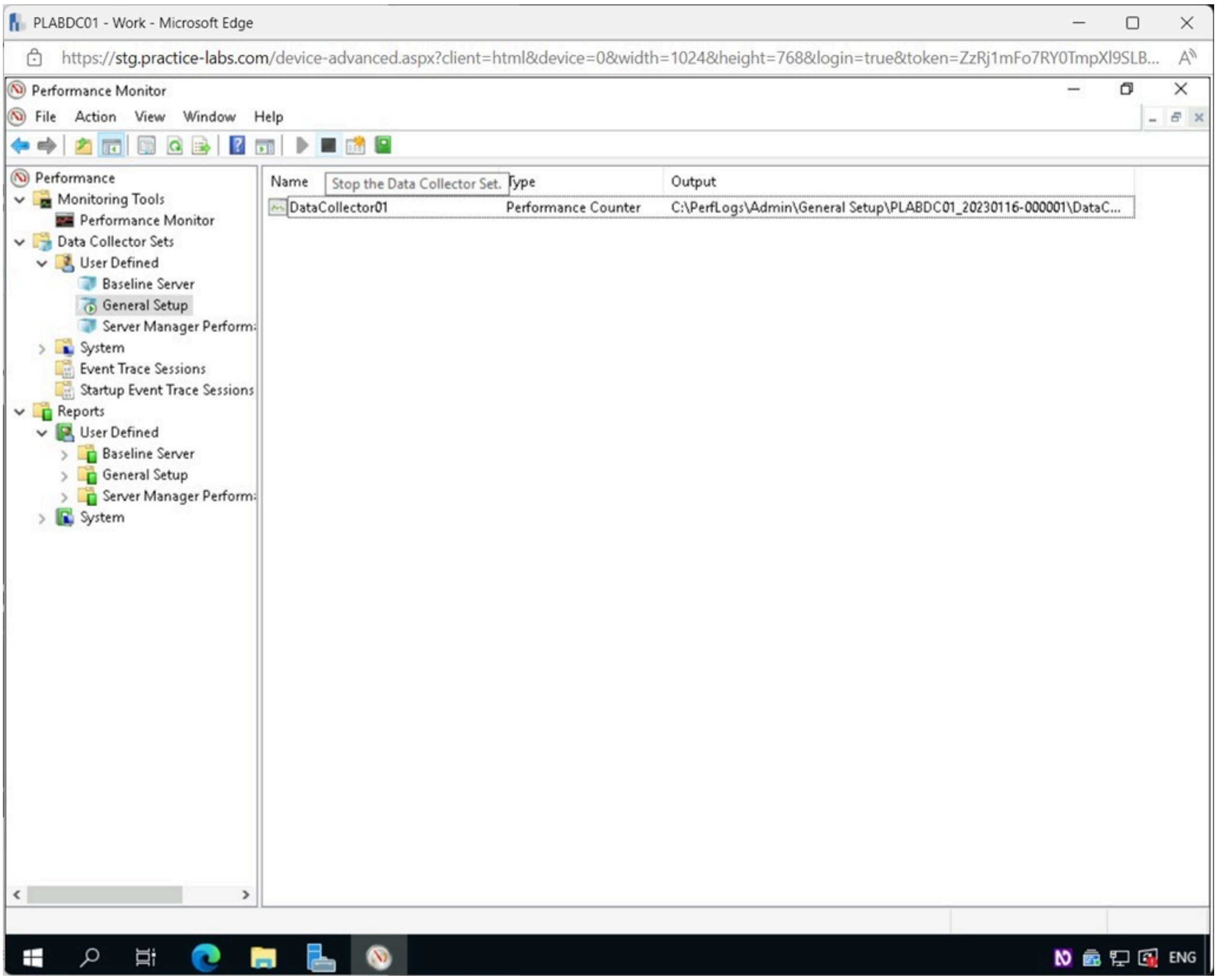

Figure 2.47 Screenshot of PLABDC01: Displaying the Performance Monitor window. Stop the Data Collector Set icon on the Toolbar is selected.

Step 20

In the **Performance Monitor** window, expand **Reports > User Defined > General Setup** on the left pane.

Select **PLABDC01_2023…**

Notice the output from the data set. There is still a lot of information in this template, but it is more refined than the predefined template previously used.

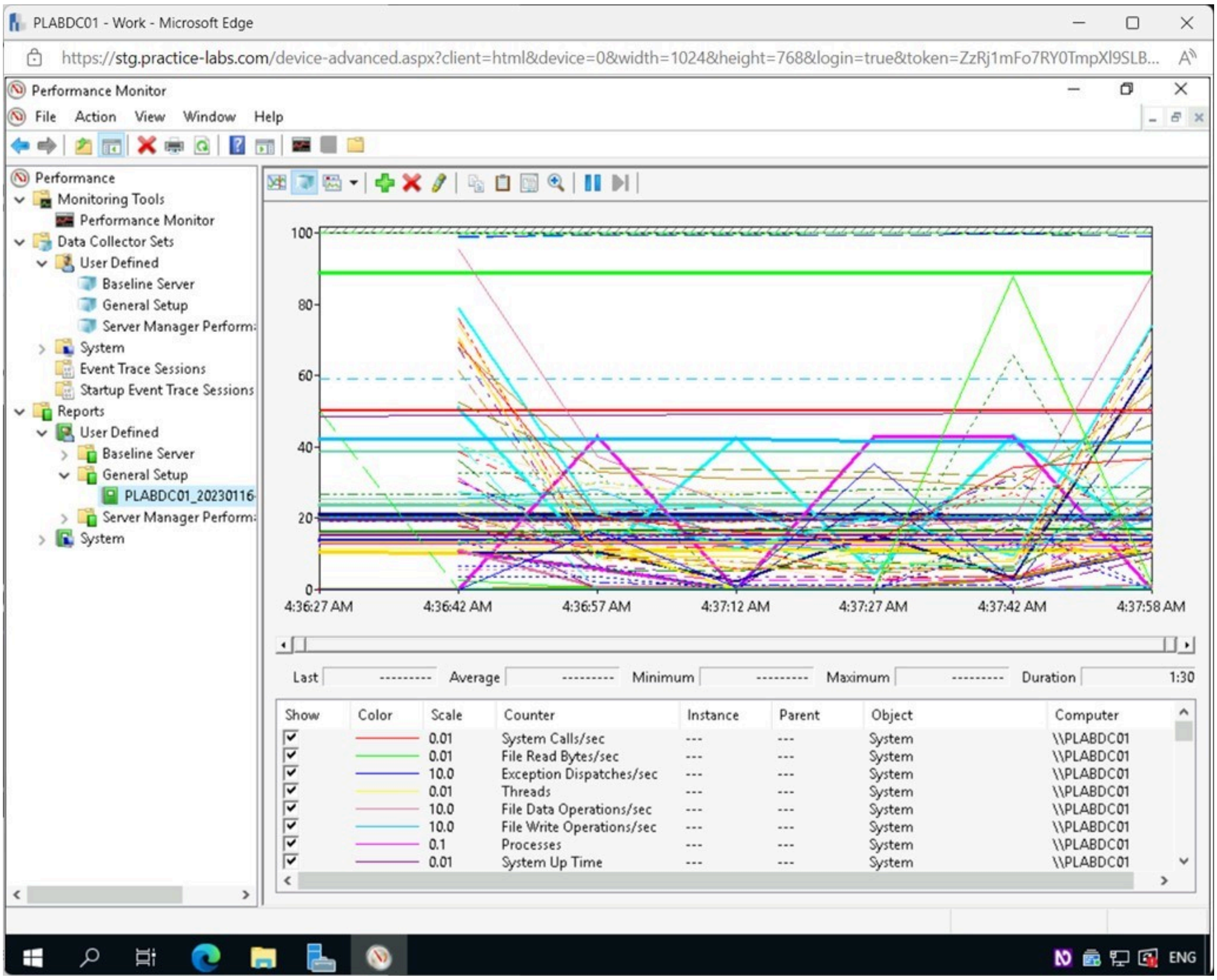

Figure 2.48 Screenshot of PLABDC01: Displaying the Performance Monitor window, showing the Reports for PLABDC01_2022 on the right pane.

Step 21

Right-click **User Defined** under the **Data Collector Sets** node and select **New** > **Data Collector Set**.

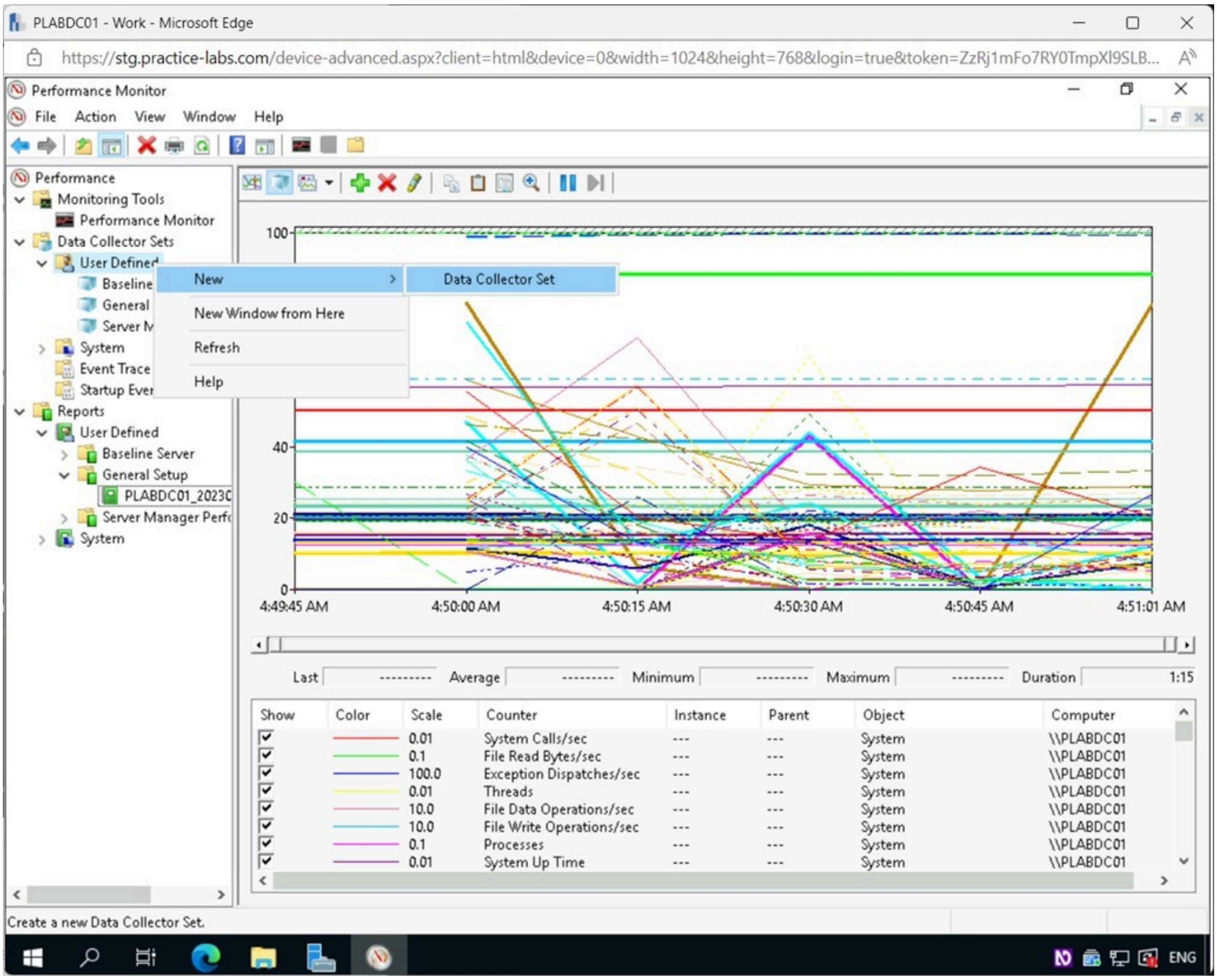

Figure 2.49 Screenshot of PLABDC01: Displaying the Performance Monitor window. User Defined is right-clicked, and New > Data Collector Set menu options are selected.

Step 22

In the **Create new Data Collector Set - How would you like to create this new data collector set?** window, type the following for the **Name** field:

```
Common Performance Counters
```

Choose the **Create manually(Advanced)** option.

Click **Next**.

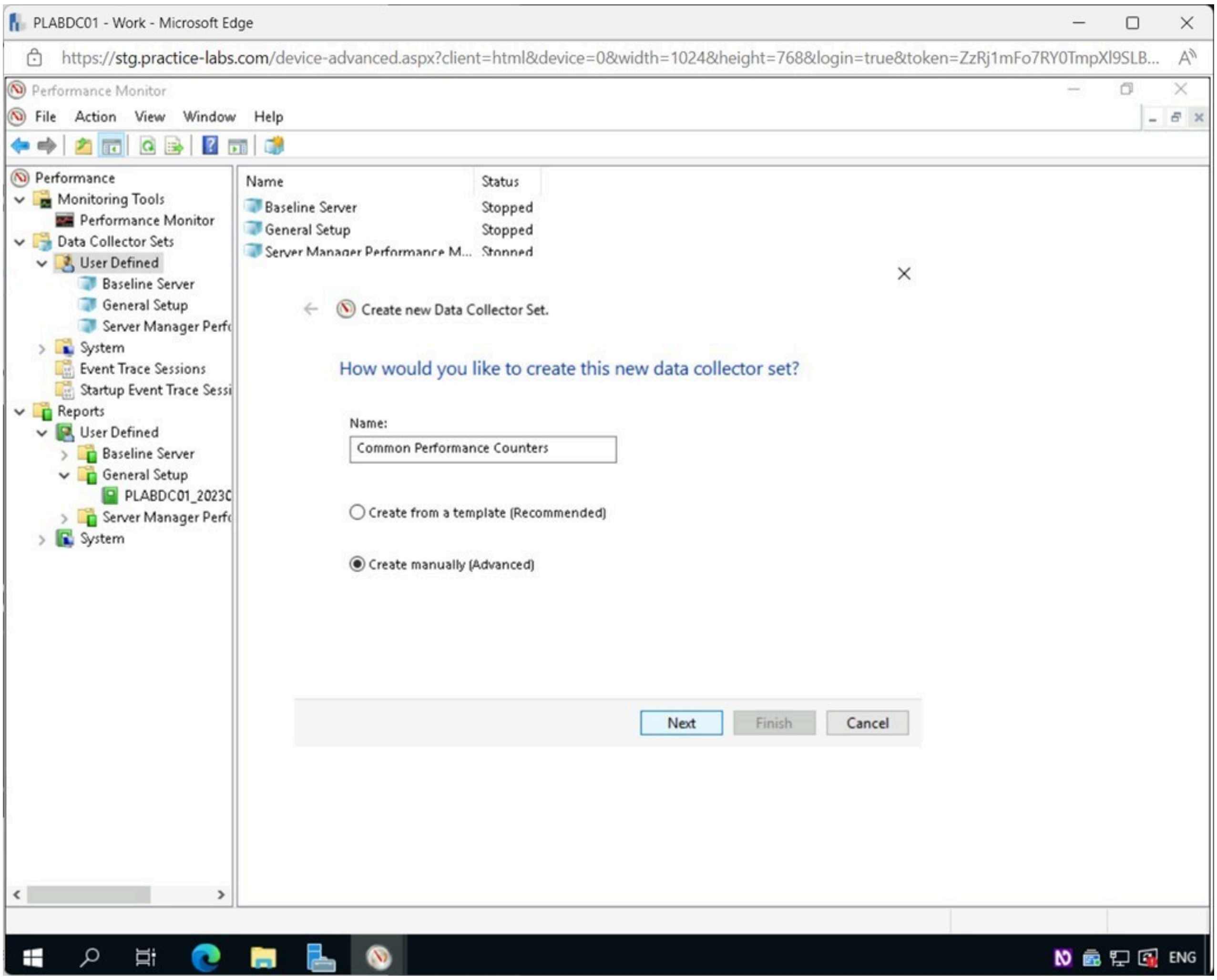

Figure 2.50 Screenshot of PLABDC01: Displaying the Create new Data Collector Set - How would you like to create this new data collector set? window, showing the required settings performed and the Next button selected.

Step 23

In the **What type of data do you want to include?** page, tick the **Performance counter** checkbox.

Click **Next**.

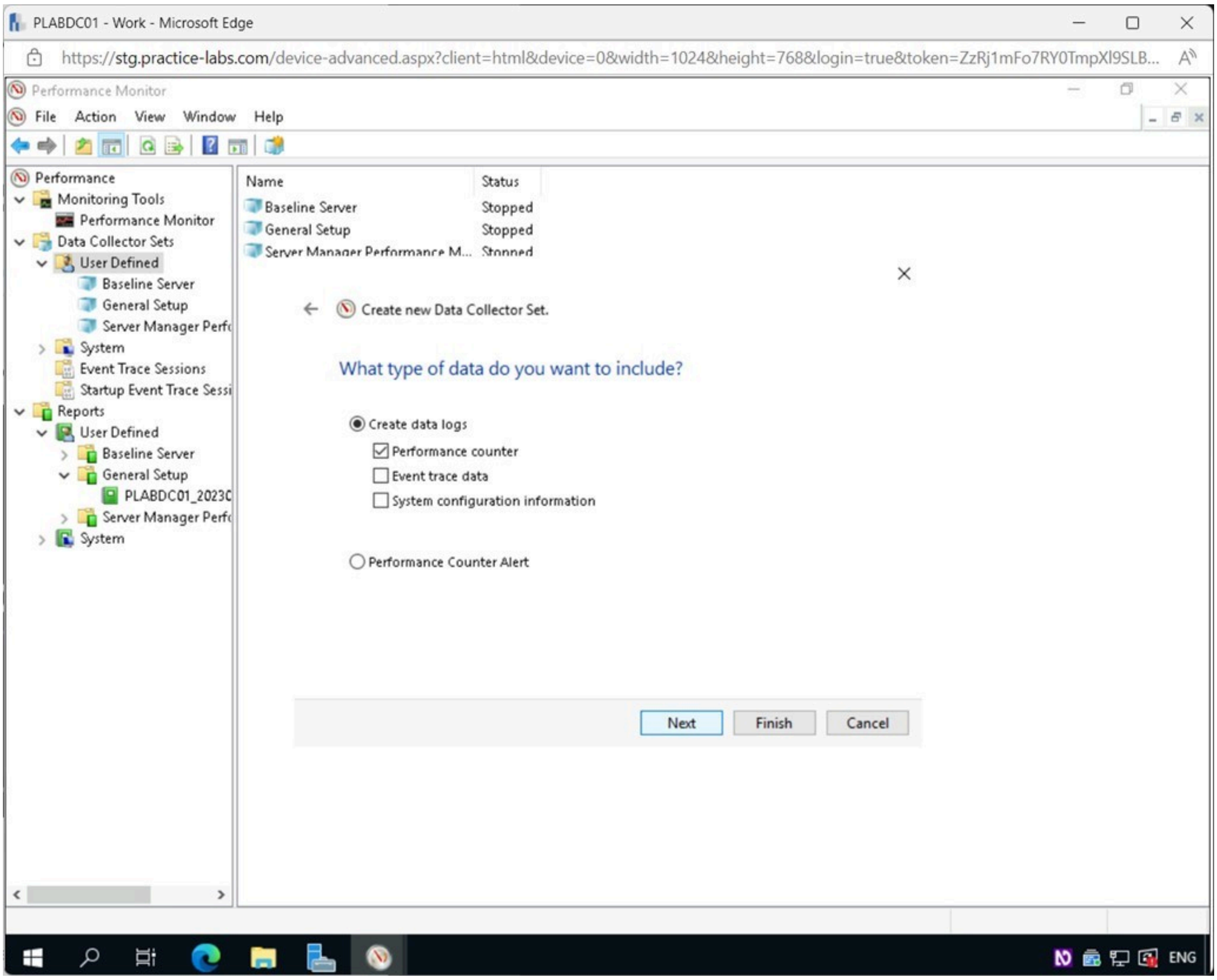

Figure 2.51 Screenshot of PLABDC01: Displaying the What type of data do you want to include? page, showing the required settings performed and the Next button selected.

Step 24

In the **Which performance counters would you like to log?** page, click **Add.**

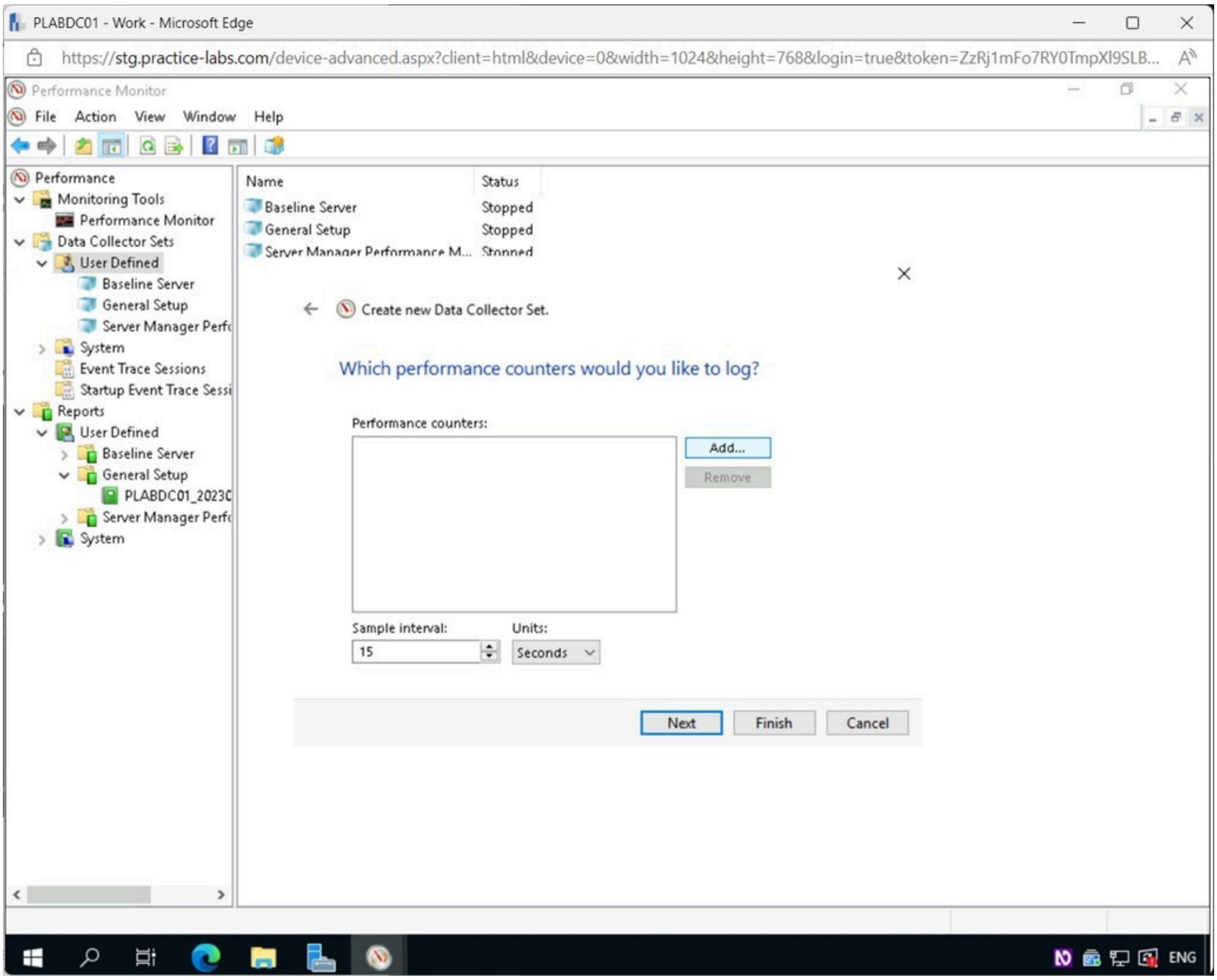

Figure 2.52 Screenshot of PLABDC01: Displaying the Which performance counters would you like to log? page, showing the Add button selected.

Step 25

Select the following categories and sub-categories and click **Add**>>:

- Expand **Memory** and select **Pages per Second:** This tracks the number of times a hard page fault is resolved by reading or writing a page to disk. More than 1,000 is considered excessive paging and may indicate issues with the memory.
- Expand **Network Interface** and select **Bytes Total/Sec:** This keeps track of the total number of bytes sent across each network interface at any given time. If more than 70% of the interface is being used, the network is at capacity.
- Expand **PhysicalDisk** and select **Avg. Disk Queue Length:** To see how many I/O operations are queued up, look at this counter. It's possible that the disk is the bottleneck.
- Expand **PhysicalDisk** and select **% Disk Time:** This displays how much of the sample interval the disk was really used for. The disk system is at capacity if this counter reaches 85%.
- Expand **Processor** and select **% Processor Time:** This keeps track of how much of the time the CPU is really doing something other than sitting idle. When the number goes

beyond 85%, it means the server might benefit from a more powerful CPU.

- Expand **System** and select **Processor Queue Length:** The number of threads in the processor's queue is displayed here. If the value exceeds double the number of CPUs for a lengthy period, the server can crash.

Click **OK**.

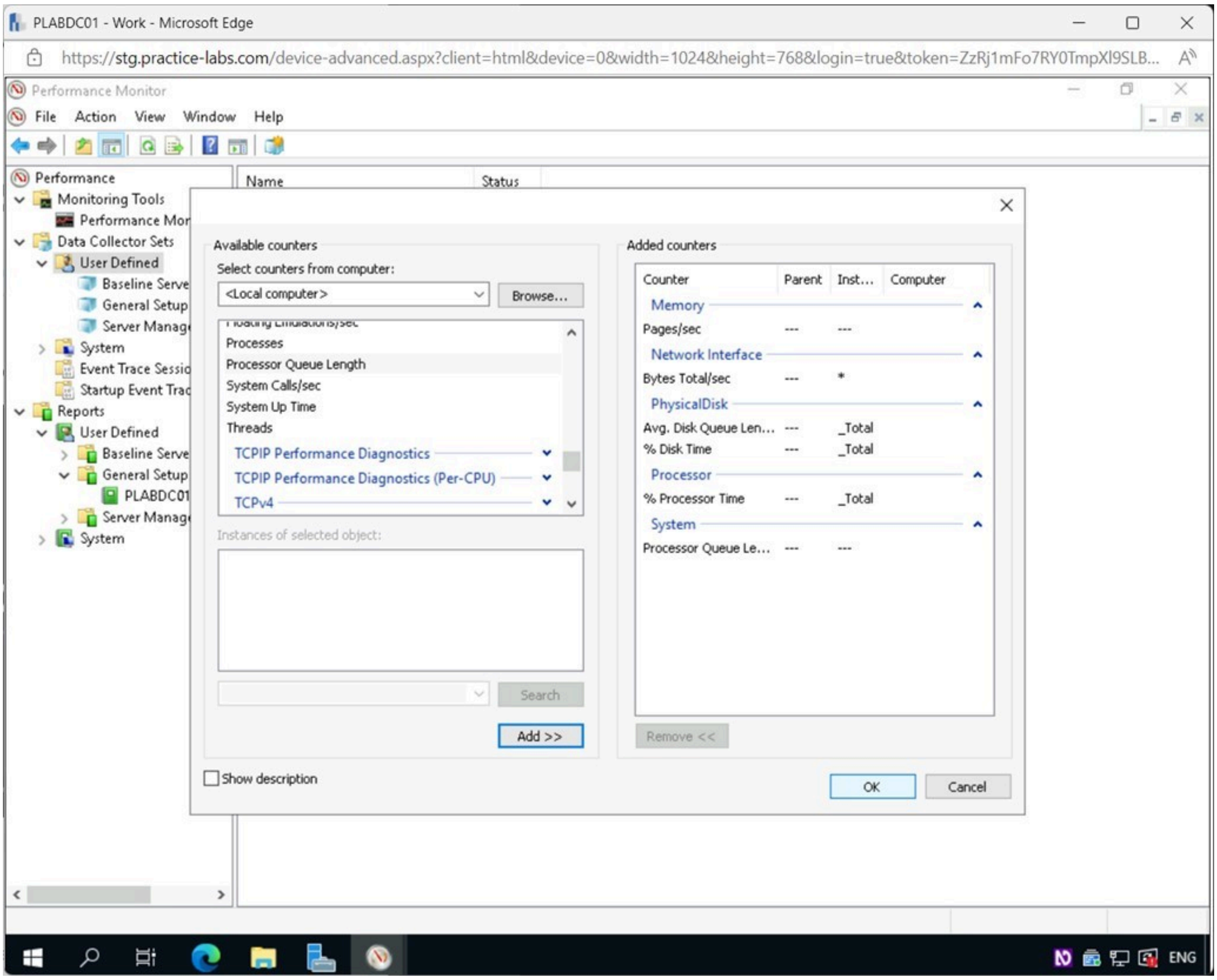

Figure 2.53 Screenshot of PLABDC01: Displaying the Available counters window, showing the required settings performed and the OK button selected.

Step 26

Back on the **Which performance counters would you like to log?** page, click **Finish**.

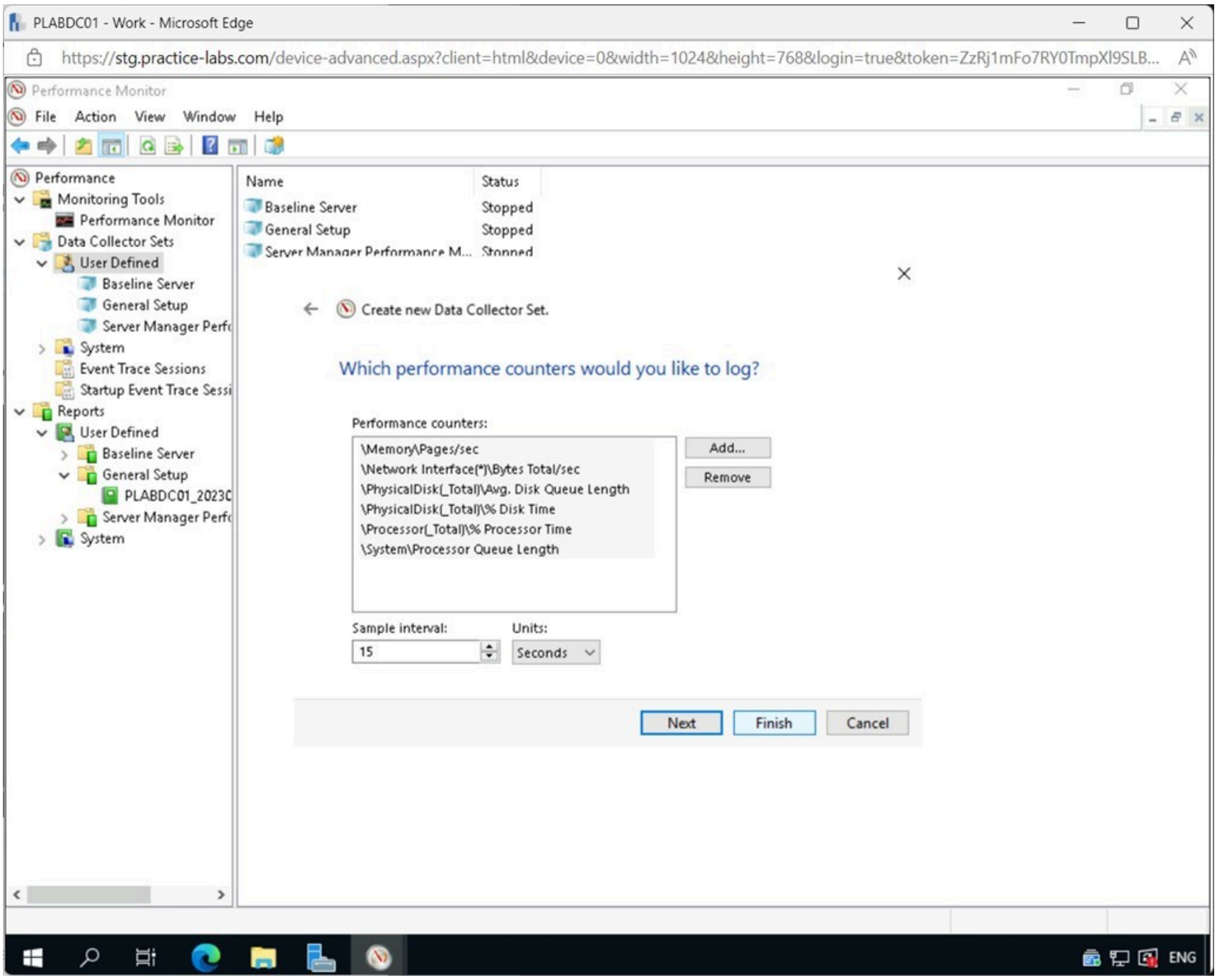

Figure 2.54 Screenshot of PLABDC01: Displaying selecting Finish in the Which performance counters would you like to log? page.

Step 27

Back on the **Performance Monitor** window, notice **Common Performance Counters** appear on the right pane.

Select **Common Performance Counters** and then click the **Start the Data Collector Set** icon on the **Toolbar**.

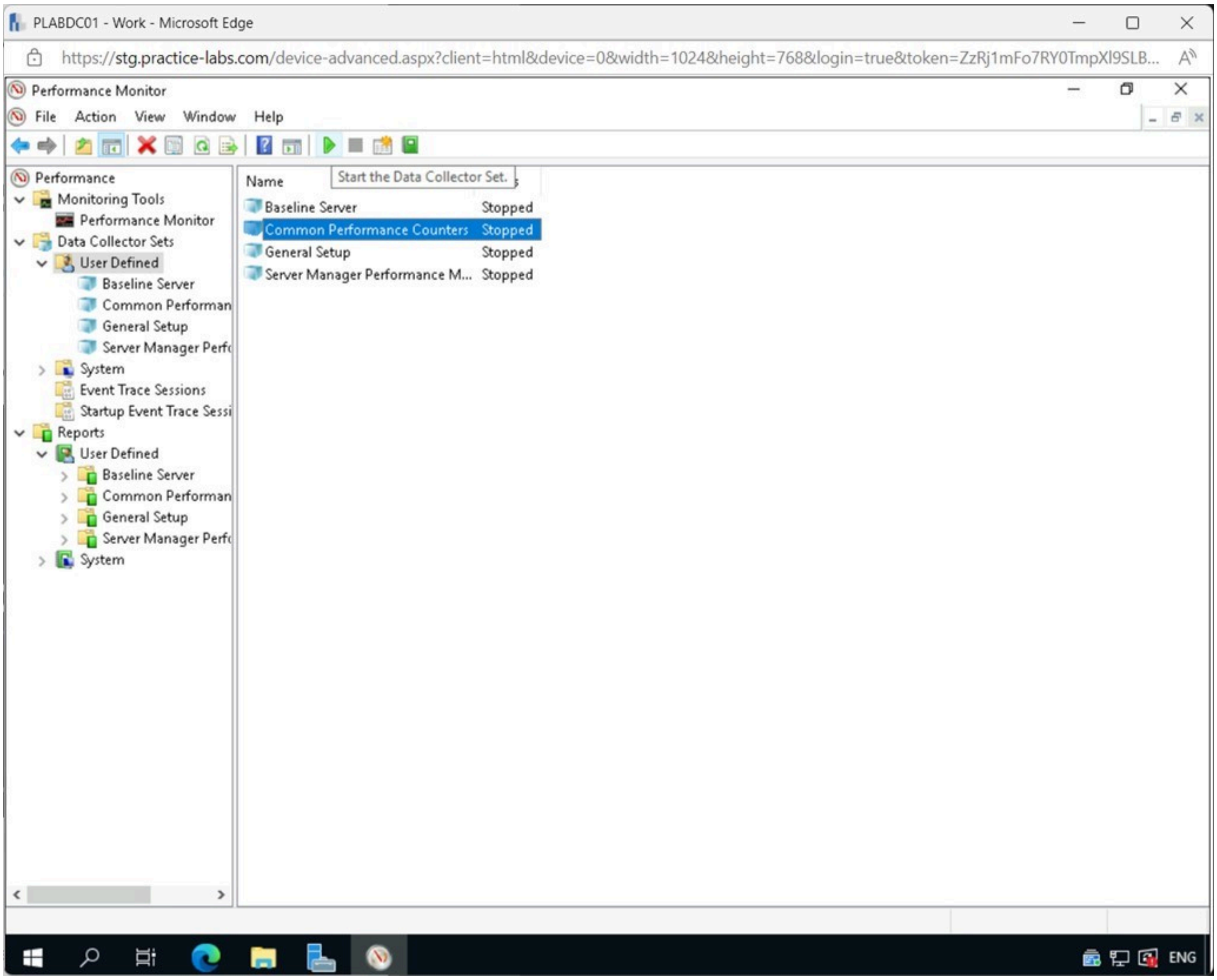

Figure 2.55 Screenshot of PLABDC01: Displaying the Performance Monitor window. Start the Data Collector Set icon on the Toolbar is selected.

Step 28

Let the process run for around one minute, and then press the **Stop the Data Collector Set** icon on the **Toolbar**.

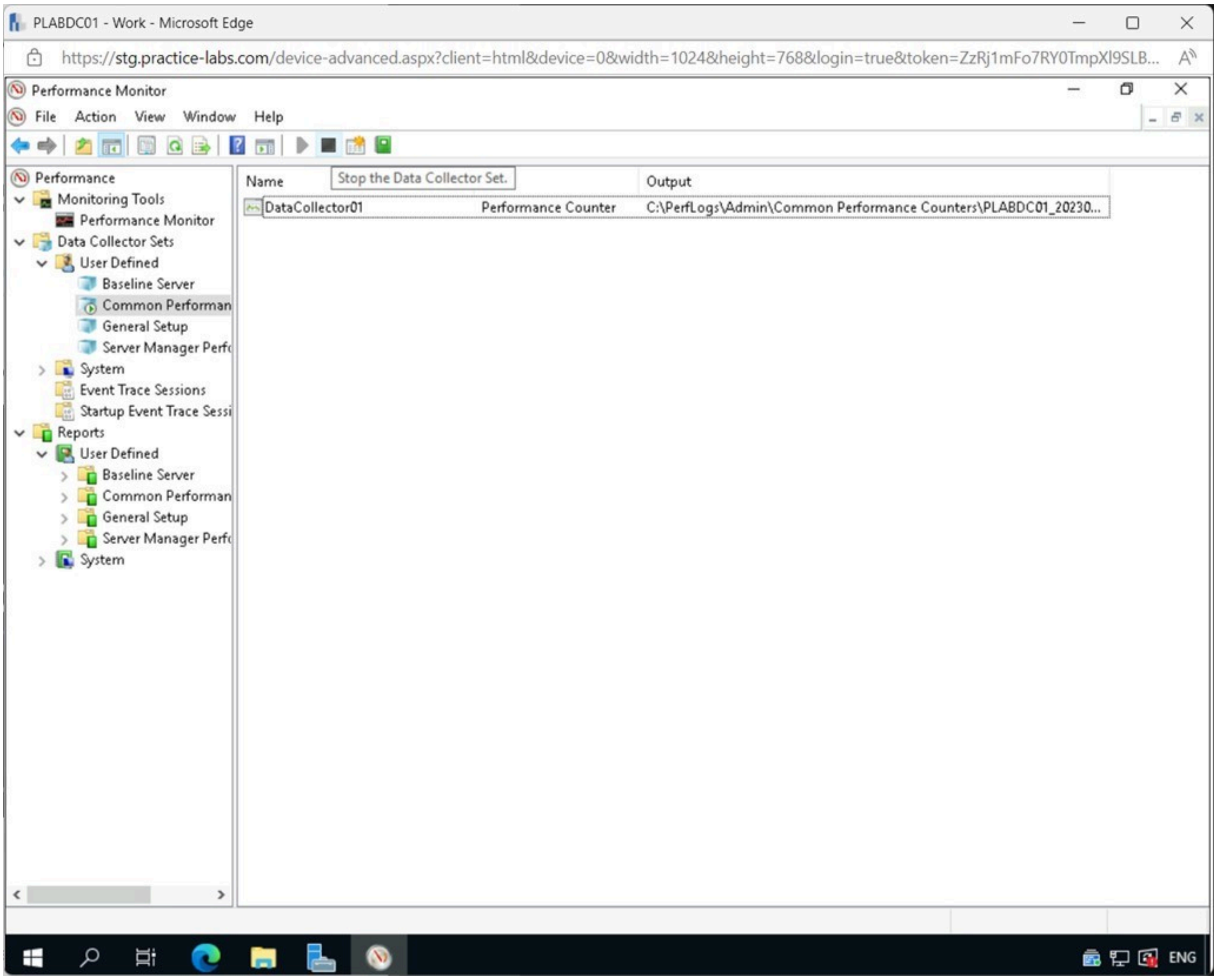

Figure 2.56 Screenshot of PLABDC01: Displaying the Performance Monitor window. Stop the Data Collector Set icon on the Toolbar is selected.

Step 29

In the **Performance Monitor** window, expand **Reports > User Defined > Common Performance Counters** on the left pane.

Select **PLABDC01_2023…**

Notice the output from the data set. This report is much more manageable and has some actionable items we want to keep an eye on.

Close the **Performance Monitor** window.

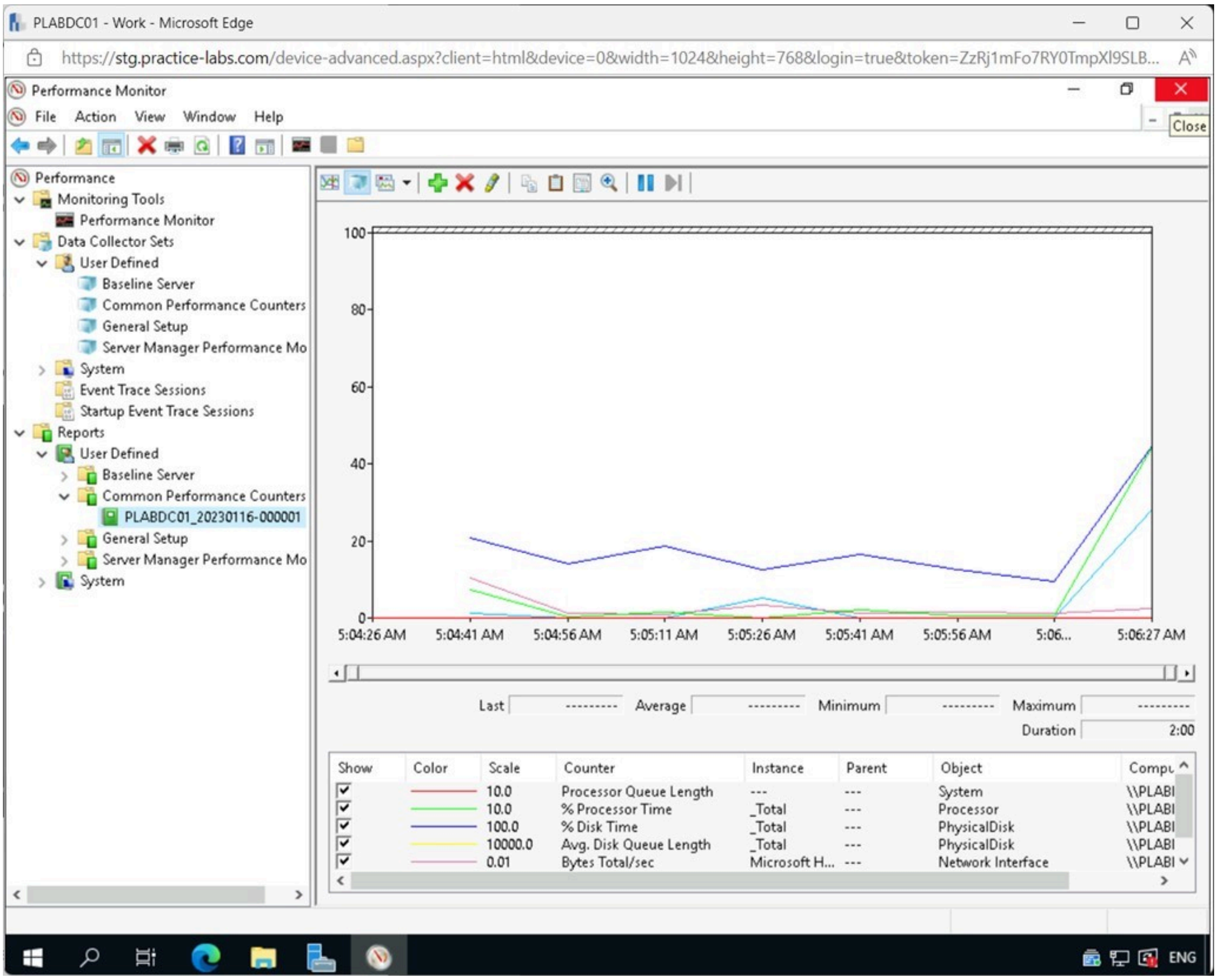

Figure 2.57 Screenshot of PLABDC01: Displaying the Performance Monitor window, showing the Reports for PLABDC01_2022 on the right pane. Close icon on the top-right corner of the window is selected

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Troubleshooting Server Storage Related Issues** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Common Storage Problems

- Exercise 2 - Tools and Techniques for Troubleshooting

You should now be able to:

- Know about Read/Write Errors due to Permissions
- Illustrate the Use of Disk Cleanup for Insufficient Disk Space Errors
- Use Defragment and Optimize Drives for Slow Performance of Disks
- Use Component Services for Unable to Access Logical Drive Errors
- Explore Page / Swap / Scratch File or Partition Issues
- Describe Caching Issues
- Use the Check Disk (chkdsk) Tool
- Explore the System File Checker (SFC) Tool
- Use the Event Viewer
- Know about Monitoring Tools

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.